

Structural and Metrological Catalogue of Particles

States, Routes, Masses, and Widths

Oumar Haidara Fall · Rubén Ramos Balsa

Coauthorship without precedence of contribution

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Presentation

This catalogue brings together the structural inventory of 324 oriented routes and 855 metrological records of masses and widths. Its organisation makes each route's identity and fields consultable and the metrological observations locatable, without equating the number of documentary records with that of individualised predictions. It accompanies *Particles, Persistence and Derivation of the Mass Spectrum*, which develops the definitions and proofs constructing the particle, the atlas, and the mass operator. The complete structural and comparison data domain used by that development is expressed here.

The first section distinguishes 13 multisections, 56 route collections, and 23 inventory classes. Section 3 presents the 324 routes, preserving every field of their two concordant records. Section 4 brings together 471 masses and 384 widths from the fixed metrological edition, with their identifiers, statuses, units, and sources. Central values, intervals, bounds, and widths retain their own meanings. An observation's presence in the inventory makes it locatable for comparison; its physical identification requires the reader and chart specified in the article.

Each route is presented as a complete record, with aligned fields and unabridged sequences. The linked index and bookmarks locate each cell and orientation. A technical appendix in this same volume preserves provenance hashes, ordinals from both sources and their original codes. Metrological tables group identity, species, value, uncertainty and documentary status.

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1 Structural length, action, and gravitational response

The catalogue brings together 324 routes and 855 metrological records. Their interpretation follows the construction of the same HMT–MD state, in the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\text{enr}} \longrightarrow \mathcal{C}_{\text{joint}}^{\text{disc}} \longrightarrow L_{\alpha} \longrightarrow (H\Lambda, R\Lambda) \longrightarrow G. \quad (\text{A.1})$$

The coordinates π , φ , e , and α_{HMT} are correlated outputs of the preceding typed nonadic image; they are not entered as conventional values to select this realization. For $N = 12 \cdot 4 \cdot 120 = 5760$, the marked length is

$$L_{\alpha} = \alpha_{\text{HMT}}^{16} \frac{N-1}{4N} L_{*} = \frac{5759}{23040} \alpha_{\text{HMT}}^{16} L_{*}. \quad (\text{A.2})$$

In this section we abbreviate $\hbar = \hbar_{\text{ret}}$ and $c = c_{\text{int}}$. On the nonzero positive fibre, the energy H , reduced action length Λ , and conjugate radial readout R satisfy the following two operator relations. We also define $M := H/c^2$ and declare the physical radial identification $R := GM/c^2$. Comparing it, on the invertible fibre, with the operator relation $R = L_{\alpha}^2 H/(\hbar c)$ gives

$$H\Lambda = \hbar c I, \quad R\Lambda = L_{\alpha}^2 I, \quad R = \frac{L_{\alpha}^2}{\hbar c} H, \quad \boxed{G = \frac{c^3 L_{\alpha}^2}{\hbar}}. \quad (\text{A.3})$$

Metrological recognition of G is posterior and takes no part in selecting the state, the readout operator, or its coefficients.

1.1 Record M10 and preservation of the mass law

Code M10 in the documentary repertoire refers to the typed realization in (A.3). M10 neither postulates a gravitational particle nor turns the external record retained by the inventory under GRAVITATIONAL_OR_HYPOTHETICAL into an HMT output: that record remains external metrological data, separate from the $L_{\alpha} \mapsto G$ construction.

The energy scale induced by the same length is

$$Q_L := \frac{\hbar c}{L_{\alpha}}. \quad (\text{A.4})$$

If an equivalent chart replaces Q_L by Q'_L , the electronic coefficient co-transforms according to

$$b'_e = b_e \frac{Q_L}{Q'_L}, \quad Q'_L b'_e = Q_L b_e. \quad (\text{A.5})$$

The definitions of the clock energy and electronic factor give

$$E_{\text{clk}} = Q_L, \quad m_{\text{clk}} = \frac{E_{\text{clk}}}{c^2}, \quad b_e = \frac{E_e^{(0)}}{E_{\text{clk}}}, \quad m_{\text{clk}} b_e = \frac{E_e^{(0)}}{c^2}.$$

The exact cancellation of Q_L proves that the realization of G does not derive the masses anew, select routes, or modify the tables already constructed. The specialization $\Xi_{\Gamma} = 1$ in the compatibility theorem is equivalent to $\bar{\lambda}_{\Gamma} = R_{108} = L_{\alpha}$: it identifies a coincidence of lengths and neither asserts nor creates the existence of a Planck-mass species in the catalogue.

The 10^{-34} decade belongs exclusively to the absolute action–clock–mass branch, $\hbar = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S$; it is not reinserted into the relative exponent, the route character, or a species correction.

1.2 Constitutive channels and the oriented Catalan moment

Let $0 < q_+, q_- < 1$ be the labelled transport channels and P_+, P_- their orthogonal sheet projectors. For $s \in (0, 1)$, define

$$\mathcal{C}_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad \mathcal{D} = s \frac{d}{ds}, \quad f(s) = \frac{1-s^3}{1-s^4}. \quad (\text{A.6})$$

Local uniform convergence of the series and its derivatives permits two applications of $\mathcal{D}$. The resulting geometric series gives

$$\mathcal{D}^2 \mathcal{C}_2(s) = \frac{s}{1+s^2}, \quad f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 \mathcal{C}_2(s).$$

The identities $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$ determine

$$s_{\pm} = q_{\pm}^{30}, \quad r_{\pm} = f(s_{\pm}) = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}.$$

Thus $V = r_+ P_+ + r_- P_-$ combines both responses while retaining their labels. The $\mathcal{C}_2$ series converges absolutely at $s = 1$; its value is G_{Cat} , distinguished from the gravitational coefficient G .

The full function is recovered from f by

$$\mathcal{C}_2(s) = \int_0^s \log\left(\frac{s}{t}\right) \frac{1+t}{1+t+t^2} f(t) dt.$$

Indeed, $(1+t)f(t)/(1+t+t^2) = 1/(1+t^2)$; applying $\mathcal{D}$ twice gives $s/(1+s^2)$. Homogeneous solutions have the form $a \log s + b$, and $\mathcal{C}_2(s) \rightarrow 0$ as $s \downarrow 0$ forces both coefficients to vanish. The response, its regular recovery, and the meaning of the Catalan coordinate used in the catalogue entries are thereby explicit.

2 Catalogues of multisections, collections and classes

This appendix organizes three distinct levels of the internal inventory. The 13 multisections group atlas cells; the 56 collections classify internal routes; the 23 classes document their articulation with physical sectors and representations. These cardinalities do not count the same object and cannot be added to obtain a number of particles. The tables accompany the definitions and proofs of the article: their purpose is to locate each record, its incidences and its readers, not to replace a proof with a census.

The documentary order of the rows, their identifiers and all repetitions appearing in the sources are preserved. Literal identifiers allow the tables to be checked against the JSON files. A dash denotes an empty source cell; it does not represent zero. The files also preserve every field and digit, the original statuses and verifiable provenance. Sections 3 and 4 of this catalogue express the 324 oriented routes and the 855 metrological records, respectively.

2.1 The 13 multisections of the atlas

For a multisection M , let $\mathcal{C}(M)$ be its set of cells and let $\gamma_{B_1}(c)$ be the primary route preserved by the atlas in cell c . In this structural appendix, the fiber is defined by

$$B_1(M) := \text{span}\{e_{\gamma_{B_1}(c)} : c \in \mathcal{C}(M)\} \subseteq \ell^2(\mathcal{R}_{\text{adm}}).$$

The primary routes of distinct cells determine distinct basis vectors; consequently, $\text{rank } B_1(M) = \dim B_1(M) = \#\mathcal{C}(M)$. This definition fixes the object to which the table's rank refers; the collections column counts the route collections present, not physical species. The integer spectra of the readers K_D and K_Ω are preserved separately. In the literal notation, $\times$ indicates multiplicity. The 13 rows have the documentary status of an exact diagonal operator on the fiber B_1 . Their publication as a physical scalar requires the declared transport, except for the rank-one selection provided for in the record itself. This statement reproduces the recorded scope and does not turn every spectrum into a unique mass.

Table 1: The 13 multisections, without row selection.

Multisection	Rank and incidence	Classification and depth	Integer readers
charged_lepton_G0	<i>Rank:</i> 1 <i>Cells (identifiers):</i> 41 <i>Collections:</i> 1	<i>Collection:</i> charged_lepton <i>Depth:</i> G0_center <i>Color:</i> none	<i>K D:</i> (80,54,6) $\times$ 1 <i>K Ω:</i> (80,54,6) $\times$ 1
charged_lepton_G2	<i>Rank:</i> 8 <i>Cells (identifiers):</i> 21, 23,25,39,43,57,59,61 <i>Collections:</i> 8	<i>Collection:</i> charged_lepton <i>Depth:</i> G2 <i>Color:</i> none	<i>K D:</i> (0,24,0) $\times$ 1; (40,78,27) $\times$ 2; (80, 54,7) $\times$ 1; (80,66,2) $\times$ 1; (80,66,3) $\times$ 1; (100,66,0) $\times$ 1; (100,66,2) $\times$ 1 <i>K Ω:</i> (80,54,6) $\times$ 8
charged_lepton_G4	<i>Rank:</i> 16 <i>Cells (identifiers):</i> 1,3, 5,7,9,19,27,37,45,55,63, 73,75,77,79,81 <i>Collections:</i> 16	<i>Collection:</i> charged_lepton <i>Depth:</i> G4 <i>Color:</i> none	<i>K D:</i> (0,24,0) $\times$ 1; (40,24,2) $\times$ 2; (40,54, 6) $\times$ 2; (40,84,30) $\times$ 2; (60,78,25) $\times$ 3; (80,66,2) $\times$ 2; (80,66,3) $\times$ 3; (80,66,4) $\times$ 1 <i>K Ω:</i> (24,54,2) $\times$ 1; (56,54,5) $\times$ 4; (80, 54,6) $\times$ 11
neutrino_G1	<i>Rank:</i> 2 <i>Cells (identifiers):</i> 40, 42 <i>Collections:</i> 2	<i>Collection:</i> neutrino <i>Depth:</i> G1 <i>Color:</i> none	<i>K D:</i> (40,24,2) $\times$ 1; (40,78,27) $\times$ 1 <i>K Ω:</i> (40,54,4) $\times$ 1; (80,54,6) $\times$ 1
neutrino_G2	<i>Rank:</i> 4 <i>Cells (identifiers):</i> 22, 24,58,60 <i>Collections:</i> 4	<i>Collection:</i> neutrino <i>Depth:</i> G2 <i>Color:</i> none	<i>K D:</i> (0,24,0) $\times$ 1; (40,84,30) $\times$ 1; (60, 78,25) $\times$ 1; (80,66,3) $\times$ 1 <i>K Ω:</i> (4,54,2) $\times$ 1; (80,54,6) $\times$ 3

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Continuation

Multisection	Rank and incidence	Classification and depth	Integer readers
neutrino_G3	<i>Rank:</i> 6 <i>Cells (identifiers):</i> 20, 26,38,44,56,62 <i>Collections:</i> 6	<i>Collection:</i> neutrino <i>Depth:</i> G3 <i>Color:</i> none	<i>K D:</i> (40,24,2)x2; (40,78,27)x1; (60,84,28)x1; (80,54,6)x1; (80,66,3)x1 <i>K Ω:</i> (36,54,4)x1; (56,54,5)x1; (80,54,6)x4
neutrino_G4	<i>Rank:</i> 8 <i>Cells (identifiers):</i> 2,4, 6,8,74,76,78,80 <i>Collections:</i> 8	<i>Collection:</i> neutrino <i>Depth:</i> G4 <i>Color:</i> none	<i>K D:</i> (0,24,0)x1; (40,24,2)x1; (40,78,27)x2; (40,84,30)x1; (80,54,7)x1; (80,66,2)x1; (100,66,0)x1 <i>K Ω:</i> (36,54,4)x1; (56,54,5)x1; (80,54,6)x6
quark_down_G1	<i>Rank:</i> 2 <i>Cells (identifiers):</i> 32, 50 <i>Collections:</i> 2	<i>Collection:</i> quark_down <i>Depth:</i> G1 <i>Color:</i> B G	<i>K D:</i> (40,24,2)x1; (40,78,27)x1 <i>K Ω:</i> (40,54,4)x1; (80,54,6)x1
quark_down_G2	<i>Rank:</i> 4 <i>Cells (identifiers):</i> 30, 34,48,52 <i>Collections:</i> 4	<i>Collection:</i> quark_down <i>Depth:</i> G2 <i>Color:</i> B G R	<i>K D:</i> (0,24,0)x1; (40,84,30)x1; (60,78,25)x1; (80,66,2)x1 <i>K Ω:</i> (4,54,2)x1; (80,54,6)x3
quark_down_G3	<i>Rank:</i> 6 <i>Cells (identifiers):</i> 12, 14,16,66,68,70 <i>Collections:</i> 6	<i>Collection:</i> quark_down <i>Depth:</i> G3 <i>Color:</i> B G R	<i>K D:</i> (40,24,2)x2; (40,78,27)x1; (60,84,28)x1; (80,54,6)x1; (80,66,4)x1 <i>K Ω:</i> (36,54,4)x1; (80,54,6)x5
quark_down_G4	<i>Rank:</i> 8 <i>Cells (identifiers):</i> 10, 18,28,36,46,54,64,72 <i>Collections:</i> 8	<i>Collection:</i> quark_down <i>Depth:</i> G4 <i>Color:</i> B G R	<i>K D:</i> (0,24,0)x1; (40,24,2)x1; (40,78,27)x2; (40,84,30)x1; (80,54,7)x1; (80,66,3)x1; (100,66,2)x1 <i>K Ω:</i> (36,54,4)x1; (80,54,6)x7
quark_up_G1	<i>Rank:</i> 4 <i>Cells (identifiers):</i> 31, 33,49,51 <i>Collections:</i> 3	<i>Collection:</i> quark_up <i>Depth:</i> G1 <i>Color:</i> B G R	<i>K D:</i> (40,54,6)x1; (60,78,25)x1; (80,66,2)x1; (80,66,3)x1 <i>K Ω:</i> (80,54,6)x4
quark_up_G3	<i>Rank:</i> 12 <i>Cells (identifiers):</i> 11, 13,15,17,29,35,47,53, 65,67,69,71 <i>Collections:</i> 12	<i>Collection:</i> quark_up <i>Depth:</i> G3 <i>Color:</i> B G R	<i>K D:</i> (40,24,2)x2; (40,78,27)x2; (60,84,28)x1; (80,54,6)x3; (80,66,3)x1; (80,66,4)x1; (100,66,0)x1; (100,66,2)x1 <i>K Ω:</i> (56,54,5)x3; (80,54,6)x9

2.2 The 56 route collections

The signature identifying a collection is preserved literally, with its signs and separators. The cells and multisections make explicit that a route collection is not, by definition, equivalent to a physical species. The joint profiles count the combinations recorded within the collection. The decimal spectra of the character and the complete signature spectra remain intact in `catalogo_familias.json`; the incidence and integer readers published here provide a means of navigating that information.

Table 2: The 56 internal route collections.

Collection and signature	Incidence	K_D	K_Ω
1 24 0 12 0 -12 +++++-----	<i>Number of cells:</i> 5 <i>Cells (identifiers):</i> 21,31,41,51,81 <i>Multisections:</i> charged_lepton_G0 charged_lepton_G2 charged_lepton_G4 quark_up_G1 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> R none <i>Joint profiles:</i> 5	(0,24,0)x2; (40,54,6)x1; (60,78,25)x1; (80,54,6)x1	(24,54,2)x1; (80,54,6)x4
2 24 0 4 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 11,61 <i>Multisections:</i> charged_lepton_G2 quark_up_G3 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> R none <i>Joint profiles:</i> 2	(60,84,28)x1; (80,54,7)x1	(80,54,6)x2
3 24 0 8 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 1,71 <i>Multisections:</i> charged_lepton_G4 quark_up_G3 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> R none <i>Joint profiles:</i> 2	(60,78,25)x1; (80,54,6)x1	(80,54,6)x2
4 28 -6 -2 -2 -8 0+----0+----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 53 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> G <i>Joint profiles:</i> 1	(80,66,3)x1	(56,54,5)x1
5 28 -6 -4 -2 -8 0+----0+----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 20 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(80,66,3)x1	(56,54,5)x1
6 28 -6 0 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 30,70 <i>Multisections:</i> quark_down_G2 quark_ down_G3 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 2	(40,24,2)x1; (80,66,2)x1	(80,54,6)x2
7 28 -6 10 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 33,64 <i>Multisections:</i> quark_down_G4 quark_ up_G1 <i>Collections:</i> quark_down quark_up <i>Color:</i> G <i>Joint profiles:</i> 2	(40,24,2)x1; (80,66,2)x1	(36,54,4)x1; (80,54,6)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
8 28 -6 12 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 10 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(100,66,2)x1	(80,54,6)x1
9 28 -6 2 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 3 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
10 28 -6 4 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 50 <i>Multisections:</i> quark_down_G1 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
11 28 -6 4 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 9,40 <i>Multisections:</i> charged_lepton_G4 neutrino_G1 <i>Collections:</i> charged_lepton neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,24,2)x1; (80,66,2)x1	(40,54,4)x1; (80,54,6)x1
12 28 -6 4 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 80 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
13 28 -6 6 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 74 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
14 28 -6 6 0 -12 +++++-----	<i>Number of cells:</i> 3 <i>Cells (identifiers):</i> 13,43,63 <i>Multisections:</i> charged_lepton_G2 charged_lepton_G4 quark_up_G3 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> G none <i>Joint profiles:</i> 3	(40,24,2)x1; (80,66,2)x1; (100,66,2)x1	(80,54,6)x3
15 28 -6 6 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 23 <i>Multisections:</i> charged_lepton_G2 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
16 28 6 8 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 60 <i>Multisections:</i> neutrino_G2 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
17 28 6 -10 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 19 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
18 28 6 -2 0 -12 +++++-----	<i>Number of cells:</i> 5 <i>Cells (identifiers):</i> 18,29,49,59,79 <i>Multisections:</i> charged_lepton_G2 charged_lepton_G4 quark_down_G4 quark_up_G1 quark_up_G3 <i>Collections:</i> charged_lepton quark_ down quark_up <i>Color:</i> B none <i>Joint profiles:</i> 4	(40,24,2)x1; (40,78,27)x1; (80,54,6)x5 (80,66,3)x2; (100,66,0)x1	
19 28 6 -4 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 2,42 <i>Multisections:</i> neutrino_G1 neutrino_ G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,78,27)x1; (100,66,0) x1	(80,54,6)x2
20 28 6 -6 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 69 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> B <i>Joint profiles:</i> 1	(80,66,4)x1	(80,54,6)x1
21 28 6 -8 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 52 <i>Multisections:</i> quark_down_G2 <i>Collections:</i> quark_down <i>Color:</i> B <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
22 28 6 0 0 -12 +++++-----	<i>Number of cells:</i> 4 <i>Cells (identifiers):</i> 22,32,72,73 <i>Multisections:</i> charged_lepton_G4 neutrino_G2 quark_down_G1 quark_ down_G4 <i>Collections:</i> charged_lepton neutrino quark_down <i>Color:</i> B none <i>Joint profiles:</i> 3	(40,24,2)x1; (40,78,27)x1; (40,54,4)x1; (80,54,6)x3 (80,66,3)x2	

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Continuation

Collection and signature	Incidence	K_D	K_Ω
23 28 6 2 0 -12 +++++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 8,39 <i>Multisections:</i> charged_lepton_G2 neutrino_G4 <i>Collections:</i> charged_lepton neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,24,2)x1; (40,78,27)x1	(36,54,4)x1; (80,54,6)x1
24 28 6 4 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 12 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> B <i>Joint profiles:</i> 1	(80,66,4)x1	(80,54,6)x1
25 28 6 8 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 62 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(80,54,6)x1
26 30 -6 -4 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 4 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(80,54,7)x1	(56,54,5)x1
27 30 -6 -8 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 55 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(60,78,25)x1	(80,54,6)x1
28 30 -6 0 -2 -8 ++0---++0---	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 24,34 <i>Multisections:</i> neutrino_G2 quark_ down_G2 <i>Collections:</i> neutrino quark_down <i>Color:</i> R none <i>Joint profiles:</i> 2	(0,24,0)x1; (60,78,25)x1	(4,54,2)x1; (80,54,6)x1
29 30 -6 0 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 75 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,54,6)x1	(56,54,5)x1
30 30 -6 0 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 14 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,6)x1	(80,54,6)x1

Continued on the next page

Continuation

Collection and signature	Incidence	K_D	K_Ω
31 30 -6 0 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 54 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(0,24,0)x1	(80,54,6)x1
32 30 -6 4 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 65 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,6)x1	(80,54,6)x1
33 30 -6 8 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 44 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(60,84,28)x1	(80,54,6)x1
34 30 6 -4 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 68 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(60,84,28)x1	(80,54,6)x1
35 30 6 -8 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 17 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,6)x1	(80,54,6)x1
36 30 6 0 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 38,58 <i>Multisections:</i> neutrino_G2 neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(60,78,25)x1; (80,54,6)x1	(80,54,6)x2
37 30 6 0 0 -8 +++0-+++0-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 78 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(0,24,0)x1	(80,54,6)x1
38 30 6 0 2 -8 +++0-+++0-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 27 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,54,6)x1	(56,54,5)x1

Continued on the next page

Continuation

Collection and signature	Incidence	K_D	K_Ω
39 30 6 0 2 -8 +++0---++0--	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 48 <i>Multisections:</i> quark_down_G2 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(0,24,0)x1	(4,54,2)x1
40 30 6 4 0 -12 +++-----++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 7 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(60,78,25)x1	(80,54,6)x1
41 30 6 8 0 -12 ++++-----++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 28 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,7)x1	(80,54,6)x1
42 34 0 -12 -2 -8 0+----0+----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 45 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(80,66,4)x1	(56,54,5)x1
43 34 0 -12 0 -12 ++++-----++---	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 57,76 <i>Multisections:</i> charged_lepton_G2 neutrino_G4 <i>Collections:</i> charged_lepton neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,84,30)x1; (80,66,2)x1	(80,54,6)x2
44 34 0 -16 0 -8 ++++-0++++-0-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 37 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(80,54,6)x1
45 34 0 -16 0 -8 +0+----+0+----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 15 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> B <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
46 34 0 -20 -2 -8 0+----0+----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 35 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> B <i>Joint profiles:</i> 1	(100,66,0)x1	(56,54,5)x1

Continued on the next page

Continuation

Collection and signature	Incidence	K_D	K_Ω
47 34 0 -4 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 56 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(36,54,4)x1
48 34 0 -4 0 -12 +++-----++---	<i>Number of cells:</i> 3 <i>Cells (identifiers):</i> 6,25,46 <i>Multisections:</i> charged_lepton_G2 neutrino_G4 quark_down_G4 <i>Collections:</i> charged_lepton neutrino quark_down <i>Color:</i> B none <i>Joint profiles:</i> 2	(80,66,2)x1; (80,66,3)x2	(80,54,6)x3
49 34 0 -4 2 -8 ++++-0++++-0	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 67 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> G <i>Joint profiles:</i> 1	(100,66,2)x1	(56,54,5)x1
50 34 0 -8 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 66 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> B <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
51 34 0 -8 -2 -8 0+---0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 77 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(80,66,3)x1	(56,54,5)x1
52 34 0 -8 0 -12 ++++-----++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 36 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
53 34 0 -8 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 5 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(80,54,6)x1
54 34 0 0 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 26 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1

Continued on the next page

Continuation

Collection and signature	Incidence	K_D	K_Ω
55 34 0 0 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 47 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> G <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
56 34 0 0 2 -8 +++0--++0--	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 16 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(40,24,2)x1	(36,54,4)x1

2.3 The 23 documented classes

These rows bring together public sectors, residences and representation. The status key refers to the adjacent legend and preserves the status of the source. In particular, an exact internal operator, a conditional evaluation and a closed scalar realization remain distinct assertions. The table neither updates their status nor turns a conditional record into a prediction.

Table 3: Original class statuses.

Key	Literal status	Documentary reading
C01	CLOSED_ELECTRON_TWO_WAY	Electron: documented bidirectional closure.
C02	CLOSED_NULL_CHART	Null chart closed at this stage.
C03	NOT_A_UNIVERSAL_MASS_SCALAR	The class is not published as a universal mass scalar.
C04	OPEN_FLAVOR_COLOR_SCALARIZATION	Flavor and color scalarization open at this stage.
C05	OPEN_MIXING_AND_SCALE	Mixing and scale open at this stage.
C06	OPEN_NAMED_STATE_SCALARIZATION	Scalarization of the individually identified state open at this stage.
C07	OPEN_PHYSICAL_SCALARIZATION	Physical scalarization open at this stage.
C08	OPEN_ROUTE_AND_SCALARIZATION	Route and scalarization open at this stage.

Table 4: The 23 catalogue classes, in their documentary order.

Class	Representatives and residence	Incidence and representation	Status
SM-LC-1 lepton_cargado	<i>Representatives:</i> e^- / e^+ <i>Residence:</i> center (5,5) and antisection	<i>Selected cells:</i> 1 <i>Multisections:</i> charged_lepton_G0 <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C01
SM-LC-2 lepton_cargado	<i>Representatives:</i> μ^- / μ^+ <i>Residence:</i> charged-lepton multisection	<i>Selected cells:</i> 8 <i>Multisections:</i> charged_lepton_G2 <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C07

Continued on the next page

<i>Continuation</i>			
Class	Representatives and residence	Incidence and representation	Status
SM-LC-3 lepton_cargado	<i>Representatives:</i> τ^- / τ^+ <i>Residence:</i> charged-lepton multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> charged_lepton_G4 <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C07
SM-NU-1 neutrino	<i>Representatives:</i> $\nu_e / \text{anti-}\nu_e$ <i>Residence:</i> neutrino multisections	<i>Selected cells:</i> 20 <i>Multisections:</i> neutrino_G1 neutrino_G2 neutrino_G3 neutrino_G4 <i>Charge:</i> 0 <i>Representation:</i> colorless	C05
SM-NU-2 neutrino	<i>Representatives:</i> $\nu_\mu / \text{anti-}\nu_\mu$ <i>Residence:</i> neutrino multisections	<i>Selected cells:</i> 20 <i>Multisections:</i> neutrino_G1 neutrino_G2 neutrino_G3 neutrino_G4 <i>Charge:</i> 0 <i>Representation:</i> colorless	C05
SM-NU-3 neutrino	<i>Representatives:</i> $\nu_\tau / \text{anti-}\nu_\tau$ <i>Residence:</i> neutrino multisections	<i>Selected cells:</i> 20 <i>Multisections:</i> neutrino_G1 neutrino_G2 neutrino_G3 neutrino_G4 <i>Charge:</i> 0 <i>Representation:</i> colorless	C05
SM-Q-U1 quark	<i>Representatives:</i> $u / \text{anti-}u$ <i>Residence:</i> up-type multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> quark_up_G1 quark_up_G3 <i>Charge:</i> $\pm 2/3 e$ <i>Representation:</i> $R, G, B = Z/3$ orbit	C04
SM-Q-D1 quark	<i>Representatives:</i> $d / \text{anti-}d$ <i>Residence:</i> down-type multisection	<i>Selected cells:</i> 20 <i>Multisections:</i> quark_down_G1 quark_down_G2 quark_down_G3 quark_down_G4 <i>Charge:</i> $\pm 1/3 e$ <i>Representation:</i> $R, G, B = Z/3$ orbit	C04
SM-Q-U2 quark	<i>Representatives:</i> $c / \text{anti-}c$ <i>Residence:</i> up-type multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> quark_up_G1 quark_up_G3 <i>Charge:</i> $\pm 2/3 e$ <i>Representation:</i> $R, G, B = Z/3$ orbit	C04
SM-Q-D2 quark	<i>Representatives:</i> $s / \text{anti-}s$ <i>Residence:</i> down-type multisection	<i>Selected cells:</i> 20 <i>Multisections:</i> quark_down_G1 quark_down_G2 quark_down_G3 quark_down_G4 <i>Charge:</i> $\pm 1/3 e$ <i>Representation:</i> $R, G, B = Z/3$ orbit	C04
SM-Q-U3 quark	<i>Representatives:</i> $t / \text{anti-}t$ <i>Residence:</i> up-type multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> quark_up_G1 quark_up_G3 <i>Charge:</i> $\pm 2/3 e$ <i>Representation:</i> $R, G, B = Z/3$ orbit	C04
SM-Q-D3 quark	<i>Representatives:</i> $b / \text{anti-}b$ <i>Residence:</i> down-type multisection	<i>Selected cells:</i> 20 <i>Multisections:</i> quark_down_G1 quark_down_G2 quark_down_G3 quark_down_G4 <i>Charge:</i> $\pm 1/3 e$ <i>Representation:</i> $R, G, B = Z/3$ orbit	C04
SM-GA-EM boson_gauge	<i>Representatives:</i> γ <i>Residence:</i> null gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> colorless	C02
SM-GA-GL boson_gauge	<i>Representatives:</i> g <i>Residence:</i> color gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> gauge octet	C02

Continued on the next page

<i>Continuation</i>			
Class	Representatives and residence	Incidence and representation	Status
SM-GA-W boson_gauge	<i>Representatives:</i> W^+ / W^- <i>Residence:</i> massive gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C08
SM-GA-Z boson_gauge	<i>Representatives:</i> Z^0 <i>Residence:</i> massive gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> colorless	C08
SM-H-1 escalar	<i>Representatives:</i> H <i>Residence:</i> scalar sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> colorless	C08
COMP-MESON meson	<i>Representatives:</i> $\pi^0, \pi^\pm, K^0, K^\pm, \eta, \rho, \varphi, J/\psi, Y, \dots$ <i>Residence:</i> product of section and antisection	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the composite state <i>Representation:</i> singlet or other representations	C06
COMP-BARYON barion	<i>Representatives:</i> p, n and baryonic collections <i>Residence:</i> product of three sections	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the composite state <i>Representation:</i> physical singlet	C06
TOPO-FIB anyon_no_abeliano	<i>Representatives:</i> $1, \tau$ <i>Residence:</i> topological sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> defined by the topological codomain <i>Representation:</i> not applicable	C03
TOPO-ISING sector_majorana	<i>Representatives:</i> $1, \psi, \sigma$ <i>Residence:</i> topological sector / zero	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the realization <i>Representation:</i> colorless	C03
TOPO-Z6 anyones_abelianos	<i>Representatives:</i> boson, fermion and four braided sectors <i>Residence:</i> topological sector $Z/6$	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> $Z/6$ defined by braiding <i>Representation:</i> colorless	C03
VIRT-1 virtual	<i>Representatives:</i> virtual sections <i>Residence:</i> TPK inverse system	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the prolongation sector <i>Representation:</i> according to sector	C03

3 Oriented routes: reading guide and records

Each route occupies one page and retains its identity within the 81 atlas cells. The four orientations of each cell are consulted separately. The record proceeds from position and classification to returns, memory, signature and the two readers; the complete sequences show the recorded evolution.

Class, orbit, cell and kinematic-state returns are distinct quantities. Phase repetition and memory change are likewise distinguished. The readers K_D and K_Q are displayed together for comparison, without identifying them when their records differ. Decimal numbers retain every digit in the sources.

A dash denotes an empty field, not zero. Classification descriptions are rendered in English; the technical key preserves the original codes. Provenance hashes and the two source ordinals appear in the linked appendix entry for the same route, within this volume. The two records are joined by exact identity, preserving every field.

Documentary status. The routes are internal and independent of an external target mass. Records are joined by exact key, without consulting external masses. The preserved tower status states that route memory is available and that the η transducer requires the corresponding certificate. These three statements apply to every record; Pareto persistence, which can vary, is recorded individually.

Route index

Route	Page	Route	Page	Route	Page
r1c1_h-1v-1	20	r1c1_h-1v+1	21	r1c1_h+1v-1	22
r1c1_h+1v+1	23	r1c2_h-1v-1	24	r1c2_h-1v+1	25
r1c2_h+1v-1	26	r1c2_h+1v+1	27	r1c3_h-1v-1	28
r1c3_h-1v+1	29	r1c3_h+1v-1	30	r1c3_h+1v+1	31
r1c4_h-1v-1	32	r1c4_h-1v+1	33	r1c4_h+1v-1	34
r1c4_h+1v+1	35	r1c5_h-1v-1	36	r1c5_h-1v+1	37
r1c5_h+1v-1	38	r1c5_h+1v+1	39	r1c6_h-1v-1	40
r1c6_h-1v+1	41	r1c6_h+1v-1	42	r1c6_h+1v+1	43
r1c7_h-1v-1	44	r1c7_h-1v+1	45	r1c7_h+1v-1	46
r1c7_h+1v+1	47	r1c8_h-1v-1	48	r1c8_h-1v+1	49
r1c8_h+1v-1	50	r1c8_h+1v+1	51	r1c9_h-1v-1	52
r1c9_h-1v+1	53	r1c9_h+1v-1	54	r1c9_h+1v+1	55
r2c1_h-1v-1	56	r2c1_h-1v+1	57	r2c1_h+1v-1	58
r2c1_h+1v+1	59	r2c2_h-1v-1	60	r2c2_h-1v+1	61
r2c2_h+1v-1	62	r2c2_h+1v+1	63	r2c3_h-1v-1	64
r2c3_h-1v+1	65	r2c3_h+1v-1	66	r2c3_h+1v+1	67
r2c4_h-1v-1	68	r2c4_h-1v+1	69	r2c4_h+1v-1	70
r2c4_h+1v+1	71	r2c5_h-1v-1	72	r2c5_h-1v+1	73
r2c5_h+1v-1	74	r2c5_h+1v+1	75	r2c6_h-1v-1	76
r2c6_h-1v+1	77	r2c6_h+1v-1	78	r2c6_h+1v+1	79
r2c7_h-1v-1	80	r2c7_h-1v+1	81	r2c7_h+1v-1	82
r2c7_h+1v+1	83	r2c8_h-1v-1	84	r2c8_h-1v+1	85
r2c8_h+1v-1	86	r2c8_h+1v+1	87	r2c9_h-1v-1	88
r2c9_h-1v+1	89	r2c9_h+1v-1	90	r2c9_h+1v+1	91
r3c1_h-1v-1	92	r3c1_h-1v+1	93	r3c1_h+1v-1	94
r3c1_h+1v+1	95	r3c2_h-1v-1	96	r3c2_h-1v+1	97
r3c2_h+1v-1	98	r3c2_h+1v+1	99	r3c3_h-1v-1	100
r3c3_h-1v+1	101	r3c3_h+1v-1	102	r3c3_h+1v+1	103
r3c4_h-1v-1	104	r3c4_h-1v+1	105	r3c4_h+1v-1	106
r3c4_h+1v+1	107	r3c5_h-1v-1	108	r3c5_h-1v+1	109
r3c5_h+1v-1	110	r3c5_h+1v+1	111	r3c6_h-1v-1	112
r3c6_h-1v+1	113	r3c6_h+1v-1	114	r3c6_h+1v+1	115
r3c7_h-1v-1	116	r3c7_h-1v+1	117	r3c7_h+1v-1	118
r3c7_h+1v+1	119	r3c8_h-1v-1	120	r3c8_h-1v+1	121
r3c8_h+1v-1	122	r3c8_h+1v+1	123	r3c9_h-1v-1	124
r3c9_h-1v+1	125	r3c9_h+1v-1	126	r3c9_h+1v+1	127
r4c1_h-1v-1	128	r4c1_h-1v+1	129	r4c1_h+1v-1	130
r4c1_h+1v+1	131	r4c2_h-1v-1	132	r4c2_h-1v+1	133
r4c2_h+1v-1	134	r4c2_h+1v+1	135	r4c3_h-1v-1	136
r4c3_h-1v+1	137	r4c3_h+1v-1	138	r4c3_h+1v+1	139
r4c4_h-1v-1	140	r4c4_h-1v+1	141	r4c4_h+1v-1	142
r4c4_h+1v+1	143	r4c5_h-1v-1	144	r4c5_h-1v+1	145
r4c5_h+1v-1	146	r4c5_h+1v+1	147	r4c6_h-1v-1	148
r4c6_h-1v+1	149	r4c6_h+1v-1	150	r4c6_h+1v+1	151
r4c7_h-1v-1	152	r4c7_h-1v+1	153	r4c7_h+1v-1	154
r4c7_h+1v+1	155	r4c8_h-1v-1	156	r4c8_h-1v+1	157
r4c8_h+1v-1	158	r4c8_h+1v+1	159	r4c9_h-1v-1	160
r4c9_h-1v+1	161	r4c9_h+1v-1	162	r4c9_h+1v+1	163
r5c1_h-1v-1	164	r5c1_h-1v+1	165	r5c1_h+1v-1	166
r5c1_h+1v+1	167	r5c2_h-1v-1	168	r5c2_h-1v+1	169
r5c2_h+1v-1	170	r5c2_h+1v+1	171	r5c3_h-1v-1	172
r5c3_h-1v+1	173	r5c3_h+1v-1	174	r5c3_h+1v+1	175
r5c4_h-1v-1	176	r5c4_h-1v+1	177	r5c4_h+1v-1	178
r5c4_h+1v+1	179	r5c5_h-1v-1	180	r5c5_h-1v+1	181
r5c5_h+1v-1	182	r5c5_h+1v+1	183	r5c6_h-1v-1	184
r5c6_h-1v+1	185	r5c6_h+1v-1	186	r5c6_h+1v+1	187
r5c7_h-1v-1	188	r5c7_h-1v+1	189	r5c7_h+1v-1	190

Route	Page	Route	Page	Route	Page
r5c7_h+1v+1	191	r5c8_h-1v-1	192	r5c8_h-1v+1	193
r5c8_h+1v-1	194	r5c8_h+1v+1	195	r5c9_h-1v-1	196
r5c9_h-1v+1	197	r5c9_h+1v-1	198	r5c9_h+1v+1	199
r6c1_h-1v-1	200	r6c1_h-1v+1	201	r6c1_h+1v-1	202
r6c1_h+1v+1	203	r6c2_h-1v-1	204	r6c2_h-1v+1	205
r6c2_h+1v-1	206	r6c2_h+1v+1	207	r6c3_h-1v-1	208
r6c3_h-1v+1	209	r6c3_h+1v-1	210	r6c3_h+1v+1	211
r6c4_h-1v-1	212	r6c4_h-1v+1	213	r6c4_h+1v-1	214
r6c4_h+1v+1	215	r6c5_h-1v-1	216	r6c5_h-1v+1	217
r6c5_h+1v-1	218	r6c5_h+1v+1	219	r6c6_h-1v-1	220
r6c6_h-1v+1	221	r6c6_h+1v-1	222	r6c6_h+1v+1	223
r6c7_h-1v-1	224	r6c7_h-1v+1	225	r6c7_h+1v-1	226
r6c7_h+1v+1	227	r6c8_h-1v-1	228	r6c8_h-1v+1	229
r6c8_h+1v-1	230	r6c8_h+1v+1	231	r6c9_h-1v-1	232
r6c9_h-1v+1	233	r6c9_h+1v-1	234	r6c9_h+1v+1	235
r7c1_h-1v-1	236	r7c1_h-1v+1	237	r7c1_h+1v-1	238
r7c1_h+1v+1	239	r7c2_h-1v-1	240	r7c2_h-1v+1	241
r7c2_h+1v-1	242	r7c2_h+1v+1	243	r7c3_h-1v-1	244
r7c3_h-1v+1	245	r7c3_h+1v-1	246	r7c3_h+1v+1	247
r7c4_h-1v-1	248	r7c4_h-1v+1	249	r7c4_h+1v-1	250
r7c4_h+1v+1	251	r7c5_h-1v-1	252	r7c5_h-1v+1	253
r7c5_h+1v-1	254	r7c5_h+1v+1	255	r7c6_h-1v-1	256
r7c6_h-1v+1	257	r7c6_h+1v-1	258	r7c6_h+1v+1	259
r7c7_h-1v-1	260	r7c7_h-1v+1	261	r7c7_h+1v-1	262
r7c7_h+1v+1	263	r7c8_h-1v-1	264	r7c8_h-1v+1	265
r7c8_h+1v-1	266	r7c8_h+1v+1	267	r7c9_h-1v-1	268
r7c9_h-1v+1	269	r7c9_h+1v-1	270	r7c9_h+1v+1	271
r8c1_h-1v-1	272	r8c1_h-1v+1	273	r8c1_h+1v-1	274
r8c1_h+1v+1	275	r8c2_h-1v-1	276	r8c2_h-1v+1	277
r8c2_h+1v-1	278	r8c2_h+1v+1	279	r8c3_h-1v-1	280
r8c3_h-1v+1	281	r8c3_h+1v-1	282	r8c3_h+1v+1	283
r8c4_h-1v-1	284	r8c4_h-1v+1	285	r8c4_h+1v-1	286
r8c4_h+1v+1	287	r8c5_h-1v-1	288	r8c5_h-1v+1	289
r8c5_h+1v-1	290	r8c5_h+1v+1	291	r8c6_h-1v-1	292
r8c6_h-1v+1	293	r8c6_h+1v-1	294	r8c6_h+1v+1	295
r8c7_h-1v-1	296	r8c7_h-1v+1	297	r8c7_h+1v-1	298
r8c7_h+1v+1	299	r8c8_h-1v-1	300	r8c8_h-1v+1	301
r8c8_h+1v-1	302	r8c8_h+1v+1	303	r8c9_h-1v-1	304
r8c9_h-1v+1	305	r8c9_h+1v-1	306	r8c9_h+1v+1	307
r9c1_h-1v-1	308	r9c1_h-1v+1	309	r9c1_h+1v-1	310
r9c1_h+1v+1	311	r9c2_h-1v-1	312	r9c2_h-1v+1	313
r9c2_h+1v-1	314	r9c2_h+1v+1	315	r9c3_h-1v-1	316
r9c3_h-1v+1	317	r9c3_h+1v-1	318	r9c3_h+1v+1	319
r9c4_h-1v-1	320	r9c4_h-1v+1	321	r9c4_h+1v-1	322
r9c4_h+1v+1	323	r9c5_h-1v-1	324	r9c5_h-1v+1	325
r9c5_h+1v-1	326	r9c5_h+1v+1	327	r9c6_h-1v-1	328
r9c6_h-1v+1	329	r9c6_h+1v-1	330	r9c6_h+1v+1	331
r9c7_h-1v-1	332	r9c7_h-1v+1	333	r9c7_h+1v-1	334
r9c7_h+1v+1	335	r9c8_h-1v-1	336	r9c8_h-1v+1	337
r9c8_h+1v-1	338	r9c8_h+1v+1	339	r9c9_h-1v-1	340
r9c9_h-1v+1	341	r9c9_h+1v-1	342	r9c9_h+1v+1	343

Route 001

r1c1_h-1v-1

Cell 1 (1, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,1;9,9
Classification signature	24 0 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 001/324

Route 002

r1c1_h-1v+1

Cell 1 (1, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,1;9,9
Classification signature	24 0 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 002/324

Route 003

r1c1_h+1v-1

Cell 1 (1, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,1;9,9
Classification signature	24 0 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 003/324

Route 004

r1c1_h+1v+1

Cell 1 (1, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,1;9,9
Classification signature	24 0 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 004/324

Route 005

r1c2_h-1v-1

Cell 2 (1, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,2;9,8
Classification signature	28 6 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 005/324

Route 006

r1c2_h-1v+1

Cell 2 (1, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,2;9,8
Classification signature	28 6 -4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (48,54,5)
κ_D	39.82508593118257305617326
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 006/324

Route 007

r1c2_h+1v-1

Cell 2 (1, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,2;9,8
Classification signature	28 6 -4 0 -12 +++----++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (48,54,5)
κ_D	39.82508593118257305617326
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 007/324

Route 008

r1c2_h+1v+1

Cell 2 (1, 2) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,2;9,8
Classification signature	28 6 -4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 008/324

Route 009

r1c3_h-1v-1

Cell 3 (1, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,3;9,7
Classification signature	28 -6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 009/324

Route 010

r1c3_h-1v+1

Cell 3 (1, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,3;9,7
Classification signature	28 -6 2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 010/324

Route 011

r1c3_h+1v-1

Cell 3 (1, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,3;9,7
Classification signature	28 -6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 011/324

Route 012

r1c3_h+1v+1

Cell 3 (1, 3) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,3;9,7
Classification signature	28 -6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 012/324

Route 013

r1c4_h-1v-1

Cell 4 (1, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,4;9,6
Classification signature	30 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	30, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 30 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,54,7) / (56,54,5)
κ_D	79.60672133621751972652496
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	26; 6.339850002884624
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 013/324

Route 014

r1c4_h-1v+1

Cell 4 (1, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,4;9,6
Classification signature	30 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 014/324

Route 015

r1c4_h+1v-1

Cell 4 (1, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,4;9,6
Classification signature	30 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 015/324

Route 016

r1c4_h+1v+1

Cell 4 (1, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,4;9,6
Classification signature	30 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 30 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,54,7) / (56,54,5)
κ_D	79.60672133621751972652496
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 016/324

Route 017

r1c5_h-1v-1

Cell 5 (1, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,5;9,5
Classification signature	34 0 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(100,66,0) / (48,54,5)
κ_D	99.51837473042726913417918
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 017/324

Route 018

r1c5_h-1v+1

Cell 5 (1, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,5;9,5
Classification signature	34 0 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 018/324

Route 019

r1c5_h+1v-1

Cell 5 (1, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,5;9,5
Classification signature	34 0 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 019/324

Route 020

r1c5_h+1v+1

Cell 5 (1, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,5;9,5
Classification signature	34 0 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(100,66,0) / (48,54,5)
κ_D	99.51837473042726913417918
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	30; 8.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 020/324

Route 021

r1c6_h-1v-1

Cell 6 (1, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,6;9,4
Classification signature	34 0 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 021/324

Route 022

r1c6_h-1v+1

Cell 6 (1, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,6;9,4
Classification signature	34 0 -4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 022/324

Route 023

r1c6_h+1v-1

Cell 6 (1, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,6;9,4
Classification signature	34 0 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 023/324

Route 024

r1c6_h+1v+1

Cell 6 (1, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,6;9,4
Classification signature	34 0 -4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 024/324

Route 025

r1c7_h-1v-1

Cell 7 (1, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,7;9,3
Classification signature	30 6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 025/324

Route 026

r1c7_h-1v+1

Cell 7 (1, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,7;9,3
Classification signature	30 6 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 026/324

Route 027

r1c7_h+1v-1

Cell 7 (1, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,7;9,3
Classification signature	30 6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 027/324

Route 028

r1c7_h+1v+1

Cell 7 (1, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,7;9,3
Classification signature	30 6 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 028/324

Route 029

r1c8_h-1v-1

Cell 8 (1, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,8;9,2
Classification signature	28 6 2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 029/324

Route 030

r1c8_h-1v+1

Cell 8 (1, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,8;9,2
Classification signature	28 6 2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	++00--++00--; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 030/324

Route 031

r1c8_h+1v-1

Cell 8 (1, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,8;9,2
Classification signature	28 6 2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+--+00+--+0; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 031/324

Route 032

r1c8_h+1v+1

Cell 8 (1, 8) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,8;9,2
Classification signature	28 6 2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 032/324

Route 033

r1c9_h-1v-1

Cell 9 (1, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,9;9,1
Classification signature	28 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 033/324

Route 034

r1c9_h-1v+1

Cell 9 (1, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,9;9,1
Classification signature	28 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 034/324

Route 035

r1c9_h+1v-1

Cell 9 (1, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,9;9,1
Classification signature	28 -6 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 035/324

Route 036

r1c9_h+1v+1

Cell 9 (1, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,9;9,1
Classification signature	28 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 036/324

Route 037

r2c1_h-1v-1

Cell 10 (2, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 2,1;8,9
Classification signature	28 -6 12 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 037/324

Route 038

r2c1_h-1v+1

Cell 10 (2, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 2,1;8,9
Classification signature	28 -6 12 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (48,54,5)
κ_D	39.82508593118257305617326
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 038/324

Route 039

r2c1_h+1v-1

Cell 10 (2, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 2,1;8,9
Classification signature	28 -6 12 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (48,54,5)
κ_D	39.82508593118257305617326
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 039/324

Route 040

r2c1_h+1v+1

Cell 10 (2, 1) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : yes 2,1;8,9
Classification signature	28 -6 12 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 040/324

Route 041

r2c2_h-1v-1

Cell 11 (2, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,2;8,8
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 9; orbit: 9; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 041/324

Route 042

r2c2_h-1v+1

Cell 11 (2, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : yes 2,2;8,8
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,84,28) / (80,54,6)
κ_D	59.39013588401554063774148
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 042/324

Route 043

r2c2_h+1v-1

Cell 11 (2, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,2;8,8
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,84,28) / (80,54,6)
κ_D	59.39013588401554063774148
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 043/324

Route 044

r2c2_h+1v+1

Cell 11 (2, 2) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,2;8,8
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 18; orbit: 18; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 044/324

Route 045

r2c3_h-1v-1

Cell 12 (2, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : yes 2,3;8,7
Classification signature	28 6 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 045/324

Route 046

r2c3_h-1v+1

Cell 12 (2, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,3;8,7
Classification signature	28 6 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 44 2 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 046/324

Route 047

r2c3_h+1v-1

Cell 12 (2, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,3;8,7
Classification signature	28 6 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 44 -2 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 047/324

Route 048

r2c3_h+1v+1

Cell 12 (2, 3) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,3;8,7
Classification signature	28 6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 048/324

Route 049

r2c4_h-1v-1

Cell 13 (2, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,4;8,6
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 049/324

Route 050

r2c4_h-1v+1

Cell 13 (2, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,4;8,6
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 050/324

Route 051

r2c4_h+1v-1

Cell 13 (2, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : yes 2,4;8,6
Classification signature	28 -6 6 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 051/324

Route 052

r2c4_h+1v+1

Cell 13 (2, 4) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,4;8,6
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 052/324

Route 053

r2c5_h-1v-1

Cell 14 (2, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,5;8,5
Classification signature	30 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 053/324

Route 054

r2c5_h-1v+1

Cell 14 (2, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,5;8,5
Classification signature	30 -6 0 0 -12 +++++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -4, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	++ +00-++ +00-; 50 4 -4 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	20, 4

Readers and action factors

K_D / K_Ω	(60,84,28) / (20,54,4)
κ_D	59.39013588401554063774148
κ_Ω	19.60638774694944330636281
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000013592751151887498

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 054/324

Route 055

r2c5_h+1v-1

Cell 14 (2, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,5;8,5
Classification signature	30 -6 0 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 4, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	00+---00+---; 50 -4 4 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	20, 4

Readers and action factors

K_D / K_Ω	(60,84,28) / (20,54,4)
κ_D	59.39013588401554063774148
κ_Ω	19.60638774694944330636281
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000013592751151887498

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 055/324

Route 056

r2c5_h+1v+1

Cell 14 (2, 5) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : yes 2,5;8,5
Classification signature	30 -6 0 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 30 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 056/324

Route 057

r2c6_h-1v-1

Cell 15 (2, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,6;8,4
Classification signature	34 0 -16 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,4) / (48,54,5)
κ_D	79.51881951611803769439539
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 057/324

Route 058

r2c6_h-1v+1

Cell 15 (2, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : yes 2,6;8,4
Classification signature	34 0 -16 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	++ +---++ +---; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 058/324

Route 059

r2c6_h+1v-1

Cell 15 (2, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,6;8,4
Classification signature	34 0 -16 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 059/324

Route 060

r2c6_h+1v+1

Cell 15 (2, 6) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,6;8,4
Classification signature	34 0 -16 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,4) / (48,54,5)
κ_D	79.51881951611803769439539
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 060/324

Route 061

r2c7_h-1v-1

Cell 16 (2, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,7;8,3
Classification signature	34 0 0 2 -8 +++0--+++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,2) / (56,54,5)
κ_D	99.51859712327265341428729
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 061/324

Route 062

r2c7_h-1v+1

Cell 16 (2, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : yes 2,7;8,3
Classification signature	34 0 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	++00--++00--; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 062/324

Route 063

r2c7_h+1v-1

Cell 16 (2, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,7;8,3
Classification signature	34 0 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+--+00+--+0; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 063/324

Route 064

r2c7_h+1v+1

Cell 16 (2, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,7;8,3
Classification signature	34 0 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0--++0--; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,2) / (56,54,5)
κ_D	99.51859712327265341428729
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 064/324

Route 065

r2c8_h-1v-1

Cell 17 (2, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : yes 2,8;8,2
Classification signature	30 6 -8 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 30 -6 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 065/324

Route 066

r2c8_h-1v+1

Cell 17 (2, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,8;8,2
Classification signature	30 6 -8 0 -12 +++++----

Returns, memory and transport

Returns with phase	class: 18; orbit: 18; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(60,84,28) / (56,54,5)
κ_D	59.39013588401554063774148
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 066/324

Route 067

r2c8_h+1v-1

Cell 17 (2, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,8;8,2
Classification signature	30 6 -8 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 9; orbit: 9; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(60,84,28) / (56,54,5)
κ_D	59.39013588401554063774148
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 067/324

Route 068

r2c8_h+1v+1

Cell 17 (2, 8) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,8;8,2
Classification signature	30 6 -8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 6 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 068/324

Route 069

r2c9_h-1v-1

Cell 18 (2, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 2,9;8,1
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 069/324

Route 070

r2c9_h-1v+1

Cell 18 (2, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : yes 2,9;8,1
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 070/324

Route 071

r2c9_h+1v-1

Cell 18 (2, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 2,9;8,1
Classification signature	28 6 -2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 071/324

Route 072

r2c9_h+1v+1

Cell 18 (2, 9) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 2,9;8,1
Classification signature	28 6 -2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 072/324

Route 073

r3c1_h-1v-1

Cell 19 (3, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,1;7,9
Classification signature	28 6 -10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 073/324

Route 074

r3c1_h-1v+1

Cell 19 (3, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 3,1;7,9
Classification signature	28 6 -10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 074/324

Route 075

r3c1_h+1v-1

Cell 19 (3, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,1;7,9
Classification signature	28 6 -10 0 -12 +++-+++-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 075/324

Route 076

r3c1_h+1v+1

Cell 19 (3, 1) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,1;7,9
Classification signature	28 6 -10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 076/324

Route 077

r3c2_h-1v-1

Cell 20 (3, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,2;7,8
Classification signature	28 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---+++0--; 28 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 077/324

Route 078

r3c2_h-1v+1

Cell 20 (3, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,2;7,8
Classification signature	28 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 2 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 078/324

Route 079

r3c2_h+1v-1

Cell 20 (3, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,2;7,8
Classification signature	28 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 -2 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 079/324

Route 080

r3c2_h+1v+1

Cell 20 (3, 2) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : yes 3,2;7,8
Classification signature	28 -6 -4 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 28 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 080/324

Route 081

r3c3_h-1v-1

Cell 21 (3, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,3;7,7
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,54,6) / (80,54,6)
κ_D	39.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 081/324

Route 082

r3c3_h-1v+1

Cell 21 (3, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,3;7,7
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(0,24,0) / (80,54,6)
κ_D	-0.1751364616628112239348425
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 082/324

Route 083

r3c3_h+1v-1

Cell 21 (3, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,3;7,7
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(0,24,0) / (80,54,6)
κ_D	-0.1751364616628112239348425
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 083/324

Route 084

r3c3_h+1v+1

Cell 21 (3, 3) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,3;7,7
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,54,6) / (80,54,6)
κ_D	39.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 084/324

Route 085

r3c4_h-1v-1

Cell 22 (3, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,4;7,6
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 085/324

Route 086

r3c4_h-1v+1

Cell 22 (3, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,4;7,6
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 086/324

Route 087

r3c4_h+1v-1

Cell 22 (3, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,4;7,6
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 087/324

Route 088

r3c4_h+1v+1

Cell 22 (3, 4) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,4;7,6
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 088/324

Route 089

r3c5_h-1v-1

Cell 23 (3, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,5;7,5
Classification signature	28 -6 6 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 28 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (48,54,5)
κ_D	79.51870831969534555434134
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 089/324

Route 090

r3c5_h-1v+1

Cell 23 (3, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,5;7,5
Classification signature	28 -6 6 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 090/324

Route 091

r3c5_h+1v-1

Cell 23 (3, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,5;7,5
Classification signature	28 -6 6 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 091/324

Route 092

r3c5_h+1v+1

Cell 23 (3, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,5;7,5
Classification signature	28 -6 6 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 28 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (48,54,5)
κ_D	79.51870831969534555434134
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 092/324

Route 093

r3c6_h-1v-1

Cell 24 (3, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,6;7,4
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0++++-0; 30 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 093/324

Route 094

r3c6_h-1v+1

Cell 24 (3, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,6;7,4
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, 0
τ_{12} ; historical signature $\mathbb{Z}^5$	+0000-+0000-; 20 0 0 0 0
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	4, 2

Readers and action factors

K_D / K_Ω	(0,24,0) / (4,54,2)
κ_D	-0.1751364616628112239348425
κ_Ω	3.606165354104059026254707
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000002500088676796314

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 094/324

Route 095

r3c6_h+1v-1

Cell 24 (3, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,6;7,4
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, 0
τ_{12} ; historical signature $\mathbb{Z}^5$	00+-0000+-00; 20 0 0 0 0
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	4, 2

Readers and action factors

K_D / K_Ω	(0,24,0) / (4,54,2)
κ_D	-0.1751364616628112239348425
κ_Ω	3.606165354104059026254707
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000002500088676796314

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 095/324

Route 096

r3c6_h+1v+1

Cell 24 (3, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,6;7,4
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 30 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 096/324

Route 097

r3c7_h-1v-1

Cell 25 (3, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,7;7,3
Classification signature	34 0 -4 0 -12 +++----++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++----++----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 097/324

Route 098

r3c7_h-1v+1

Cell 25 (3, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,7;7,3
Classification signature	34 0 -4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 098/324

Route 099

r3c7_h+1v-1

Cell 25 (3, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,7;7,3
Classification signature	34 0 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 099/324

Route 100

r3c7_h+1v+1

Cell 25 (3, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,7;7,3
Classification signature	34 0 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 100/324

Route 101

r3c8_h-1v-1

Cell 26 (3, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,8;7,2
Classification signature	34 0 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 101/324

Route 102

r3c8_h-1v+1

Cell 26 (3, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : yes 3,8;7,2
Classification signature	34 0 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 102/324

Route 103

r3c8_h+1v-1

Cell 26 (3, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,8;7,2
Classification signature	34 0 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 103/324

Route 104

r3c8_h+1v+1

Cell 26 (3, 8) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,8;7,2
Classification signature	34 0 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 104/324

Route 105

r3c9_h-1v-1

Cell 27 (3, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 3,9;7,1
Classification signature	30 6 0 2 -8 +++-0++++-0

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 30 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 105/324

Route 106

r3c9_h-1v+1

Cell 27 (3, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,9;7,1
Classification signature	30 6 0 2 -8 +++-0++++-0

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(0,24,0) / (56,54,5)
κ_D	-0.1751364616628112239348425
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	108; 46.52932501298079
First separation block	orbit: —; cell: —; route: —
Final ambiguity	6

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 106/324

Route 107

r3c9_h+1v-1

Cell 27 (3, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,9;7,1
Classification signature	30 6 0 2 -8 +++-0++++-0

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-0++++-0; 50 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(0,24,0) / (56,54,5)
κ_D	-0.1751364616628112239348425
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	108; 46.52932501298079
First separation block	orbit: —; cell: —; route: —
Final ambiguity	6

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 107/324

Route 108

r3c9_h+1v+1

Cell 27 (3, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,9;7,1
Classification signature	30 6 0 2 -8 +++-0++++-0

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-0++++-0; 30 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 108/324

Route 109

r4c1_h-1v-1

Cell 28 (4, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : yes 4,1;6,9
Classification signature	30 6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 109/324

Route 110

r4c1_h-1v+1

Cell 28 (4, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : no 4,1;6,9
Classification signature	30 6 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 110/324

Route 111

r4c1_h+1v-1

Cell 28 (4, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : no 4,1;6,9
Classification signature	30 6 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 111/324

Route 112

r4c1_h+1v+1

Cell 28 (4, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : no 4,1;6,9
Classification signature	30 6 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	26; 6.339850002884624
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 112/324

Route 113

r4c2_h-1v-1

Cell 29 (4, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 4,2;6,8
Classification signature	28 6 -2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 113/324

Route 114

r4c2_h-1v+1

Cell 29 (4, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 4,2;6,8
Classification signature	28 6 -2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 114/324

Route 115

r4c2_h+1v-1

Cell 29 (4, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : yes 4,2;6,8
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 115/324

Route 116

r4c2_h+1v+1

Cell 29 (4, 2) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 4,2;6,8
Classification signature	28 6 -2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 116/324

Route 117

r4c3_h-1v-1

Cell 30 (4, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; G
Primary route and inversion orbit	B_1 : no 4,3;6,7
Classification signature	28 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 117/324

Route 118

r4c3_h-1v+1

Cell 30 (4, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; G
Primary route and inversion orbit	B_1 : no 4,3;6,7
Classification signature	28 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 118/324

Route 119

r4c3_h+1v-1

Cell 30 (4, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; G
Primary route and inversion orbit	B_1 : no 4,3;6,7
Classification signature	28 -6 0 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 119/324

Route 120

r4c3_h+1v+1

Cell 30 (4, 3) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; G
Primary route and inversion orbit	B_1 : yes 4,3;6,7
Classification signature	28 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 120/324

Route 121

r4c4_h-1v-1

Cell 31 (4, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : no 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 121/324

Route 122

r4c4_h-1v+1

Cell 31 (4, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : yes 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 122/324

Route 123

r4c4_h+1v-1

Cell 31 (4, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : no 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 123/324

Route 124

r4c4_h+1v+1

Cell 31 (4, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : no 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 124/324

Route 125

r4c5_h-1v-1

Cell 32 (4, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection quark_down_G1 / quark_down_G1**Collection and sector** down-type quark; quark**Physical classification** down-type quark**Generation, depth and colour** 1; G1; B**Primary route and inversion orbit** B_1 : no 4,5;6,5**Classification signature** 28|6|0|0|-12|+++---++----

Returns, memory and transport

Returns with phase class: 27; orbit: 27; cell: 27**Kinematic return** within 108: 54; extension: 216**Turns and sheet switches** 71; 107**Carry and deferral** total debt: 1; final: 0; deferred blocks: 1**Final triads; Pareto persistence** 17; yes $K \longrightarrow K + 9$ repetitions: 9; memory changes: 9

Signature, phase and differences

 $n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$ 28, -6, 0, 0, -12 τ_{12} ; **historical signature** $\mathbb{Z}^5$ +++---++----; 28|-6|0|0|-12 $D_1, D_3, D_4, D_{27}, D_{36}$ 66, 64, 64, 60, 40 $\Omega_{90/120}, P_6$ 80, 6

Readers and action factors

 K_D / K_Ω (100,66,0) / (80,54,6) κ_D 99.51837473042726913417918 κ_Ω 79.60661013979482758647091 $R_{\text{act},D}$ 1.000000068994276645021895 $R_{\text{act},\Omega}$ 1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area 18; 0.0**First separation block** orbit: 1; cell: 1; route: 1**Final ambiguity** 1

Complete sequences

 w_6 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657**Stable triads per block** 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 125/324

Route 126

r4c5_h-1v+1

Cell 32 (4, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection quark_down_G1 / quark_down_G1**Collection and sector** down-type quark; quark**Physical classification** down-type quark**Generation, depth and colour** 1; G1; B**Primary route and inversion orbit** B_1 : no 4,5;6,5**Classification signature** 28|6|0|0|-12|+++---++++---

Returns, memory and transport

Returns with phase class: 27; orbit: 27; cell: 27**Kinematic return** within 108: 54; extension: 216**Turns and sheet switches** 71; 107**Carry and deferral** total debt: 2; final: 0; deferred blocks: 2**Final triads; Pareto persistence** 17; yes $K \longrightarrow K + 9$ repetitions: 9; memory changes: 9

Signature, phase and differences

 $n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$ 20, 0, 0, 0, -8 τ_{12} ; **historical signature** $\mathbb{Z}^5$ +0+-0-+0+-0-; 20|0|0|0|-8 $D_1, D_3, D_4, D_{27}, D_{36}$ 24, 20, 24, 24, 16 $\Omega_{90/120}, P_6$ 40, 4

Readers and action factors

 K_D / K_Ω (40,24,2) / (40,54,4) κ_D 39.82508593118257305617326 κ_Ω 39.60638774694944330636281 $R_{\text{act},D}$ 1.000000027610006159514215 $R_{\text{act},\Omega}$ 1.000000027458386692651400

Prefix separation

Ambiguity area; $\log_2$ area 18; 0.0**First separation block** orbit: 1; cell: 1; route: 1**Final ambiguity** 1

Complete sequences

 w_6 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3**Stable triads per block** 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 126/324

Route 127

r4c5_h+1v-1

Cell 32 (4, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G1 / quark_down_G1
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	1; G1; B
Primary route and inversion orbit	B_1 : yes 4,5;6,5
Classification signature	28 6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+-0-+0+-0-; 20 0 0 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	40, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (40,54,4)
κ_D	39.82508593118257305617326
κ_Ω	39.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000027458386692651400

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 127/324

Route 128

r4c5_h+1v+1

Cell 32 (4, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G1 / quark_down_G1
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	1; G1; B
Primary route and inversion orbit	B_1 : no 4,5;6,5
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 128/324

Route 129

r4c6_h-1v-1

Cell 33 (4, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; G
Primary route and inversion orbit	B_1 : no 4,6;6,4
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 129/324

Route 130

r4c6_h-1v+1

Cell 33 (4, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; G
Primary route and inversion orbit	B_1 : no 4,6;6,4
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 130/324

Route 131

r4c6_h+1v-1

Cell 33 (4, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; G
Primary route and inversion orbit	B_1 : no 4,6;6,4
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 131/324

Route 132

r4c6_h+1v+1

Cell 33 (4, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; G
Primary route and inversion orbit	B_1 : yes 4,6;6,4
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 132/324

Route 133

r4c7_h-1v-1

Cell 34 (4, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : no 4,7;6,3
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 30 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,54,7) / (56,54,5)
κ_D	79.60672133621751972652496
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 133/324

Route 134

r4c7_h-1v+1

Cell 34 (4, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : yes 4,7;6,3
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 134/324

Route 135

r4c7_h+1v-1

Cell 34 (4, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : no 4,7;6,3
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 135/324

Route 136

r4c7_h+1v+1

Cell 34 (4, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : no 4,7;6,3
Classification signature	30 -6 0 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 30 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,54,7) / (56,54,5)
κ_D	79.60672133621751972652496
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 136/324

Route 137

r4c8_h-1v-1

Cell 35 (4, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : yes 4,8;6,2
Classification signature	34 0 -20 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,0) / (56,54,5)
κ_D	99.51837473042726913417918
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 137/324

Route 138

r4c8_h-1v+1

Cell 35 (4, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 4,8;6,2
Classification signature	34 0 -20 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 50 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 138/324

Route 139

r4c8_h+1v-1

Cell 35 (4, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 4,8;6,2
Classification signature	34 0 -20 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---+++0--; 50 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 139/324

Route 140

r4c8_h+1v+1

Cell 35 (4, 8) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 4,8;6,2
Classification signature	34 0 -20 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,0) / (56,54,5)
κ_D	99.51837473042726913417918
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 140/324

Route 141

r4c9_h-1v-1

Cell 36 (4, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 4,9;6,1
Classification signature	34 0 -8 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 141/324

Route 142

r4c9_h-1v+1

Cell 36 (4, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : yes 4,9;6,1
Classification signature	34 0 -8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 142/324

Route 143

r4c9_h+1v-1

Cell 36 (4, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 4,9;6,1
Classification signature	34 0 -8 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 143/324

Route 144

r4c9_h+1v+1

Cell 36 (4, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 4,9;6,1
Classification signature	34 0 -8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 144/324

Route 145

r5c1_h-1v-1

Cell 37 (5, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 5,1;5,9
Classification signature	34 0 -16 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(100,66,2) / (48,54,5)
κ_D	99.51859712327265341428729
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	30; 8.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 145/324

Route 146

r5c1_h-1v+1

Cell 37 (5, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 5,1;5,9
Classification signature	34 0 -16 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 146/324

Route 147

r5c1_h+1v-1

Cell 37 (5, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 5,1;5,9
Classification signature	34 0 -16 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 147/324

Route 148

r5c1_h+1v+1

Cell 37 (5, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 5,1;5,9
Classification signature	34 0 -16 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(100,66,2) / (48,54,5)
κ_D	99.51859712327265341428729
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 148/324

Route 149

r5c2_h-1v-1

Cell 38 (5, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : yes 5,2;5,8
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 149/324

Route 150

r5c2_h-1v+1

Cell 38 (5, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 5,2;5,8
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -4, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	+++00+++00; 50 4 -4 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	20, 4

Readers and action factors

K_D / K_Ω	(60,84,28) / (20,54,4)
κ_D	59.39013588401554063774148
κ_Ω	19.60638774694944330636281
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000013592751151887498

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 150/324

Route 151

r5c2_h+1v-1

Cell 38 (5, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 5,2;5,8
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 4, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	+00---+00---; 50 -4 4 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	20, 4

Readers and action factors

K_D / K_Ω	(60,84,28) / (20,54,4)
κ_D	59.39013588401554063774148
κ_Ω	19.60638774694944330636281
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000013592751151887498

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 151/324

Route 152

r5c2_h+1v+1

Cell 38 (5, 2) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 5,2;5,8
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 152/324

Route 153

r5c3_h-1v-1

Cell 39 (5, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 5,3;5,7
Classification signature	28 6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 153/324

Route 154

r5c3_h-1v+1

Cell 39 (5, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 5,3;5,7
Classification signature	28 6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 154/324

Route 155

r5c3_h+1v-1

Cell 39 (5, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 5,3;5,7
Classification signature	28 6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 155/324

Route 156

r5c3_h+1v+1

Cell 39 (5, 3) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 5,3;5,7
Classification signature	28 6 2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 156/324

Route 157

r5c4_h-1v-1

Cell 40 (5, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : no 5,4;5,6
Classification signature	28 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	++-+---++-+---; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 157/324

Route 158

r5c4_h-1v+1

Cell 40 (5, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : no 5,4;5,6
Classification signature	28 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+-0-+0+-0-; 20 0 0 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	40, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (40,54,4)
κ_D	39.82508593118257305617326
κ_Ω	39.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000027458386692651400

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 158/324

Route 159

r5c4_h+1v-1

Cell 40 (5, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : yes 5,4;5,6
Classification signature	28 -6 4 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+-0-+0+-0-; 20 0 0 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	40, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (40,54,4)
κ_D	39.82508593118257305617326
κ_Ω	39.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000027458386692651400

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 159/324

Route 160

r5c4_h+1v+1

Cell 40 (5, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : no 5,4;5,6
Classification signature	28 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 160/324

Route 161

r5c5_h-1v-1

Cell 41 (5, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G0 / charged_lepton_G0
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	0; G0, centre; —
Primary route and inversion orbit	B_1 : no 5,5
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 161/324

Route 162

r5c5_h-1v+1

Cell 41 (5, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G0 / charged_lepton_G0
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	0; G0, centre; —
Primary route and inversion orbit	B_1 : no 5,5
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(60,84,28) / (48,54,5)
κ_D	59.39013588401554063774148
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 162/324

Route 163

r5c5_h+1v-1

Cell 41 (5, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G0 / charged_lepton_G0
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	0; G0, centre; —
Primary route and inversion orbit	B_1 : no 5,5
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(60,84,28) / (48,54,5)
κ_D	59.39013588401554063774148
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 163/324

Route 164

r5c5_h+1v+1

Cell 41 (5, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G0 / charged_lepton_G0
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	0; G0, centre; —
Primary route and inversion orbit	B_1 : yes 5,5
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 164/324

Route 165

r5c6_h-1v-1

Cell 42 (5, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : no 5,4;5,6
Classification signature	28 6 -4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 165/324

Route 166

r5c6_h-1v+1

Cell 42 (5, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : yes 5,4;5,6
Classification signature	28 6 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 166/324

Route 167

r5c6_h+1v-1

Cell 42 (5, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : no 5,4;5,6
Classification signature	28 6 -4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 167/324

Route 168

r5c6_h+1v+1

Cell 42 (5, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G1 / neutrino_G1
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	1; G1; —
Primary route and inversion orbit	B_1 : no 5,4;5,6
Classification signature	28 6 -4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 168/324

Route 169

r5c7_h-1v-1

Cell 43 (5, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 5,3;5,7
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	++---++---; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 169/324

Route 170

r5c7_h-1v+1

Cell 43 (5, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 5,3;5,7
Classification signature	28 -6 6 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 50 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 170/324

Route 171

r5c7_h+1v-1

Cell 43 (5, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 5,3;5,7
Classification signature	28 -6 6 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 50 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 171/324

Route 172

r5c7_h+1v+1

Cell 43 (5, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 5,3;5,7
Classification signature	28 -6 6 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 172/324

Route 173

r5c8_h-1v-1

Cell 44 (5, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 5,2;5,8
Classification signature	30 -6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 173/324

Route 174

r5c8_h-1v+1

Cell 44 (5, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : yes 5,2;5,8
Classification signature	30 -6 8 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,84,28) / (80,54,6)
κ_D	59.39013588401554063774148
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 174/324

Route 175

r5c8_h+1v-1

Cell 44 (5, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 5,2;5,8
Classification signature	30 -6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,84,28) / (80,54,6)
κ_D	59.39013588401554063774148
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 175/324

Route 176

r5c8_h+1v+1

Cell 44 (5, 8) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 5,2;5,8
Classification signature	30 -6 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 176/324

Route 177

r5c9_h-1v-1

Cell 45 (5, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 5,1;5,9
Classification signature	34 0 -12 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,4) / (56,54,5)
κ_D	79.51881951611803769439539
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 177/324

Route 178

r5c9_h-1v+1

Cell 45 (5, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 5,1;5,9
Classification signature	34 0 -12 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 178/324

Route 179

r5c9_h+1v-1

Cell 45 (5, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 5,1;5,9
Classification signature	34 0 -12 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 179/324

Route 180

r5c9_h+1v+1

Cell 45 (5, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 5,1;5,9
Classification signature	34 0 -12 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,4) / (56,54,5)
κ_D	79.51881951611803769439539
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	30; 8.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 180/324

Route 181

r6c1_h-1v-1

Cell 46 (6, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : yes 4,9;6,1
Classification signature	34 0 -4 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 181/324

Route 182

r6c1_h-1v+1

Cell 46 (6, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 4,9;6,1
Classification signature	34 0 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++++-0++++-0; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 182/324

Route 183

r6c1_h+1v-1

Cell 46 (6, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 4,9;6,1
Classification signature	34 0 -4 0 -12 +++++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 183/324

Route 184

r6c1_h+1v+1

Cell 46 (6, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 4,9;6,1
Classification signature	34 0 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 184/324

Route 185

r6c2_h-1v-1

Cell 47 (6, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 4,8;6,2
Classification signature	34 0 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (48,54,5)
κ_D	79.51870831969534555434134
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 185/324

Route 186

r6c2_h-1v+1

Cell 47 (6, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : yes 4,8;6,2
Classification signature	34 0 0 0 -8 +0+----+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 186/324

Route 187

r6c2_h+1v-1

Cell 47 (6, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 4,8;6,2
Classification signature	34 0 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 187/324

Route 188

r6c2_h+1v+1

Cell 47 (6, 2) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 4,8;6,2
Classification signature	34 0 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (48,54,5)
κ_D	79.51870831969534555434134
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 188/324

Route 189

r6c3_h-1v-1

Cell 48 (6, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : no 4,7;6,3
Classification signature	30 6 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 30 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 189/324

Route 190

r6c3_h-1v+1

Cell 48 (6, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : yes 4,7;6,3
Classification signature	30 6 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, 0
τ_{12} ; historical signature $\mathbb{Z}^5$	00+-0000+-00; 20 0 0 0 0
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	4, 2

Readers and action factors

K_D / K_Ω	(0,24,0) / (4,54,2)
κ_D	-0.1751364616628112239348425
κ_Ω	3.606165354104059026254707
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000002500088676796314

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 190/324

Route 191

r6c3_h+1v-1

Cell 48 (6, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : no 4,7;6,3
Classification signature	30 6 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, 0
τ_{12} ; historical signature $\mathbb{Z}^5$	+0000-+0000-; 20 0 0 0 0
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	4, 2

Readers and action factors

K_D / K_Ω	(0,24,0) / (4,54,2)
κ_D	-0.1751364616628112239348425
κ_Ω	3.606165354104059026254707
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000002500088676796314

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 191/324

Route 192

r6c3_h+1v+1

Cell 48 (6, 3) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; R
Primary route and inversion orbit	B_1 : no 4,7;6,3
Classification signature	30 6 0 2 -8 +++0--++0--

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0--++0--; 30 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 192/324

Route 193

r6c4_h-1v-1

Cell 49 (6, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; B
Primary route and inversion orbit	B_1 : yes 4,6;6,4
Classification signature	28 6 -2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 193/324

Route 194

r6c4_h-1v+1

Cell 49 (6, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; B
Primary route and inversion orbit	B_1 : no 4,6;6,4
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 194/324

Route 195

r6c4_h+1v-1

Cell 49 (6, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; B
Primary route and inversion orbit	B_1 : no 4,6;6,4
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 195/324

Route 196

r6c4_h+1v+1

Cell 49 (6, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; B
Primary route and inversion orbit	B_1 : no 4,6;6,4
Classification signature	28 6 -2 0 -12 +++----++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++----++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 196/324

Route 197

r6c5_h-1v-1

Cell 50 (6, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection quark_down_G1 / quark_down_G1

Collection and sector down-type quark; quark

Physical classification down-type quark

Generation, depth and colour 1; G1; G

Primary route and inversion orbit B_1 : no 4,5;6,5

Classification signature 28|-6|4|-2|-8|++0---++0---

Returns, memory and transport

Returns with phase class: 27; orbit: 27; cell: 27

Kinematic return within 108: 54; extension: 216

Turns and sheet switches 71; 107

Carry and deferral total debt: 1; final: 0; deferred blocks: 1

Final triads; Pareto persistence 17; yes

 $K \longrightarrow K + 9$ repetitions: 9; memory changes: 9

Signature, phase and differences

 $n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$ 28, 6, -2, 0, -8 τ_{12} ; historical signature $\mathbb{Z}^5$ +++-0+++--0; 28|6|-2|0|-8 $D_1, D_3, D_4, D_{27}, D_{36}$ 66, 60, 66, 48, 32 $\Omega_{90/120}, P_6$ 56, 5

Readers and action factors

 K_D / K_Ω (80,66,3) / (56,54,5) κ_D 79.51870831969534555434134 κ_Ω 55.60649894337213544641686 $R_{\text{act},D}$ 1.000000055128871799705072 $R_{\text{act},\Omega}$ 1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area 18; 0.0

First separation block orbit: 1; cell: 1; route: 1

Final ambiguity 1

Complete sequences

 w_6 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327

Stable triads per block 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 197/324

Route 198

r6c5_h-1v+1

Cell 50 (6, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection quark_down_G1 / quark_down_G1**Collection and sector** down-type quark; quark**Physical classification** down-type quark**Generation, depth and colour** 1; G1; G**Primary route and inversion orbit** B_1 : yes 4,5;6,5**Classification signature** 28|-6|4|-2|-8|++0---++0---

Returns, memory and transport

Returns with phase class: 27; orbit: 27; cell: 27**Kinematic return** within 108: 54; extension: 216**Turns and sheet switches** 71; 107**Carry and deferral** total debt: 1; final: 0; deferred blocks: 1**Final triads; Pareto persistence** 17; yes $K \longrightarrow K + 9$ repetitions: 9; memory changes: 9

Signature, phase and differences

 $n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$ 14, -4, 0, 0, -12 τ_{12} ; **historical signature** $\mathbb{Z}^5$ ++-+---++-+---; 14|-4|0|0|-12 $D_1, D_3, D_4, D_{27}, D_{36}$ 78, 24, 78, 24, 16 $\Omega_{90/120}, P_6$ 80, 6

Readers and action factors

 K_D / K_Ω (40,78,27) / (80,54,6) κ_D 39.43380880300855130367114 κ_Ω 79.60661013979482758647091 $R_{\text{act},D}$ 1.000000027338740854894358 $R_{\text{act},\Omega}$ 1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area 18; 0.0**First separation block** orbit: 1; cell: 1; route: 1**Final ambiguity** 1

Complete sequences

 w_6 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672**Stable triads per block** 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 198/324

Route 199

r6c5_h+1v-1

Cell 50 (6, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G1 / quark_down_G1
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	1; G1; G
Primary route and inversion orbit	B_1 : no 4,5;6,5
Classification signature	28 -6 4 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 199/324

Route 200

r6c5_h+1v+1

Cell 50 (6, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G1 / quark_down_G1
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	1; G1; G
Primary route and inversion orbit	B_1 : no 4,5;6,5
Classification signature	28 -6 4 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 28 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 200/324

Route 201

r6c6_h-1v-1

Cell 51 (6, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : no 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,54,6) / (80,54,6)
κ_D	39.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 201/324

Route 202

r6c6_h-1v+1

Cell 51 (6, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : no 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(0,24,0) / (48,54,5)
κ_D	-0.1751364616628112239348425
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 202/324

Route 203

r6c6_h+1v-1

Cell 51 (6, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : no 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(0,24,0) / (48,54,5)
κ_D	-0.1751364616628112239348425
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 203/324

Route 204

r6c6_h+1v+1

Cell 51 (6, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G1 / quark_up_G1
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	1; G1; R
Primary route and inversion orbit	B_1 : yes 4,4;6,6
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,54,6) / (80,54,6)
κ_D	39.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 204/324

Route 205

r6c7_h-1v-1

Cell 52 (6, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; B
Primary route and inversion orbit	B_1 : no 4,3;6,7
Classification signature	28 6 -8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 205/324

Route 206

r6c7_h-1v+1

Cell 52 (6, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; B
Primary route and inversion orbit	B_1 : yes 4,3;6,7
Classification signature	28 6 -8 0 -12 +++++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 206/324

Route 207

r6c7_h+1v-1

Cell 52 (6, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; B
Primary route and inversion orbit	B_1 : no 4,3;6,7
Classification signature	28 6 -8 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 207/324

Route 208

r6c7_h+1v+1

Cell 52 (6, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G2 / quark_down_G2
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	2; G2; B
Primary route and inversion orbit	B_1 : no 4,3;6,7
Classification signature	28 6 -8 0 -12 +++----++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++----++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 208/324

Route 209

r6c8_h-1v-1

Cell 53 (6, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 4,2;6,8
Classification signature	28 -6 -2 -2 -8 0++++0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 28 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 209/324

Route 210

r6c8_h-1v+1

Cell 53 (6, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 4,2;6,8
Classification signature	28 -6 -2 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 210/324

Route 211

r6c8_h+1v-1

Cell 53 (6, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 4,2;6,8
Classification signature	28 -6 -2 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 211/324

Route 212

r6c8_h+1v+1

Cell 53 (6, 8) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : yes 4,2;6,8
Classification signature	28 -6 -2 -2 -8 0+ +---0+ +---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+ +---0+ +---; 28 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 212/324

Route 213

r6c9_h-1v-1

Cell 54 (6, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : no 4,1;6,9
Classification signature	30 -6 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 30 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (48,54,5)
κ_D	39.60661013979482758647091
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	26; 6.339850002884624
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 213/324

Route 214

r6c9_h-1v+1

Cell 54 (6, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : no 4,1;6,9
Classification signature	30 -6 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(0,24,0) / (80,54,6)
κ_D	-0.1751364616628112239348425
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	0.999999998785810851337881
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 214/324

Route 215

r6c9_h+1v-1

Cell 54 (6, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : yes 4,1;6,9
Classification signature	30 -6 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(0,24,0) / (80,54,6)
κ_D	-0.1751364616628112239348425
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 215/324

Route 216

r6c9_h+1v+1

Cell 54 (6, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; R
Primary route and inversion orbit	B_1 : no 4,1;6,9
Classification signature	30 -6 0 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	30, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 30 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (48,54,5)
κ_D	39.60661013979482758647091
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 216/324

Route 217

r7c1_h-1v-1

Cell 55 (7, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,9;7,1
Classification signature	30 -6 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 30 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,54,7) / (48,54,5)
κ_D	79.60672133621751972652496
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 217/324

Route 218

r7c1_h-1v+1

Cell 55 (7, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 3,9;7,1
Classification signature	30 -6 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 218/324

Route 219

r7c1_h+1v-1

Cell 55 (7, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,9;7,1
Classification signature	30 -6 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 219/324

Route 220

r7c1_h+1v+1

Cell 55 (7, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,9;7,1
Classification signature	30 -6 -8 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 30 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,54,7) / (48,54,5)
κ_D	79.60672133621751972652496
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 220/324

Route 221

r7c2_h-1v-1

Cell 56 (7, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,8;7,2
Classification signature	34 0 -4 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,0) / (56,54,5)
κ_D	99.51837473042726913417918
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 221/324

Route 222

r7c2_h-1v+1

Cell 56 (7, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : yes 3,8;7,2
Classification signature	34 0 -4 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+--+00+--+0; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 222/324

Route 223

r7c2_h+1v-1

Cell 56 (7, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,8;7,2
Classification signature	34 0 -4 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	++00--++00--; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 223/324

Route 224

r7c2_h+1v+1

Cell 56 (7, 2) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,8;7,2
Classification signature	34 0 -4 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,0) / (56,54,5)
κ_D	99.51837473042726913417918
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 224/324

Route 225

r7c3_h-1v-1

Cell 57 (7, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,7;7,3
Classification signature	34 0 -12 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 225/324

Route 226

r7c3_h-1v+1

Cell 57 (7, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,7;7,3
Classification signature	34 0 -12 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++ + 0 -- ++ + 0 --; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 226/324

Route 227

r7c3_h+1v-1

Cell 57 (7, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,7;7,3
Classification signature	34 0 -12 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 227/324

Route 228

r7c3_h+1v+1

Cell 57 (7, 3) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,7;7,3
Classification signature	34 0 -12 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 228/324

Route 229

r7c4_h-1v-1

Cell 58 (7, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,6;7,4
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 229/324

Route 230

r7c4_h-1v+1

Cell 58 (7, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,6;7,4
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 230/324

Route 231

r7c4_h+1v-1

Cell 58 (7, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,6;7,4
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 231/324

Route 232

r7c4_h+1v+1

Cell 58 (7, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,6;7,4
Classification signature	30 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 30 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 232/324

Route 233

r7c5_h-1v-1

Cell 59 (7, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,5;7,5
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 233/324

Route 234

r7c5_h-1v+1

Cell 59 (7, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,5;7,5
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 234/324

Route 235

r7c5_h+1v-1

Cell 59 (7, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,5;7,5
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 50 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 235/324

Route 236

r7c5_h+1v+1

Cell 59 (7, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,5;7,5
Classification signature	28 6 -2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 236/324

Route 237

r7c6_h-1v-1

Cell 60 (7, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,4;7,6
Classification signature	28 -6 8 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 237/324

Route 238

r7c6_h-1v+1

Cell 60 (7, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,4;7,6
Classification signature	28 -6 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 238/324

Route 239

r7c6_h+1v-1

Cell 60 (7, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,4;7,6
Classification signature	28 -6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 239/324

Route 240

r7c6_h+1v+1

Cell 60 (7, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G2 / neutrino_G2
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,4;7,6
Classification signature	28 -6 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 240/324

Route 241

r7c7_h-1v-1

Cell 61 (7, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : yes 3,3;7,7
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 241/324

Route 242

r7c7_h-1v+1

Cell 61 (7, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,3;7,7
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 242/324

Route 243

r7c7_h+1v-1

Cell 61 (7, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,3;7,7
Classification signature	24 0 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 28, 78, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,78,25) / (80,54,6)
κ_D	59.43358641016316702356304
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041204222405344905
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17

Provenance and original codes

Complete record · 243/324

Route 244

r7c7_h+1v+1

Cell 61 (7, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G2 / charged_lepton_G2
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	2; G2; —
Primary route and inversion orbit	B_1 : no 3,3;7,7
Classification signature	24 0 4 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 70, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,7) / (80,54,6)
κ_D	79.60672133621751972652496
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189889622315337
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	22; 4.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17

Provenance and original codes

Complete record · 244/324

Route 245

r7c8_h-1v-1

Cell 62 (7, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,2;7,8
Classification signature	28 6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Provenance and original codes

Complete record · 245/324

Route 246

r7c8_h-1v+1

Cell 62 (7, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,2;7,8
Classification signature	28 6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 246/324

Route 247

r7c8_h+1v-1

Cell 62 (7, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : yes 3,2;7,8
Classification signature	28 6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 247/324

Route 248

r7c8_h+1v+1

Cell 62 (7, 8) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G3 / neutrino_G3
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	3; G3; —
Primary route and inversion orbit	B_1 : no 3,2;7,8
Classification signature	28 6 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 64, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,0) / (80,54,6)
κ_D	99.51837473042726913417918
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994276645021895
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 248/324

Route 249

r7c9_h-1v-1

Cell 63 (7, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,1;7,9
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 249/324

Route 250

r7c9_h-1v+1

Cell 63 (7, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,1;7,9
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 250/324

Route 251

r7c9_h+1v-1

Cell 63 (7, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 3,1;7,9
Classification signature	28 -6 6 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 251/324

Route 252

r7c9_h+1v+1

Cell 63 (7, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 3,1;7,9
Classification signature	28 -6 6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 64, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,2) / (80,54,6)
κ_D	79.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128794709248880
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 252/324

Route 253

r8c1_h-1v-1

Cell 64 (8, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 2,9;8,1
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 253/324

Route 254

r8c1_h-1v+1

Cell 64 (8, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 2,9;8,1
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+--+00+--+0; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 254/324

Route 255

r8c1_h+1v-1

Cell 64 (8, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : yes 2,9;8,1
Classification signature	28 -6 10 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	++00--++00--; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	36, 4

Readers and action factors

K_D / K_Ω	(40,24,2) / (36,54,4)
κ_D	39.82508593118257305617326
κ_Ω	35.60638774694944330636281
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000024685259569118152

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 255/324

Route 256

r8c1_h+1v+1

Cell 64 (8, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; G
Primary route and inversion orbit	B_1 : no 2,9;8,1
Classification signature	28 -6 10 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 256/324

Route 257

r8c2_h-1v-1

Cell 65 (8, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,8;8,2
Classification signature	30 -6 4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 30 6 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 257/324

Route 258

r8c2_h-1v+1

Cell 65 (8, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,8;8,2
Classification signature	30 -6 4 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 9; orbit: 9; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(60,84,28) / (56,54,5)
κ_D	59.39013588401554063774148
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 258/324

Route 259

r8c2_h+1v-1

Cell 65 (8, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,8;8,2
Classification signature	30 -6 4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 18; orbit: 18; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(60,84,28) / (56,54,5)
κ_D	59.39013588401554063774148
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 259/324

Route 260

r8c2_h+1v+1

Cell 65 (8, 2) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : yes 2,8;8,2
Classification signature	30 -6 4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 -6 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 260/324

Route 261

r8c3_h-1v-1

Cell 66 (8, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,7;8,3
Classification signature	34 0 -8 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,4) / (56,54,5)
κ_D	79.51881951611803769439539
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 261/324

Route 262

r8c3_h-1v+1

Cell 66 (8, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : yes 2,7;8,3
Classification signature	34 0 -8 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 262/324

Route 263

r8c3_h+1v-1

Cell 66 (8, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,7;8,3
Classification signature	34 0 -8 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 263/324

Route 264

r8c3_h+1v+1

Cell 66 (8, 3) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,7;8,3
Classification signature	34 0 -8 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,4) / (56,54,5)
κ_D	79.51881951611803769439539
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 264/324

Route 265

r8c4_h-1v-1

Cell 67 (8, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,6;8,4
Classification signature	34 0 -4 2 -8 ++++-0++++-0

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,2) / (56,54,5)
κ_D	99.51859712327265341428729
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 265/324

Route 266

r8c4_h-1v+1

Cell 67 (8, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,6;8,4
Classification signature	34 0 -4 2 -8 ++++-0++++-0

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+ + 0 --- + + 0 ---; 50 -6 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 266/324

Route 267

r8c4_h+1v-1

Cell 67 (8, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,6;8,4
Classification signature	34 0 -4 2 -8 ++++-0++++-0

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++++-0++++-0; 50 6 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,24,2) / (56,54,5)
κ_D	39.82508593118257305617326
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 267/324

Route 268

r8c4_h+1v+1

Cell 67 (8, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : yes 2,6;8,4
Classification signature	34 0 -4 2 -8 ++++-0++++-0

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++++-0++++-0; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(100,66,2) / (56,54,5)
κ_D	99.51859712327265341428729
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 268/324

Route 269

r8c5_h-1v-1

Cell 68 (8, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,5;8,5
Classification signature	30 6 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 30 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 269/324

Route 270

r8c5_h-1v+1

Cell 68 (8, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : yes 2,5;8,5
Classification signature	30 6 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,84,28) / (80,54,6)
κ_D	59.39013588401554063774148
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 270/324

Route 271

r8c5_h+1v-1

Cell 68 (8, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,5;8,5
Classification signature	30 6 -4 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(60,84,28) / (80,54,6)
κ_D	59.39013588401554063774148
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 271/324

Route 272

r8c5_h+1v+1

Cell 68 (8, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,5;8,5
Classification signature	30 6 -4 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 30 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 272/324

Route 273

r8c6_h-1v-1

Cell 69 (8, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : yes 2,4;8,6
Classification signature	28 6 -6 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 28 -6 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 273/324

Route 274

r8c6_h-1v+1

Cell 69 (8, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,4;8,6
Classification signature	28 6 -6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 274/324

Route 275

r8c6_h+1v-1

Cell 69 (8, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,4;8,6
Classification signature	28 6 -6 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 275/324

Route 276

r8c6_h+1v+1

Cell 69 (8, 6) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; B
Primary route and inversion orbit	B_1 : no 2,4;8,6
Classification signature	28 6 -6 0 -12 +++----++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++----++----; 28 6 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 276/324

Route 277

r8c7_h-1v-1

Cell 70 (8, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,3;8,7
Classification signature	28 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16

Provenance and original codes

Complete record · 277/324

Route 278

r8c7_h-1v+1

Cell 70 (8, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,3;8,7
Classification signature	28 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++-; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 278/324

Route 279

r8c7_h+1v-1

Cell 70 (8, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : yes 2,3;8,7
Classification signature	28 -6 0 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++-----; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 20, 24, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,24,2) / (80,54,6)
κ_D	39.82508593118257305617326
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027610006159514215
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3
Stable triads per block	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 279/324

Route 280

r8c7_h+1v+1

Cell 70 (8, 7) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G3 / quark_down_G3
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	3; G3; G
Primary route and inversion orbit	B_1 : no 2,3;8,7
Classification signature	28 -6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 64, 68, 60, 40
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(100,66,2) / (80,54,6)
κ_D	99.51859712327265341428729
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000068994430825936417
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 280/324

Route 281

r8c8_h-1v-1

Cell 71 (8, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,2;8,8
Classification signature	24 0 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 18; orbit: 18; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Provenance and original codes

Complete record · 281/324

Route 282

r8c8_h-1v+1

Cell 71 (8, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,2;8,8
Classification signature	24 0 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -4, 2, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-0+++0-0; 50 4 -4 2 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	24, 4

Readers and action factors

K_D / K_Ω	(60,84,28) / (24,54,4)
κ_D	59.39013588401554063774148
κ_Ω	23.60638774694944330636281
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000016365878244659810

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 282/324

Route 283

r8c8_h+1v-1

Cell 71 (8, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : no 2,2;8,8
Classification signature	24 0 8 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 0; deferred blocks: 2
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 4, -2, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+0---0+0---; 50 -4 4 -2 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 28, 84, 36, 24
$\Omega_{90/120}, P_6$	24, 4

Readers and action factors

K_D / K_Ω	(60,84,28) / (24,54,4)
κ_D	59.39013588401554063774148
κ_Ω	23.60638774694944330636281
$R_{\text{act},D}$	1.000000041174098946741575
$R_{\text{act},\Omega}$	1.000000016365878244659810

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17

Provenance and original codes

Complete record · 283/324

Route 284

r8c8_h+1v+1

Cell 71 (8, 8) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_up_G3 / quark_up_G3
Collection and sector	up-type quark; quark
Physical classification	up-type quark
Generation, depth and colour	3; G3; R
Primary route and inversion orbit	B_1 : yes 2,2;8,8
Classification signature	24 0 8 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 9; orbit: 9; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, v_{120}, v_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 56, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,54,6) / (80,54,6)
κ_D	79.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055189812531859140
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	72; 36.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	4

Complete sequences

w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Provenance and original codes

Complete record · 284/324

Route 285

r8c9_h-1v-1

Cell 72 (8, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 2,1;8,9
Classification signature	28 6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243
Stable triads per block	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 285/324

Route 286

r8c9_h-1v+1

Cell 72 (8, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : yes 2,1;8,9
Classification signature	28 6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 286/324

Route 287

r8c9_h+1v-1

Cell 72 (8, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 2,1;8,9
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 287/324

Route 288

r8c9_h+1v+1

Cell 72 (8, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	quark_down_G4 / quark_down_G4
Collection and sector	down-type quark; quark
Physical classification	down-type quark
Generation, depth and colour	4; G4; B
Primary route and inversion orbit	B_1 : no 2,1;8,9
Classification signature	28 6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 68, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,4) / (80,54,6)
κ_D	79.51881951611803769439539
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128948890161264
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 288/324

Route 289

r9c1_h-1v-1

Cell 73 (9, 1) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,9;9,1
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 -6 0 -2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 289/324

Route 290

r9c1_h-1v+1

Cell 73 (9, 1) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,9;9,1
Classification signature	28 6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 290/324

Route 291

r9c1_h+1v-1

Cell 73 (9, 1) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,9;9,1
Classification signature	28 6 0 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (48,54,5)
κ_D	39.39035827686092491784959
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 291/324

Route 292

r9c1_h+1v+1

Cell 73 (9, 1) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,9;9,1
Classification signature	28 6 0 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 2, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++----; 28 6 0 2 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 292/324

Route 293

r9c2_h-1v-1

Cell 74 (9, 2) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,8;9,2
Classification signature	28 -6 6 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++--0+++--0; 28 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 293/324

Route 294

r9c2_h-1v+1

Cell 74 (9, 2) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,8;9,2
Classification signature	28 -6 6 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 294/324

Route 295

r9c2_h+1v-1

Cell 74 (9, 2) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,8;9,2
Classification signature	28 -6 6 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 295/324

Route 296

r9c2_h+1v+1

Cell 74 (9, 2) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,8;9,2
Classification signature	28 -6 6 -2 -8 ++0---++0---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++0---++0---; 28 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 296/324

Route 297

r9c3_h-1v-1

Cell 75 (9, 3) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,7;9,3
Classification signature	30 -6 0 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 30 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 297/324

Route 298

r9c3_h-1v+1

Cell 75 (9, 3) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,7;9,3
Classification signature	30 -6 0 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 50 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(0,24,0) / (56,54,5)
κ_D	-0.1751364616628112239348425
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	108; 46.52932501298079
First separation block	orbit: —; cell: —; route: —
Final ambiguity	6

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 298/324

Route 299

r9c3_h+1v-1

Cell 75 (9, 3) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,7;9,3
Classification signature	30 -6 0 -2 -8 0+ +---0+ +---

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 6, -2, 4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	++ +0--++ +0--; 50 6 -2 4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(0,24,0) / (56,54,5)
κ_D	-0.1751364616628112239348425
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	108; 46.52932501298079
First separation block	orbit: —; cell: —; route: —
Final ambiguity	6

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 299/324

Route 300

r9c3_h+1v+1

Cell 75 (9, 3) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,7;9,3
Classification signature	30 -6 0 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, -4, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 30 -6 2 -4 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (56,54,5)
κ_D	39.60661013979482758647091
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 300/324

Route 301

r9c4_h-1v-1

Cell 76 (9, 4) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,6;9,4
Classification signature	34 0 -12 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 301/324

Route 302

r9c4_h-1v+1

Cell 76 (9, 4) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,6;9,4
Classification signature	34 0 -12 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 8 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 302/324

Route 303

r9c4_h+1v-1

Cell 76 (9, 4) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,6;9,4
Classification signature	34 0 -12 0 -12 +++-+++-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	8, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 8 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,84,30) / (80,54,6)
κ_D	39.39035827686092491784959
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 303/324

Route 304

r9c4_h+1v+1

Cell 76 (9, 4) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,6;9,4
Classification signature	34 0 -12 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 34 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 304/324

Route 305

r9c5_h-1v-1

Cell 77 (9, 5) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,5;9,5
Classification signature	34 0 -8 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---+++0--; 34 0 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	30; 8.0
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 305/324

Route 306

r9c5_h-1v+1

Cell 77 (9, 5) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,5;9,5
Classification signature	34 0 -8 -2 -8 0++++-0+----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, 2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++----; 44 2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 306/324

Route 307

r9c5_h+1v-1

Cell 77 (9, 5) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,5;9,5
Classification signature	34 0 -8 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	44, -2, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 44 -2 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 307/324

Route 308

r9c5_h+1v+1

Cell 77 (9, 5) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,5;9,5
Classification signature	34 0 -8 -2 -8 0++++-0++++

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	34, 0, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0++++-0++++; 34 0 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (56,54,5)
κ_D	79.51870831969534555434134
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 308/324

Route 309

r9c6_h-1v-1

Cell 78 (9, 6) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,4;9,6
Classification signature	30 6 0 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 30 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (48,54,5)
κ_D	39.60661013979482758647091
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	18; 0.0
First separation block	orbit: 1; cell: 1; route: 1
Final ambiguity	1

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 309/324

Route 310

r9c6_h-1v+1

Cell 78 (9, 6) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,4;9,6
Classification signature	30 6 0 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 14 6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(0,24,0) / (80,54,6)
κ_D	-0.1751364616628112239348425
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 310/324

Route 311

r9c6_h+1v-1

Cell 78 (9, 6) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,4;9,6
Classification signature	30 6 0 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -6, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++; 14 -6 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(0,24,0) / (80,54,6)
κ_D	-0.1751364616628112239348425
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	0.999999998785810851337881
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: 1; cell: 1; route: —
Final ambiguity	2

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 311/324

Route 312

r9c6_h+1v+1

Cell 78 (9, 6) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,4;9,6
Classification signature	30 6 0 0 -8 +++0-+++0-

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	30, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 30 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(40,54,6) / (48,54,5)
κ_D	39.60661013979482758647091
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	26; 6.339850002884624
First separation block	orbit: 5; cell: 5; route: 5
Final ambiguity	1

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 312/324

Route 313

r9c7_h-1v-1

Cell 79 (9, 7) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,3;9,7
Classification signature	28 6 -2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 18; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 1
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 0, -4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++++-; 28 -6 0 -4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16

Provenance and original codes

Complete record · 313/324

Route 314

r9c7_h-1v+1

Cell 79 (9, 7) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,3;9,7
Classification signature	28 6 -2 0 -12 +++---++----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, 4, -2, 2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0---++0--; 50 4 -2 2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Provenance and original codes

Complete record · 314/324

Route 315

r9c7_h+1v-1

Cell 79 (9, 7) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,3;9,7
Classification signature	28 6 -2 0 -12 +++++-----

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 3; final: 1; deferred blocks: 3
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	50, -4, 2, -2, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	0+----0+----; 50 -4 2 -2 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	84, 24, 84, 24, 16
$\Omega_{90/120}, P_6$	56, 5

Readers and action factors

K_D / K_Ω	(40,84,30) / (56,54,5)
κ_D	39.39035827686092491784959
κ_Ω	55.60649894337213544641686
$R_{\text{act},D}$	1.000000027308617396708705
$R_{\text{act},\Omega}$	1.000000038550972354141645

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16

Provenance and original codes

Complete record · 315/324

Route 316

r9c7_h+1v+1

Cell 79 (9, 7) Orientation $h = 1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,3;9,7
Classification signature	28 6 -2 0 -12 +++---++---

Returns, memory and transport

Returns with phase	class: 9; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, 0, 4, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++---++---; 28 6 0 4 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(80,66,3) / (80,54,6)
κ_D	79.51870831969534555434134
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	54; 28.5293250129808
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 316/324

Route 317

r9c8_h-1v-1

Cell 80 (9, 8) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,2;9,8
Classification signature	28 -6 4 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, 6, -2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+++0-+++0-; 28 6 -2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (48,54,5)
κ_D	79.51870831969534555434134
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	40; 20.339850002884624
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Provenance and original codes

Complete record · 317/324

Route 318

r9c8_h-1v+1

Cell 80 (9, 8) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,2;9,8
Classification signature	28 -6 4 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, -4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 -4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Provenance and original codes

Complete record · 318/324

Route 319

r9c8_h+1v-1

Cell 80 (9, 8) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,2;9,8
Classification signature	28 -6 4 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	14, 4, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++-+++-; 14 4 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	78, 24, 78, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,78,27) / (80,54,6)
κ_D	39.43380880300855130367114
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027338740854894358
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 319/324

Route 320

r9c8_h+1v+1

Cell 80 (9, 8) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	neutrino_G4 / neutrino_G4
Collection and sector	neutrino; lepton
Physical classification	neutrino
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,2;9,8
Classification signature	28 -6 4 0 -8 +0+---+0+---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; yes
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	28, -6, 2, 0, -8
τ_{12} ; historical signature $\mathbb{Z}^5$	+0+---+0+---; 28 -6 2 0 -8
$D_1, D_3, D_4, D_{27}, D_{36}$	66, 60, 66, 48, 32
$\Omega_{90/120}, P_6$	48, 5

Readers and action factors

K_D / K_Ω	(80,66,3) / (48,54,5)
κ_D	79.51870831969534555434134
κ_Ω	47.60649894337213544641686
$R_{\text{act},D}$	1.000000055128871799705072
$R_{\text{act},\Omega}$	1.000000033004718053243084

Prefix separation

Ambiguity area; $\log_2$ area	36; 18.0
First separation block	orbit: —; cell: —; route: —
Final ambiguity	2

Complete sequences

w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 320/324

Route 321

r9c9_h-1v-1

Cell 81 (9, 9) Orientation $h = -1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,1;9,9
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 1; final: 0; deferred blocks: 1
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,54,6) / (80,54,6)
κ_D	39.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17

Provenance and original codes

Complete record · 321/324

Route 322

r9c9_h-1v+1

Cell 81 (9, 9) Orientation $h = -1$, $v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : yes 1,1;9,9
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+00-00+00-0; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	24, 2

Readers and action factors

K_D / K_Ω	(0,24,0) / (24,54,2)
κ_D	-0.1751364616628112239348425
κ_Ω	23.60616535410405902625471
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000016365724063753403

Prefix separation

Ambiguity area; $\log_2$ area	108; 46.52932501298079
First separation block	orbit: —; cell: —; route: —
Final ambiguity	6

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 322/324

Route 323

r9c9_h+1v-1

Cell 81 (9, 9) Orientation $h = 1$, $v = -1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,1;9,9
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 0; final: 0; deferred blocks: 0
Final triads; Pareto persistence	17; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	20, 0, 0, 0, -4
τ_{12} ; historical signature $\mathbb{Z}^5$	0+00-00+00-0; 20 0 0 0 -4
$D_1, D_3, D_4, D_{27}, D_{36}$	24, 24, 24, 0, 0
$\Omega_{90/120}, P_6$	24, 2

Readers and action factors

K_D / K_Ω	(0,24,0) / (24,54,2)
κ_D	-0.1751364616628112239348425
κ_Ω	23.60616535410405902625471
$R_{\text{act},D}$	0.9999999998785810851337881
$R_{\text{act},\Omega}$	1.000000016365724063753403

Prefix separation

Ambiguity area; $\log_2$ area	108; 46.52932501298079
First separation block	orbit: —; cell: —; route: —
Final ambiguity	6

Complete sequences

w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Provenance and original codes

Complete record · 323/324

Route 324

r9c9_h+1v+1

Cell 81 (9, 9) Orientation $h = 1, v = 1$

Index

Structural identity

Class and multisection	charged_lepton_G4 / charged_lepton_G4
Collection and sector	charged lepton; lepton
Physical classification	charged lepton
Generation, depth and colour	4; G4; —
Primary route and inversion orbit	B_1 : no 1,1;9,9
Classification signature	24 0 12 0 -12 +++++---

Returns, memory and transport

Returns with phase	class: 27; orbit: 27; cell: 27
Kinematic return	within 108: 54; extension: 216
Turns and sheet switches	71; 107
Carry and deferral	total debt: 2; final: 1; deferred blocks: 2
Final triads; Pareto persistence	16; no
$K \longrightarrow K + 9$	repetitions: 9; memory changes: 9

Signature, phase and differences

$n_A, s_C, \kappa_0, \nu_{120}, \nu_{270}$	24, 0, 0, 0, -12
τ_{12} ; historical signature $\mathbb{Z}^5$	+++++---; 24 0 0 0 -12
$D_1, D_3, D_4, D_{27}, D_{36}$	54, 60, 72, 24, 16
$\Omega_{90/120}, P_6$	80, 6

Readers and action factors

K_D / K_Ω	(40,54,6) / (80,54,6)
κ_D	39.60661013979482758647091
κ_Ω	79.60661013979482758647091
$R_{\text{act},D}$	1.000000027458540873559518
$R_{\text{act},\Omega}$	1.000000055189812531859140

Prefix separation

Ambiguity area; $\log_2$ area	58; 30.189475010096174
First separation block	orbit: —; cell: —; route: —
Final ambiguity	3

Complete sequences

w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423
Stable triads per block	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16

Provenance and original codes

Complete record · 324/324

4 Metrological inventory: 471 masses and 384 widths

The following tables catalogue external observations; they do not count HMT predictions. They preserve the PDG 2026 documentary snapshot dated 2026-01-15, as recorded in the preserved files. The consultation date recorded in the integral is 2026-07-26. Neither the edition nor the interpretation of its observables has been updated during this transcription. The stable identifier distinguishes different records of the same species; repetitions are not removed, and mass, pole, width and estimate are not merged.

The value column preserves both the literal expression and, where it differs, its original display. An uncertainty is not reconstructed from decimal digits, nor is an interval converted into a standard deviation. Positive and negative uncertainties remain separate, together with the unit, limit type and available confidence level. ND abbreviates UNAVAILABLE; NA abbreviates NOT_APPLICABLE. A dash denotes an empty field. The original types NONE, R and U are preserved as source codes alongside the value expression: respectively, no limit flag, range and upper limit. No confidence level is inferred where the record does not provide one.

In this snapshot, all 855 records indicate that scheme and scale are unavailable. All preserve the condition that comparison follows the freezing of the internal result, and state that the external value was not used as a selector. The status keys *E* and map keys *M* are reversible editorial abbreviations of the source statuses and maps, described in the next two legends. The word “open” in those legends reproduces the documentary status of that snapshot; it is not a new mathematical revision.

Table 6: Preserved statuses of the external inventory.

Key	Literal status	Documentary reading
E01	CONDITIONED_NOMINAL_SIGNATURE__ROUTE_AND_POLE_REALIZATION_OPEN	Conditional nominal signature; route and pole realization open.
E02	EXACT_COLOR_FIBER_OPERATOR__NOMINAL_SIGNATURE_CONDITIONED__SCHEME_OPEN	Exact color-fiber operator; conditional nominal signature; scheme open.
E03	EXACT_FIBER_OPERATOR__NOMINAL_SIGNATURE_CONDITIONED__SCALARIZATION_OPEN	Exact fiber operator; conditional nominal signature; scalarization open.
E04	EXTERNAL_RECORD_ACCOUNTED__INDIVIDUAL_HMT_REALIZATION_NOT_YET_JOINED	External record inventoried; individual HMT realization not yet linked at this stage.
E05	NULL_SECTION_CLOSED	Null section closed.
E06	SCALAR_CLOSED_ELECTRON_BETA	Electronic beta scalar closed.

Table 7: Realization maps recorded in the inventory.

Key	Literal status	Documentary reading
M01	NONE	None, according to the record.
M02	NONE_FOR_MASS_NULLITY	None for mass nullity, according to the record.
M03	flavor_color_coupling_to_spectral_projection_then_scheme_scale_transport	Flavor-color coupling to the spectral projection and scheme-and-scale transport.
M04	generation_coupling_to_spectral_projection_without_mass_target	Generational coupling to the spectral projection without a target mass.
M05	identity_to_public_sector_or_new_typed_sector	Identity and identified sector or new typed sector.

Continued on the next page

Continuation

Key	Literal status	Documentary reading
M06	join_named_boson_to_enriched_route_then_complex_pole_operator	Individually identified boson, memory-bearing route and complex-pole operator.
M07	named_baryon_binding_map	Binding map for the individually identified baryon.
M08	named_meson_binding_map	Binding map for the individually identified meson.
M09	quark_identity_scheme_scale_map	Quark identity, scheme and scale.
M10	typed_gravitational_representation_map	Typed gravitational representation.

4.1 Masses: 471 records

Identity and species	Value and unit	Uncertainty	Sector and status
1. Sigma(1580) MASS Sigma(1580)- Sigmar(1580)+ Sigma(1580)0 Sigmar(1580)0 Sigma(1580)+ Sigmar(1580)- PDG2026:observable:B000:B000M:349891	~1580 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
2. N(1895) BREIT-WIGNER MASS N(1895)0 Nbar(1895)0 N(1895)+ Nbar(1895)- PDG2026:observable:B004:B004M:348705	1870 to 1895 to 1920 MeV Display: 1870 to 1920 (~1895)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
3. N(2060) BREIT-WIGNER MASS N(2060)0 Nbar(2060)0 N(2060)+ Nbar(2060)- PDG2026:observable:B005:B005M:348836	2030 to 2100 to 2200 MeV Display: 2030 to 2200 (~2100)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
4. Lambda(2585) MASS (BUMPS) Lambda(2585)0 Lambdabar(2585)0 PDG2026:observable:B007:B007M:349780	~2585 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
5. Delta(1700) BREIT-WIGNER MASS Delta(1700)- Deltabar(1700)+ Delta(1700)0 Deltabar(1700)0 Delta(1700)+ Deltabar(1700)- Delta(1700)++ Deltabar(1700)-- PDG2026:observable:B010:B010M:349097	1690 to 1710 to 1730 MeV Display: 1690 to 1730 (~1710)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
6. Delta(1905) BREIT-WIGNER MASS Delta(1905)- Deltabar(1905)+ Delta(1905)0 Deltabar(1905)0 Delta(1905)+ Deltabar(1905)- Delta(1905)++ Deltabar(1905)-- PDG2026:observable:B011:B011M:349151	1855 to 1880 to 1910 MeV Display: 1855 to 1910 (~1880)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
7. Delta(1910) BREIT-WIGNER MASS Delta(1910)- Deltabar(1910)+ Delta(1910)0 Deltabar(1910)0 Delta(1910)+ Deltabar(1910)- Delta(1910)++ Deltabar(1910)-- PDG2026:observable:B012:B012M:349182	1850 to 1900 to 1950 MeV Display: 1850 to 1950 (~1900)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
8. Delta(1930) BREIT-WIGNER MASS Delta(1930)- Deltabar(1930)+ Delta(1930)0 Deltabar(1930)0 Delta(1930)+ Deltabar(1930)- Delta(1930)++ Deltabar(1930)-- PDG2026:observable:B013:B013M:349249	1900 to 1950 to 2000 MeV Display: 1900 to 2000 (~1950)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
9. N(1710) BREIT-WIGNER MASS N(1710)0 Nbar(1710)0 N(1710)+ Nbar(1710)- PDG2026:observable:B014:B014M:348535	1680 TO 1710 TO 1740 MeV Display: 1680 to 1740 (~1710)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
10. N(1720) BREIT-WIGNER MASS N(1720)0 Nbar(1720)0 N(1720)+ Nbar(1720)- PDG2026:observable:B015:B015M:348569	1680 to 1720 to 1750 MeV Display: 1680 to 1750 (~1720)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
11. N(1875) BREIT-WIGNER MASS N(1875)0 Nbar(1875)0 N(1875)+ Nbar(1875)- PDG2026:observable:B016:B016M:348633	1850 to 1875 to 1920 MeV Display: 1850 to 1920 (~1875)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
12. N(1990) BREIT-WIGNER MASS N(1990)0 Nbar(1990)0 N(1990)+ Nbar(1990)- PDG2026:observable:B017:B017M:348778	1950 to 2020 to 2100 MeV Display: 1950 to 2100 (~2020)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
13. N(1700) BREIT-WIGNER MASS N(1700)0 Nbar(1700)0 N(1700)+ Nbar(1700)- PDG2026:observable:B018:B018M:348499	1650 to 1720 to 1800 MeV Display: 1650 to 1800 (~1720)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
14. Delta(1600) BREIT-WIGNER MASS Delta(1600)- Deltabar(1600)+ Delta(1600)0 Deltabar(1600)0 Delta(1600)+ Deltabar(1600)- Delta(1600)++ Deltabar(1600)-- PDG2026:observable:B019:B019M:349050	1500 to 1570 to 1640 MeV Display: 1500 to 1640 (~1570)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
15. Xi(1620) MASS Xi(1620)- Xibar(1620)+ Xi(1620)0 Xibar(1620)0 PDG2026:observable:B021:B021M:350410	~1620 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
16. Xi(2250) MASS Xi(2250)- Xibar(2250)+ Xi(2250)0 Xibar(2250)0 PDG2026:observable:B022:B022M:350457	~2250 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
17. N(2120) BREIT-WIGNER MASS N(2120)0 Nbar(2120)0 N(2120)+ Nbar(2120)- PDG2026:observable:B024:B024M:348906	2060 to 2120 to 2160 MeV Display: 2060 to 2160 (~2120)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
18. Sigma(2100) MASS Sigma(2100)- Sigmabar(2100)+ Sigma(2100)0 Sigmabar(2100)0 Sigma(2100)+ Sigmabar(2100)- PDG2026:observable:B026:B026M:350202	~2100 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
19. Lambda(2085) MASS Lambda(2085)0 Lambdabar(2085)0 PDG2026:observable:B027:B027M:349731	~2020 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
20. Delta(1900) BREIT-WIGNER MASS Delta(1900)- Deltabar(1900)+ Delta(1900)0 Deltabar(1900)0 Delta(1900)+ Deltabar(1900)- Delta(1900)++ Deltabar(1900)-- PDG2026:observable:B030:B030M:349126	1840 TO 1860 TO 1920 MeV Display: 1840 to 1920 (~1860)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
21. Sigma(1620) MASS Sigma(1620)- Sigmabar(1620)+ Sigma(1620)0 Sigmabar(1620)0 Sigma(1620)+ Sigmabar(1620)- PDG2026:observable:B032:B032M:349902	1600 to 1620 to 1650 MeV Display: 1600 to 1650 (~1620)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
22. MIXED CHARGES Delta(1232)- Deltabar(1232)+ Delta(1232)0 Deltabar(1232)0 Delta(1232)+ Deltabar(1232)- Delta(1232)++ Deltabar(1232)-- PDG2026:observable:B033:B033M:349034	1230 TO 1232 TO 1234 MeV Display: 1230 to 1234 (~1232)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
23. Sigma(2070) MASS Sigma(2070)- Sigmabar(2070)+ Sigma(2070)0 Sigmabar(2070)0 Sigma(2070)+ Sigmabar(2070)- PDG2026:observable:B034:B034M:350186	2020 to 2060 to 2100 MeV Display: 2020 to 2100 (~2060)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
24. Lambda(2110) MASS Lambda(2110)0 Lambdabar(2110)0 PDG2026:observable:B035:B035M:349763	2050 to 2090 to 2130 MeV Display: 2050 to 2130 (~2090)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
25. Lambda(1800) MASS Lambda(1800)0 Lambdabar(1800)0 PDG2026:observable:B036:B036M:349555	1750 to 1800 to 1850 MeV Display: 1750 to 1850 (~1800)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
26. PRODUCTION EXPERIMENTS Lambda(1405)0 Lambdabar(1405)0 PDG2026:observable:B037:B037M1:349427	1405.1+1.3-1.0 MeV	u_+ : 1.31196935395758 21 u_- : 0.96455487987763 788 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
27. Lambda(1520) MASS Lambda(1520)0 Lambdabar(1520)0 PDG2026:observable:B038:B038M:349437	1518 to 1519 to 1520 MeV Display: 1518 to 1520 (~1519)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
28. Lambda(1820) MASS Lambda(1820)0 Lambdabar(1820)0 PDG2026:observable:B039:B039M:349594	1815 TO 1820 TO 1825 MeV Display: 1815 to 1825 (~1820)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
29. Lambda(1670) MASS Lambda(1670)0 Lambdabar(1670)0 PDG2026:observable:B040:B040M:349494	1670 to 1674 to 1678 MeV Display: 1670 to 1678 (~1674)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
30. Lambda(2100) MASS Lambda(2100)0 Lambdabar(2100)0 PDG2026:observable:B041:B041M:349741	2090 TO 2100 TO 2110 MeV Display: 2090 to 2110 (~2100)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
31. Lambda(2350) MASS Lambda(2350)0 Lambdabar(2350)0 PDG2026:observable:B042:B042M:349775	2340 TO 2350 TO 2370 MeV Display: 2340 to 2370 (~2350)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
32. Sigma(1385)+ MASS Sigma(1385)- Sigmapar(1385)+ Sigma(1385)0 Sigmapar(1385)0 Sigma(1385)+ Sigmapar(1385)- PDG2026:observable:B043:B043M+:349876	1382.83+-0.34 MeV	u_+ : 0.33905435517601 018 u_- : 0.34419099315609 542 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
33. Sigma(1385)0 MASS Sigma(1385)- Sigmapar(1385)+ Sigma(1385)0 Sigmapar(1385)0 Sigma(1385)+ Sigmapar(1385)- PDG2026:observable:B043:B043M0:349877	1383.8+-0.9 MeV	u_+ : 0.93514404403237 228 u_- : 0.93514404403237 228 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
34. Sigma(1385)- MASS Sigma(1385)- Sigmapar(1385)+ Sigma(1385)0 Sigmapar(1385)0 Sigma(1385)+ Sigmapar(1385)- PDG2026:observable:B043:B043M-:349878	1387.2+-0.5 MeV	u_+ : 0.54653703280513 61 u_- : 0.54653703280513 61 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
35. Sigma(1670) MASS Sigma(1670)- Sigmabar(1670)+ Sigma(1670)0 Sigmabar(1670)0 Sigma(1670)+ Sigmabar(1670)- PDG2026:observable:B044:B044M:349946	1665 to 1675 to 1685 MeV Display: 1665 to 1685 (~1675)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
36. Sigma(1775) MASS Sigma(1775)- Sigmabar(1775)+ Sigma(1775)0 Sigmabar(1775)0 Sigma(1775)+ Sigmabar(1775)- PDG2026:observable:B045:B045M:349983	1770 TO 1775 TO 1780 MeV Display: 1770 to 1780 (~1775)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
37. Sigma(1915) MASS Sigma(1915)- Sigmabar(1915)+ Sigma(1915)0 Sigmabar(1915)0 Sigma(1915)+ Sigmabar(1915)- PDG2026:observable:B046:B046M:350080	1900 TO 1915 TO 1935 MeV Display: 1900 to 1935 (~1915)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
38. Sigma(2030) MASS Sigma(2030)- Sigmabar(2030)+ Sigma(2030)0 Sigmabar(2030)0 Sigma(2030)+ Sigmabar(2030)- PDG2026:observable:B047:B047M:350166	2025 TO 2030 TO 2040 MeV Display: 2025 to 2040 (~2030)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
39. Sigma(2250) MASS Sigma(2250)- Sigmabar(2250)+ Sigma(2250)0 Sigmabar(2250)0 Sigma(2250)+ Sigmabar(2250)- PDG2026:observable:B048:B048M:350277	2210 TO 2250 TO 2280 MeV Display: 2210 to 2280 (~2250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
40. Xi(1530)0 MASS Xi(1530)- Xibar(1530)+ Xi(1530)0 Xibar(1530)0 PDG2026:observable:B049:B049M0:350399	1531.80+-0.32 MeV	u_+ : 0.31575272391181 958 u_- : 0.31575272391181 958 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
41. Xi(1530)- MASS Xi(1530)- Xibar(1530)+ Xi(1530)0 Xibar(1530)0 PDG2026:observable:B049:B049M-:350401	1535.0+-0.6 MeV	u_+ : 0.63202166316087 116 u_- : 0.63202166316087 116 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
42. Xi(1820) MASS Xi(1820)- Xibar(1820)+ Xi(1820)0 Xibar(1820)0 PDG2026:observable:B050:B050M:350420	1823+-5 MeV	u_+ : 5 u_- : 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
43. Xi(1950) MASS Xi(1950)- Xibar(1950)+ Xi(1950)0 Xibar(1950)0 PDG2026:observable:B052:B052M:350437	1950+-15 MeV	u_+ : 15 u_- : 15 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
44. Sigma(2455) MASS Sigma(2455)- Sigmabar(2455)+ Sigma(2455)0 Sigmabar(2455)0 Sigma(2455)+ Sigmabar(2455)- PDG2026:observable:B053:B053M:350283	~2455 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
45. Sigma(2620) MASS Sigma(2620)- Sigmabar(2620)+ Sigma(2620)0 Sigmabar(2620)0 Sigma(2620)+ Sigmabar(2620)- PDG2026:observable:B054:B054M:350284	~2620 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
46. Lambda(1690) MASS Lambda(1690)0 Lambdabar(1690)0 PDG2026:observable:B055:B055M:349518	1685 to 1690 to 1695 MeV Display: 1685 to 1695 (~1690)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
47. Lambda(1830) MASS Lambda(1830)0 Lambdabar(1830)0 PDG2026:observable:B056:B056M:349614	1820 to 1825 to 1830 MeV Display: 1820 to 1830 (~1825)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
48. Sigma(1750) MASS Sigma(1750)- Sigmapar(1750)+ Sigma(1750)0 Sigmapar(1750)0 Sigma(1750)+ Sigmapar(1750)- PDG2026:observable:B057:B057M:349966	1700 to 1750 to 1800 MeV Display: 1700 to 1800 (~1750)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
49. Sigma(3000) MASS Sigma(3000)- Sigmapar(3000)+ Sigma(3000)0 Sigmapar(3000)0 Sigma(3000)+ Sigmapar(3000)- PDG2026:observable:B059:B059M:350285	~3000 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
50. Lambda(1890) MASS Lambda(1890)0 Lambdabar(1890)0 PDG2026:observable:B060:B060M:349635	1870 to 1890 to 1910 MeV Display: 1870 to 1910 (~1890)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
51. N(1440) BREIT-WIGNER MASS N(1440)0 Nbar(1440)0 N(1440)+ Nbar(1440)- PDG2026:observable:B061:B061M:348326	1410 to 1440 to 1470 MeV Display: 1410 to 1470 (~1440)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
52. N(1520) BREIT-WIGNER MASS N(1520)0 Nbar(1520)0 N(1520)+ Nbar(1520)- PDG2026:observable:B062:B062M:348349	1510 TO 1515 TO 1520 MeV Display: 1510 to 1520 (~1515)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
53. N(1535) BREIT-WIGNER MASS N(1535)0 Nbar(1535)0 N(1535)+ Nbar(1535)- PDG2026:observable:B063:B063M:348383	1515 to 1530 to 1545 MeV Display: 1515 to 1545 (~1530)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
54. N(1675) BREIT-WIGNER MASS N(1675)0 Nbar(1675)0 N(1675)+ Nbar(1675)- PDG2026:observable:B064:B064M:348435	1665 to 1675 to 1680 MeV Display: 1665 to 1680 (~1675)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
55. N(1680) BREIT-WIGNER MASS N(1680)0 Nbar(1680)0 N(1680)+ Nbar(1680)- PDG2026:observable:B065:B065M:348467	1680 TO 1685 TO 1690 MeV Display: 1680 to 1690 (~1685)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
56. N(1650) BREIT-WIGNER MASS N(1650)0 Nbar(1650)0 N(1650)+ Nbar(1650)- PDG2026:observable:B066:B066M:348410	1635 to 1650 to 1665 MeV Display: 1635 to 1665 (~1650)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
57. Sigma(1880) MASS Sigma(1880)- Sigmapar(1880)+ Sigma(1880)0 Sigmapar(1880)0 Sigma(1880)+ Sigmapar(1880)- PDG2026:observable:B067:B067M:349999	1820 to 1880 to 1940 MeV Display: 1820 to 1940 (~1880)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
58. Xi(2030) MASS Xi(2030)- Xibar(2030)+ Xi(2030)0 Xibar(2030)0 PDG2026:observable:B068:B068M:350443	2025+-5 MeV	u_+ : 5 u_- : 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
59. N(2190) BREIT-WIGNER MASS N(2190)0 Nbar(2190)0 N(2190)+ Nbar(2190)- PDG2026:observable:B071:B071M:348953	2140 to 2180 to 2220 MeV Display: 2140 to 2220 (~2180)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
60. Lambda(1810) MASS Lambda(1810)0 Lambdabar(1810)0 PDG2026:observable:B077:B077M:349576	1740 to 1790 to 1840 MeV Display: 1740 to 1840 (~1790)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
61. Sigma(1660) MASS Sigma(1660)- Sigmabar(1660)+ Sigma(1660)0 Sigmabar(1660)0 Sigma(1660)+ Sigmabar(1660)- PDG2026:observable:B079:B079M:349920	1640 to 1660 to 1680 MeV Display: 1640 to 1680 (~1660)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
62. Delta(1620) BREIT-WIGNER MASS Delta(1620)- Deltabar(1620)+ Delta(1620)0 Deltabar(1620)0 Delta(1620)+ Deltabar(1620)- Delta(1620)++ Deltabar(1620)-- PDG2026:observable:B082:B082M:349078	1590 to 1610 to 1630 MeV Display: 1590 to 1630 (~1610)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
63. Delta(1950) BREIT-WIGNER MASS Delta(1950)- Deltabar(1950)+ Delta(1950)0 Deltabar(1950)0 Delta(1950)+ Deltabar(1950)- Delta(1950)++ Deltabar(1950)-- PDG2026:observable:B083:B083M:349308	1915 to 1930 to 1950 MeV Display: 1915 to 1950 (~1930)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
64. Delta(2420) BREIT-WIGNER MASS Delta(2420)- Deltabar(2420)+ Delta(2420)0 Deltabar(2420)0 Delta(2420)+ Deltabar(2420)- Delta(2420)++ Deltabar(2420)-- PDG2026:observable:B084:B084M:349359	2300 to 2450 to 2600 MeV Display: 2300 to 2600 (~2450)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
65. N(1880) BREIT-WIGNER MASS N(1880)0 Nbar(1880)0 N(1880)+ Nbar(1880)- PDG2026:observable:B087:B087M:348671	1830 to 1880 to 1930 MeV Display: 1830 to 1930 (~1880)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
66. Sigma(2080) MASS Sigma(2080)- Sigmabar(2080)+ Sigma(2080)0 Sigmabar(2080)0 Sigma(2080)+ Sigmabar(2080)- PDG2026:observable:B088:B088M:350188	2060 to 2090 to 2120 MeV Display: 2060 to 2120 (~2090)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
67. Lambda(2000) MASS Lambda(2000)0 Lambdabar(2000)0 PDG2026:observable:B089:B089M:349646	~2000 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
68. N(2220) BREIT-WIGNER MASS N(2220)0 Nbar(2220)0 N(2220)+ Nbar(2220)- PDG2026:observable:B090:B090M:348986	2200 TO 2250 TO 2300 MeV Display: 2200 to 2300 (~2250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
69. Sigma(1910) MASS Sigma(1910)- Sigmabar(1910)+ Sigma(1910)0 Sigmabar(1910)0 Sigma(1910)+ Sigmabar(1910)- PDG2026:observable:B098:B098M:350050	1870 to 1910 to 1950 MeV Display: 1870 to 1950 (~1910)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
70. Xi(2500) MASS Xi(2500)- Xibar(2500)+ Xi(2500)0 Xibar(2500)0 PDG2026:observable:B099:B099M:350464	~2500 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
71. Lambda(1600) MASS Lambda(1600)0 Lambdabar(1600)0 PDG2026:observable:B101:B101M:349472	1570 to 1600 to 1630 MeV Display: 1570 to 1630 (~1600)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
72. Lambda_c(2625)+ MASS Lambda_c(2625)+ Lambdabar_c(2625)- PDG2026:observable:B102:B102M:350934	2628.00+-0.15 MeV	u_+ : 0.14735968814977 221 u_- : 0.14735968814977 221 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
73. Xi(2120) MASS Xi(2120)- Xibar(2120)+ Xi(2120)0 Xibar(2120)0 PDG2026:observable:B103:B103M:350454	~2120 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
74. Sigma_c(2455)++ MASS Sigma_c(2455)0 Sigmapar_c(2455)0 Sigma_c(2455)+ Sigmapar_c(2455)- Sigma_c(2455)++ Sigmapar_c(2455)-- PDG2026:observable:B104:B104M++:350985	2453.97+-0.14 MeV	u_+ : 0.14100978554010 379 u_- : 0.14100978554010 379 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
75. Sigma_c(2455)+ MASS Sigma_c(2455)0 Sigmapar_c(2455)0 Sigma_c(2455)+ Sigmapar_c(2455)- Sigma_c(2455)++ Sigmapar_c(2455)-- PDG2026:observable:B104:B104M+:350986	2452.65+0.22-0.16 MeV	u_+ : 0.21793257206126 77 u_- : 0.15750625023659 159 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
76. Sigma_c(2455)0 MASS Sigma_c(2455)0 Sigmapar_c(2455)0 Sigma_c(2455)+ Sigmapar_c(2455)- Sigma_c(2455)++ Sigmapar_c(2455)-- PDG2026:observable:B104:B104M0:350987	2453.75+-0.14 MeV	u_+ : 0.14100937832457 261 u_- : 0.14100937832457 261 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
77. MIXED CHARGES Xi(1690)- Xibar(1690)+ Xi(1690)0 Xibar(1690)0 PDG2026:observable:B105:B105M:350412	1690+-10 MeV	u_+ : 10 u_- : 10 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
78. Lambda(2325) MASS Lambda(2325)0 Lambdabar(2325)0 PDG2026:observable:B112:B112M:349774	~2325 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
79. N(2250) BREIT-WIGNER MASS N(2250)0 Nbar(2250)0 N(2250)+ Nbar(2250)- PDG2026:observable:B113:B113M:349004	2250 to 2280 to 2320 MeV Display: 2250 to 2320 (~2280)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
80. Sigma_c(2520)++ MASS Sigma_c(2520)0 Sigmapar_c(2520)0 Sigma_c(2520)+ Sigmapar_c(2520)- Sigma_c(2520)++ Sigmapar_c(2520)-- PDG2026:observable:B115:B115M++:351001	2518.41+-0.22 MeV	u_+ : 0.22441420873039 311 u_- : 0.22441420873039 311 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
81. Sigma_c(2520)+ MASS Sigma_c(2520)0 Sigmapar_c(2520)0 Sigma_c(2520)+ Sigmapar_c(2520)- Sigma_c(2520)++ Sigmapar_c(2520)-- PDG2026:observable:B115:B115M+:351002	2517.4+0.7-0.5 MeV	u_+ : 0.68505102093303 338 u_- : 0.52419384686487 569 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
82. Sigma_c(2520)0 MASS Sigma_c(2520)0 Sigambar_c(2520)0 Sigma_c(2520)+ Sigambar_c(2520)- Sigma_c(2520)++ Sigambar_c(2520)-- PDG2026:observable:B115:B115M0:351003	2518.48+-0.21 MeV	u_+ : 0.20795724066290 969 u_- : 0.20795724066290 969 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
83. Delta(1920) BREIT-WIGNER MASS Delta(1920)- Deltabar(1920)+ Delta(1920)0 Deltabar(1920)0 Delta(1920)+ Deltabar(1920)- Delta(1920)++ Deltabar(1920)-- PDG2026:observable:B117:B117M:349208	1870 to 1920 to 1970 MeV Display: 1870 to 1970 (~1920)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
84. Sigma(3170) MASS (PRODUCTION EXPERIMENTS) Sigma(3170)- Sigambar(3170)+ Sigma(3170)0 Sigambar(3170)0 Sigma(3170)+ Sigambar(3170)- PDG2026:observable:B118:B118M:350286	~3170 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
85. Lambda_c(2595)+ MASS Lambda_c(2595)+ Lambdabar_c(2595)- PDG2026:observable:B119:B119M:350917	2592.25+-0.28 MeV	u_+ : 0.28142494558940 517 u_- : 0.28142494558940 517 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
86. N(2600) BREIT-WIGNER MASS N(2600)0 Nbar(2600)0 N(2600)+ Nbar(2600)- PDG2026:observable:B120:B120M:349022	2550 TO 2600 TO 2750 MeV Display: 2550 to 2750 (~2600)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
87. Lambda_c(2940)+ MASS Lambda_c(2940)0 Lambdabar_c(2940)0 PDG2026:observable:B122:B122M:350979	2939.6+-1.3-1.5 MeV	u_+ : 1.26187668447521 3 u_- : 1.46718498507142 8 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
88. N(~3000) BREIT-WIGNER MASS N(~3000)0 Nbar(~3000)0 N(~3000)+ Nbar(~3000)- PDG2026:observable:B128:B128M:349027	~3000 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
89. Xi_c(2970)+ MASS Xi_c(2970)0 Xibar_c(2970)0 PDG2026:observable:B130:B130M+:351271	2965.2+-1.0 MeV	u_+ : 1.01568106038590 41 u_- : 1.01568106038590 41 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
90. Xi_c(2970)0 MASS Xi_c(2970)0 Xibar_c(2970)0 PDG2026:observable:B130:B130M0:351273	2967.3+-1.9 MeV	u_+ : 1.85008439480852 7 u_- : 1.85008439480852 7 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
91. Xi(2370) MASS Xi(2370)- Xibar(2370)+ Xi(2370)0 Xibar(2370)0 PDG2026:observable:B131:B131M:350458	~2370 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
92. N(2100) BREIT-WIGNER MASS N(2100)0 Nbar(2100)0 N(2100)+ Nbar(2100)- PDG2026:observable:B132:B132M:348871	2050 to 2100 to 2150 MeV Display: 2050 to 2150 (~2100)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
93. Delta(2200) BREIT-WIGNER MASS Delta(2200)- Deltabar(2200)+ Delta(2200)0 Deltabar(2200)0 Delta(2200)+ Deltabar(2200)- Delta(2200)++ Deltabar(2200)-- PDG2026:observable:B135:B135M:349339	2150 to 2200 to 2250 MeV Display: 2150 to 2250 (~2200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
94. Delta(1940) BREIT-WIGNER MASS Delta(1940)- Deltabar(1940)+ Delta(1940)0 Deltabar(1940)0 Delta(1940)+ Deltabar(1940)- Delta(1940)++ Deltabar(1940)-- PDG2026:observable:B136:B136M:349272	1940 to 2000 to 2060 MeV Display: 1940 to 2060 (~2000)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
95. Omega(2250)- MASS Omega(2250)- Omegabar(2250)+ PDG2026:observable:B141:B141M:350516	2252+-9 MeV	u_+ : 8.83410444096544 19 u_- : 8.83410444096544 19 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
96. Omega(2380)- MASS Omega(2380)- Omegabar(2380)+ PDG2026:observable:B142:B142M:350520	~2380 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-BARYON E04 / M07
97. Omega(2470)- MASS Omega(2470)- Omegabar(2470)+ PDG2026:observable:B143:B143M:350523	2474+-12 MeV	u_+ : 12 u_- : 12 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
98. N(1900) BREIT-WIGNER MASS N(1900)0 Nbar(1900)0 N(1900)+ Nbar(1900)- PDG2026:observable:B144:B144M:348740	1890 to 1920 to 1950 MeV Display: 1890 to 1950 (~1920)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
99. Xi_c(2645)+ MASS Xi_c(2645)0 Xibar_c(2645)0 Xi_c(2645)+ Xibar_c(2645)- PDG2026:observable:B146:B146M+:351204	2645.17+-0.27 MeV	u_+ : 0.26281213522405 311 u_- : 0.27522452658037 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
100. Xi_c(2645)0 MASS Xi_c(2645)0 Xibar_c(2645)0 Xi_c(2645)+ Xibar_c(2645)- PDG2026:observable:B146:B146M0:351206	2646.24+-0.18 MeV	u_+ : 0.17792728963027 479 u_- : 0.18097989449585 48 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
101. Xi_c(3080)+ MASS Xi_c(3080)0 Xibar_c(3080)0 PDG2026:observable:B147:B147M+:351307	3077.2+-0.4 MeV	u_+ : 0.38652749298648 748 u_- : 0.38652749298648 748 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
102. Xi_c(3080)0 MASS Xi_c(3080)0 Xibar_c(3080)0 PDG2026:observable:B147:B147M0:351308	3079.9+-1.4 MeV	u_+ : 1.36034585012566 09 u_- : 1.36034585012566 09 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
103. Xi_c(2815)+ MASS Xi_c(2815)0 Xibar_c(2815)0 Xi_c(2815)+ Xibar_c(2815)- PDG2026:observable:B148:B148M+:351232	2816.60+-0.16 MeV	u_+ : 0.15517126784005 561 u_- : 0.15767230758639 7 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
104. Xi_c(2815)0 MASS Xi_c(2815)0 Xibar_c(2815)0 Xi_c(2815)+ Xibar_c(2815)- PDG2026:observable:B148:B148M0:351234	2819.85+-0.27 MeV	u_+ : 0.26809134636591 098 u_- : 0.28025492882096 131 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
105. Xi_c(2790)+ MASS Xi_c(2790)0 Xibar_c(2790)0 Xi_c(2790)+ Xibar_c(2790)- PDG2026:observable:B149:B149M+:351217	2792.0+-0.5 MeV	u_+ : 0.52973897333377 062 u_- : 0.53585270924116 923 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
106. Xi_c(2790)0 MASS Xi_c(2790)0 Xibar_c(2790)0 Xi_c(2790)+ Xibar_c(2790)- PDG2026:observable:B149:B149M0:351218	2794.0+-0.5 MeV	u_+ : 0.49234423994343 779 u_- : 0.49234423994343 779 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
107. Lambda_c(2765)+ MASS Lambda_c(2765)0 Lambdabar_c(2765)0 PDG2026:observable:B150:B150M:350951	2766.6+-2.4 MeV	u_+ : 2.40407986556187 4 u_- : 2.40407986556187 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
108. Lambda_c(2880)+ MASS Lambda_c(2880)+ Lambdabar_c(2880)- PDG2026:observable:B151:B151M:350960	2881.63+-0.24 MeV	u_+ : 0.23613203643473 499 u_- : 0.24555868414433 36 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
109. Sigma_c(2800)++ MASS Sigma_c(2800)0 Sigmapar_c(2800)0 PDG2026:observable:B155:B155M+:351015	2801+4-6 MeV	u_+ : 4.40676752279935 26 u_- : 5.79996551713898 52 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
110. Sigma_c(2800)+ MASS Sigma_c(2800)0 Sigmapar_c(2800)0 PDG2026:observable:B155:B155M+:351016	2792+14-5 MeV	u_+ : 13.6901278299361 1 u_- : 5.01792785918669 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
111. Sigma_c(2800)0 MASS Sigma_c(2800)0 Sigmabar_c(2800)0 PDG2026:observable:B155:B155M0:351017	2806+5-7 MeV	u_+ : 4.8590019921640533 u_- : 7.3871442628294739 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
112. Xi_c(2930)+ MASS Xi_c(2930)0 Xibar_c(2930)0 PDG2026:observable:B156:B156A03:351263	2942+-5 MeV	u_+ : 4.6486557196677838 u_- : 4.6486557196677838 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
113. Xi_c(2930)0 MASS Xi_c(2930)0 Xibar_c(2930)0 PDG2026:observable:B156:B156A04:351264	2938.55+-0.30 MeV	u_+ : 0.30185805977630881 u_- : 0.30185805977630881 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
114. Xi_c(3055)+ MASS Xi_c(3055)0 Xibar_c(3055)0 PDG2026:observable:B157:B157M+:351297	3055.2+-0.7 MeV	u_+ : 0.68999235826894623 u_- : 0.68999235826894623 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
115. Xi_c(3055)0 MASS Xi_c(3055)0 Xibar_c(3055)0 PDG2026:observable:B157:B157M:351298	3061.0+-0.8 MeV	u_+ : 0.83240615086627034 u_- : 0.83240615086627034 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
116. Xi_c(3123)+ MASS Xi_c(3123)0 Xibar_c(3123)0 PDG2026:observable:B158:B158M+:351325	3122.9+-1.3 MeV	u_+ : 1.334166406412634 u_- : 1.334166406412634 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
117. Xi_b(5945)0 MASS Xi_b(5945)0 Xibar_b(5945)0 PDG2026:observable:B161:B161M:351844	5952.3+-0.6 MeV	u_+ : 0.57765272773906229 u_- : 0.57765272773906229 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
118. Lambda_b(5912)0 MASS Lambda_b(5912)0 Lambdabar_b(5912)0 PDG2026:observable:B162:B162M:351694	5912.16+-0.16 MeV	u_+ : 0.16007003100330261 u_- : 0.16007003100330261 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
119. Lambda_b(5920)0 MASS Lambda_b(5920)0 Lambdabar_b(5920)0 PDG2026:observable:B163:B163M:351698	5920.07+-0.16 MeV	u_+ : 0.15862170338385659 u_- : 0.15862170338385659 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
120. Lambda(1710) MASS Lambda(1710)0 Lambdabar(1710)0 PDG2026:observable:B164:B164M:349532	1713+-13 MeV	u_+ : 13 u_- : 13 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
121. Lambda(2050) MASS Lambda(2050)0 Lambdabar(2050)0 PDG2026:observable:B165:B165M:349651	2056+-22 MeV	u_+ : 22 u_- : 22 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
122. Sigma(1780) MASS Sigma(1780)- Sigmabar(1780)+ Sigma(1780)0 Sigmabar(1780)0 Sigma(1780)+ Sigmabar(1780)- PDG2026:observable:B166:B166M:349991	1730 to 1780 to 1830 MeV Display: 1730 to 1830 (~1780)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
123. Sigma(1900) MASS Sigma(1900)- Sigmabar(1900)+ Sigma(1900)0 Sigmabar(1900)0 Sigma(1900)+ Sigmabar(1900)- PDG2026:observable:B167:B167M:350019	1900 to 1925 to 1950 MeV Display: 1900 to 1950 (~1925)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
124. Sigma(1940) MASS Sigma(1940)- Sigmabar(1940)+ Sigma(1940)0 Sigmabar(1940)0 Sigma(1940)+ Sigmabar(1940)- PDG2026:observable:B168:B168M:350102	1920 to 1940 to 1960 MeV Display: 1920 to 1960 (~1940)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
125. Xi_b'(5935)- MASS Xi_b^(5935)- Xibar_b^(5935)+ PDG2026:observable:B169:B169M:351840	5934.9+-0.4 MeV	u_+ : 0.41352429707823 54 u_- : 0.41352429707823 54 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
126. Xi_b(5955)- MASS Xi_b(5955)- Xibar_b(5955)+ PDG2026:observable:B170:B170M:351848	5955.5+-0.4 MeV	u_+ : 0.41457381951292 599 u_- : 0.41457381951292 599 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
127. P_c_cbar(4380)+ MASS P_c_cbar(4380)+ PDG2026:observable:B171:B171M:351951	4380+-30 MeV	u_+ : 30.0832179129826 51 u_- : 30.0832179129826 51 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
128. P_c_cbar(4457)+ MASS P_c_cbar(4457)+ PDG2026:observable:B172:B172M:351977	4457.3+4.0-1.8 MeV	u_+ : 4.14366987102013 16 u_- : 1.80277563773199 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
129. Omega_c(3000)0 MASS Omega_c(3000)0 Omegabar_c(3000)0 PDG2026:observable:B173:B173M:351376	3000.46+-0.25 MeV	u_+ : 0.24317658361921 44 u_- : 0.26399637894382 499 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
130. Omega_c(3050)0 MASS Omega_c(3050)0 Omegabar_c(3050)0 PDG2026:observable:B174:B174M:351380	3050.17+-0.19 MeV	u_+ : 0.18820356358563 539 u_- : 0.19089950264325 409 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
131. Omega_c(3065)0 MASS Omega_c(3065)0 Omegabar_c(3065)0 PDG2026:observable:B175:B175M:351384	3065.58+-0.21 MeV	u_+ : 0.21345733449248 469 u_- : 0.21391475011375 18 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
132. Omega_c(3090)0 MASS Omega_c(3090)0 Omegabar_c(3090)0 PDG2026:observable:B176:B176M:351388	3090.15+-0.26 MeV	u_+ : 0.25239209275859 231 u_- : 0.26305043429981 828 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
133. Omega_c(3120)0 MASS Omega_c(3120)0 Omegabar_c(3120)0 PDG2026:observable:B177:B177M:351392	3118.98+0.27-0.35 MeV	u_+ : 0.27459060435491 961 u_- : 0.34669871646719 441 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
134. Lambda_c(2860)+ MASS Lambda_c(2860)0 Lambdabar_c(2860)0 PDG2026:observable:B178:B178M:350956	2856.1+2.3-6.0 MeV	u_+ : 2.33666428910958 5 u_- : 5.87367006223536 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
135. Omega(2012)- MASS Omega(2012)- Omegabar(2012)+ PDG2026:observable:B179:B179M:350506	2012.9+-0.4 MeV	u_+ : 0.44536259664740 679 u_- : 0.44536259664740 679 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
136. Xi_b(6227)- MASS Xi_b(6227)- Xi_b(6227)+ PDG2026:observable:B180:B180M:351864	6227.9+-0.9 MeV	u_+ : 0.90815128746094 542 u_- : 0.90815128746094 542 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
137. Sigma_b(6097)+ MASS Sigma_b(6097)+ Sigmapar_b(6097)- PDG2026:observable:B181:B181M:351734	6095.8+-1.7 MeV	u_+ : 1.74642491965729 81 u_- : 1.74642491965729 81 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
138. Sigma_b(6097)- MASS Sigma_b(6097)- Sigmapar_b(6097)+ PDG2026:observable:B182:B182M:351738	6098.0+-1.8 MeV	u_+ : 1.77200451466693 5 u_- : 1.77200451466693 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
139. Sigma(2110) MASS Sigma(2110)- Sigmabar(2110)+ Sigma(2110)0 Sigmabar(2110)0 Sigma(2110)+ Sigmabar(2110)- PDG2026:observable:B183:B183M:350228	2110+-50 MeV	u_+ : 51.9913875255207 71 u_- : 51.9913875255207 71 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
140. Sigma(2230) MASS Sigma(2230)- Sigmabar(2230)+ Sigma(2230)0 Sigmabar(2230)0 Sigma(2230)+ Sigmabar(2230)- PDG2026:observable:B184:B184M:350253	2240+-27 MeV	u_+ : 27 u_- : 27 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
141. P_c_cbar(4312)+ MASS P_c_cbar(4312)+ PDG2026:observable:B185:B185M:351935	4311.9+7.0-0.9 MeV	u_+ : 6.83593446428503 97 u_- : 0.92195444572928 864 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
142. P_c_cbar(4440)+ MASS P_c_cbar(4440)+ PDG2026:observable:B186:B186M:351955	4440+4-5 MeV	u_+ : 4.30116263352131 33 u_- : 4.87647413609464 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
143. Lambda_b(6146)0 MASS Lambda_b(6146)0 Lambdabar_b(6146)0 PDG2026:observable:B187:B187M:351706	6146.2+-0.4 MeV	u_+ : 0.41759268734261 917 u_- : 0.41759268734261 917 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
144. Lambda_b(6152)0 MASS Lambda_b(6152)0 Lambdabar_b(6152)0 PDG2026:observable:B188:B188M:351711	6152.5+-0.4 MeV	u_+ : 0.35727006911778 508 u_- : 0.35727006911778 508 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
145. Lambda(2070) MASS Lambda(2070)0 Lambdabar(2070)0 PDG2026:observable:B190:B190M:349676	2070+-24 MeV	u_+ : 24 u_- : 24 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
146. Lambda(2080) MASS Lambda(2080)0 Lambdabar(2080)0 PDG2026:observable:B191:B191M:349711	2082+-13 MeV	u_+ : 13 u_- : 13 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
147. Xi_c(2923)0 MASS Xi_c(2923)0 Xibar_c(2923)0 PDG2026:observable:B192:B192M0:351255	2923.2+-0.4 MeV	u_+ : 0.39743785839914 991 u_- : 0.39743785839914 991 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
148. Xi_c(2923)+ MASS Xi_c(2923)0 Xibar_c(2923)0 PDG2026:observable:B192:B192MP:351256	2922.8+-0.6 MeV	u_+ : 0.61644140029689 765 u_- : 0.61644140029689 765 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
149. Omega_b(6316)- MASS Omega_b(6316)- Omegabar_b(6316)+ PDG2026:observable:B193:B193M:351905	6315.6+-0.6 MeV	u_+ : 0.58048724929847 628 u_- : 0.58048724929847 628 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
150. Omega_b(6330)- MASS Omega_b(6330)- Omegabar_b(6330)+ PDG2026:observable:B194:B194M:351909	6330.3+-0.6 MeV	u_+ : 0.56503579231594 825 u_- : 0.56503579231594 825 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
151. Omega_b(6340)- MASS Omega_b(6340)- Omegabar_b(6340)+ PDG2026:observable:B195:B195M:351913	6339.7+-0.6 MeV	u_+ : 0.55323181994360 315 u_- : 0.55323181994360 315 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
152. Omega_b(6350)- MASS Omega_b(6350)- Omegabar_b(6350)+ PDG2026:observable:B196:B196M:351917	6349.8+-0.6 MeV	u_+ : 0.60080400015155 633 u_- : 0.60080400015155 633 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
153. Lambda_b(6070)0 MASS Lambda_b(6070)0 Lambdabar_b(6070)0 PDG2026:observable:B197:B197M:351702	6072.3+-2.9 MeV	u_+ : 2.94698808292754 59 u_- : 2.94698808292754 59 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
154. Xi_b(6227)0 MASS Xi_b(6227)0 Xibar_b(6227)0 PDG2026:observable:B199:B199M:351870	6226.8+-1.6 MeV	u_+ : 1.54377784025284 7 u_- : 1.63500765137529 09 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
155. Xi_b(6100)- MASS Xi_b(6100)- Xibar_b(6100)+ PDG2026:observable:B200:B200M:351860	6099.8+-0.4 MeV	u_+ : 0.44830769038414 819 u_- : 0.44830769038414 819 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
156. Xi_b(6327)0 MASS Xi_b(6327)0 Xibar_b(6327)0 PDG2026:observable:B201:B201M:351874	6327.28+-0.35 MeV	u_+ : 0.35468295701936 398 u_- : 0.34205262752974 142 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
157. Xi_b(6333)0 MASS Xi_b(6333)0 Xibar_b(6333)0 PDG2026:observable:B202:B202M:351878	6332.69+-0.28 MeV	u_+ : 0.27802877548915689 u_- : 0.28425340807103788 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
158. Xi_c(2882) MASS Xi_c(2882)0 Xibar_c(2882)0 PDG2026:observable:B203:B203M0:351251	2882+-9 MeV	u_+ : 9.0476516290140179 u_- : 9.0476516290140179 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
159. P_c_cbar_s(4338)0 MASS P_c_cbar_s(4338)+ PDG2026:observable:B204:B204M:351945	4338.2+-0.8 MeV	u_+ : 0.80622577482985491 u_- : 0.80622577482985491 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
160. P_c_cbar(4312)+ MASS P_c_cbar(4312)+ PDG2026:observable:B205:B205M:351999	4466+6-5 MeV	u_+ : 6.3937962412886078 u_- : 4.5877765420030174 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
161. Lambda_c(2910)+ MASS Lambda_c(2910)0 Lambdabar_c(2910)0 PDG2026:observable:B206:B206M:350973	2914+-7 MeV	u_+ : 6.7675697262754522 u_- : 6.7675697262754522 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
162. Xi_b(6087)0 MASS Xi_b(6087) PDG2026:observable:B207:B207M:351852	6087.0+-0.5 MeV	u_+ : 0.463167553682577 u_- : 0.463167553682577 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
163. Xi_b(6095)0 MASS Xi_b(6095) PDG2026:observable:B208:B208M:351856	6095.1+-0.4 MeV	u_+ : 0.44082216684770159 u_- : 0.44082216684770159 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
164. Omega_c(3185)0 MASS Omega_c(3185)0 Omegabar_c(3185)0 PDG2026:observable:B209:B209M:351396	3185+7.6-1.9 MeV	u_+ : 7.5953933407032981 u_- : 1.933907960581372 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
165. Omega_c(3327)0 MASS Omega_c(3327)0 Omegabar_c(3327)0 PDG2026:observable:B210:B210M:351400	3327.1+1.2-1.8 MeV	u_+ : 1.2206555615733701 u_- : 1.7804493814764859 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
166. Omega(2109)- MASS Omega(2109)- Omegabar(2109)+ PDG2026:observable:B211:B211M:350514	2108+-5 MeV	u_+ : 5.2773099207835052 u_- : 5.2773099207835052 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
167. graviton MASS graviton PDG2026:observable:G033:G033M:332463	<1.76 E-23 eV Display: <1.76E-23	u_+ : NA u_- : NA Type: U Confidence: ND	GRAVITATIONAL_OR_HYPOTHETICAL E04 / M10
168. omega(782) MASS omega(782)0 PDG2026:observable:M001:M001M:333932	782.66+-0.13 MeV	u_+ : 0.12901268855075981 u_- : 0.12899704603475121 Type: NONE Confidence: ND	COMP-MESON E04 / M08
169. eta'(958) MASS eta^(958)0 PDG2026:observable:M002:M002M:334006	957.78+-0.06 MeV	u_+ : 0.062631506813104018 u_- : 0.062631506813104018 Type: NONE Confidence: ND	COMP-MESON E04 / M08
170. f_0(980) MASS f_0(980)0 PDG2026:observable:M003:M003M1:334163	990+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08
171. phi(1020) MASS phi(1020)0 PDG2026:observable:M004:M004M:334182	1019.460+-0.016 MeV	u_+ : 0.0155652966373344 u_- : 0.0155652966373344 Type: NONE Confidence: ND	COMP-MESON E04 / M08
172. f_2(1270) MASS f_2(1270)0 PDG2026:observable:M005:M005M:334354	1275.4+-0.8 MeV	u_+ : 0.83226160812299765 u_- : 0.83243720122632603 Type: NONE Confidence: ND	COMP-MESON E04 / M08
173. f_1(1420) MASS f_1(1420)0 PDG2026:observable:M006:M006M2:334554	1427.7+1.7-1.4 MeV	u_+ : 1.669286801641032 u_- : 1.4373229857894489 Type: NONE Confidence: ND	COMP-MESON E04 / M08
174. f_1(1285) MASS f_1(1285)0 PDG2026:observable:M008:M008M:334403	1281.8+-0.5 MeV	u_+ : 0.45740251642395002 u_- : 0.45001100251184373 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
175. NEUTRAL ONLY, e+ e- rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009M0:333862	775.26+-0.23 MeV	u_+ : 0.23429187483587 821 u_- : 0.23429187483587 821 Type: NONE Confidence: ND	COMP-MESON E04 / M08
176. CHARGED ONLY, tau DECAYS and e+ e- rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009M5:333863	775.11+-0.34 MeV	u_+ : 0.34073877998562 979 u_- : 0.34073877998562 979 Type: NONE Confidence: ND	COMP-MESON E04 / M08
177. MIXED CHARGES, OTHER REACTIONS rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009M7:333864	763.0+-1.2 MeV	u_+ : 1.23693168768529 8 u_- : 1.23693168768529 8 Type: NONE Confidence: ND	COMP-MESON E04 / M08
178. CHARGED ONLY, HADROPRODUCED rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009M2:333865	766.5+-1.1 MeV	u_+ : 1.10685352334429 6 u_- : 1.10685352334429 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08
179. NEUTRAL ONLY, PHOTOPRODUCED rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009M0P:333866	769.6+-0.8 MeV	u_+ : 0.84746260647359 306 u_- : 0.82807525956483 763 Type: NONE Confidence: ND	COMP-MESON E04 / M08
180. NEUTRAL ONLY, OTHER REACTIONS rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009M0R:333867	769.0+-0.9 MeV	u_+ : 0.85061998301285 813 u_- : 0.85061998301285 813 Type: NONE Confidence: ND	COMP-MESON E04 / M08
181. a_1(1260) MASS a_1(1260)0 a_1(1260)+ a_1(1260)- PDG2026:observable:M010:M010M:334331	1230+-40 MeV	u_+ : 40 u_- : 40 Type: NONE Confidence: ND	COMP-MESON E04 / M08
182. b_1(1235) MASS b_1(1235)0 b_1(1235)+ b_1(1235)- PDG2026:observable:M011:M011M:334309	1229.5+-3.2 MeV	u_+ : 3.16916129668438 5 u_- : 3.16916129668438 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
183. a_2(1320) MASS a_2(1320)0 a_2(1320)+ a_2(1320)- PDG2026:observable:M012:M012M0:334469	1318.2+-0.6 MeV	u_+ : 0.60920186906481 122 u_- : 0.60699292355954 471 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
184. 3pi MODE a_2(1320)0 a_2(1320)+ a_2(1320)- PDG2026:observable:M012:M012M1:334470	1318.6+-1.3 MeV	u_+ : 1.32538345937344 39 u_- : 1.30846795325698 1 Type: NONE Confidence: ND	COMP-MESON E04 / M08
185. K Kbar MODE a_2(1320)0 a_2(1320)+ a_2(1320)- PDG2026:observable:M012:M012M2:334471	1318.1+-0.7 MeV	u_+ : 0.69776876905725 704 u_- : 0.69776876905725 704 Type: NONE Confidence: ND	COMP-MESON E04 / M08
186. eta pi MODE a_2(1320)0 a_2(1320)+ a_2(1320)- PDG2026:observable:M012:M012M3:334472	1317.7+-1.4 MeV	u_+ : 1.44182788627946 u_- : 1.44182788627946 Type: NONE Confidence: ND	COMP-MESON E04 / M08
187. eta' pi MODE a_2(1320)0 a_2(1320)+ a_2(1320)- PDG2026:observable:M012:M012M4:334473	1322+-7 MeV	u_+ : 6.67661222337908 37 u_- : 7.07631643322348 49 Type: NONE Confidence: ND	COMP-MESON E04 / M08
188. f_2'(1525) MASS f_2'(1525)0 PDG2026:observable:M013:M013MX:334665	1517.3+-2.4 MeV	u_+ : 2.43466563247887 4 u_- : 2.32415269962939 91 Type: NONE Confidence: ND	COMP-MESON E04 / M08
189. PRODUCED BY K+- BEAM f_2'(1525)0 PDG2026:observable:M013:M013M2:334666	1518.0+-2.7 MeV	u_+ : 2.75752274044423 6 u_- : 2.63263276117844 79 Type: NONE Confidence: ND	COMP-MESON E04 / M08
190. PRODUCED IN e+ e- ANNIHILATION AND PARTICLE DECAYS f_2'(1525)0 PDG2026:observable:M013:M013M3:334667	1514+-4 MeV	u_+ : 4.42458598817588 47 u_- : 4.08971545863226 64 Type: NONE Confidence: ND	COMP-MESON E04 / M08
191. PRODUCED IN pbar p ANNIHILATION f_2'(1525)0 PDG2026:observable:M013:M013M9:334668	1512+-4 MeV	u_+ : 3.65524619448102 91 u_- : 3.65524619448102 91 Type: NONE Confidence: ND	COMP-MESON E04 / M08
192. CENTRAL PRODUCTION f_2'(1525)0 PDG2026:observable:M013:M013M4:334669	1515+-15 MeV	u_+ : 15 u_- : 15 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
193. PRODUCED IN e p COLLISIONS $f_2^{(1525)}0$ PDG2026:observable:M013:M013M10:334670	1512+-3.2 MeV	u_+ : 3.31058907144937 02 u_- : 3.04138126514911 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
194. $f_0(500)$ BREIT-WIGNER MASS $f_0(500)0$ PDG2026:observable:M014:M014M:333857	400 to 800 MeV	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
195. $\rho_3(1690)$ MASS $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015M:334842	1688.8+-2.1 MeV	u_+ : 2.09855167236953 61 u_- : 2.09855167236953 61 Type: NONE Confidence: ND	COMP-MESON E04 / M08
196. 2pi MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015M1:334843	1686+-4 MeV	u_+ : 3.97500251085426 28 u_- : 3.97500251085426 28 Type: NONE Confidence: ND	COMP-MESON E04 / M08
197. K Kbar AND K Kbar pi MODES $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015M2:334844	1696+-4 MeV	u_+ : 3.56623601326023 2 u_- : 3.56623601326023 2 Type: NONE Confidence: ND	COMP-MESON E04 / M08
198. (4pi)+- MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015M3:334845	1686+-5 MeV	u_+ : 4.64284850304302 04 u_- : 4.64284850304302 04 Type: NONE Confidence: ND	COMP-MESON E04 / M08
199. omega pi MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015M5:334846	1681+-7 MeV	u_+ : 6.52436854824307 93 u_- : 6.52436854824307 93 Type: NONE Confidence: ND	COMP-MESON E04 / M08
200. eta pi+ pi- MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015M6:334847	1682+-12 MeV	u_+ : 12.4568219780610 01 u_- : 12.4568219780610 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
201. $f_4(2050)$ MASS $f_4(2050)0$ PDG2026:observable:M016:M016M:335130	2018+-11 MeV	u_+ : 11.3698561841563 4 u_- : 11.5056263269526 21 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
202. $a_4(1970)$ MASS $a_4(1970)0 \mid a_4(1970)+ \mid a_4(1970)-$ PDG2026:observable:M017:M017M:335090	1967+-16 MeV	u_+ : 16.0388762689347 11 u_- : 15.8917126074592 4 Type: NONE Confidence: ND	COMP-MESON E04 / M08
203. $K^*(892)$ T-Matrix Pole $\sqrt{s}$ $K^*(892)0 \mid \bar{K}^{*}(892)0 \mid K^*(892)+ \mid K^*(892)-$ PDG2026:observable:M018:M018TMP:335789	(890+-14) - i (26+-6) MeV Display: 890+-14	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-MESON E04 / M08
204. CHARGED ONLY, HADROPRODUCED $K^*(892)0 \mid \bar{K}^{*}(892)0 \mid K^*(892)+ \mid K^*(892)-$ PDG2026:observable:M018:M018M1:335790	891.88+-0.23 MeV	u_+ : 0.23110722559041 66 u_- : 0.23110722559041 66 Type: NONE Confidence: ND	COMP-MESON E04 / M08
205. CHARGED ONLY, PRODUCED IN τ LEPTON DECAYS $K^*(892)0 \mid \bar{K}^{*}(892)0 \mid K^*(892)+ \mid K^*(892)-$ PDG2026:observable:M018:M018MCT:335791	895.5+-0.8 MeV	u_+ : 0.76655071586947 199 u_- : 0.76655071586947 199 Type: NONE Confidence: ND	COMP-MESON E04 / M08
206. NEUTRAL ONLY $K^*(892)0 \mid \bar{K}^{*}(892)0 \mid K^*(892)+ \mid K^*(892)-$ PDG2026:observable:M018:M018M2:335792	895.56+-0.20 MeV	u_+ : 0.20357938811155 91 u_- : 0.20506335702397 691 Type: NONE Confidence: ND	COMP-MESON E04 / M08
207. $K_0^*(1430)$ MASS $K_0^*(1430)0 \mid \bar{K}^{*}_0(1430)0 \mid K_0^*(1430)+ \mid K_0^*(1430)-$ PDG2026:observable:M019:M019M:335878	1425+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-MESON E04 / M08
208. $\pi_2(2100)$ MASS $\pi_2(2100)0 \mid \pi_2(2100)+ \mid \pi_2(2100)-$ PDG2026:observable:M020:M020M:335146	2090+-29 MeV	u_+ : 29.4174202707276 u_- : 29.4174202707276 Type: NONE Confidence: ND	COMP-MESON E04 / M08
209. CHARGED ONLY, WITH FINAL STATE $K \pi$ $K_2^{*}(1430)0 \mid \bar{K}^{*}_2(1430)0 \mid K_2^{*}(1430)+ \mid K_2^{*}(1430)-$ PDG2026:observable:M022:M022M1:335888	1427.3+-1.5 MeV	u_+ : 1.50306501170133 u_- : 1.51764648227118 19 Type: NONE Confidence: ND	COMP-MESON E04 / M08
210. NEUTRAL ONLY $K_2^{*}(1430)0 \mid \bar{K}^{*}_2(1430)0 \mid K_2^{*}(1430)+ \mid K_2^{*}(1430)-$ PDG2026:observable:M022:M022M4:335889	1432.4+-1.3 MeV	u_+ : 1.25769408082424 21 u_- : 1.25769408082424 21 Type: NONE Confidence: ND	COMP-MESON E04 / M08
211. $K_2(1770)$ MASS $K_2(1770)0 \mid \bar{K}^{*}_2(1770)0 \mid K_2(1770)+ \mid K_2(1770)-$ PDG2026:observable:M023:M023M:335963	1773+-8 MeV	u_+ : 7.98415545885385 04 u_- : 7.96445414241581 86 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
212. psi(4160) MASS psi(4160) PDG2026:observable:M025:M025M:346343	4191+-5 MeV	u_+ : 5.1959983332711133 u_- : 4.9802084626429108 Type: NONE Confidence: ND	COMP-MESON E04 / M08
213. eta_c(1S) MASS eta_c(1S) PDG2026:observable:M026:M026M:343286	2984.09+-0.30 MeV	u_+ : 0.29677104771232732 u_- : 0.30021031998551229 Type: NONE Confidence: ND	COMP-MESON E04 / M08
214. eta(1405) MASS eta(1405)0 PDG2026:observable:M027:M027MX:334525	1408.0+1.8-1.2 MeV	u_+ : 1.817274601784945 u_- : 1.231728539767738 Type: NONE Confidence: ND	COMP-MESON E04 / M08
215. eta pi pi MODE eta(1405)0 PDG2026:observable:M027:M027M1:334526	1405.5+-2.3 MeV	u_+ : 2.2617401948054172 u_- : 2.353950701102808 Type: NONE Confidence: ND	COMP-MESON E04 / M08
216. K Kbar pi MODE (a_0(980) pi or direct K Kbar pi) eta(1405)0 PDG2026:observable:M027:M027M4:334527	1413.5+2.0-0.9 MeV	u_+ : 1.9652283255103979 u_- : 0.88864461966865371 Type: NONE Confidence: ND	COMP-MESON E04 / M08
217. pi pi gamma MODE eta(1405)0 PDG2026:observable:M027:M027M2:334528	1403+-17 MeV	u_+ : 16.617414327881129 u_- : 16.617414327881129 Type: NONE Confidence: ND	COMP-MESON E04 / M08
218. phi gamma MODE eta(1405)0 PDG2026:observable:M027:M027M7:334529	1422+6-8 MeV	u_+ : 6.2625873247404709 u_- : 8.0777472107017552 Type: NONE Confidence: ND	COMP-MESON E04 / M08
219. K_1(1270) MASS K_1(1270)0 Kbar_1(1270)0 K_1(1270)+ K_1(1270)- PDG2026:observable:M028:M028MX:335817	1256+-6 MeV	u_+ : 6.4239824760278736 u_- : 6.4239824760278736 Type: NONE Confidence: ND	COMP-MESON E04 / M08
220. PRODUCED BY K-, BACKWARD SCATTERING, HYPERON EXCHANGE K_1(1270)0 Kbar_1(1270)0 K_1(1270)+ K_1(1270)- PDG2026:observable:M028:M028M2:335818	1275+-10 MeV	u_+ : 10 u_- : 10 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
221. PRODUCED BY K BEAMS $K_1(1270)0 \mid \bar{K}_1(1270)0 \mid K_1(1270)+ \mid K_1(1270)-$ PDG2026:observable:M028:M028M3:335819	1270+-10 MeV	u_+ : 10 u_- : 10 Type: NONE Confidence: ND	COMP-MESON E04 / M08
222. PRODUCED BY BEAMS OTHER THAN K MESONS $K_1(1270)0 \mid \bar{K}_1(1270)0 \mid K_1(1270)+ \mid K_1(1270)-$ PDG2026:observable:M028:M028M1:335820	1252+-9 MeV	u_+ : 8.8239123358673819 u_- : 8.8239123358673819 Type: NONE Confidence: ND	COMP-MESON E04 / M08
223. PRODUCED IN tau LEPTON DECAYS $K_1(1270)0 \mid \bar{K}_1(1270)0 \mid K_1(1270)+ \mid K_1(1270)-$ PDG2026:observable:M028:M028MT:335821	1250+-50 MeV	u_+ : 47.381430961928537 u_- : 47.381430961928537 Type: NONE Confidence: ND	COMP-MESON E04 / M08
224. X(3940) MASS $X(3940)$ PDG2026:observable:M029:M029M:346244	3942+-9 MeV	u_+ : 9.0427689971016019 u_- : 8.2141639308556176 Type: NONE Confidence: ND	COMP-MESON E04 / M08
225. h_1(1170) MASS $h_1(1170)0$ PDG2026:observable:M030:M030M:334306	1166+-6 MeV	u_+ : 5.8309518948453007 u_- : 5.8309518948453007 Type: NONE Confidence: ND	COMP-MESON E04 / M08
226. pi- p --> omega pi0 n $\rho_5(2350)0 \mid \rho_5(2350)+ \mid \rho_5(2350)-$ PDG2026:observable:M033:M033M3:335259	2330+-35 MeV	u_+ : 35 u_- : 35 Type: NONE Confidence: ND	COMP-MESON E04 / M08
227. pi_2(1670) MASS $\pi_2(1670)0 \mid \pi_2(1670)+ \mid \pi_2(1670)-$ PDG2026:observable:M034:M034M:334783	1670.6+2.9-1.2 MeV	u_+ : 2.9248728518866489 u_- : 1.1987692041830089 Type: NONE Confidence: ND	COMP-MESON E04 / M08
228. K_4*(2045) MASS $K_4^{*}(2045)0 \mid \bar{K}_4^{*}(2045)0 \mid K_4^{*}(2045)+ \mid K_4^{*}(2045)-$ PDG2026:observable:M035:M035M:336016	2048+8-9 MeV	u_+ : 8.0644653972191538 u_- : 8.9483400524220826 Type: NONE Confidence: ND	COMP-MESON E04 / M08
229. a_0(980) MASS $a_0(980)0 \mid a_0(980)+ \mid a_0(980)-$ PDG2026:observable:M036:M036MX:334171	980+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08
230. eta(1295) MASS $\eta(1295)0$ PDG2026:observable:M037:M037M:334451	1294+-4 MeV	u_+ : 3.892110114334967 u_- : 3.892110114334967 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
231. f₂(1810) MASS f ₂ (1810)0 PDG2026:observable:M038:M038M:335002	1815+-12 MeV	u_+ : 12.52864229025908 u_- : 11.9803740701169 Type: NONE Confidence: ND	COMP-MESON E04 / M08
232. K₂(2250) MASS K ₂ (2250)0 Kbar ₂ (2250)0 K ₂ (2250)+ K ₂ (2250)- PDG2026:observable:M040:M040M:336032	2247+-17 MeV	u_+ : 16.84303842133038 u_- : 16.84303842133038 Type: NONE Confidence: ND	COMP-MESON E04 / M08
233. p p CENTRAL PRODUCTION f ₄ (2300)0 PDG2026:observable:M041:M041M4:335244	2320+-60 MeV	u_+ : 60 u_- : 60 Type: NONE Confidence: ND	COMP-MESON E04 / M08
234. f₂(2150) MASS, COMBINED MODES f ₂ (2150)0 PDG2026:observable:M042:M042M:335158	2157+-12 MeV	u_+ : 11.76667948645094 u_- : 11.76667948645094 Type: NONE Confidence: ND	COMP-MESON E04 / M08
235. eta eta MODE f ₂ (2150)0 PDG2026:observable:M042:M042M3:335159	2157+-12 MeV	u_+ : 11.76667948645094 u_- : 11.76667948645094 Type: NONE Confidence: ND	COMP-MESON E04 / M08
236. omega₃(1670) MASS omega ₃ (1670)0 PDG2026:observable:M045:M045M:334778	1667+-4 MeV	u_+ : 3.9911056080338438 u_- : 3.9911056080338438 Type: NONE Confidence: ND	COMP-MESON E04 / M08
237. Upsilon(4S) MASS Upsilon(4S) PDG2026:observable:M047:M047M:347744	10579.4+-1.2 MeV	u_+ : 1.1896060082538451 u_- : 1.1896060082538451 Type: NONE Confidence: ND	COMP-MESON E04 / M08
238. Upsilon(3S) MASS Upsilon(3S) PDG2026:observable:M048:M048M:347659	10355.1+-0.5 MeV	u_+ : 0.5 u_- : 0.5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
239. Upsilon(1S) MASS Upsilon(1S) PDG2026:observable:M049:M049M:346837	9460.40+-0.10 MeV	u_+ : 0.098429823160610927 u_- : 0.098429823160610927 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
240. chi_c2(3930) MASS chi_c2(3930) PDG2026:observable:M050:M050M:346228	3922.5+-1.0 MeV	u_+ : 1.01989448998559 71 u_- : 1.01989448998559 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
241. Upsilon(2S) MASS Upsilon(2S) PDG2026:observable:M052:M052M:347286	10023.4+-0.5 MeV	u_+ : 0.5 u_- : 0.5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
242. psi(3770) MASS psi(3770) PDG2026:observable:M053:M053M:345866	3773.7+-0.7 MeV	u_+ : 0.69575116407149 418 u_- : 0.69575116407149 418 Type: NONE Confidence: ND	COMP-MESON E04 / M08
243. phi_3(1850) MASS phi_3(1850)0 PDG2026:observable:M054:M054M:335023	1854+-7 MeV	u_+ : 6.88247201611685 27 u_- : 6.66666666666666 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
244. chi_c1(1P) MASS chi_c1(1P) PDG2026:observable:M055:M055M:344598	3510.67+-0.05 MeV	u_+ : 0.05073975077067 1888 u_- : 0.05073975077067 1888 Type: NONE Confidence: ND	COMP-MESON E04 / M08
245. chi_c0(1P) MASS chi_c0(1P) PDG2026:observable:M056:M056M:344304	3415.50+-0.19 MeV	u_+ : 0.19311829210333 57 u_- : 0.19240527880424 149 Type: NONE Confidence: ND	COMP-MESON E04 / M08
246. chi_c2(1P) MASS chi_c2(1P) PDG2026:observable:M057:M057M:344926	3556.17+-0.07 MeV	u_+ : 0.07161203246291 1895 u_- : 0.07161203246291 1895 Type: NONE Confidence: ND	COMP-MESON E04 / M08
247. pi(1300) MASS pi(1300)0 pi(1300)+ pi(1300)- PDG2026:observable:M058:M058M:334463	1300+-100 MeV	u_+ : 100 u_- : 100 Type: NONE Confidence: ND	COMP-MESON E04 / M08
248. eta_c(2S) MASS eta_c(2S) PDG2026:observable:M059:M059M:345228	3637.8+-0.6 MeV	u_+ : 0.61759763649141 219 u_- : 0.62004876788434 371 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
249. $K_3^*(1780)$ MASS $K_3^{*}(1780)0 \mid \bar{K}3^{*}(1780)0 \mid K_3^{*}(1780)+ \mid K_3^{*}(1780)-$ PDG2026:observable:M060:M060M:335975	1779+-8 MeV	u_+ : 7.821524428845243 u_- : 7.545019387046446 Type: NONE Confidence: ND	COMP-MESON E04 / M08
250. $D^*(2007)0$ MASS $D^{*}(2007)0 \mid \bar{D}^{*}(2007)0$ PDG2026:observable:M061:M061M:337777	2006.86+-0.05 MeV	u_+ : 0.053769014750598178 u_- : 0.053769014750598178 Type: NONE Confidence: ND	COMP-MESON E04 / M08
251. $D^*(2010)++$ MASS $D^{*}(2010)+ \mid \bar{D}^{*}(2010)-$ PDG2026:observable:M062:M062M:337794	2010.27+-0.04 MeV	u_+ : 0.044491245593188788 u_- : 0.044491245593188788 Type: NONE Confidence: ND	COMP-MESON E04 / M08
252. $K_1(1400)$ MASS $K_1(1400)0 \mid \bar{K}1(1400)0 \mid K_1(1400)+ \mid K_1(1400)-$ PDG2026:observable:M064:M064M:335845	1403+-7 MeV	u_+ : 6.8731168328374306 u_- : 6.8731168328374306 Type: NONE Confidence: ND	COMP-MESON E04 / M08
253. η ρ^0 AND π^+ π^- MODES $\rho(1700)0 \mid \rho(1700)+ \mid \rho(1700)-$ PDG2026:observable:M065:M065M0:334880	1720+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08
254. $f_2(1430)$ MASS $f_2(1430)0$ PDG2026:observable:M066:M066M1:334577	~1430 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-MESON E04 / M08
255. $e^+ e^-$ PRODUCTION $\phi(1680)0$ PDG2026:observable:M067:M067M1:334825	1680+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08
256. $f_0(1710)$ Breit-Wigner MASS $f_0(1710)0$ PDG2026:observable:M068:M068M:334927	1723+7-6 MeV	u_+ : 6.7390813885535801 u_- : 5.5617775425136911 Type: NONE Confidence: ND	COMP-MESON E04 / M08
257. $J/\psi(1S)$ MASS $J/\psi(1S)$ PDG2026:observable:M070:M070M:343505	3096.900+-0.006 MeV	u_+ : 0.0061858811299374487 u_- : 0.0061858811299374487 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
258. psi(2S) MASS psi(2S) PDG2026:observable:M071:M071M:345299	3686.097+-0.011 MeV	u_+ : 0.01055310115798 376 u_- : 0.01055310115798 376 Type: NONE Confidence: ND	COMP-MESON E04 / M08
259. psi(4040) MASS psi(4040) PDG2026:observable:M072:M072M:346249	4040+-4 MeV	u_+ : 4.299999999999999 98 u_- : 4.299999999999999 98 Type: NONE Confidence: ND	COMP-MESON E04 / M08
260. psi(4415) MASS psi(4415) PDG2026:observable:M073:M073M:346684	4415+-5 MeV	u_+ : 4.52788325337319 54 u_- : 4.52788325337319 54 Type: NONE Confidence: ND	COMP-MESON E04 / M08
261. psi(4230) MASS psi(4230) PDG2026:observable:M074:M074M:346460	4221.7+-2.5 MeV	u_+ : 2.45686850851105 82 u_- : 2.45440062347709 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
262. pi(1800) MASS pi(1800)0 pi(1800)+ pi(1800)- PDG2026:observable:M075:M075M:334966	1810+9-11 MeV	u_+ : 9.40459492210335 31 u_- : 11.0275139992719 19 Type: NONE Confidence: ND	COMP-MESON E04 / M08
263. chi_b0(1P) MASS chi_b0(1P) PDG2026:observable:M076:M076M:347129	9859.44+-0.42+-0.31 MeV	u_+ : 0.419999999999999 998 u_- : 0.419999999999999 998 Type: NONE Confidence: ND	COMP-MESON E04 / M08
264. chi_b1(1P) MASS chi_b1(1P) PDG2026:observable:M077:M077M:347175	9892.78+-0.26+-0.31 MeV	u_+ : 0.260000000000000 001 u_- : 0.260000000000000 001 Type: NONE Confidence: ND	COMP-MESON E04 / M08
265. chi_b2(1P) MASS chi_b2(1P) PDG2026:observable:M078:M078M:347234	9912.21+-0.26+-0.31 MeV	u_+ : 0.260000000000000 001 u_- : 0.260000000000000 001 Type: NONE Confidence: ND	COMP-MESON E04 / M08
266. chi_b0(2P) MASS chi_b0(2P) PDG2026:observable:M079:M079M:347497	10232.5+-0.4+-0.5 MeV	u_+ : 0.400000000000000 002 u_- : 0.400000000000000 002 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
267. chi_b1(2P) MASS chi_b1(2P) PDG2026:observable:M080:M080M:347543	10255.46+-0.22+-0.50 MeV	u_+ : 0.22 u_- : 0.22 Type: NONE Confidence: ND	COMP-MESON E04 / M08
268. chi_b2(2P) MASS chi_b2(2P) PDG2026:observable:M081:M081M:347610	10268.65+-0.22+-0.50 MeV	u_+ : 0.22 u_- : 0.22 Type: NONE Confidence: ND	COMP-MESON E04 / M08
269. f_J(2220) MASS f_J(2220)0 PDG2026:observable:M082:M082M:335203	2231.1+-3.5 MeV	u_+ : 3.53352481075623 18 u_- : 3.52146107791007 88 Type: NONE Confidence: ND	COMP-MESON E04 / M08
270. f_1(1510) MASS f_1(1510)0 PDG2026:observable:M084:M084M:334658	1518+-5 MeV	u_+ : 5.21629534269634 03 u_- : 5.21629534269634 03 Type: NONE Confidence: ND	COMP-MESON E04 / M08
271. X(1835) MASS X(1835)0 PDG2026:observable:M085:M085M:335011	1834.0+-4.0-3.0 MeV	u_+ : 4.11787074570238 51 u_- : 2.98440759509600 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
272. f_6(2510) MASS f_6(2510)0 PDG2026:observable:M089:M089M:335277	2470+-50 MeV	u_+ : 44.9999979654523 51 u_- : 44.9999979654523 51 Type: NONE Confidence: ND	COMP-MESON E04 / M08
273. K_3(2320) MASS K_3(2320)0 Kbar_3(2320)0 K_3(2320)+ K_3(2320)- PDG2026:observable:M090:M090M:336034	2324+-24 MeV	u_+ : 24 u_- : 24 Type: NONE Confidence: ND	COMP-MESON E04 / M08
274. K_4(2500) MASS K_4(2500)0 Kbar_4(2500)0 K_4(2500)+ K_4(2500)- PDG2026:observable:M091:M091M:336040	2490+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08
275. Upsilon(10860) MASS Upsilon(10860) PDG2026:observable:M092:M092M:347804	10885.2+-2.6-1.6 MeV	u_+ : 2.56584711714697 32 u_- : 1.56861014024743 19 Type: NONE Confidence: ND	COMP-MESON E04 / M08
276. Upsilon(11020) MASS Upsilon(11020) PDG2026:observable:M093:M093M:347905	11000+-4 MeV	u_+ : 4.02342815892643 29 u_- : 4.02423073877622 44 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
277. $K^*(1410)$ MASS $K^*(1410)0 \mid \bar{K}^*(1410)0 \mid K^*(1410)+ \mid K^*(1410)-$ PDG2026:observable:M094:M094M:335865	1414+-15 MeV	u_+ : 14.5398454347451 6 u_- : 14.5398454347451 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08
278. $K^*(1680)$ MASS $K^*(1680)0 \mid \bar{K}^*(1680)0 \mid K^*(1680)+ \mid K^*(1680)-$ PDG2026:observable:M095:M095M:335946	1718+-18 MeV	u_+ : 17.0787269165451 9 u_- : 18.6981626725943 39 Type: NONE Confidence: ND	COMP-MESON E04 / M08
279. $K_5^*(2380)$ MASS $K_5^*(2380)0 \mid \bar{K}_5^*(2380)0 \mid K_5^*(2380)+ \mid K_5^*(2380)-$ PDG2026:observable:M098:M098M:336036	2382+-24 MeV	u_+ : 23.6008474424118 9 u_- : 23.6008474424118 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
280. $K_1(1650)$ MASS $K_1(1650)0 \mid \bar{K}_1(1650)0 \mid K_1(1650)+ \mid K_1(1650)-$ PDG2026:observable:M099:M099M:335944	1650+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-MESON E04 / M08
281. $\eta_2(1870)$ MASS $\eta_2(1870)0$ PDG2026:observable:M101:M101M:335032	1842+-8 MeV	u_+ : 8.04307530759601 09 u_- : 8.04307530759601 09 Type: NONE Confidence: ND	COMP-MESON E04 / M08
282. $\phi(2170)$ MASS $\phi(2170)0 \mid \phi(2170)+ \mid \phi(2170)-$ PDG2026:observable:M103:M103M:335180	2164+-5 MeV	u_+ : 5.11327360016105 46 u_- : 5.11327360016105 46 Type: NONE Confidence: ND	COMP-MESON E04 / M08
283. $K_2^*(1980)$ MASS $K_2^*(1980)0 \mid \bar{K}_2^*(1980)0 \mid K_2^*(1980)+ \mid K_2^*(1980)-$ PDG2026:observable:M104:M104M:336003	1990+60-50 MeV	u_+ : 59.1051525697060 07 u_- : 45.4057722472393 12 Type: NONE Confidence: ND	COMP-MESON E04 / M08
284. $\rho(1450)$ MASS $\rho(1450)0 \mid \rho(1450)+ \mid \rho(1450)-$ PDG2026:observable:M105:M105M0:334592	1465 +-25 MeV Display: 1465+-25	u_+ : 25 u_- : 25 Type: NONE Confidence: ND	COMP-MESON E04 / M08
285. $f_2(2010)$ MASS $f_2(2010)0$ PDG2026:observable:M106:M106M:335112	2010+60-80 MeV	u_+ : 62 u_- : 76 Type: NONE Confidence: ND	COMP-MESON E04 / M08
286. $f_2(2300)$ MASS $f_2(2300)0$ PDG2026:observable:M107:M107M:335234	2297+-28 MeV	u_+ : 28 u_- : 28 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
287. f₂(2340) MASS f ₂ (2340)0 PDG2026:observable:M108:M108M:335252	2346+21-10 MeV	u_+ : 21.300562489710181 u_- : 9.7423326379963484 Type: NONE Confidence: ND	COMP-MESON E04 / M08
288. h₁(1415) MASS h ₁ (1415)0 PDG2026:observable:M109:M109M:334552	1409+9-8 MeV	u_+ : 9.4194068122733494 u_- : 7.776226726894202 Type: NONE Confidence: ND	COMP-MESON E04 / M08
289. f₀(2200) MASS f ₀ (2200)0 PDG2026:observable:M112:M112M:335199	2187+-14 MeV	u_+ : 13.53304175951927 u_- : 14.057750535181979 Type: NONE Confidence: ND	COMP-MESON E04 / M08
290. eta(1760) MASS eta(1760)0 PDG2026:observable:M114:M114M:334950	1751+-15 MeV	u_+ : 14.814874939752929 u_- : 14.980145590239159 Type: NONE Confidence: ND	COMP-MESON E04 / M08
291. eta(2225) MASS eta(2225)0 PDG2026:observable:M115:M115M:335232	2221+13-10 MeV	u_+ : 13.27133387053455 u_- : 9.5202460778307483 Type: NONE Confidence: ND	COMP-MESON E04 / M08
292. f₂(1640) MASS f ₂ (1640)0 PDG2026:observable:M117:M117M:334752	1639+-6 MeV	u_+ : 5.8313784211153239 u_- : 5.8313784211153239 Type: NONE Confidence: ND	COMP-MESON E04 / M08
293. D_{s1}(2536)+- MASS D _{s1} (2536)+ D _{s1} (2536)- PDG2026:observable:M121:M121M:338481	2535.12+-0.06 MeV	u_+ : 0.060350977470898563 u_- : 0.060350977470898563 Type: NONE Confidence: ND	COMP-MESON E04 / M08
294. f₂(1565) MASS f ₂ (1565)0 PDG2026:observable:M123:M123M:334703	1571+-13 MeV	u_+ : 13.379294632448991 u_- : 13.379294632448991 Type: NONE Confidence: ND	COMP-MESON E04 / M08
295. omega(1420) MASS omega(1420)0 PDG2026:observable:M125:M125M:334567	1410+-60 MeV	u_+ : 60 u_- : 60 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
296. omega(1650) MASS omega(1650)0 PDG2026:observable:M126:M126M:334768	1670+-30 MeV	u_+ : 30 u_- : 30 Type: NONE Confidence: ND	COMP-MESON E04 / M08
297. K(3100) MASS K(3100)0 Kbar(3100)0 K(3100)+ K(3100)- PDG2026:observable:M129:M129M:336041	~3100 MeV	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-MESON E04 / M08
298. 3-BODY DECAYS K(3100)0 Kbar(3100)0 K(3100)+ K(3100)- PDG2026:observable:M129:M129M1:336042	3054+-11 MeV	u_+ : 10.68148842445026 u_- : 10.68148842445026 Type: NONE Confidence: ND	COMP-MESON E04 / M08
299. 4-BODY DECAYS K(3100)0 Kbar(3100)0 K(3100)+ K(3100)- PDG2026:observable:M129:M129M2:336043	3059+-11 MeV	u_+ : 10.641252264801439 u_- : 10.641252264801439 Type: NONE Confidence: ND	COMP-MESON E04 / M08
300. K_0^*(1950) MASS K_0^*(1950)0 Kbar_0^*(1950)0 K_0^*(1950)+ K_0^*(1950)- PDG2026:observable:M134:M134M:335999	1957+-14 MeV	u_+ : 14.320221981073329 u_- : 14.320221981073329 Type: NONE Confidence: ND	COMP-MESON E04 / M08
301. f_2(1950) MASS f_2(1950)0 PDG2026:observable:M135:M135M:335075	1936+-12 MeV	u_+ : 11.73086748275699 u_- : 11.73086748275699 Type: NONE Confidence: ND	COMP-MESON E04 / M08
302. f_2(1910) omega omega MODE f_2(1910)0 PDG2026:observable:M142:M142M2:335057	1900+-9 MeV	u_+ : 9.266557494580514 u_- : 9.266557494580514 Type: NONE Confidence: ND	COMP-MESON E04 / M08
303. h_c(1P) MASS h_c(1P) PDG2026:observable:M144:M144M:344851	3525.37+-0.14 MeV	u_+ : 0.14473463841322931 u_- : 0.14473463841322931 Type: NONE Confidence: ND	COMP-MESON E04 / M08
304. K_2(1820) MASS K_2(1820)0 Kbar_2(1820)0 K_2(1820)+ K_2(1820)- PDG2026:observable:M146:M146M:335990	1819+-12 MeV	u_+ : 12.067628581227879 u_- : 12.471842992923371 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
305. Mass (Breit-Wigner) $f_0(1370)0$ PDG2026:observable:M147:M147M:334506	1200 to 1500 MeV	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
306. $D_{s2}^*(2573)$ MASS $D_{s2}^{*+}(2573) + D_{s2}^{*-}(2573) -$ PDG2026:observable:M148:M148M:338509	2569.1+-0.8 MeV	u_+ : 0.82780630176758 352 u_- : 0.82649831496169 768 Type: NONE Confidence: ND	COMP-MESON E04 / M08
307. $a_0(1450)$ MASS $a_0(1450)0 a_0(1450)+ a_0(1450)-$ PDG2026:observable:M149:M149M:334579	1439+-34 MeV	u_+ : 34.4262301167357 38 u_- : 34.4262301167357 38 Type: NONE Confidence: ND	COMP-MESON E04 / M08
308. $B_J^*(5732)$ MASS $B_J^{*+}(5732)0 B_{\bar{B}J}^{*+}(5732)0 B_J^{*+}(5732)+ B_J^{*-}(5732)-$ PDG2026:observable:M151:M151M:342551	5698+-8 MeV	u_+ : 7.58694682434463 72 u_- : 7.69302546888399 34 Type: NONE Confidence: ND	COMP-MESON E04 / M08
309. $f_0(1500)$ MASS $f_0(1500)0$ PDG2026:observable:M152:M152M:334628	1522+-25 MeV	u_+ : 25 u_- : 25 Type: NONE Confidence: ND	COMP-MESON E04 / M08
310. $B_{sJ}^*(5850)$ MASS $B_{sJ}^{*+}(5850)0 B_{\bar{B}sJ}^{*+}(5850)0$ PDG2026:observable:M153:M153M:343151	5853+-15 MeV	u_+ : 15 u_- : 15 Type: NONE Confidence: ND	COMP-MESON E04 / M08
311. $\eta_2(1645)$ MASS $\eta_2(1645)0$ PDG2026:observable:M154:M154M:334758	1617+-5 MeV	u_+ : 5.26131479626401 97 u_- : 5.26131479626401 97 Type: NONE Confidence: ND	COMP-MESON E04 / M08
312. $f_0(2020)$ MASS $f_0(2020)0$ PDG2026:observable:M156:M156M:335120	1982+54.1-3.0 MeV	u_+ : 54.0832691319598 38 u_- : 3 Type: NONE Confidence: ND	COMP-MESON E04 / M08
313. $D^*(2640)^{-}$ MASS $D^{*+}(2640)0 D_{\bar{B}}^{*+}(2640)0 D^{*+}(2640)+ D^{*+}(2640)-$ PDG2026:observable:M158:M158M:337854	2637+-6 MeV	u_+ : 6.32455532033675 9 u_- : 6.32455532033675 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
314. $\chi_{c0}(3915)$ MASS $\chi_{c0}(3915)$ PDG2026:observable:M159:M159M:346204	3922.1+-1.8 MeV	u_+ : 1.77307050384254 29 u_- : 1.76036183430800 81 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
315. K(1630) MASS K(1630)0 Kbar(1630)0 K(1630)+ K(1630)- PDG2026:observable:M160:M160M:335942	1629+-7 MeV	u_+ : 7 u_- : 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
316. a_1(1640) MASS a_1(1640)0 a_1(1640)+ a_1(1640)- PDG2026:observable:M161:M161M:334740	1655+-16 MeV	u_+ : 15.6153624587929 11 u_- : 16.7429048405724 61 Type: NONE Confidence: ND	COMP-MESON E04 / M08
317. a_2(1700) MASS a_2(1700)0 a_2(1700)+ a_2(1700)- PDG2026:observable:M162:M162M:334903	1706+-14 MeV	u_+ : 13.5014409315998 2 u_- : 14.8390744674554 29 Type: NONE Confidence: ND	COMP-MESON E04 / M08
318. pi_1(1600) T-Matrix Pole sqrt(s) pi_1(1600)0 pi_1(1600)+ pi_1(1600)- PDG2026:observable:M164:M164TMP:334723	(1480 -- 1680) - i (150 -- 300) MeV Display: (1480--1680) -i(150--300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
319. pi_1(1600) MASS (eta pi mode) pi_1(1600)0 pi_1(1600)+ pi_1(1600)- PDG2026:observable:M164:M164MEP:334724	1354+-25 MeV	u_+ : 25.6203714275794 11 u_- : 24.3777786955451 Type: NONE Confidence: ND	COMP-MESON E04 / M08
320. pi_1(1600) MASS (non-eta pi mode) pi_1(1600)0 pi_1(1600)+ pi_1(1600)- PDG2026:observable:M164:M164M:334725	1645+-40-17 MeV	u_+ : 39.7197562267507 18 u_- : 16.5756798763040 68 Type: NONE Confidence: ND	COMP-MESON E04 / M08
321. h_1(1595) MASS h_1(1595)0 PDG2026:observable:M166:M166M:334720	1594+-18-60 MeV	u_+ : 18.0277563773199 5 u_- : 61.8465843842649 08 Type: NONE Confidence: ND	COMP-MESON E04 / M08
322. f_0(2100) MASS f_0(2100)0 PDG2026:observable:M168:M168M:335156	2095+-17-19 MeV	u_+ : 17.0604966887379 81 u_- : 18.9796610074380 18 Type: NONE Confidence: ND	COMP-MESON E04 / M08
323. eta_b(1S) MASS eta_b(1S) PDG2026:observable:M171:M171M:346822	9398.7+-2.0 MeV	u_+ : 2.08161014572945 2 u_- : 2.00237037046477 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
324. $D_s^0(2317)^+$ MASS $D_s^0(2317)^+ D_s^0(2317)^-$ PDG2026:observable:M172:M172M:338429	2317.8+-0.5 MeV	u_+ : 0.51003296350202 032 u_- : 0.51003296350202 032 Type: NONE Confidence: ND	COMP-MESON E04 / M08
325. $D_s1(2460)^+$ MASS $D_s1(2460)^+ D_s1(2460)^-$ PDG2026:observable:M173:M173M:338448	2459.5+-0.6 MeV	u_+ : 0.63443027023618 292 u_- : 0.63443027023618 292 Type: NONE Confidence: ND	COMP-MESON E04 / M08
326. $K_0^*(700)$ T-Matrix Pole sqrt(s) $K_0^*(700)0 K_{\bar{0}}^*(700)0 K_0^*(700)^+ K_0^*(700)^-$ PDG2026:observable:M174:M174TMP:335785	(630 -- 730) - i (260 -- 340) MeV Display: (630--730) -i(260--340)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
327. $K_0^*(700)$ Breit-Wigner Mass $K_0^*(700)0 K_{\bar{0}}^*(700)0 K_0^*(700)^+ K_0^*(700)^-$ PDG2026:observable:M174:M174M:335786	838+-11 MeV	u_+ : 11.3879040790194 1 u_- : 11.2973395209359 Type: NONE Confidence: ND	COMP-MESON E04 / M08
328. $\eta(1475)$ MASS $\eta(1475)0$ PDG2026:observable:M175:M175M5:334616	1476+-4 MeV	u_+ : 3.92907502750456 58 u_- : 4.01031273484770 26 Type: NONE Confidence: ND	COMP-MESON E04 / M08
329. $\chi_{c1}(3872)$ MASS FROM J/psi X MODE $\chi_{c1}(3872)$ PDG2026:observable:M176:M176M:346130	3871.64+-0.06 MeV	u_+ : 0.05709885902177 2717 u_- : 0.05709583843793 0757 Type: NONE Confidence: ND	COMP-MESON E04 / M08
330. $Upsilon_2(1D)$ MASS $Upsilon_2(1D)$ PDG2026:observable:M177:M177M:347489	10163.7+-1.4 MeV	u_+ : 1.43860105887897 99 u_- : 1.43860105887897 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
331. $D_1(2430)0$ MASS $D_1(2430)0 D_{\bar{1}}(2430)0$ PDG2026:observable:M180:M180M:337827	2412+-9 MeV	u_+ : 9.17479434026708 77 u_- : 9.17479434026708 77 Type: NONE Confidence: ND	COMP-MESON E04 / M08
332. $\psi(4360)$ MASS $\psi(4360)$ PDG2026:observable:M181:M181M:346635	4373+-7 MeV	u_+ : 7.02670299984180 07 u_- : 7.12633898599779 33 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
333. D_{s1}[*](2700)⁺ MASS D _{s1} [*] (2700) ⁺ D _{s1} [*] (2700) ⁻ PDG2026:observable:M182:M182M:338524	2714⁺-5 MeV	u_+ : 5.33752618609189 34 u_- : 4.57804726750868 96 Type: NONE Confidence: ND	COMP-MESON E04 / M08
334. pi₂(1880)0 MASS pi ₂ (1880)0 PDG2026:observable:M185:M185M:335043	1874⁺+26-5 MeV	u_+ : 25.8979460588110 09 u_- : 4.79881566383704 16 Type: NONE Confidence: ND	COMP-MESON E04 / M08
335. B_{s2}[*](5840)0 MASS B _{s2} [*] (5840)0 Bbar _{s2} [*] (5840)0 PDG2026:observable:M186:M186M:343137	5839.88⁺-0.12 MeV	u_+ : 0.11598422775051 18 u_- : 0.11598422775051 18 Type: NONE Confidence: ND	COMP-MESON E04 / M08
336. B_{s1}(5830)0 MASS B _{s1} (5830)0 Bbar _{s1} (5830)0 PDG2026:observable:M187:M187M:343129	5828.73⁺-0.20 MeV	u_+ : 0.19651551951131 191 u_- : 0.19651551951131 191 Type: NONE Confidence: ND	COMP-MESON E04 / M08
337. rho(1570)0 MASS rho(1570)0 PDG2026:observable:M188:M188M:334714	1570⁺-70 MeV	u_+ : 71.6937933157396 83 u_- : 71.6937933157396 83 Type: NONE Confidence: ND	COMP-MESON E04 / M08
338. psi(4660) MASS psi(4660) PDG2026:observable:M189:M189M:346767	4623⁺-10 MeV	u_+ : 10.2090449422513 1 u_- : 10.1279727634031 61 Type: NONE Confidence: ND	COMP-MESON E04 / M08
339. X(4160) MASS X(4160) PDG2026:observable:M190:M190M:346451	4153⁺+23-21 MeV	u_+ : 23.0376548187802 9 u_- : 20.8165617804000 72 Type: NONE Confidence: ND	COMP-MESON E04 / M08
340. T_ccbar(4050)⁺ MASS T _c cbar(4050) ⁺ PDG2026:observable:M191:M191M:347990	4051⁺+24-40 MeV	u_+ : 24.4131112314674 01 u_- : 43.3243580448689 37 Type: NONE Confidence: ND	COMP-MESON E04 / M08
341. T_ccbar(4250)⁺ MASS T _c cbar(4250) ⁺ PDG2026:observable:M192:M192M:348026	4250⁺+190-50 MeV	u_+ : 185.299757150407 49 u_- : 45.4532727094540 48 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
342. chi_c1(4140) MASS chi_c1(4140) PDG2026:observable:M193:M193M:346336	4146.5+-3.0 MeV	u_+ : 2.98854241183424 7 u_- : 3.00161533279208 29 Type: NONE Confidence: ND	COMP-MESON E04 / M08
343. X(4350) MASS X(4350) PDG2026:observable:M194:M194M:346628	4351+-5 MeV	u_+ : 4.65295604965273 67 u_- : 5.14781507049350 04 Type: NONE Confidence: ND	COMP-MESON E04 / M08
344. T_c_cbar_1(4430)+ MASS T_c_cbar_1(4430)+ PDG2026:observable:M195:M195M:348030	4478+15-17 MeV	u_+ : 14.7691701609154 1 u_- : 17.4534435216461 89 Type: NONE Confidence: ND	COMP-MESON E04 / M08
345. D_s1*(2860)+ MASS D_s1^*(2860)+ D_s1^*(2860)- PDG2026:observable:M196:M196M:338533	2859+-27 MeV	u_+ : 26.8328157299974 81 u_- : 26.8328157299974 81 Type: NONE Confidence: ND	COMP-MESON E04 / M08
346. D_sJ(3040)+ MASS D_sJ(3040)+ D_sJ(3040)- PDG2026:observable:M197:M197M:338544	3044+31-9 MeV	u_+ : 31.0483493925200 48 u_- : 9.43398113205660 32 Type: NONE Confidence: ND	COMP-MESON E04 / M08
347. D_0(2550)0 MASS D_0(2550)0 Dbar_0(2550)0 D_0(2550)+ D_0(2550)- PDG2026:observable:M198:M198M:337843	2549+-19 MeV	u_+ : 19.0040028413363 41 u_- : 19.0040028413363 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
348. D_1*(2600)0 MASS D_1^*(2600)0 Dbar_1^*(2600)0 D_1^*(2600)+ D_1^*(2600)- PDG2026:observable:M199:M199M:337846	2627+-10 MeV	u_+ : 9.81282444363102 79 u_- : 9.81282444363102 79 Type: NONE Confidence: ND	COMP-MESON E04 / M08
349. eta_b(2S) MASS eta_b(2S) PDG2026:observable:M200:M200M:347281	9999+-4 MeV	u_+ : 4.48218696620299 36 u_- : 3.98246155034797 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
350. D_3*(2750) MASS D_3^*(2750)0 Dbar_3^*(2750)0 D_3^*(2750)+ D_3^*(2750)- PDG2026:observable:M203:M203M:337860	2763.1+-3.2 MeV	u_+ : 3.19085331467817 88 u_- : 3.19085331467817 88 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
351. h_b(1P) MASS h _b (1P) PDG2026:observable:M204:M204M:347229	9899.3+-0.8 MeV	u_+ : 0.75076896116034 797 u_- : 0.75076896116034 797 Type: NONE Confidence: ND	COMP-MESON E04 / M08
352. h_b(2P) MASS h _b (2P) PDG2026:observable:M205:M205M:347593	10259.8+-1.2 MeV	u_+ : 1.20830459735945 71 u_- : 1.20830459735945 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
353. chi_b1(3P) MASS chi _b 1(3P) PDG2026:observable:M206:M206M:347734	10513.4+-0.7 MeV	u_+ : 0.67007462271003 815 u_- : 0.67007462271003 815 Type: NONE Confidence: ND	COMP-MESON E04 / M08
354. T_bbbar₁(10610)+- MASS T _b bbar ₁ (10610)+ PDG2026:observable:M207:M207M:348052	10607.2+-2.0 MeV	u_+ : 2 u_- : 2 Type: NONE Confidence: ND	COMP-MESON E04 / M08
355. T_bbbar₁(10610)0 MASS T _b bbar ₁ (10610)+ PDG2026:observable:M207:M207M0:348053	10609+-6 MeV	u_+ : 5.65685424949238 06 u_- : 5.65685424949238 06 Type: NONE Confidence: ND	COMP-MESON E04 / M08
356. T_bbbar₁(10650)+ MASS T _b bbar ₁ (10650)+ PDG2026:observable:M208:M208M:348075	10652.2+-1.5 MeV	u_+ : 1.5 u_- : 1.5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
357. T_ccbar₁(3900) MASS T _c cbar ₁ (3900)+ PDG2026:observable:M210:M210M:347938	3886.7+-2.3 MeV	u_+ : 2.29336919741447 7 u_- : 2.27328228957628 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
358. psi₂(3823) MASS psi ₂ (3823) PDG2026:observable:M212:M212M:346100	3823.51+-0.34 MeV	u_+ : 0.33904706932929 501 u_- : 0.33904706932929 501 Type: NONE Confidence: ND	COMP-MESON E04 / M08
359. T_ccbar(4020) MASS T _c cbar(4020)+ PDG2026:observable:M213:M213M:347976	4024.1+-1.9 MeV	u_+ : 1.82469063465848 7 u_- : 1.96716232389574 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
360. T_c_cbar_0(4240)+ MASS T_c_cbar_0(4240)0 PDG2026:observable:M216:M216M:348022	4239+50-21 MeV	u_+ : 48.466483264210538 u_- : 20.591260281974002 Type: NONE Confidence: ND	COMP-MESON E04 / M08
361. B_c(2S)+- MASS B_c(2S)+ B_c(2S)- PDG2026:observable:M217:M217M:343283	6871.2+-1.0 MeV	u_+ : 1.018454581927803 u_- : 1.018454581927803 Type: NONE Confidence: ND	COMP-MESON E04 / M08
362. T_c_cbar(4055)+ MASS T_c_cbar(4055)+ PDG2026:observable:M223:M223M:348000	4054+-3.2 MeV	u_+ : 3.16227766016838 u_- : 3.16227766016838 Type: NONE Confidence: ND	COMP-MESON E04 / M08
363. D_s3*(2860)+ MASS D_s3^*(2860)+ D_s3^*(2860)- PDG2026:observable:M226:M226M:338535	2860+-7 MeV	u_+ : 7.0007142492748553 u_- : 7.0007142492748553 Type: NONE Confidence: ND	COMP-MESON E04 / M08
364. a_0(1950) MASS a_0(1950)0 PDG2026:observable:M227:M227M:335070	1931+-26 MeV	u_+ : 26.0768096208106 u_- : 26.0768096208106 Type: NONE Confidence: ND	COMP-MESON E04 / M08
365. D_2(2740)0 MASS D_2(2740)0 Dbar_2(2740)0 D_2(2740)+ D_2(2740)- PDG2026:observable:M228:M228M:337857	2747+-6 MeV	u_+ : 6.3932930732238127 u_- : 6.3932930732238127 Type: NONE Confidence: ND	COMP-MESON E04 / M08
366. D_2*(3000)0 MASS D_2^*(3000)0 Dbar_2^*(3000)0 D_2^*(3000)+ D_2^*(3000)- PDG2026:observable:M229:M229M:337872	3210+-60 MeV	u_+ : 56.938563381947027 u_- : 56.938563381947027 Type: NONE Confidence: ND	COMP-MESON E04 / M08
367. T_c_cbar_1(4200)+ MASS T_c_cbar_1(4200)+ PDG2026:observable:M231:M231M:348012	4242+-26 MeV	u_+ : 26.306839009567629 u_- : 25.567261845351091 Type: NONE Confidence: ND	COMP-MESON E04 / M08
368. T_b_sbar(5568)+ MASS T_b_sbar(5568)+ PDG2026:observable:M232:M232M:348036	5566.9+-3.3 MeV	u_+ : 3.255764119219942 u_- : 3.324154027718933 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
369. chi_c1(4274) MASS chi_c1(4274) PDG2026:observable:M233:M233M:346624	4290+-6 MeV	u_+ : 5.68239790637520 86 u_- : 6.46108198239139 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
370. chi_c0(4500) MASS chi_c0(4500) PDG2026:observable:M234:M234M:346755	4484+-12 MeV	u_+ : 12.0028200724577 39 u_- : 11.9852845503755 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
371. chi_c0(4700) MASS chi_c0(4700) PDG2026:observable:M235:M235M:346816	4708+6-4 MeV	u_+ : 5.96903618444770 42 u_- : 3.94085054655756 12 Type: NONE Confidence: ND	COMP-MESON E04 / M08
372. chi_c0(3860) MASS chi_c0(3860) PDG2026:observable:M237:M237M:346122	3862+50-35 MeV	u_+ : 47.7074417675062 53 u_- : 34.5398320783410 92 Type: NONE Confidence: ND	COMP-MESON E04 / M08
373. chi_b2(3P) MASS chi_b2(3P) PDG2026:observable:M238:M238M:347741	10524.0+-0.8 MeV	u_+ : 0.77833154889160 183 u_- : 0.77833154889160 183 Type: NONE Confidence: ND	COMP-MESON E04 / M08
374. pi_2(2005) MASS pi_2(2005)0 PDG2026:observable:M239:M239M:335106	1963+17-27 MeV	u_+ : 16.6635405259114 29 u_- : 27.4183231822566 39 Type: NONE Confidence: ND	COMP-MESON E04 / M08
375. T_c_cbar(4100)+ MASS T_c_cbar(4100)+ PDG2026:observable:M240:M240M:348006	4096+-28 MeV	u_+ : 26.9072480941474 2 u_- : 29.7321374946370 09 Type: NONE Confidence: ND	COMP-MESON E04 / M08
376. psi_3(3842) MASS psi_3(3842) PDG2026:observable:M241:M241M:346116	3842.71+-0.20 MeV	u_+ : 0.200000000000000 001 u_- : 0.200000000000000 001 Type: NONE Confidence: ND	COMP-MESON E04 / M08
377. Upsilon(10753) MASS Upsilon(10753) PDG2026:observable:M243:M243M:347800	10756.6+-2.8 MeV	u_+ : 2.84604989415154 2 u_- : 2.84604989415154 2 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
378. B₁(5721)+ mass B ₁ (5721)0 Bbar ₁ (5721)0 B ₁ (5721)+ B ₁ (5721)- PDG2026:observable:M244:M244M+:342539	5726.0+2.5-2.7 MeV	u_+ : 2.45596777983158 4 u_- : 2.72915140828024 59 Type: NONE Confidence: ND	COMP-MESON E04 / M08
379. B₁(5721)0 mass B ₁ (5721)0 Bbar ₁ (5721)0 B ₁ (5721)+ B ₁ (5721)- PDG2026:observable:M244:M244M0:342542	5726.1+-1.2 MeV	u_+ : 1.23302959923312 99 u_- : 1.26292541194705 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
380. B₂[*](5747)+ mass B ₂ [*] (5747)0 Bbar ₂ [*] (5747)0 B ₂ [*] (5747)+ B ₂ [*] (5747)- PDG2026:observable:M245:M245M+:342556	5737.3+-0.7 MeV	u_+ : 0.71060165855206 67 u_- : 0.73206597699796 105 Type: NONE Confidence: ND	COMP-MESON E04 / M08
381. B₂[*](5747)0 mass B ₂ [*] (5747)0 Bbar ₂ [*] (5747)0 B ₂ [*] (5747)+ B ₂ [*] (5747)- PDG2026:observable:M245:M245M0:342559	5739.6+-0.7 MeV	u_+ : 0.65346568058668 519 u_- : 0.65346568058668 519 Type: NONE Confidence: ND	COMP-MESON E04 / M08
382. B_J(5840)+ MASS B _J (5840)0 Bbar _J (5840)0 B _J (5840)+ B _J (5840)- PDG2026:observable:M246:M246M+:342571	5851+-19 MeV	u_+ : 19.1051594213077 78 u_- : 19.1051594213077 78 Type: NONE Confidence: ND	COMP-MESON E04 / M08
383. B_J(5840)0 MASS B _J (5840)0 Bbar _J (5840)0 B _J (5840)+ B _J (5840)- PDG2026:observable:M246:M246M0:342574	5863+-9 MeV	u_+ : 8.60261899078296 47 u_- : 8.60261899078296 47 Type: NONE Confidence: ND	COMP-MESON E04 / M08
384. eta(2370) MASS eta(2370)0 PDG2026:observable:M247:M247M:335261	2377+-9 MeV	u_+ : 8.80721158281389 9 u_- : 9.65401633399925 Type: NONE Confidence: ND	COMP-MESON E04 / M08
385. B_J(5970)+ MASS B _J (5970)0 Bbar _J (5970)0 B _J (5970)+ B _J (5970)- PDG2026:observable:M248:M248M+:342583	5965+-5 MeV	u_+ : 4.50537231097705 78 u_- : 4.50537231097705 78 Type: NONE Confidence: ND	COMP-MESON E04 / M08
386. B_J(5970)0 MASS B _J (5970)0 Bbar _J (5970)0 B _J (5970)+ B _J (5970)- PDG2026:observable:M248:M248M0:342586	5971+-5 MeV	u_+ : 5.34798685867918 88 u_- : 5.34798685867918 88 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
387. D₁[*](2760)0 MASS D ₁ [*] (2760)0 Dbar ₁ [*] (2760)0 D ₁ [*] (2760)+ D ₁ [*] (2760)- PDG2026:observable:M249:M249M:337868	2781+-22 MeV	u_+ : 22.2036033111745 18 u_- : 22.2036033111745 18 Type: NONE Confidence: ND	COMP-MESON E04 / M08
388. T_{cbar}sbar₀[*](2870)0 MASS T _{cbar} sbar ₀ [*] (2870)0 PDG2026:observable:M250:M250M:347916	2874+-11 MeV	u_+ : 10.7371274384021 71 u_- : 10.7371274384021 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
389. T_{cbar}sbar₁[*](2900)0 MASS T _{cbar} sbar ₁ [*] (2900)0 PDG2026:observable:M251:M251M:347922	2904+-5 MeV	u_+ : 5.09901951359278 36 u_- : 5.09901951359278 36 Type: NONE Confidence: ND	COMP-MESON E04 / M08
390. D₀[*](2300) MASS D ₀ [*] (2300)0 Dbar ₀ [*] (2300)0 D ₀ [*] (2300)+ D ₀ [*] (2300)- PDG2026:observable:M252:M252M:337813	2343+-10 MeV	u_+ : 9.65444207944618 43 u_- : 9.65444207944618 43 Type: NONE Confidence: ND	COMP-MESON E04 / M08
391. D₁(2420) MASS D ₁ (2420)0 Dbar ₁ (2420)0 D ₁ (2420)+ D ₁ (2420)- PDG2026:observable:M253:M253M:337817	2422.1+-0.6 MeV	u_+ : 0.57502282665337 301 u_- : 0.57502282665337 301 Type: NONE Confidence: ND	COMP-MESON E04 / M08
392. D₂[*](2460) MASS D ₂ [*] (2460)0 Dbar ₂ [*] (2460)0 D ₂ [*] (2460)+ D ₂ [*] (2460)- PDG2026:observable:M254:M254M:337830	2461.1+-0.8 MeV	u_+ : 0.79751842909338 577 u_- : 0.79751842909338 577 Type: NONE Confidence: ND	COMP-MESON E04 / M08
393. X(1750) MASS X(1750)0 PDG2026:observable:M255:M255M:334941	1753.8+-2.7 MeV	u_+ : 2.67672205618767 3 u_- : 2.73412427080383 49 Type: NONE Confidence: ND	COMP-MESON E04 / M08
394. D_{s0}(2590)+ MASS D _{s0} (2590)+ D _{s0} (2590)- PDG2026:observable:M256:M256M:338520	2591+-9 MeV	u_+ : 9.21954445729288 71 u_- : 9.21954445729288 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
395. B_{sJ}(6063)0 MASS B _{sJ} (6063)0 Bbar _{sJ} (6063)0 PDG2026:observable:M257:M257M:343153	6063.5+-1.4 MeV	u_+ : 1.44222051018559 6 u_- : 1.44222051018559 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
396. B_sJ(6114)0 MASS B _s J(6114)0 B _{bar} _s J(6114)0 PDG2026:observable:M258:M258M:343157	6114+-6 MeV	u_+ : 5.83095189484530 07 u_- : 5.83095189484530 07 Type: NONE Confidence: ND	COMP-MESON E04 / M08
397. T_c c_{bar} s_{bar} 1(4000) MASS T _c c _{bar} s _{bar} 1(4000)0 PDG2026:observable:M259:M259M:347963	3980 -- 4010 MeV Display: 3980--4010	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
398. T_c c_{bar} s_{bar} 1(4220)+ MASS T _c c _{bar} s _{bar} 1(4220)+ PDG2026:observable:M260:M260M:348018	4220+50-40 MeV	u_+ : 49.2442890089805 23 u_- : 38.4187454245970 91 Type: NONE Confidence: ND	COMP-MESON E04 / M08
399. chi_{c1}(4685) MASS chi _{c1} (4685) PDG2026:observable:M261:M261M:346810	4677+13-15 MeV	u_+ : 13.2823829672178 6 u_- : 15.1449634763239 69 Type: NONE Confidence: ND	COMP-MESON E04 / M08
400. X(4630) MASS X(4630) PDG2026:observable:M262:M262M:346761	4636+29-100 MeV	u_+ : 29.1909074314939 4 u_- : 102.406425525287 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
401. a₀(1710) MASS a ₀ (1710)0 PDG2026:observable:M263:M263M:334917	1713+-19 MeV	u_+ : 18.6275746229363 29 u_- : 18.6275746229363 29 Type: NONE Confidence: ND	COMP-MESON E04 / M08
402. f₀(1810) Breit-Wigner MASS f ₀ (1810)0 PDG2026:observable:M264:M264M:334992	1785+-13 MeV	u_+ : 13.4781230224536 11 u_- : 13.3053237583047 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
403. T_c c_c(3875)+ T-Matrix Pole sqrt(s) T _c c _c (3875)+ PDG2026:observable:M265:M265TMG:347934	(3874.75+-0.10) - i (0.024+0.001-0.007) MeV Display: 3874.75+-0.10	u_+ : ND u_- : ND Type: NONE Confidence: ND	COMP-MESON E04 / M08
404. f₀(2470) MASS f ₀ (2470)0 PDG2026:observable:M266:M266M:335273	2470+6-7 MeV	u_+ : 5.65685424949238 06 u_- : 7.21110255092797 82 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
405. eta_1(1855) MASS eta_1(1855)0 PDG2026:observable:M267:M267M:335028	1855+11-9 MeV	u_+ : 10.81665382639197 u_- : 9.0553851381374173 Type: NONE Confidence: ND	COMP-MESON E04 / M08
406. T_c_c_cbar_cbar(6900)0 MASS T_c_c_cbar_cbar(6900)+ PDG2026:observable:M268:M268M:348044	6898+-12 MeV	u_+ : 11.88793882015602 u_- : 11.51955409419207 Type: NONE Confidence: ND	COMP-MESON E04 / M08
407. T_c_sbar_0*(2900)0 MASS T_c_sbar_0^*(2900)+ PDG2026:observable:M269:M269M:347926	2892+-21 MeV	u_+ : 20.518284528683189 u_- : 20.518284528683189 Type: NONE Confidence: ND	COMP-MESON E04 / M08
408. omega(2220) MASS omega(2220)0 PDG2026:observable:M270:M270M:335221	2188+-21 MeV	u_+ : 20.57966470955094 u_- : 20.57966470955094 Type: NONE Confidence: ND	COMP-MESON E04 / M08
409. T_c c cbar cbar(6600) MASS T_c c cbar cbar(6600)+ PDG2026:observable:M272:M272M:348040	6646+-28-25 MeV	u_+ : 28.34909296526099 u_- : 24.501265528141591 Type: NONE Confidence: ND	COMP-MESON E04 / M08
410. T_c c cbar cbar(7100) MASS T_c c cbar cbar(7100)+ PDG2026:observable:M273:M273M:348048	7134+-60-29 MeV	u_+ : 63.126856408346519 u_- : 29.154759474226498 Type: NONE Confidence: ND	COMP-MESON E04 / M08
411. d-QUARK MASS d dbar PDG2026:observable:Q001:Q001M:333589	4.70+-0.07 MeV	u_+ : 0.07000000000000007 u_- : 0.07000000000000007 Type: NONE Confidence: 90	SM-Q-D1 E02 / M03
412. u-QUARK MASS u ubar PDG2026:observable:Q002:Q002M:333588	2.16+-0.07 MeV	u_+ : 0.07000000000000007 u_- : 0.07000000000000007 Type: NONE Confidence: 90	SM-Q-U1 E02 / M03
413. s-QUARK MASS s sbar PDG2026:observable:Q003:Q003M:333590	92.9 +-0.7 MeV Display: 92.9+-0.7	u_+ : 0.699999999999999996 u_- : 0.699999999999999996 Type: NONE Confidence: 90	SM-Q-D2 E02 / M03

Identity and species	Value and unit	Uncertainty	Sector and status
414. c-QUARK MASS c cbar PDG2026:observable:Q004:Q004M:333604	1.2729+-0.0045 GeV	u_+ : 0.0044999999999999 99997 u_- : 0.0044999999999999 99997 Type: NONE Confidence: 90	SM-Q-U2 E02 / M03
415. b-QUARK MASS b bbar PDG2026:observable:Q005:Q005M:333611	4.186+-0.006 GeV	u_+ : 0.0060000000000000 00001 u_- : 0.0060000000000000 00001 Type: NONE Confidence: 90	SM-Q-D3 E02 / M03
416. t-Quark Mass (Direct Measurements) t tbar PDG2026:observable:Q007:Q007TP:333614	172.60+-0.27 GeV	u_+ : 0.27137500202757 259 u_- : 0.27304780674185 603 Type: NONE Confidence: ND	SM-Q-U3 E02 / M03
417. t-Quark Mass from Cross-Section Measurements t tbar PDG2026:observable:Q007:Q007TP2:333615	162.5+2.1-1.5 GeV	u_+ : 2.13161370878597 99 u_- : 1.53243343055240 01 Type: NONE Confidence: ND	QUARK_ EXTENDED E04 / M09
418. t-Quark Pole Mass from Cross-Section Measurements t tbar PDG2026:observable:Q007:Q007TP4:333616	172.1+-0.6 GeV	u_+ : 0.61408644700147 419 u_- : 0.58967762064252 371 Type: NONE Confidence: ND	QUARK_ EXTENDED E04 / M09
419. gamma MASS gamma PDG2026:observable:S000:S000M:332461	<1 E-18 eV Display: <1E-18	u_+ : NA u_- : NA Type: U Confidence: ND	SM-GA-EM E05 / M02
420. e MASS (atomic mass units u) e- e+ PDG2026:observable:S003:S003AMU:332837	(5.485799 09044+-0.00000000010) E-4 u Display: 548.5799 09044+-0.000000010	u_+ : 9.699999999999999 91e-15 u_- : 9.699999999999999 91e-15 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
421. e MASS e- e+ PDG2026:observable:S003:S003M:332838	0.510998 95069+-0.00000000016 MeV	u_+ : 1.599999999999999 99e-10 u_- : 1.599999999999999 99e-10 Type: NONE Confidence: ND	SM-LC-1 E06 / M01
422. mu MASS (atomic mass units u) mu- mu+ PDG2026:observable:S004:S004AMU:332848	0.1134289257+-0.000000 0025 u	u_+ : 2.500000000000000 01e-09 u_- : 2.500000000000000 01e-09 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

Identity and species	Value and unit	Uncertainty	Sector and status
423. mu MASS mu- mu+ PDG2026:observable:S004:S004M:332849	105.6583755+-0.0000023 MeV	u_+ : 2.3e-06 u_- : 2.3e-06 Type: NONE Confidence: ND	SM-LC-2 E03 / M04
424. pi+- MASS pi+ pi- PDG2026:observable:S008:S008M:333660	139.57039+-0.00018 MeV	u_+ : 0.00018200716040 826009 u_- : 0.00018200716040 826009 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
425. pi0 MASS pi0 PDG2026:observable:S009:S009M:333693	134.9768+-0.0005 MeV	u_+ : 0.00048567928228 423351 u_- : 0.00048567928228 423351 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
426. K+- MASS K+ K- PDG2026:observable:S010:S010M:335281	493.677+-0.015 MeV	u_+ : 0.01540566522607 1509 u_- : 0.01540566522607 1509 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
427. K0 MASS K0 Kbar0 PDG2026:observable:S011:S011M:335501	497.611+-0.013 MeV	u_+ : 0.01299019975642 42 u_- : 0.01299019975642 42 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
428. eta MASS eta PDG2026:observable:S014:S014M:333729	547.862+-0.017 MeV	u_+ : 0.01733421371191 4789 u_- : 0.01733421371191 4789 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
429. p MASS (atomic mass units u) p pbar PDG2026:observable:S016:S016AMU:348093	1.007276 4665789+-0.000000 0000083 u	u_+ : 8.299999999999999 98e-12 u_- : 8.299999999999999 98e-12 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
430. p MASS (MeV) p pbar PDG2026:observable:S016:S016M:348095	938.27208943+-0.000000 29 MeV	u_+ : 2.899999999999999 98e-07 u_- : 2.899999999999999 98e-07 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
431. n MASS (atomic mass units u) n nbar PDG2026:observable:S017:S017AMU:348291	1.0086649161+-0.000000 0004 u	u_+ : 4.000000000000000 01e-10 u_- : 4.000000000000000 01e-10 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

Identity and species	Value and unit	Uncertainty	Sector and status
432. n MASS (MeV) n nbar PDG2026:observable:S017:S017M:348292	939.5654219+-0.0000005 MeV	u_+ : 4.799999999999999 96e-07 u_- : 4.799999999999999 96e-07 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
433. Lambda MASS Lambda0 Lambdabar0 PDG2026:observable:S018:S018M:349367	1115.683+-0.006 MeV	u_+ : 0.00643763777862 19867 u_- : 0.00643763777862 19867 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
434. Sigma+ MASS Sigma+ Sigmabar- PDG2026:observable:S019:S019M:349781	1189.37+-0.07 MeV	u_+ : 0.06962288561220 7835 u_- : 0.06962288561220 7835 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
435. Sigma- MASS Sigma- Sigmabar+ PDG2026:observable:S020:S020M:349844	1197.449+-0.029 MeV	u_+ : 0.02915850307660 851 u_- : 0.02915850307660 851 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
436. Sigma0 MASS Sigma0 Sigmabar0 PDG2026:observable:S021:S021M:349828	1192.642+-0.024 MeV	u_+ : 0.02359569142575 4781 u_- : 0.02359569142575 4781 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
437. Xi- MASS Xi- Xibar+ PDG2026:observable:S022:S022M:350344	1321.71+-0.07 MeV	u_+ : 0.06653527362254 3412 u_- : 0.06653527362254 3412 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
438. Xibar+ MASS Xi- Xibar+ PDG2026:observable:S022:S022M1:350346	1321.71+-0.07 MeV	u_+ : 0.06653527362254 3412 u_- : 0.06653527362254 3412 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
439. Xi0 MASS Xi0 Xibar0 PDG2026:observable:S023:S023M:350293	1314.86+-0.20 MeV	u_+ : 0.20343194946851 781 u_- : 0.20343194946851 781 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
440. Omega- MASS Omega- Omegabar+ PDG2026:observable:S024:S024M:350469	1672.45+-0.29 MeV	u_+ : 0.29314631976651 251 u_- : 0.29314631976651 251 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

Identity and species	Value and unit	Uncertainty	Sector and status
441. Sigma_b+ MASS Sigma_b()- Sigambar_b()+ Sigma_b()0 Sigambar_b()0 Sigma_b()+ Sigambar_b()- PDG2026:observable:S026:S026M+:351717	5810.56+-0.25 MeV	u_+ : 0.25274062207162 651 u_- : 0.25263562022485 619 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
442. Sigma_b- MASS Sigma_b()- Sigambar_b()+ Sigma_b()0 Sigambar_b()0 Sigma_b()+ Sigambar_b()- PDG2026:observable:S026:S026M-:351718	5815.64+-0.27 MeV	u_+ : 0.27460654053367 178 u_- : 0.27449500386905 717 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
443. D+- MASS D+ D- PDG2026:observable:S031:S031M:336044	1869.66+-0.05 MeV	u_+ : 0.04680602241160 9027 u_- : 0.04680602241160 9027 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
444. D0 MASS D0 Dbar0 PDG2026:observable:S032:S032M:336689	1864.84+-0.04 MeV	u_+ : 0.04445870561980 1281 u_- : 0.04445870561980 1281 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
445. Lambda_c+ MASS Lambda_c()+ Lambdabar_c()- PDG2026:observable:S033:S033M:350527	2286.46+-0.14 MeV	u_+ : 0.13999999999999 89 u_- : 0.13999999999999 89 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
446. D_s+- MASS D_s()+ D_s()- PDG2026:observable:S034:S034M:337875	1968.35+-0.07 MeV	u_+ : 0.06786120952959 6483 u_- : 0.06786120952959 6483 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
447. tau MASS tau- tau+ PDG2026:observable:S035:S035M:332893	1776.93+-0.09 MeV	u_+ : 0.08491495391323 0548 u_- : 0.08758019173411 4063 Type: NONE Confidence: ND	SM-LC-3 E03 / M04
448. mass(Lambda_b0) Lambda_b()0 Lambdabar_b()0 PDG2026:observable:S040:S040M:351420	5619.57+-0.16 MeV	u_+ : 0.15728560301875 261 u_- : 0.15728560301875 261 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
449. B+- MASS B+ B- PDG2026:observable:S041:S041M:338546	5279.41+-0.07 MeV	u_+ : 0.07108797774343 7336 u_- : 0.07108797774343 7336 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

Identity and species	Value and unit	Uncertainty	Sector and status
450. B0 MASS B0 Bbar0 PDG2026:observable:S042:S042M:340312	5279.72+-0.08 MeV	u_+ : 0.08435943092202 2555 u_- : 0.08435943092202 2555 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
451. W MASS W+ W- PDG2026:observable:S043:S043M:332464	80.3625 +-0.0077 GeV Display: 80.3625+-0.0077	u_+ : 0.00770000000000 00002 u_- : 0.00770000000000 00002 Type: NONE Confidence: ND	SM-GA-W E01 / M06
452. Z MASS Z0 PDG2026:observable:S044:S044M:332513	91.1879+-0.0020 GeV	u_+ : 0.00196689903262 91501 u_- : 0.00196689903262 91501 Type: NONE Confidence: ND	SM-GA-Z E01 / M06
453. Xi_c+ MASS Xi_c()+ Xibar_c()- PDG2026:observable:S045:S045M:351029	2467.79+-0.15 MeV	u_+ : 0.14783797163412 299 u_- : 0.15152111693025 43 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
454. Omega_c0 MASS Omega_c()0 Omegabar_c()0 PDG2026:observable:S047:S047M:351327	2695.3+-0.4 MeV	u_+ : 0.40393344242041 429 u_- : 0.40393344242041 429 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
455. Xi_c0 MASS Xi_c()0 Xibar_c()0 PDG2026:observable:S048:S048M:351116	2470.50+-0.25 MeV	u_+ : 0.24609003107560 39 u_- : 0.25936293704487 828 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
456. Omega_c(2770)0 MASS Omega_c(2770)0 Omegabar_c(2770)0 PDG2026:observable:S053:S053M:351372	2766.0+0.9-1.0 MeV	u_+ : 0.90442879377866 114 u_- : 1.02466734765194 1 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
457. Xi_c'+ MASS Xi_c^{}()+ Xibar_c^{}()- PDG2026:observable:S058:S058M:351195	2578.3+-0.4 MeV	u_+ : 0.43863752697991 182 u_- : 0.43863752697991 182 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
458. Xi_c'0 MASS Xi_c^{}()0 Xibar_c^{}()0 PDG2026:observable:S059:S059M:351200	2578.8+-0.5 MeV	u_+ : 0.48022871688352 092 u_- : 0.48697631770634 259 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

Identity and species	Value and unit	Uncertainty	Sector and status
459. Sigma_b*+ MASS Sigma_b^*(0)- Sigmabar_b^*(0)+ Sigma_b^*(0) Sigmabar_b^*(0) Sigma_b^*(0)+ Sigmabar_b^*(0)- PDG2026:observable:S062:S062M+:351724	5830.32+-0.27 MeV	u_+ : 0.27472915359118 399 u_- : 0.27501743053510 241 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
460. Sigma_b*- MASS Sigma_b^*(0)- Sigmabar_b^*(0)+ Sigma_b^*(0) Sigmabar_b^*(0) Sigma_b^*(0)+ Sigmabar_b^*(0)- PDG2026:observable:S062:S062M-:351725	5834.74+-0.30 MeV	u_+ : 0.29816079945753 06 u_- : 0.29855805603668 217 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
461. Omega_b- MASS Omega_b(0)- Omegabar_b(0)+ PDG2026:observable:S063:S063M:351884	6045.8+-0.8 MeV	u_+ : 0.76541057086097 497 u_- : 0.76541057086097 497 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
462. Xi_cc+ MASS Xi_cc(0)+ Xibar_cc(0)- PDG2026:observable:S065:S065M:351404	3518.9+-0.9 MeV	u_+ : 0.94868329805051 377 u_- : 0.94868329805051 377 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
463. Xi_cc++ MASS Xi_cc(0)++ Xi_cc(0)-- PDG2026:observable:S068:S068M:351409	3621.6+-0.4 MeV	u_+ : 0.37802116342871 6 u_- : 0.37802116342871 6 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
464. Xi_b- MASS Xi_b(0)- Xibar_b(0)+ PDG2026:observable:S069:S069M-:351742	5797.0+-0.4 MeV	u_+ : 0.43674265363571 568 u_- : 0.43674265363571 568 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
465. Xi_b0 MASS Xi_b(0)0 Xibar_b(0)0 PDG2026:observable:S070:S070M0:351793	5791.7+-0.4 MeV	u_+ : 0.41342961949753 559 u_- : 0.41342961949753 559 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
466. D_s*+- MASS D_s^*(0)+ D_s^*(0)- PDG2026:observable:S074:S074M:338415	2112.2+-0.4 MeV	u_+ : 0.39518670911609 538 u_- : 0.39518670911609 538 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
467. B* MASS B^*0 Bbar^*0 B^*+ B^*- PDG2026:observable:S085:S085M:342531	5324.75+-0.20 MeV	u_+ : 0.20338563693989 911 u_- : 0.20338563693989 911 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

Identity and species	Value and unit	Uncertainty	Sector and status
468. B_s0 MASS B _s ()0 Bbar _s ()0 PDG2026:observable:S086:S086M:342595	5366.93+-0.10 MeV	u_+ : 0.09935167389588 262 u_- : 0.09935167389588 262 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
469. B_s* MASS B _s ^*0 Bbar _s ^*0 PDG2026:observable:S087:S087M:343123	5415.4+-1.4 MeV	u_+ : 1.43538923919178 2 u_- : 1.43538923919178 2 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
470. B_c+ MASS B _c ()+ B _c ()- PDG2026:observable:S091:S091M:343161	6274.47+-0.32 MeV	u_+ : 0.31906112267087 638 u_- : 0.31906112267087 638 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
471. H MASS H PDG2026:observable:S126:S126M:332733	125.13+-0.11 GeV	u_+ : 0.11172144776548 799 u_- : 0.11172144776548 799 Type: NONE Confidence: ND	SM-H-1 E01 / M06

4.2 Widths: 384 records

Identity and species	Value and unit	Uncertainty	Sector and status
1. N(1895) BREIT-WIGNER WIDTH N(1895)0 Nbar(1895)0 N(1895)+ Nbar(1895)- PDG2026:observable:B004:B004W:348706	80 to 120 to 200 MeV Display: 80 to 200 (~120)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
2. N(2060) BREIT-WIGNER WIDTH N(2060)0 Nbar(2060)0 N(2060)+ Nbar(2060)- PDG2026:observable:B005:B005W:348837	300 to 400 to 450 MeV Display: 300 to 450 (~400)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
3. Delta(1700) BREIT-WIGNER WIDTH Delta(1700)- Deltabar(1700)+ Delta(1700)0 Deltabar(1700)0 Delta(1700)+ Deltabar(1700)- Delta(1700)++ Deltabar(1700)-- PDG2026:observable:B010:B010W:349098	220 to 300 to 380 MeV Display: 220 to 380 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
4. Delta(1905) BREIT-WIGNER WIDTH Delta(1905)- Deltabar(1905)+ Delta(1905)0 Deltabar(1905)0 Delta(1905)+ Deltabar(1905)- Delta(1905)++ Deltabar(1905)-- PDG2026:observable:B011:B011W:349152	270 to 330 to 400 MeV Display: 270 to 400 (~330)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
5. Delta(1910) BREIT-WIGNER WIDTH Delta(1910)- Deltabar(1910)+ Delta(1910)0 Deltabar(1910)0 Delta(1910)+ Deltabar(1910)- Delta(1910)++ Deltabar(1910)-- PDG2026:observable:B012:B012W:349183	200 to 300 to 400 MeV Display: 200 to 400 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
6. Delta(1930) BREIT-WIGNER WIDTH Delta(1930)- Deltabar(1930)+ Delta(1930)0 Deltabar(1930)0 Delta(1930)+ Deltabar(1930)- Delta(1930)++ Deltabar(1930)-- PDG2026:observable:B013:B013W:349250	200 to 300 to 400 MeV Display: 200 to 400 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
7. N(1710) BREIT-WIGNER WIDTH N(1710)0 Nbar(1710)0 N(1710)+ Nbar(1710)- PDG2026:observable:B014:B014W:348536	80 to 140 to 200 MeV Display: 80 to 200 (~140)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
8. N(1720) BREIT-WIGNER WIDTH N(1720)0 Nbar(1720)0 N(1720)+ Nbar(1720)- PDG2026:observable:B015:B015W:348570	150 to 250 to 400 MeV Display: 150 to 400 (~250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
9. N(1875) BREIT-WIGNER WIDTH N(1875)0 Nbar(1875)0 N(1875)+ Nbar(1875)- PDG2026:observable:B016:B016W:348634	120 to 200 to 250 MeV Display: 120 to 250 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
10. N(1990) BREIT-WIGNER WIDTH N(1990)0 Nbar(1990)0 N(1990)+ Nbar(1990)- PDG2026:observable:B017:B017W:348779	200 to 300 to 400 MeV Display: 200 to 400 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
11. N(1700) BREIT-WIGNER WIDTH N(1700)0 Nbar(1700)0 N(1700)+ Nbar(1700)- PDG2026:observable:B018:B018W:348500	100 to 200 to 300 MeV Display: 100 to 300 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
12. Delta(1600) BREIT-WIGNER WIDTH Delta(1600)- Deltabar(1600)+ Delta(1600)0 Deltabar(1600)0 Delta(1600)+ Deltabar(1600)- Delta(1600)++ Deltabar(1600)-- PDG2026:observable:B019:B019W:349051	200 to 250 to 300 MeV Display: 200 to 300 (~250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
13. Xi(1620) WIDTH Xi(1620)- Xibar(1620)+ Xi(1620)0 Xibar(1620)0 PDG2026:observable:B021:B021W:350411	32+8-9 MeV	u_+ : 8.1499070124541788 u_- : 9.4119789269834886 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
14. N(2120) BREIT-WIGNER WIDTH N(2120)0 Nbar(2120)0 N(2120)+ Nbar(2120)- PDG2026:observable:B024:B024W:348907	260 to 300 to 360 MeV Display: 260 to 360 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
15. Delta(1900) BREIT-WIGNER WIDTH Delta(1900)- Deltabar(1900)+ Delta(1900)0 Deltabar(1900)0 Delta(1900)+ Deltabar(1900)- Delta(1900)++ Deltabar(1900)-- PDG2026:observable:B030:B030W:349127	180 to 250 to 320 MeV Display: 180 to 320 (~250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
16. Sigma(1620) WIDTH Sigma(1620)- Sigmabar(1620)+ Sigma(1620)0 Sigmabar(1620)0 Sigma(1620)+ Sigmabar(1620)- PDG2026:observable:B032:B032W:349903	40 to 70 to 100 MeV Display: 40 to 100 (~70)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
17. MIXED CHARGES Delta(1232)- Deltabar(1232)+ Delta(1232)0 Deltabar(1232)0 Delta(1232)+ Deltabar(1232)- Delta(1232)++ Deltabar(1232)-- PDG2026:observable:B033:B033W:349035	114 TO 117 TO 120 MeV Display: 114 to 120 (~117)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
18. Sigma(2070) WIDTH Sigma(2070)- Sigmabar(2070)+ Sigma(2070)0 Sigmabar(2070)0 Sigma(2070)+ Sigmabar(2070)- PDG2026:observable:B034:B034W:350187	100 to 200 to 300 MeV Display: 100 to 300 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
19. Lambda(2110) WIDTH Lambda(2110)0 Lambdabar(2110)0 PDG2026:observable:B035:B035W:349764	200 to 250 to 300 MeV Display: 200 to 300 (~250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
20. Lambda(1800) WIDTH Lambda(1800)0 Lambdabar(1800)0 PDG2026:observable:B036:B036W:349556	150 to 200 to 250 MeV Display: 150 to 250 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
21. PRODUCTION EXPERIMENTS Lambda(1405)0 Lambdabar(1405)0 PDG2026:observable:B037:B037W1:349428	50.5+-2.0 MeV	u_+ : 1.96116135138184 u_- : 1.96116135138184 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
22. Lambda(1520) WIDTH Lambda(1520)0 Lambdabar(1520)0 PDG2026:observable:B038:B038W:349439	15 to 16 to 17 MeV Display: 15 to 17 (~16)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
23. Lambda(1820) WIDTH Lambda(1820)0 Lambdabar(1820)0 PDG2026:observable:B039:B039W:349595	70 TO 80 TO 90 MeV Display: 70 to 90 (~80)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
24. Lambda(1670) WIDTH Lambda(1670)0 Lambdabar(1670)0 PDG2026:observable:B040:B040W:349495	25 to 30 to 35 MeV Display: 25 to 35 (~30)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
25. Lambda(2100) WIDTH Lambda(2100)0 Lambdabar(2100)0 PDG2026:observable:B041:B041W:349742	100 TO 200 TO 250 MeV Display: 100 to 250 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
26. Lambda(2350) WIDTH Lambda(2350)0 Lambdabar(2350)0 PDG2026:observable:B042:B042W:349776	100 TO 150 TO 250 MeV Display: 100 to 250 (~150)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
27. Sigma(1385)+ WIDTH Sigma(1385)- Sigmapar(1385)+ Sigma(1385)0 Sigmapar(1385)0 Sigma(1385)+ Sigmapar(1385)- PDG2026:observable:B043:B043W+:349879	36.2+-0.7 MeV	u_+ : 0.69132395371450 484 u_- : 0.70656746664477 266 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
28. Sigma(1385)0 WIDTH Sigma(1385)- Sigmapar(1385)+ Sigma(1385)0 Sigmapar(1385)0 Sigma(1385)+ Sigmapar(1385)- PDG2026:observable:B043:B043W0:349880	44+-8 MeV	u_+ : 8.16804215502619 97 u_- : 8.16804215502619 97 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
29. Sigma(1385)- WIDTH Sigma(1385)- Sigmapar(1385)+ Sigma(1385)0 Sigmapar(1385)0 Sigma(1385)+ Sigmapar(1385)- PDG2026:observable:B043:B043W-:349881	39.4+-2.1 MeV	u_+ : 2.08086471886561 2 u_- : 2.08086471886561 2 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
30. Sigma(1670) WIDTH Sigma(1670)- Sigmapar(1670)+ Sigma(1670)0 Sigmapar(1670)0 Sigma(1670)+ Sigmapar(1670)- PDG2026:observable:B044:B044W:349947	40 to 70 to 100 MeV Display: 40 to 100 (~70)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
31. Sigma(1775) WIDTH Sigma(1775)- Sigmapar(1775)+ Sigma(1775)0 Sigmapar(1775)0 Sigma(1775)+ Sigmapar(1775)- PDG2026:observable:B045:B045W:349984	105 TO 120 TO 135 MeV Display: 105 to 135 (~120)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
32. Sigma(1915) WIDTH Sigma(1915)- Sigmapar(1915)+ Sigma(1915)0 Sigmapar(1915)0 Sigma(1915)+ Sigmapar(1915)- PDG2026:observable:B046:B046W:350081	80 TO 120 TO 160 MeV Display: 80 to 160 (~120)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
33. Sigma(2030) WIDTH Sigma(2030)- Sigmapar(2030)+ Sigma(2030)0 Sigmapar(2030)0 Sigma(2030)+ Sigmapar(2030)- PDG2026:observable:B047:B047W:350167	150 TO 180 TO 200 MeV Display: 150 to 200 (~180)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
34. Sigma(2250) WIDTH Sigma(2250)- Sigmabar(2250)+ Sigma(2250)0 Sigmabar(2250)0 Sigma(2250)+ Sigmabar(2250)- PDG2026:observable:B048:B048W:350278	60 TO 100 TO 150 MeV Display: 60 to 150 (~100)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
35. Xi(1530)0 WIDTH Xi(1530)- Xibar(1530)+ Xi(1530)0 Xibar(1530)0 PDG2026:observable:B049:B049W0:350405	9.1+-0.5 MeV	u_+ : 0.48058271927942953 u_- : 0.48058271927942953 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
36. Xi(1530)- WIDTH Xi(1530)- Xibar(1530)+ Xi(1530)0 Xibar(1530)0 PDG2026:observable:B049:B049W-:350406	9.9+1.7-1.9 MeV	u_+ : 1.731350237665344 u_- : 1.930203247244187 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
37. Xi(1820) WIDTH Xi(1820)- Xibar(1820)+ Xi(1820)0 Xibar(1820)0 PDG2026:observable:B050:B050W:350422	24+15-10 MeV	u_+ : 15 u_- : 10 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
38. Xi(1950) WIDTH Xi(1950)- Xibar(1950)+ Xi(1950)0 Xibar(1950)0 PDG2026:observable:B052:B052W:350438	60+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
39. Lambda(1690) WIDTH Lambda(1690)0 Lambdabar(1690)0 PDG2026:observable:B055:B055W:349519	60 to 70 to 80 MeV Display: 60 to 80 (~70)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
40. Lambda(1830) WIDTH Lambda(1830)0 Lambdabar(1830)0 PDG2026:observable:B056:B056W:349615	60 to 90 to 120 MeV Display: 60 to 120 (~90)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
41. Sigma(1750) WIDTH Sigma(1750)- Sigmabar(1750)+ Sigma(1750)0 Sigmabar(1750)0 Sigma(1750)+ Sigmabar(1750)- PDG2026:observable:B057:B057W:349967	100 to 150 to 200 MeV Display: 100 to 200 (~150)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
42. Lambda(1890) WIDTH Lambda(1890)0 Lambdabar(1890)0 PDG2026:observable:B060:B060W:349636	80 to 120 to 160 MeV Display: 80 to 160 (~120)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
43. N(1440) BREIT-WIGNER WIDTH N(1440)0 Nbar(1440)0 N(1440)+ Nbar(1440)- PDG2026:observable:B061:B061W:348327	250 to 350 to 450 MeV Display: 250 to 450 (~350)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
44. N(1520) BREIT-WIGNER WIDTH N(1520)0 Nbar(1520)0 N(1520)+ Nbar(1520)- PDG2026:observable:B062:B062W:348350	100 to 110 to 120 MeV Display: 100 to 120 (~110)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
45. N(1535) BREIT-WIGNER WIDTH N(1535)0 Nbar(1535)0 N(1535)+ Nbar(1535)- PDG2026:observable:B063:B063W:348384	125 TO 150 TO 175 MeV Display: 125 to 175 (~150)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
46. N(1675) BREIT-WIGNER WIDTH N(1675)0 Nbar(1675)0 N(1675)+ Nbar(1675)- PDG2026:observable:B064:B064W:348436	130 to 145 to 160 MeV Display: 130 to 160 (~145)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
47. N(1680) BREIT-WIGNER WIDTH N(1680)0 Nbar(1680)0 N(1680)+ Nbar(1680)- PDG2026:observable:B065:B065W:348468	115 to 120 to 130 MeV Display: 115 to 130 (~120)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
48. N(1650) BREIT-WIGNER WIDTH N(1650)0 Nbar(1650)0 N(1650)+ Nbar(1650)- PDG2026:observable:B066:B066W:348411	100 to 125 to 150 MeV Display: 100 to 150 (~125)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
49. Sigma(1880) WIDTH Sigma(1880)- Sigmapbar(1880)+ Sigma(1880)0 Sigmapbar(1880)0 Sigma(1880)+ Sigmapbar(1880)- PDG2026:observable:B067:B067W:350000	100 to 200 to 300 MeV Display: 100 to 300 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
50. Xi(2030) WIDTH Xi(2030)- Xibar(2030)+ Xi(2030)0 Xibar(2030)0 PDG2026:observable:B068:B068W:350445	20+15-5 MeV	u_+ : 15 u_- : 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
51. N(2190) BREIT-WIGNER WIDTH N(2190)0 Nbar(2190)0 N(2190)+ Nbar(2190)- PDG2026:observable:B071:B071W:348954	300 to 400 to 500 MeV Display: 300 to 500 (~400)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
52. Lambda(1810) WIDTH Lambda(1810)0 Lambdabar(1810)0 PDG2026:observable:B077:B077W:349577	50 to 110 to 170 MeV Display: 50 to 170 (~110)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
53. Sigma(1660) WIDTH Sigma(1660)- Sigmapbar(1660)+ Sigma(1660)0 Sigmapbar(1660)0 Sigma(1660)+ Sigmapbar(1660)- PDG2026:observable:B079:B079W:349921	100 to 200 to 300 MeV Display: 100 to 300 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
54. Delta(1620) BREIT-WIGNER WIDTH Delta(1620)- Deltabar(1620)+ Delta(1620)0 Deltabar(1620)0 Delta(1620)+ Deltabar(1620)- Delta(1620)++ Deltabar(1620)-- PDG2026:observable:B082:B082W:349079	110 to 130 to 150 MeV Display: 110 to 150 (~130)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
55. Delta(1950) BREIT-WIGNER WIDTH Delta(1950)- Deltabar(1950)+ Delta(1950)0 Deltabar(1950)0 Delta(1950)+ Deltabar(1950)- Delta(1950)++ Deltabar(1950)-- PDG2026:observable:B083:B083W:349309	235 to 285 to 335 MeV Display: 235 to 335 (~285)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
56. Delta(2420) BREIT-WIGNER WIDTH Delta(2420)- Deltabar(2420)+ Delta(2420)0 Deltabar(2420)0 Delta(2420)+ Deltabar(2420)- Delta(2420)++ Deltabar(2420)-- PDG2026:observable:B084:B084W:349360	300 to 500 to 700 MeV Display: 300 to 700 (~500)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
57. N(1880) BREIT-WIGNER WIDTH N(1880)0 Nbar(1880)0 N(1880)+ Nbar(1880)- PDG2026:observable:B087:B087W:348672	200 to 300 to 400 MeV Display: 200 to 400 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
58. Sigma(2080) WIDTH Sigma(2080)- Sigmabar(2080)+ Sigma(2080)0 Sigmabar(2080)0 Sigma(2080)+ Sigmabar(2080)- PDG2026:observable:B088:B088W:350189	100 to 170 to 240 MeV Display: 100 to 240 (~170)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
59. N(2220) BREIT-WIGNER WIDTH N(2220)0 Nbar(2220)0 N(2220)+ Nbar(2220)- PDG2026:observable:B090:B090W:348987	350 TO 400 TO 500 MeV Display: 350 to 500 (~400)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
60. Sigma(1910) WIDTH Sigma(1910)- Sigmabar(1910)+ Sigma(1910)0 Sigmabar(1910)0 Sigma(1910)+ Sigmabar(1910)- PDG2026:observable:B098:B098W:350051	150 to 220 to 300 MeV Display: 150 to 300 (~220)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
61. Lambda(1600) WIDTH Lambda(1600)0 Lambdabar(1600)0 PDG2026:observable:B101:B101W:349473	150 to 200 to 250 MeV Display: 150 to 250 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
62. Lambda_c(2625)+ WIDTH Lambda_c(2625)+ Lambdabar_c(2625)- PDG2026:observable:B102:B102W:350937	< 0.52 MeV Display: <0.52	u_+ : NA u_- : NA Type: U Confidence: 90	COMP-BARYON E04 / M07
63. Sigma_c(2455)++ WIDTH Sigma_c(2455)0 Sigmabar_c(2455)0 Sigma_c(2455)+ Sigmabar_c(2455)- Sigma_c(2455)++ Sigmabar_c(2455)-- PDG2026:observable:B104:B104W++:350997	1.89+0.09-0.18 MeV	u_+ : 0.08596874087768 8063 u_- : 0.17741442138916 921 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
64. Sigma_c(2455)+ WIDTH Sigma_c(2455)0 Sigmabar_c(2455)0 Sigma_c(2455)+ Sigmabar_c(2455)- Sigma_c(2455)++ Sigmabar_c(2455)-- PDG2026:observable:B104:B104W+:350998	2.3+-0.4 MeV	u_+ : 0.42426406871192 851 u_- : 0.42426406871192 851 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
65. Sigma_c(2455)0 WIDTH Sigma_c(2455)0 Sigmabar_c(2455)0 Sigma_c(2455)+ Sigmabar_c(2455)- Sigma_c(2455)++ Sigmabar_c(2455)-- PDG2026:observable:B104:B104W0:350999	1.83+0.11-0.19 MeV	u_+ : 0.11134194993665 61 u_- : 0.19347936378811 631 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
66. MIXED CHARGES Xi(1690)- Xibar(1690)+ Xi(1690)0 Xibar(1690)0 PDG2026:observable:B105:B105W:350413	20+-15 MeV	u_+ : 15 u_- : 15 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
67. N(2250) BREIT-WIGNER WIDTH N(2250)0 Nbar(2250)0 N(2250)+ Nbar(2250)- PDG2026:observable:B113:B113W:349005	300 to 500 to 600 MeV Display: 300 to 600 (~500)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
68. Sigma_c(2520)++ WIDTH Sigma_c(2520)0 Sigmabar_c(2520)0 Sigma_c(2520)+ Sigmabar_c(2520)- Sigma_c(2520)++ Sigmabar_c(2520)-- PDG2026:observable:B115:B115W++:351011	14.78+0.30-0.40 MeV	u_+ : 0.30221098294841 742 u_- : 0.37821726734358 602 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
69. Sigma_c(2520)+ WIDTH Sigma_c(2520)0 Sigambar_c(2520)0 Sigma_c(2520)+ Sigambar_c(2520)- Sigma_c(2520)++ Sigambar_c(2520)-- PDG2026:observable:B115:B115W+:351012	17.2+4.0-2.2 MeV	u_+ : 3.8600518131237571 u_- : 2.213594362117866 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
70. Sigma_c(2520)0 WIDTH Sigma_c(2520)0 Sigambar_c(2520)0 Sigma_c(2520)+ Sigambar_c(2520)- Sigma_c(2520)++ Sigambar_c(2520)-- PDG2026:observable:B115:B115W0:351013	15.3+0.4-0.5 MeV	u_+ : 0.43805327113015552 u_- : 0.49172579862174309 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
71. Delta(1920) BREIT-WIGNER WIDTH Delta(1920)- Deltabar(1920)+ Delta(1920)0 Deltabar(1920)0 Delta(1920)+ Deltabar(1920)- Delta(1920)++ Deltabar(1920)-- PDG2026:observable:B117:B117W:349209	240 to 300 to 360 MeV Display: 240 to 360 (~300)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
72. Lambda_c(2595)+ WIDTH Lambda_c(2595)+ Lambdabar_c(2595)- PDG2026:observable:B119:B119W:350920	2.6+-0.6 MeV	u_+ : 0.55758407437802593 u_- : 0.55758407437802593 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
73. N(2600) BREIT-WIGNER WIDTH N(2600)0 Nbar(2600)0 N(2600)+ Nbar(2600)- PDG2026:observable:B120:B120W:349023	500 TO 650 TO 800 MeV Display: 500 to 800 (~650)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
74. Lambda_c(2940)+ WIDTH Lambda_c(2940)0 Lambdabar_c(2940)0 PDG2026:observable:B122:B122W:350980	20+6-5 MeV	u_+ : 5.9819637952003841 u_- : 5.2285381312943766 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
75. Xi_c(2970)+ WIDTH Xi_c(2970)0 Xibar_c(2970)0 PDG2026:observable:B130:B130W+:351281	27.2+2.8-3.3 MeV	u_+ : 2.7691726997636179 u_- : 3.3147478336925231 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
76. N(2100) BREIT-WIGNER WIDTH N(2100)0 Nbar(2100)0 N(2100)+ Nbar(2100)- PDG2026:observable:B132:B132W:348872	200 to 260 to 320 MeV Display: 200 to 320 (~260)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
77. Delta(2200) BREIT-WIGNER WIDTH Delta(2200)- Deltabar(2200)+ Delta(2200)0 Deltabar(2200)0 Delta(2200)+ Deltabar(2200)- Delta(2200)++ Deltabar(2200)-- PDG2026:observable:B135:B135W:349340	200 to 350 to 500 MeV Display: 200 to 500 (~350)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
78. Delta(1940) BREIT-WIGNER WIDTH Delta(1940)- Deltabar(1940)+ Delta(1940)0 Deltabar(1940)0 Delta(1940)+ Deltabar(1940)- Delta(1940)++ Deltabar(1940)-- PDG2026:observable:B136:B136W:349273	300 to 400 to 500 MeV Display: 300 to 500 (~400)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
79. Omega(2250)- WIDTH Omega(2250)- Omegabar(2250)+ PDG2026:observable:B141:B141W:350517	55+-18 MeV	u_+ : 17.6983644476396 51 u_- : 17.6983644476396 51 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
80. Omega(2470)- WIDTH Omega(2470)- Omegabar(2470)+ PDG2026:observable:B143:B143W:350524	72+-33 MeV	u_+ : 33 u_- : 33 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
81. N(1900) BREIT-WIGNER WIDTH N(1900)0 Nbar(1900)0 N(1900)+ Nbar(1900)- PDG2026:observable:B144:B144W:348741	100 to 200 to 320 MeV Display: 100 to 320 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
82. Xi_c(2645)+ WIDTH Xi_c(2645)0 Xibar_c(2645)0 Xi_c(2645)+ Xibar_c(2645)- PDG2026:observable:B146:B146W+:351213	2.14+-0.19 MeV	u_+ : 0.18989896013117 41 u_- : 0.18989896013117 41 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
83. Xi_c(2645)0 WIDTH Xi_c(2645)0 Xibar_c(2645)0 Xi_c(2645)+ Xibar_c(2645)- PDG2026:observable:B146:B146W0:351214	2.35+-0.22 MeV	u_+ : 0.22203603311174 519 u_- : 0.22203603311174 519 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
84. Xi_c(3080)+ WIDTH Xi_c(3080)0 Xibar_c(3080)0 PDG2026:observable:B147:B147W+:351309	3.8+-0.9 MeV	u_+ : 0.93945766227499 938 u_- : 0.93945766227499 938 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
85. Xi_c(3080)0 WIDTH Xi_c(3080)0 Xibar_c(3080)0 PDG2026:observable:B147:B147W0:351310	5.6+-2.2 MeV	u_+ : 2.17986010862691 2 u_- : 2.17986010862691 2 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
86. Xi_c(2815)+ WIDTH Xi_c(2815)0 Xibar_c(2815)0 Xi_c(2815)+ Xibar_c(2815)- PDG2026:observable:B148:B148W+:351240	2.15+-0.15 MeV	u_+ : 0.15193272999421 351 u_- : 0.15193272999421 351 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
87. Xi_c(2815)0 WIDTH Xi_c(2815)0 Xibar_c(2815)0 Xi_c(2815)+ Xibar_c(2815)- PDG2026:observable:B148:B148W0:351241	2.54+-0.25 MeV	u_+ : 0.24758836806279 891 u_- : 0.24758836806279 891 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
88. Xi_c(2790)+ WIDTH Xi_c(2790)0 Xibar_c(2790)0 Xi_c(2790)+ Xibar_c(2790)- PDG2026:observable:B149:B149W+:351224	8.9+-1.0 MeV	u_+ : 1 u_- : 1 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
89. Xi_c(2790)0 WIDTH Xi_c(2790)0 Xibar_c(2790)0 Xi_c(2790)+ Xibar_c(2790)- PDG2026:observable:B149:B149W0:351225	10.0+-1.1 MeV	u_+ : 1.063014581273465 u_- : 1.063014581273465 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
90. Lambda_c(2765)+ WIDTH Lambda_c(2765)0 Lambdabar_c(2765)0 PDG2026:observable:B150:B150W:350954	50 MeV	u_+ : 0 u_- : 0 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
91. Lambda_c(2880)+ WIDTH Lambda_c(2880)+ Lambdabar_c(2880)- PDG2026:observable:B151:B151W:350964	5.6+0.8-0.6 MeV	u_+ : 0.77200097671302059 u_- : 0.62289222418902057 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
92. Sigma_c(2800)++ WIDTH Sigma_c(2800)0 Sigmabar_c(2800)0 PDG2026:observable:B155:B155W++:351025	75+22-17 MeV	u_+ : 21.633307652783941 u_- : 17.029386365926399 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
93. Sigma_c(2800)+ WIDTH Sigma_c(2800)0 Sigmabar_c(2800)0 PDG2026:observable:B155:B155W+:351026	60+60-40 MeV	u_+ : 63.820059542435402 u_- : 44.418464629025621 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
94. Sigma_c(2800)0 WIDTH Sigma_c(2800)0 Sigmabar_c(2800)0 PDG2026:observable:B155:B155W0:351027	72+22-15 MeV	u_+ : 22.093560797468029 u_- : 14.823460785121441 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
95. Xi_c(2930)+ WIDTH Xi_c(2930)0 Xibar_c(2930)0 PDG2026:observable:B156:B156A05:351265	15+-9 MeV	u_+ : 9.1482238713315276 u_- : 9.1482238713315276 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
96. Xi_c(2930)0 WIDTH Xi_c(2930)0 Xibar_c(2930)0 PDG2026:observable:B156:B156A06:351266	10.2+-1.4 MeV	u_+ : 1.360147050873544 u_- : 1.360147050873544 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
97. Xi_c(3055)+ WIDTH Xi_c(3055)0 Xibar_c(3055)0 PDG2026:observable:B157:B157W+:351299	8.0+-0.8 MeV	u_+ : 0.76391812853200 158 u_- : 0.76391812853200 158 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
98. Xi_c(3055)0 WIDTH Xi_c(3055)0 Xibar_c(3055)0 PDG2026:observable:B157:B157W:351300	12.4+-2.3 MeV	u_+ : 2.28254244210266 5 u_- : 2.28254244210266 5 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
99. Xi_c(3123)+ WIDTH Xi_c(3123)0 Xibar_c(3123)0 PDG2026:observable:B158:B158W+:351326	4+-4 MeV	u_+ : 3.80131556174964 3 u_- : 3.80131556174964 3 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
100. Xi_b(5945)0 WIDTH Xi_b(5945)0 Xibar_b(5945)0 PDG2026:observable:B161:B161W:351845	0.87+-0.07 MeV	u_+ : 0.07506967217458 3921 u_- : 0.07470630547753 7885 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
101. Lambda_b(5912)0 WIDTH Lambda_b(5912)0 Lambdabar_b(5912)0 PDG2026:observable:B162:B162W:351695	<0.25 MeV	u_+ : NA u_- : NA Type: U Confidence: 90	COMP-BARYON E04 / M07
102. Lambda_b(5920)0 WIDTH Lambda_b(5920)0 Lambdabar_b(5920)0 PDG2026:observable:B163:B163W:351699	<0.19 MeV	u_+ : NA u_- : NA Type: U Confidence: 90	COMP-BARYON E04 / M07
103. Lambda(1710) WIDTH Lambda(1710)0 Lambdabar(1710)0 PDG2026:observable:B164:B164W:349533	180+-40 MeV	u_+ : 42 u_- : 42 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
104. Lambda(2050) WIDTH Lambda(2050)0 Lambdabar(2050)0 PDG2026:observable:B165:B165W:349652	490+-60 MeV	u_+ : 61 u_- : 61 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
105. Sigma(1780) WIDTH Sigma(1780)- Sigmar(1780)+ Sigma(1780)0 Sigmar(1780)0 Sigma(1780)+ Sigmar(1780)- PDG2026:observable:B166:B166W:349992	100 to 200 to 300 MeV Display: 100 to 300 (~200)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
106. Sigma(1900) WIDTH Sigma(1900)- Sigmar(1900)+ Sigma(1900)0 Sigmar(1900)0 Sigma(1900)+ Sigmar(1900)- PDG2026:observable:B167:B167W:350020	140 to 165 to 190 MeV Display: 140 to 190 (~165)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07
107. Sigma(1940) WIDTH Sigma(1940)- Sigmar(1940)+ Sigma(1940)0 Sigmar(1940)0 Sigma(1940)+ Sigmar(1940)- PDG2026:observable:B168:B168W:350103	100 to 250 to 400 MeV Display: 100 to 400 (~250)	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
108. Xi_b'(5935)- WIDTH Xi_b^(5935)- Xibar_b^(5935)+ PDG2026:observable:B169:B169W:351841	0.03+-0.032 MeV	u_+ : 0.03162277660168 3791 u_- : 0.03162277660168 3791 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
109. Xi_b(5955)- WIDTH Xi_b(5955)- Xibar_b(5955)+ PDG2026:observable:B170:B170W:351849	1.43+-0.11 MeV	u_+ : 0.11313708498984 761 u_- : 0.11313708498984 761 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
110. P_c_cbar(4380)+ WIDTH P_c_cbar(4380)+ PDG2026:observable:B171:B171W:351952	210+-90 MeV	u_+ : 87.8635305459551 86 u_- : 87.8635305459551 86 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
111. P_c_cbar(4457)+ WIDTH P_c_cbar(4457)+ PDG2026:observable:B172:B172W:351978	6.4+6.0-2.8 MeV	u_+ : 6.04069532421558 28 u_- : 2.75862284482674 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
112. Omega_c(3000)0 WIDTH Omega_c(3000)0 Omegabar_c(3000)0 PDG2026:observable:B173:B173W:351377	3.8+1.6-0.4 MeV	u_+ : 1.60654909666651 7 u_- : 0.37013511046643 488 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
113. Omega_c(3050)0 WIDTH Omega_c(3050)0 Omegabar_c(3050)0 PDG2026:observable:B174:B174W:351381	< 1.8 MeV Display: <1.8	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
114. Omega_c(3065)0 WIDTH Omega_c(3065)0 Omegabar_c(3065)0 PDG2026:observable:B175:B175W:351385	3.4+0.7-0.8 MeV	u_+ : 0.68496637416672 579 u_- : 0.79385852123859 457 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
115. Omega_c(3090)0 WIDTH Omega_c(3090)0 Omegabar_c(3090)0 PDG2026:observable:B176:B176W:351389	8.5+0.8-1.7 MeV	u_+ : 0.75213030785895 074 u_- : 1.67868996541946 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
116. Omega_c(3120)0 WIDTH Omega_c(3120)0 Omegabar_c(3120)0 PDG2026:observable:B177:B177W:351393	<2.5 MeV	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
117. Lambda_c(2860)+ WIDTH Lambda_c(2860)0 Lambdabar_c(2860)0 PDG2026:observable:B178:B178W:350957	68+12-22 MeV	u_+ : 11.7804923496431 2 u_- : 21.6233669903648 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
118. Omega(2012)- WIDTH Omega(2012)- Omegabar(2012)+ PDG2026:observable:B179:B179W:350507	6.3+-2.0 MeV	u_+ : 2.07428954575496 5 u_- : 1.91972498587065 51 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
119. Xi_b(6227)- WIDTH Xi_b(6227)- Xi_b(6227)+ PDG2026:observable:B180:B180W:351865	19.9+-2.6 MeV	u_+ : 2.58069758011278 78 u_- : 2.58069758011278 78 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
120. Sigma_b(6097)+ WIDTH Sigma_b(6097)+ Sigmabar_b(6097)- PDG2026:observable:B181:B181W:351735	31+-6 MeV	u_+ : 5.54436651025164 46 u_- : 5.54436651025164 46 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
121. Sigma_b(6097)- WIDTH Sigma_b(6097)- Sigmabar_b(6097)+ PDG2026:observable:B182:B182W:351739	29+-4 MeV	u_+ : 4.29534631898290 62 u_- : 4.29534631898290 62 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
122. Sigma(2110) WIDTH Sigma(2110)- Sigmabar(2110)+ Sigma(2110)0 Sigmabar(2110)0 Sigma(2110)+ Sigmabar(2110)- PDG2026:observable:B183:B183W:350229	310+120-50 MeV	u_+ : 122.349630121267 89 u_- : 54.7610745236418 52 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
123. Sigma(2230) WIDTH Sigma(2230)- Sigmabar(2230)+ Sigma(2230)0 Sigmabar(2230)0 Sigma(2230)+ Sigmabar(2230)- PDG2026:observable:B184:B184W:350254	350+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
124. P_c_cbar(4312)+ WIDTH P_c_cbar(4312)+ PDG2026:observable:B185:B185W:351936	10+-5 MeV	u_+ : 4.58039299623951 47 u_- : 5.24785670536077 03 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
125. P_c_cbar(4440)+ WIDTH P_c_cbar(4440)+ PDG2026:observable:B186:B186W:351956	21+10-11 MeV	u_+ : 9.98498873309329 23 u_- : 11.2258629957789 89 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
126. Lambda_b(6146)0 WIDTH Lambda_b(6146)0 Lambdabar_b(6146)0 PDG2026:observable:B187:B187W:351708	2.9+-1.3 MeV	u_+ : 1.33416640641263 4 u_- : 1.33416640641263 4 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
127. Lambda_b(6152)0 WIDTH Lambda_b(6152)0 Lambdabar_b(6152)0 PDG2026:observable:B188:B188W:351714	2.1+-0.9 MeV	u_+ : 0.85440037453175 31 u_- : 0.85440037453175 31 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
128. Lambda(2070) WIDTH Lambda(2070)0 Lambdabar(2070)0 PDG2026:observable:B190:B190W:349677	370+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
129. Lambda(2080) WIDTH Lambda(2080)0 Lambdabar(2080)0 PDG2026:observable:B191:B191W:349712	181+-29 MeV	u_+ : 29 u_- : 29 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
130. Xi_c(2923)0 WIDTH Xi_c(2923)0 Xibar_c(2923)0 PDG2026:observable:B192:B192W0:351257	5.8+-1.3 MeV	u_+ : 1.30796716242523 u_- : 1.30796716242523 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
131. Xi_c(2923)+ WIDTH Xi_c(2923)0 Xibar_c(2923)0 PDG2026:observable:B192:B192WP:351258	5.3+-1.7 MeV	u_+ : 1.66433169770932 41 u_- : 1.66433169770932 41 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
132. Omega_b(6316)- WIDTH Omega_b(6316)- Omegabar_b(6316)+ PDG2026:observable:B193:B193W:351906	<4.2 MeV	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
133. Omega_b(6330)- WIDTH Omega_b(6330)- Omegabar_b(6330)+ PDG2026:observable:B194:B194W:351910	<4.7 MeV	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
134. Omega_b(6340)- WIDTH Omega_b(6340)- Omegabar_b(6340)+ PDG2026:observable:B195:B195W:351914	<1.8 MeV	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
135. Omega_b(6350)- WIDTH Omega_b(6350)- Omegabar_b(6350)+ PDG2026:observable:B196:B196W:351918	<3.2 MeV	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
136. Lambda_b(6070)0 WIDTH Lambda_b(6070)0 Lambdabar_b(6070)0 PDG2026:observable:B197:B197W:351703	72+-11 MeV	u_+ : 11.1803398874989 51 u_- : 11.1803398874989 51 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
137. Xi_b(6227)0 WIDTH Xi_b(6227)0 Xibar_b(6227)0 PDG2026:observable:B199:B199W:351871	19+5-4 MeV	u_+ : 5.1923019942988677 u_- : 4.3324358044868934 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
138. Xi_b(6100)- WIDTH Xi_b(6100)- Xibar_b(6100)+ PDG2026:observable:B200:B200W:351861	0.94+-0.31 MeV	u_+ : 0.31048349392520053 u_- : 0.31048349392520053 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
139. Xi_b(6327)0 WIDTH Xi_b(6327)0 Xibar_b(6327)0 PDG2026:observable:B201:B201W:351875	< 2.56 MeV Display: <2.56	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
140. Xi_b(6333)0 WIDTH Xi_b(6333)0 Xibar_b(6333)0 PDG2026:observable:B202:B202W:351879	< 1.92 MeV Display: <1.92	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-BARYON E04 / M07
141. Xi_c(2882) WIDTH Xi_c(2882)0 Xibar_c(2882)0 PDG2026:observable:B203:B203W0:351252	12+-8 MeV	u_+ : 7.8568441501661468 u_- : 7.8568441501661468 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
142. P_c_cbar_s(4338)0 WIDTH P_c_cbar_s(4338)+ PDG2026:observable:B204:B204W:351946	7.0+-1.8 MeV	u_+ : 1.769180601295413 u_- : 1.769180601295413 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
143. P_c_cbar(4312)+ WIDTH P_c_cbar(4312)+ PDG2026:observable:B205:B205W:352000	19+8-7 MeV	u_+ : 8.1568830165262654 u_- : 7.255202813989527 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
144. Lambda_c(2910)+ WIDTH Lambda_c(2910)0 Lambdabar_c(2910)0 PDG2026:observable:B206:B206W:350974	52+-27 MeV	u_+ : 27.448861542876418 u_- : 27.448861542876418 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
145. Xi_b(6087)0 WIDTH Xi_b(6087) PDG2026:observable:B207:B207W:351853	2.4+-0.5 MeV	u_+ : 0.51971145840745137 u_- : 0.51971145840745137 Type: NONE Confidence: ND	COMP-BARYON E04 / M07

Identity and species	Value and unit	Uncertainty	Sector and status
146. Xi_b(6095)0 WIDTH Xi_b(6095) PDG2026:observable:B208:B208W:351857	0.50+-0.35 MeV	u_+ : 0.34785054261852 172 u_- : 0.34785054261852 172 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
147. Omega_c(3185)0 WIDTH Omega_c(3185)0 Omegabar_c(3185)0 PDG2026:observable:B209:B209W:351397	50+12-21 MeV	u_+ : 12.2065556157337 u_- : 21.1896201004170 91 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
148. Omega_c(3327)0 WIDTH Omega_c(3327)0 Omegabar_c(3327)0 PDG2026:observable:B210:B210W:351401	20+14-5 MeV	u_+ : 13.9283882771841 19 u_- : 5.09901951359278 36 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
149. Omega(2109)- WIDTH Omega(2109)- Omegabar(2109)+ PDG2026:observable:B211:B211W:350515	18+-17 MeV	u_+ : 17.3623155137786 79 u_- : 17.3623155137786 79 Type: NONE Confidence: ND	COMP-BARYON E04 / M07
150. omega(782) WIDTH omega(782)0 PDG2026:observable:M001:M001W:333933	8.68+-0.13 MeV	u_+ : 0.13200008707764 46 u_- : 0.13200008707764 46 Type: NONE Confidence: ND	COMP-MESON E04 / M08
151. eta'(958) WIDTH eta^(958)0 PDG2026:observable:M002:M002W:334007	0.188+-0.006 MeV	u_+ : 0.00644422122653 40033 u_- : 0.00644422122653 40033 Type: NONE Confidence: ND	COMP-MESON E04 / M08
152. f_0(980) WIDTH f_0(980)0 PDG2026:observable:M003:M003W1:334164	10 to 100 MeV	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
153. phi(1020) WIDTH phi(1020)0 PDG2026:observable:M004:M004W:334183	4.249+-0.013 MeV	u_+ : 0.01311655783805 327 u_- : 0.01311655783805 327 Type: NONE Confidence: ND	COMP-MESON E04 / M08
154. f_2(1270) WIDTH f_2(1270)0 PDG2026:observable:M005:M005W:334355	186.6+2.8-2.2 MeV	u_+ : 2.80231263767857 5 u_- : 2.19611562862221 81 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
155. f₁(1420) WIDTH f ₁ (1420)0 PDG2026:observable:M006:M006W:334555	55.4+-3.2 MeV	u_+ : 3.1572143311904868 u_- : 3.199679790915769 Type: NONE Confidence: ND	COMP-MESON E04 / M08
156. f₁(1285) WIDTH f ₁ (1285)0 PDG2026:observable:M008:M008W:334404	23.0+-1.0 MeV	u_+ : 1.045581016397475 u_- : 1.0256159102073861 Type: NONE Confidence: ND	COMP-MESON E04 / M08
157. NEUTRAL ONLY, e+ e- rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009W0:333870	147.4+-0.8 MeV	u_+ : 0.82096324211264493 u_- : 0.82096324211264493 Type: NONE Confidence: ND	COMP-MESON E04 / M08
158. CHARGED ONLY, tau DECAYS and e+ e- rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009W5:333871	149.1+-0.8 MeV	u_+ : 0.84906762313788198 u_- : 0.84906762313788198 Type: NONE Confidence: ND	COMP-MESON E04 / M08
159. MIXED CHARGES, OTHER REACTIONS rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009W7:333873	149.5+-1.3 MeV	u_+ : 1.3 u_- : 1.3 Type: NONE Confidence: ND	COMP-MESON E04 / M08
160. CHARGED ONLY, HADROPRODUCED rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009W2:333874	150.2+-2.4 MeV	u_+ : 2.3925241925869849 u_- : 2.3925241925869849 Type: NONE Confidence: ND	COMP-MESON E04 / M08
161. NEUTRAL ONLY, PHOTOPRODUCED rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009W0P:333876	152.3+-1.7 MeV	u_+ : 1.6144196138431619 u_- : 1.7476550950891081 Type: NONE Confidence: ND	COMP-MESON E04 / M08
162. NEUTRAL ONLY, OTHER REACTIONS rho(770)0 rho(770)+ rho(770)- PDG2026:observable:M009:M009W0R:333877	150.9+-1.7 MeV	u_+ : 1.7341880321909591 u_- : 1.7341880321909591 Type: NONE Confidence: ND	COMP-MESON E04 / M08
163. a₁(1260) WIDTH a ₁ (1260)0 a ₁ (1260)+ a ₁ (1260)- PDG2026:observable:M010:M010W:334333	250 to 600 MeV	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
164. b₁(1235) WIDTH b ₁ (1235)0 b ₁ (1235)+ b ₁ (1235)- PDG2026:observable:M011:M011W:334310	142+-9 MeV	u_+ : 8.50576301057597 03 u_- : 8.50576301057597 03 Type: NONE Confidence: ND	COMP-MESON E04 / M08
165. a₂(1320) WIDTH a ₂ (1320)0 a ₂ (1320)+ a ₂ (1320)- PDG2026:observable:M012:M012W:334474	107+-5 MeV	u_+ : 5 u_- : 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
166. 3pi MODE a ₂ (1320)0 a ₂ (1320)+ a ₂ (1320)- PDG2026:observable:M012:M012W1:334476	105.0+1.7-1.9 MeV	u_+ : 1.67242394191988 7 u_- : 1.85239615833847 09 Type: NONE Confidence: ND	COMP-MESON E04 / M08
167. K Kbar MODE a ₂ (1320)0 a ₂ (1320)+ a ₂ (1320)- PDG2026:observable:M012:M012W2:334477	109.8+-2.4 MeV	u_+ : 2.37489024273604 12 u_- : 2.37489024273604 12 Type: NONE Confidence: ND	COMP-MESON E04 / M08
168. eta pi MODE a ₂ (1320)0 a ₂ (1320)+ a ₂ (1320)- PDG2026:observable:M012:M012W3:334478	111.1+-2.4 MeV	u_+ : 2.44896593502306 5 u_- : 2.44896593502306 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
169. eta' pi MODE a ₂ (1320)0 a ₂ (1320)+ a ₂ (1320)- PDG2026:observable:M012:M012W4:334479	119+-25 MeV	u_+ : 25.0631516454720 01 u_- : 25.0631516454720 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
170. f₂'(1525) WIDTH f ₂ '(1525)0 PDG2026:observable:M013:M013WX:334671	72+7-6 MeV	u_+ : 6.88285745387930 53 u_- : 5.78868904167363 58 Type: NONE Confidence: ND	COMP-MESON E04 / M08
171. PRODUCED BY K+- BEAM f ₂ '(1525)0 PDG2026:observable:M013:M013W2:334673	82+-6 MeV	u_+ : 6.06900312634116 46 u_- : 5.74369964556114 8 Type: NONE Confidence: ND	COMP-MESON E04 / M08
172. PRODUCED IN e+ e- ANNIHILATION AND PARTICLE DECAYS f ₂ '(1525)0 PDG2026:observable:M013:M013W3:334674	86+-4 MeV	u_+ : 4.27241300467022 71 u_- : 3.95789839152937 39 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
173. PRODUCED IN $p\bar{p}$ ANNIHILATION $f_2^{\prime}(1525)0$ PDG2026:observable:M013:M013W9:334675	77+-5 MeV	u_+ : 4.799999999999999 98 u_- : 4.799999999999999 98 Type: NONE Confidence: ND	COMP-MESON E04 / M08
174. CENTRAL PRODUCTION $f_2^{\prime}(1525)0$ PDG2026:observable:M013:M013W4:334676	70+-25 MeV	u_+ : 25 u_- : 25 Type: NONE Confidence: ND	COMP-MESON E04 / M08
175. PRODUCED IN $e p$ COLLISIONS $f_2^{\prime}(1525)0$ PDG2026:observable:M013:M013W10:334677	83+-10 MeV	u_+ : 10.2956301409870 01 u_- : 9.84885780179610 39 Type: NONE Confidence: ND	COMP-MESON E04 / M08
176. $f_0(500)$ BREIT-WIGNER WIDTH $f_0(500)0$ PDG2026:observable:M014:M014W:333858	100 to 800 MeV	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
177. 2π, $K\bar{K}$, AND $K\bar{K}\pi$ MODES $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015W:334848	161+-10 MeV	u_+ : 9.70148009186499 04 u_- : 9.65373212649761 39 Type: NONE Confidence: ND	COMP-MESON E04 / M08
178. 2π MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015W1:334849	186+-14 MeV	u_+ : 14.0467249736341 89 u_- : 14.0467249736341 89 Type: NONE Confidence: ND	COMP-MESON E04 / M08
179. $K\bar{K}$ AND $K\bar{K}\pi$ MODES $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015W2:334850	204+-18 MeV	u_+ : 17.8885438199983 18 u_- : 17.8885438199983 18 Type: NONE Confidence: ND	COMP-MESON E04 / M08
180. $(4\pi)^+$ MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015W3:334851	129+-10 MeV	u_+ : 10.2844009679589 39 u_- : 10.1615727369938 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
181. $\omega\pi$ MODE $\rho_3(1690)0 \mid \rho_3(1690)+ \mid \rho_3(1690)-$ PDG2026:observable:M015:M015W5:334852	190+-40 MeV	u_+ : 35.5278384233964 18 u_- : 35.5278384233964 18 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
182. eta pi+ pi- MODE rho_3(1690)0 rho_3(1690)+ rho_3(1690)- PDG2026:observable:M015:M015W6:334853	130+-40 MeV	u_+ : 43.467351876049257 u_- : 43.467351876049257 Type: NONE Confidence: ND	COMP-MESON E04 / M08
183. f_4(2050) WIDTH f_4(2050)0 PDG2026:observable:M016:M016W:335131	237+-18 MeV	u_+ : 17.88029943252717 u_- : 18.179532837552909 Type: NONE Confidence: ND	COMP-MESON E04 / M08
184. a_4(1970) WIDTH a_4(1970)0 a_4(1970)+ a_4(1970)- PDG2026:observable:M017:M017W:335091	324+-15-18 MeV	u_+ : 14.533393315985499 u_- : 18.100116797853431 Type: NONE Confidence: ND	COMP-MESON E04 / M08
185. CHARGED ONLY, HADROPRODUCED K^*(892)0 Kbar^*(892)0 K^*(892)+ K^*(892)- PDG2026:observable:M018:M018W1:335795	48.5+-1.2 MeV	u_+ : 1.189202181804293 u_- : 1.189202181804293 Type: NONE Confidence: ND	COMP-MESON E04 / M08
186. CHARGED ONLY, PRODUCED IN tau LEPTON DECAYS K^*(892)0 Kbar^*(892)0 K^*(892)+ K^*(892)- PDG2026:observable:M018:M018W5:335797	46.2+-1.3 MeV	u_+ : 1.3416407864998741 u_- : 1.3416407864998741 Type: NONE Confidence: ND	COMP-MESON E04 / M08
187. NEUTRAL ONLY K^*(892)0 Kbar^*(892)0 K^*(892)+ K^*(892)- PDG2026:observable:M018:M018W2:335798	47.1+-0.5 MeV	u_+ : 0.49347496850837202 u_- : 0.49347496850837202 Type: NONE Confidence: ND	COMP-MESON E04 / M08
188. K_0^*(1430) WIDTH K_0^*(1430)0 Kbar_0^*(1430)0 K_0^*(1430)+ K_0^*(1430)- PDG2026:observable:M019:M019W:335879	270+-80 MeV	u_+ : 80 u_- : 80 Type: NONE Confidence: ND	COMP-MESON E04 / M08
189. pi_2(2100) WIDTH pi_2(2100)0 pi_2(2100)+ pi_2(2100)- PDG2026:observable:M020:M020W:335147	620+-50 MeV	u_+ : 52.400001025245267 u_- : 52.400001025245267 Type: NONE Confidence: ND	COMP-MESON E04 / M08
190. CHARGED ONLY, WITH FINAL STATE K pi K_2^*(1430)0 Kbar_2^*(1430)0 K_2^*(1430)+ K_2^*(1430)- PDG2026:observable:M022:M022W1:335890	100.0+-2.2 MeV	u_+ : 2.2364281468619742 u_- : 2.1785326327209291 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
191. NEUTRAL ONLY $K_2^{*}(1430)0 \mid \bar{K}2^{*}(1430)0 \mid K_2^{*}(1430)+ \mid K_2^{*}(1430)-$ PDG2026:observable:M022:M022W4:335892	109+-5 MeV	u_+ : 5.47781713955507 94 u_- : 5.47781713955507 94 Type: NONE Confidence: ND	COMP-MESON E04 / M08
192. $K_2(1770)$ WIDTH $K_2(1770)0 \mid \bar{K}2(1770)0 \mid K_2(1770)+ \mid K_2(1770)-$ PDG2026:observable:M023:M023W:335964	186+-14 MeV	u_+ : 13.9780282998288 69 u_- : 13.9632358217417 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
193. $\psi(4160)$ WIDTH $\psi(4160)$ PDG2026:observable:M025:M025W:346344	69+-10 MeV	u_+ : 10.5376852320708 19 u_- : 9.60222338741453 46 Type: NONE Confidence: ND	COMP-MESON E04 / M08
194. $\eta_c(1S)$ WIDTH $\eta_c(1S)$ PDG2026:observable:M026:M026W:343287	30.0+-0.5 MeV	u_+ : 0.46062178744488 053 u_- : 0.46062178744488 053 Type: NONE Confidence: ND	COMP-MESON E04 / M08
195. $\eta(1405)$ WIDTH $\eta(1405)0$ PDG2026:observable:M027:M027WX:334530	49.8+2.3-2.0 MeV	u_+ : 2.29806458315265 21 u_- : 1.98212074261206 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
196. η pi pi MODE $\eta(1405)0$ PDG2026:observable:M027:M027W1:334531	51.1+2.8-2.1 MeV	u_+ : 2.79792580324660 28 u_- : 2.05813227941554 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
197. K $\bar{K}$ pi MODE ($a_0(980)$ pi or direct K $\bar{K}$ pi) $\eta(1405)0$ PDG2026:observable:M027:M027W4:334532	49+-4 MeV	u_+ : 3.91295895972736 62 u_- : 4.10896396175021 12 Type: NONE Confidence: ND	COMP-MESON E04 / M08
198. pi pi gamma MODE $\eta(1405)0$ PDG2026:observable:M027:M027W2:334533	89+-17 MeV	u_+ : 17.3079325452340 81 u_- : 17.3079325452340 81 Type: NONE Confidence: ND	COMP-MESON E04 / M08
199. phi gamma MODE $\eta(1405)0$ PDG2026:observable:M027:M027W7:334534	86+7-18 MeV	u_+ : 7.13091859440282 77 u_- : 17.6082367089950 3 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
200. K₁(1270) WIDTH K ₁ (1270)0 Kbar ₁ (1270)0 K ₁ (1270)+ K ₁ (1270)- PDG2026:observable:M028:M028WX:335822	90+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08
201. PRODUCED BY K-, BACKWARD SCATTERING, HYPERON EXCHANGE K ₁ (1270)0 Kbar ₁ (1270)0 K ₁ (1270)+ K ₁ (1270)- PDG2026:observable:M028:M028W2:335824	75+-15 MeV	u_+ : 15 u_- : 15 Type: NONE Confidence: ND	COMP-MESON E04 / M08
202. PRODUCED BY K BEAMS K ₁ (1270)0 Kbar ₁ (1270)0 K ₁ (1270)+ K ₁ (1270)- PDG2026:observable:M028:M028W3:335825	90+-8 MeV	u_+ : 8 u_- : 8 Type: NONE Confidence: ND	COMP-MESON E04 / M08
203. PRODUCED BY BEAMS OTHER THAN K MESONS K ₁ (1270)0 Kbar ₁ (1270)0 K ₁ (1270)+ K ₁ (1270)- PDG2026:observable:M028:M028W1:335826	126+-16 MeV	u_+ : 16.0819577830664 29 u_- : 16.0819577830664 29 Type: NONE Confidence: ND	COMP-MESON E04 / M08
204. PRODUCED IN tau LEPTON DECAYS K ₁ (1270)0 Kbar ₁ (1270)0 K ₁ (1270)+ K ₁ (1270)- PDG2026:observable:M028:M028WT:335827	260+120-110 MeV	u_+ : 120.415945787922 99 u_- : 106.301458127346 51 Type: NONE Confidence: ND	COMP-MESON E04 / M08
205. X(3940) WIDTH X(3940) PDG2026:observable:M029:M029W:346245	43+28-18 MeV	u_+ : 28.1024381828863 21 u_- : 17.5626552105673 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08
206. h₁(1170) WIDTH h ₁ (1170)0 PDG2026:observable:M030:M030W:334307	375+-35 MeV	u_+ : 34.5253530032641 36 u_- : 34.5253530032641 36 Type: NONE Confidence: ND	COMP-MESON E04 / M08
207. pi- p --> omega pi0 n rho ₅ (2350)0 rho ₅ (2350)+ rho ₅ (2350)- PDG2026:observable:M033:M033W3:335260	400+-100 MeV	u_+ : 100 u_- : 100 Type: NONE Confidence: ND	COMP-MESON E04 / M08
208. pi₂(1670) WIDTH pi ₂ (1670)0 pi ₂ (1670)+ pi ₂ (1670)- PDG2026:observable:M034:M034W:334784	258+8-9 MeV	u_+ : 7.99452319640628 42 u_- : 9.03265163231475 12 Type: NONE Confidence: ND	COMP-MESON E04 / M08
209. K₄[*](2045) WIDTH K ₄ [*] (2045)0 Kbar ₄ [*] (2045)0 K ₄ [*] (2045)+ K ₄ [*] (2045)- PDG2026:observable:M035:M035W:336017	199+27-19 MeV	u_+ : 26.5088710269003 51 u_- : 19.3005075635472 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
210. a₀(980) WIDTH a ₀ (980)0 a ₀ (980)+ a ₀ (980)- PDG2026:observable:M036:M036W1:334172	50 to 100 MeV	u ₊ : NA u ₋ : NA Type: R Confidence: ND	COMP-MESON E04 / M08
211. eta(1295) WIDTH eta(1295)0 PDG2026:observable:M037:M037W:334452	55+-5 MeV	u ₊ : 5.36628163064819 13 u ₋ : 5.36628163064819 13 Type: NONE Confidence: ND	COMP-MESON E04 / M08
212. f₂(1810) WIDTH f ₂ (1810)0 PDG2026:observable:M038:M038W:335003	197+-22 MeV	u ₊ : 21.9794266885127 31 u ₋ : 22.1539703002476 11 Type: NONE Confidence: ND	COMP-MESON E04 / M08
213. K₂(2250) WIDTH K ₂ (2250)0 Kbar ₂ (2250)0 K ₂ (2250)+ K ₂ (2250)- PDG2026:observable:M040:M040W:336033	180+-30 MeV	u ₊ : 29.9999994865718 7 u ₋ : 29.9999994865718 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
214. p p CENTRAL PRODUCTION f ₄ (2300)0 PDG2026:observable:M041:M041W4:335245	250+-80 MeV	u ₊ : 80 u ₋ : 80 Type: NONE Confidence: ND	COMP-MESON E04 / M08
215. f₂(2150) WIDTH, COMBINED MODES f ₂ (2150)0 PDG2026:observable:M042:M042W:335160	152+-30 MeV	u ₊ : 30.1870023487088 41 u ₋ : 30.1870023487088 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
216. eta eta MODE f ₂ (2150)0 PDG2026:observable:M042:M042W3:335161	152+-30 MeV	u ₊ : 30.1870023487088 41 u ₋ : 30.1870023487088 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
217. pbar p --> pi pi f ₂ (2150)0 PDG2026:observable:M042:M042W1:335162	250 MeV	u ₊ : ND u ₋ : ND Type: NONE Confidence: ND	COMP-MESON E04 / M08
218. omega₃(1670) WIDTH omega ₃ (1670)0 PDG2026:observable:M045:M045W:334779	168+-10 MeV	u ₊ : 10.0179720379382 u ₋ : 10.0179720379382 Type: NONE Confidence: ND	COMP-MESON E04 / M08
219. Upsilon(4S) WIDTH Upsilon(4S) PDG2026:observable:M047:M047W:347745	20.5+-2.5 MeV	u ₊ : 2.47304103647041 1 u ₋ : 2.47304103647041 1 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
220. Upsilon(3S) WIDTH Upsilon(3S) PDG2026:observable:M048:M048W:347661	20.32+-1.85 keV	u_+ : 1.85000000000000 01 u_- : 1.85000000000000 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
221. Upsilon(1S) WIDTH Upsilon(1S) PDG2026:observable:M049:M049W:346838	54.02+-1.25 keV	u_+ : 1.25 u_- : 1.25 Type: NONE Confidence: ND	COMP-MESON E04 / M08
222. chi_c2(3930) WIDTH chi_c2(3930) PDG2026:observable:M050:M050W:346229	35.2+-2.2 MeV	u_+ : 2.22867305911869 32 u_- : 2.22867305911869 32 Type: NONE Confidence: ND	COMP-MESON E04 / M08
223. Upsilon(2S) WIDTH Upsilon(2S) PDG2026:observable:M052:M052W:347288	31.98+-2.63 keV	u_+ : 2.62999999999999 99 u_- : 2.62999999999999 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
224. psi(3770) WIDTH psi(3770) PDG2026:observable:M053:M053W:345870	27.2+-1.0 MeV	u_+ : 0.95924822707886 17 u_- : 0.95924822707886 17 Type: NONE Confidence: ND	COMP-MESON E04 / M08
225. phi_3(1850) WIDTH phi_3(1850)0 PDG2026:observable:M054:M054W:335024	87+28-23 MeV	u_+ : 27.5787929492030 59 u_- : 23.0924992747863 09 Type: NONE Confidence: ND	COMP-MESON E04 / M08
226. chi_c1(1P) WIDTH chi_c1(1P) PDG2026:observable:M055:M055W:344599	0.82+-0.04 MeV	u_+ : 0.04162341733038 1482 u_- : 0.04162341733038 1482 Type: NONE Confidence: ND	COMP-MESON E04 / M08
227. chi_c0(1P) WIDTH chi_c0(1P) PDG2026:observable:M056:M056W:344305	12.40+-0.27 MeV	u_+ : 0.26507716522265 817 u_- : 0.26507716522265 817 Type: NONE Confidence: ND	COMP-MESON E04 / M08
228. chi_c2(1P) WIDTH chi_c2(1P) PDG2026:observable:M057:M057W:344927	1.94+-0.09 MeV	u_+ : 0.08904841015736 6226 u_- : 0.08904841015736 6226 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
229. pi(1300) WIDTH pi(1300)0 pi(1300)+ pi(1300)- PDG2026:observable:M058:M058W:334464	200 TO 600 MeV Display: 200 to 600	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
230. eta_c(2S) WIDTH eta_c(2S) PDG2026:observable:M059:M059W:345229	11.6+-1.4 MeV	u_+ : 1.46140521435830 21 u_- : 1.42482273720458 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
231. K_3*(1780) WIDTH K_3^*(1780)0 Kbar_3^*(1780)0 K_3^*(1780)+ K_3^*(1780)- PDG2026:observable:M060:M060W:335976	161+-17 MeV	u_+ : 16.7757777591419 21 u_- : 17.6873693367902 58 Type: NONE Confidence: ND	COMP-MESON E04 / M08
232. D*(2007)0 WIDTH D^*(2007)0 Dbar^*(2007)0 PDG2026:observable:M061:M061W:337780	<2.1 MeV	u_+ : NA u_- : NA Type: U Confidence: 90	COMP-MESON E04 / M08
233. D*(2010)+- WIDTH D^*(2010)+ D^*(2010)- PDG2026:observable:M062:M062W:337799	83.4+-1.8 keV	u_+ : 1.83767139449678 71 u_- : 1.83767139449678 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
234. K_1(1400) WIDTH K_1(1400)0 Kbar_1(1400)0 K_1(1400)+ K_1(1400)- PDG2026:observable:M064:M064W:335846	174+-13 MeV	u_+ : 12.6111236551536 99 u_- : 12.6011609689773 61 Type: NONE Confidence: ND	COMP-MESON E04 / M08
235. eta rho0 AND pi+ pi- MODES rho(1700)0 rho(1700)+ rho(1700)- PDG2026:observable:M065:M065W0:334881	250+-100 MeV	u_+ : 100 u_- : 100 Type: NONE Confidence: ND	COMP-MESON E04 / M08
236. e+ e- PRODUCTION phi(1680)0 PDG2026:observable:M067:M067W1:334826	150+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-MESON E04 / M08
237. f_0(1710) Breit-Wigner WIDTH f_0(1710)0 PDG2026:observable:M068:M068W:334928	149+-10 MeV	u_+ : 10.9253240627161 7 u_- : 9.89172341764963 65 Type: NONE Confidence: ND	COMP-MESON E04 / M08
238. J/psi(1S) WIDTH J/psi(1S) PDG2026:observable:M070:M070W:343506	92.6+-1.7 keV	u_+ : 1.71825116057207 09 u_- : 1.71191328371000 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
239. psi(2S) WIDTH psi(2S) PDG2026:observable:M071:M071W:345302	293+-9 keV	u_+ : 9.12161713271473 04 u_- : 9.12161713271473 04 Type: NONE Confidence: ND	COMP-MESON E04 / M08
240. psi(4040) WIDTH psi(4040) PDG2026:observable:M072:M072W:346250	84+-12 MeV	u_+ : 12.30000000000000 01 u_- : 12.30000000000000 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
241. psi(4415) WIDTH psi(4415) PDG2026:observable:M073:M073W:346685	110+-13 MeV	u_+ : 13.2878278993015 7 u_- : 13.2878278993015 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
242. psi(4230) WIDTH psi(4230) PDG2026:observable:M074:M074W:346461	51+-7 MeV	u_+ : 6.99729038980041 02 u_- : 6.88241490298323 64 Type: NONE Confidence: ND	COMP-MESON E04 / M08
243. pi(1800) WIDTH pi(1800)0 pi(1800)+ pi(1800)- PDG2026:observable:M075:M075W:334967	215+-7-8 MeV	u_+ : 6.70657742096917 8 u_- : 8.19989930002816 08 Type: NONE Confidence: ND	COMP-MESON E04 / M08
244. f_J(2220) WIDTH f_J(2220)0 PDG2026:observable:M082:M082W:335204	23+-8-7 MeV	u_+ : 8.29072339682046 79 u_- : 6.99539815418370 2 Type: NONE Confidence: ND	COMP-MESON E04 / M08
245. f_1(1510) WIDTH f_1(1510)0 PDG2026:observable:M084:M084W:334659	73+-25 MeV	u_+ : 25.1244904562855 5 u_- : 25.1244904562855 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
246. X(1835) WIDTH X(1835)0 PDG2026:observable:M085:M085W:335012	130+-50 MeV	u_+ : 46.5214248579634 67 u_- : 48.8057618821165 28 Type: NONE Confidence: ND	COMP-MESON E04 / M08
247. f_6(2510) WIDTH f_6(2510)0 PDG2026:observable:M089:M089W:335278	260+-40 MeV	u_+ : 42.4264068711928 53 u_- : 42.4264068711928 53 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
248. K₃(2320) WIDTH K ₃ (2320)0 Kbar ₃ (2320)0 K ₃ (2320)+ K ₃ (2320)- PDG2026:observable:M090:M090W:336035	150+-30 MeV	u_+ : 30 u_- : 30 Type: NONE Confidence: ND	COMP-MESON E04 / M08
249. Upsilon(10860) WIDTH Upsilon(10860) PDG2026:observable:M092:M092W:347805	37+-4 MeV	u_+ : 4.26656207180538 96 u_- : 3.97673653123974 89 Type: NONE Confidence: ND	COMP-MESON E04 / M08
250. Upsilon(11020) WIDTH Upsilon(11020) PDG2026:observable:M093:M093W:347906	24+-8-6 MeV	u_+ : 7.70770656457082 83 u_- : 6.45715582578219 66 Type: NONE Confidence: ND	COMP-MESON E04 / M08
251. K[*](1410) WIDTH K [*] (1410)0 Kbar [*] (1410)0 K [*] (1410)+ K [*] (1410)- PDG2026:observable:M094:M094W:335866	232+-21 MeV	u_+ : 21.3823721605395 6 u_- : 21.3823721605395 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08
252. K[*](1680) WIDTH K [*] (1680)0 Kbar [*] (1680)0 K [*] (1680)+ K [*] (1680)- PDG2026:observable:M095:M095W:335947	320+-110 MeV	u_+ : 107.338345980695 71 u_- : 107.807566482925 91 Type: NONE Confidence: ND	COMP-MESON E04 / M08
253. K₅[*](2380) WIDTH K ₅ [*] (2380)0 Kbar ₅ [*] (2380)0 K ₅ [*] (2380)+ K ₅ [*] (2380)- PDG2026:observable:M098:M098W:336037	180+-50 MeV	u_+ : 48.9182992345400 36 u_- : 48.9182992345400 36 Type: NONE Confidence: ND	COMP-MESON E04 / M08
254. K₁(1650) WIDTH K ₁ (1650)0 Kbar ₁ (1650)0 K ₁ (1650)+ K ₁ (1650)- PDG2026:observable:M099:M099W:335945	150+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-MESON E04 / M08
255. eta₂(1870) WIDTH eta ₂ (1870)0 PDG2026:observable:M101:M101W:335033	225+-14 MeV	u_+ : 14.0650075480096 5 u_- : 14.0650075480096 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
256. phi(2170) WIDTH phi(2170)0 phi(2170)+ phi(2170)- PDG2026:observable:M103:M103W:335181	88+-26-21 MeV	u_+ : 25.8456182632216 7 u_- : 20.6358651464737 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
257. $K_2^*(1980)$ WIDTH $K_2^{*0}(1980) \bar{K}^{*0}(1980) K_2^{*+}(1980) K_2^{*-}(1980)$ PDG2026:observable:M104:M104W:336004	348+50-30 MeV	u_+ : 47.9950158521243 11 u_- : 29.8227003927434 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
258. $\rho(1450)$ WIDTH $\rho(1450) \rho(1450)^+ \rho(1450)^-$ PDG2026:observable:M105:M105W0:334593	400 +-60 MeV Display: 400+-60	u_+ : 60 u_- : 60 Type: NONE Confidence: ND	COMP-MESON E04 / M08
259. $f_2(2010)$ WIDTH $f_2(2010)0$ PDG2026:observable:M106:M106W:335113	200+-60 MeV	u_+ : 67 u_- : 62 Type: NONE Confidence: ND	COMP-MESON E04 / M08
260. $f_2(2300)$ WIDTH $f_2(2300)0$ PDG2026:observable:M107:M107W:335235	150+-40 MeV	u_+ : 41 u_- : 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
261. $f_2(2340)$ WIDTH $f_2(2340)0$ PDG2026:observable:M108:M108W:335253	331+27-18 MeV	u_+ : 27.4060949578306 5 u_- : 17.5990897768665 38 Type: NONE Confidence: ND	COMP-MESON E04 / M08
262. $h_1(1415)$ WIDTH $h_1(1415)0$ PDG2026:observable:M109:M109W:334553	78+-11 MeV	u_+ : 10.8947684184199 31 u_- : 10.3547610268449 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08
263. $f_0(2200)$ WIDTH $f_0(2200)0$ PDG2026:observable:M112:M112W:335200	210+-40 MeV	u_+ : 41.6386565682062 37 u_- : 42.1732516055459 43 Type: NONE Confidence: ND	COMP-MESON E04 / M08
264. $\eta(1760)$ WIDTH $\eta(1760)0$ PDG2026:observable:M114:M114W:334951	240+-30 MeV	u_+ : 30.7670000039266 5 u_- : 28.8198435999457 98 Type: NONE Confidence: ND	COMP-MESON E04 / M08
265. $\eta(2225)$ WIDTH $\eta(2225)0$ PDG2026:observable:M115:M115W:335233	185+40-20 MeV	u_+ : 36.8940365233648 68 u_- : 19.6088060392476 Type: NONE Confidence: ND	COMP-MESON E04 / M08
266. $f_2(1640)$ WIDTH $f_2(1640)0$ PDG2026:observable:M117:M117W:334753	100+60-40 MeV	u_+ : 55.0072709658098 73 u_- : 40.9999990454338 03 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
267. D_{s1}(2536)⁺- WIDTH D _{s1} (2536) ⁺ D _{s1} (2536) ⁻ PDG2026:observable:M121:M121W:338489	0.92+-0.05 MeV	u_+ : 0.04996531390494 7829 u_- : 0.04996531390494 7829 Type: NONE Confidence: ND	COMP-MESON E04 / M08
268. f₂(1565) WIDTH f ₂ (1565)0 PDG2026:observable:M123:M123W:334704	132+-23 MeV	u_+ : 22.4999992121598 28 u_- : 22.4999992121598 28 Type: NONE Confidence: ND	COMP-MESON E04 / M08
269. omega(1420) WIDTH omega(1420)0 PDG2026:observable:M125:M125W:334568	290+-190 MeV	u_+ : 190 u_- : 190 Type: NONE Confidence: ND	COMP-MESON E04 / M08
270. omega(1650) WIDTH omega(1650)0 PDG2026:observable:M126:M126W:334769	150+-50 MeV	u_+ : 50 u_- : 50 Type: NONE Confidence: ND	COMP-MESON E04 / M08
271. K₀[*](1950) WIDTH K ₀ [*] (1950)0 Kbar ₀ [*] (1950)0 K ₀ [*] (1950) ⁺ K ₀ [*] (1950) ⁻ PDG2026:observable:M134:M134W:336000	170+-50 MeV	u_+ : 49.9776346249792 67 u_- : 49.9776346249792 67 Type: NONE Confidence: ND	COMP-MESON E04 / M08
272. f₂(1950) WIDTH f ₂ (1950)0 PDG2026:observable:M135:M135W:335076	464+-24 MeV	u_+ : 24.2336083935200 51 u_- : 23.8581751115928 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
273. f₂(1910) omega omega MODE f ₂ (1910)0 PDG2026:observable:M142:M142W2:335059	167+-21 MeV	u_+ : 20.5932107794648 79 u_- : 20.5932107794648 79 Type: NONE Confidence: ND	COMP-MESON E04 / M08
274. f₂(1910) eta eta' MODE f ₂ (1910)0 PDG2026:observable:M142:M142W3:335060	140+-40 MeV	u_+ : 41 u_- : 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
275. h_c(1P) WIDTH h _c (1P) PDG2026:observable:M144:M144W:344852	0.78+-0.28 MeV	u_+ : 0.29546573405388 321 u_- : 0.26832815729997 478 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
276. $K_2(1820)$ WIDTH $K_2(1820)0 \mid \bar{K}2(1820)0 \mid K_2(1820)+ \mid K_2(1820)-$ PDG2026:observable:M146:M146W:335991	264+-34 MeV	u_+ : 33.9111628155620 37 u_- : 33.5851541639118 88 Type: NONE Confidence: ND	COMP-MESON E04 / M08
277. Full width (Breit-Wigner) $f_0(1370)0$ PDG2026:observable:M147:M147W:334507	200 to 500 MeV	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
278. $D_{s2}^*(2573)$ WIDTH $D_{s2}^{*+}(2573) \mid D_{s2}^{*-}(2573)-$ PDG2026:observable:M148:M148W:338511	16.9+-0.7 MeV	u_+ : 0.73972013426443 928 u_- : 0.73758653054042 789 Type: NONE Confidence: ND	COMP-MESON E04 / M08
279. $a_0(1450)$ WIDTH $a_0(1450)0 \mid a_0(1450)+ \mid a_0(1450)-$ PDG2026:observable:M149:M149W:334580	258+-14 MeV	u_+ : 14.0000000989268 51 u_- : 14.0000000989268 51 Type: NONE Confidence: ND	COMP-MESON E04 / M08
280. $B_J^*(5732)$ WIDTH $B_J^{*+}(5732)0 \mid \bar{B}J^{*+}(5732)0 \mid B_J^{*+}(5732)+ \mid B_J^{*+}(5732)-$ PDG2026:observable:M151:M151W:342552	128+-18 MeV	u_+ : 18.2221584567671 19 u_- : 18.2221584567671 19 Type: NONE Confidence: ND	COMP-MESON E04 / M08
281. $f_0(1500)$ WIDTH $f_0(1500)0$ PDG2026:observable:M152:M152W:334629	108+-33 MeV	u_+ : 33 u_- : 33 Type: NONE Confidence: ND	COMP-MESON E04 / M08
282. $B_{sJ}^*(5850)$ WIDTH $B_{sJ}^{*+}(5850)0 \mid \bar{B}B_{sJ}^{*+}(5850)0$ PDG2026:observable:M153:M153W:343152	47+-22 MeV	u_+ : 22 u_- : 22 Type: NONE Confidence: ND	COMP-MESON E04 / M08
283. $\eta_2(1645)$ WIDTH $\eta_2(1645)0$ PDG2026:observable:M154:M154W:334759	181+-11 MeV	u_+ : 10.7857447992021 21 u_- : 10.4916641064320 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
284. $f_0(2020)$ WIDTH $f_0(2020)0$ PDG2026:observable:M156:M156W:335121	440+-50 MeV	u_+ : 46.1735855224607 83 u_- : 49.1629942131274 19 Type: NONE Confidence: ND	COMP-MESON E04 / M08
285. $D^*(2640)+-$ WIDTH $D^{*+}(2640)0 \mid \bar{D}^{*+}(2640)0 \mid D^{*+}(2640)+ \mid D^{*+}(2640)-$ PDG2026:observable:M158:M158W:337855	<15 MeV	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
286. chi_c0(3915) WIDTH chi_c0(3915) PDG2026:observable:M159:M159W:346205	20+-4 MeV	u_+ : 3.92370931837659 9 u_- : 3.84857327141186 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
287. K(1630) WIDTH K(1630)0 Kbar(1630)0 K(1630)+ K(1630)- PDG2026:observable:M160:M160W:335943	16+19-16 MeV	u_+ : 19 u_- : 16 Type: NONE Confidence: ND	COMP-MESON E04 / M08
288. a_1(1640) WIDTH a_1(1640)0 a_1(1640)+ a_1(1640)- PDG2026:observable:M161:M161W:334741	250+-40 MeV	u_+ : 38.7680261069732 67 u_- : 38.0044644472388 31 Type: NONE Confidence: ND	COMP-MESON E04 / M08
289. a_2(1700) WIDTH a_2(1700)0 a_2(1700)+ a_2(1700)- PDG2026:observable:M162:M162W:334904	380+60-50 MeV	u_+ : 56.8021566161437 33 u_- : 49.9197868473076 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
290. pi_1(1600) WIDTH (eta pi mode) pi_1(1600)0 pi_1(1600)+ pi_1(1600)- PDG2026:observable:M164:M164WEP:334726	330+-35 MeV	u_+ : 34.7276893152504 51 u_- : 34.6931516296829 87 Type: NONE Confidence: ND	COMP-MESON E04 / M08
291. pi_1(1600) WIDTH (non-eta pi mode) pi_1(1600)0 pi_1(1600)+ pi_1(1600)- PDG2026:observable:M164:M164W:334727	370+50-60 MeV	u_+ : 50.3245163918421 17 u_- : 56.4547140483811 28 Type: NONE Confidence: ND	COMP-MESON E04 / M08
292. h_1(1595) WIDTH h_1(1595)0 PDG2026:observable:M166:M166W:334721	380+90-120 MeV	u_+ : 92.1954445729288 78 u_- : 116.619037896905 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
293. f_0(2100) WIDTH f_0(2100)0 PDG2026:observable:M168:M168W:335157	287+32-24 MeV	u_+ : 31.5949157013644 21 u_- : 24.0486752700559 91 Type: NONE Confidence: ND	COMP-MESON E04 / M08
294. eta_b(1S) WIDTH eta_b(1S) PDG2026:observable:M171:M171W:346826	10+5-4 MeV	u_+ : 4.76841579595203 06 u_- : 3.61481987601074 91 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
295. $D_{s0}^*(2317)^+$ WIDTH $D_{s0}^*(2317)^+ D_{s0}^*(2317)^-$ PDG2026:observable:M172:M172W:338433	< 3.8 MeV Display: <3.8	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-MESON E04 / M08
296. $D_{s1}(2460)^+$ WIDTH $D_{s1}(2460)^+ D_{s1}(2460)^-$ PDG2026:observable:M173:M173W:338454	< 3.5 MeV Display: <3.5	u_+ : NA u_- : NA Type: U Confidence: 95	COMP-MESON E04 / M08
297. $K_0^*(700)$ Breit-Wigner Width $K_0^*(700)0 \bar{K}_0^*(700)0 K_0^*(700)^+ K_0^*(700)^-$ PDG2026:observable:M174:M174W:335787	463$\pm$27 MeV	u_+ : 28.0180200105868 49 u_- : 25.9692055984802 12 Type: NONE Confidence: ND	COMP-MESON E04 / M08
298. $\eta(1475)$ WIDTH $\eta(1475)0$ PDG2026:observable:M175:M175W5:334617	96$\pm$9 MeV	u_+ : 9.16477812816133 59 u_- : 8.65891056010132 05 Type: NONE Confidence: ND	COMP-MESON E04 / M08
299. $\chi_{c1}(3872)$ WIDTH $\chi_{c1}(3872)$ PDG2026:observable:M176:M176W:346132	1.19$\pm$0.21 MeV	u_+ : 0.21446141015052 739 u_- : 0.21212796092607 489 Type: NONE Confidence: ND	COMP-MESON E04 / M08
300. $D_1(2430)0$ WIDTH $D_1(2430)0 \bar{D}_1(2430)0$ PDG2026:observable:M180:M180W:337828	314$\pm$29 MeV	u_+ : 28.6869527683639 7 u_- : 28.3279146860413 21 Type: NONE Confidence: ND	COMP-MESON E04 / M08
301. $\psi(4360)$ WIDTH $\psi(4360)$ PDG2026:observable:M181:M181W:346636	124$\pm$13 MeV	u_+ : 13.0611247068081 21 u_- : 13.2865041822744 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
302. $D_{s1}^*(2700)^+$ WIDTH $D_{s1}^*(2700)^+ D_{s1}^*(2700)^-$ PDG2026:observable:M182:M182W:338525	122$\pm$10 MeV	u_+ : 10.5434175204121 29 u_- : 10.0609470530140 1 Type: NONE Confidence: ND	COMP-MESON E04 / M08
303. $\pi_2(1880)$ WIDTH $\pi_2(1880)0$ PDG2026:observable:M185:M185W:335044	237$\pm$33-30 MeV	u_+ : 33.3171974639905 17 u_- : 29.5026580378861 11 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
304. B_{s2}[*](5840)0 WIDTH B _{s2} [*] (5840)0 Bbar _{s2} [*] (5840)0 PDG2026:observable:M186:M186W:343141	1.49+-0.27 MeV	u_+ : 0.26660479598163 422 u_- : 0.26660479598163 422 Type: NONE Confidence: ND	COMP-MESON E04 / M08
305. B_{s1}(5830)0 WIDTH B _{s1} (5830)0 Bbar _{s1} (5830)0 PDG2026:observable:M187:M187W:343133	0.5+-0.4 MeV	u_+ : 0.42426406871192 851 u_- : 0.42426406871192 851 Type: NONE Confidence: ND	COMP-MESON E04 / M08
306. rho(1570) WIDTH rho(1570)0 PDG2026:observable:M188:M188W:334715	140+-90 MeV	u_+ : 86.4522989862039 42 u_- : 86.4522989862039 42 Type: NONE Confidence: ND	COMP-MESON E04 / M08
307. psi(4660) WIDTH psi(4660) PDG2026:observable:M189:M189W:346768	55+-9 MeV	u_+ : 8.88972291869921 7 u_- : 8.40991396186550 06 Type: NONE Confidence: ND	COMP-MESON E04 / M08
308. X(4160) WIDTH X(4160) PDG2026:observable:M190:M190W:346452	136+-60-35 MeV	u_+ : 56.5392209238173 54 u_- : 34.6252307256621 42 Type: NONE Confidence: ND	COMP-MESON E04 / M08
309. T_ccbar(4050)+ WIDTH T _c cbar(4050)+ PDG2026:observable:M191:M191W:347991	82+-50-28 MeV	u_+ : 51.4781507049349 97 u_- : 27.8028775489156 89 Type: NONE Confidence: ND	COMP-MESON E04 / M08
310. T_ccbar(4250)+ WIDTH T _c cbar(4250)+ PDG2026:observable:M192:M192W:348027	180+-320-70 MeV	u_+ : 320.580723063630 27 u_- : 72.4016574395918 16 Type: NONE Confidence: ND	COMP-MESON E04 / M08
311. chi_{c1}(4140) WIDTH chi _{c1} (4140) PDG2026:observable:M193:M193W:346337	19+-7-5 MeV	u_+ : 6.55866734077530 64 u_- : 5.27582025104883 37 Type: NONE Confidence: ND	COMP-MESON E04 / M08
312. X(4350) WIDTH X(4350) PDG2026:observable:M194:M194W:346629	13+-18-10 MeV	u_+ : 18.4390889145857 71 u_- : 9.84885780179610 39 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
313. T_c_cbar_1(4430)+ WIDTH T_c_cbar_1(4430)+ PDG2026:observable:M195:M195W:348031	189+-29 MeV	u_+ : 28.9608626207506 99 u_- : 29.2122816924321 89 Type: NONE Confidence: ND	COMP-MESON E04 / M08
314. D_s1*(2860)+ WIDTH D_s1^*(2860)+ D_s1^*(2860)- PDG2026:observable:M196:M196W:338534	160+-80 MeV	u_+ : 80.3616824114577 64 u_- : 80.3616824114577 64 Type: NONE Confidence: ND	COMP-MESON E04 / M08
315. D_sJ(3040)+ WIDTH D_sJ(3040)+ D_sJ(3040)- PDG2026:observable:M197:M197W:338545	240+-60 MeV	u_+ : 57.8013840664737 02 u_- : 54.6717477313465 78 Type: NONE Confidence: ND	COMP-MESON E04 / M08
316. D_0(2550)0 WIDTH D_0(2550)0 Dbar_0(2550)0 D_0(2550)+ D_0(2550)- PDG2026:observable:M198:M198W:337844	165+-24 MeV	u_+ : 23.7475533977095 49 u_- : 23.7475533977095 49 Type: NONE Confidence: ND	COMP-MESON E04 / M08
317. D_1*(2600)0 WIDTH D_1^*(2600)0 Dbar_1^*(2600)0 D_1^*(2600)+ D_1^*(2600)- PDG2026:observable:M199:M199W:337847	141+-23 MeV	u_+ : 22.8365240441833 88 u_- : 22.8365240441833 88 Type: NONE Confidence: ND	COMP-MESON E04 / M08
318. eta_b(2S) WIDTH eta_b(2S) PDG2026:observable:M200:M200W:347283	< 24 MeV Display: <24	u_+ : NA u_- : NA Type: U Confidence: 90	COMP-MESON E04 / M08
319. D_3*(2750) WIDTH D_3^*(2750)0 Dbar_3^*(2750)0 D_3^*(2750)+ D_3^*(2750)- PDG2026:observable:M203:M203W:337861	66+-5 MeV	u_+ : 4.60269309173125 41 u_- : 4.60269309173125 41 Type: NONE Confidence: ND	COMP-MESON E04 / M08
320. T_b_bbar_1(10610)+- WIDTH T_b_bbar_1(10610)+ PDG2026:observable:M207:M207W:348054	18.4+-2.4 MeV	u_+ : 2.3999999999999999 99 u_- : 2.3999999999999999 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
321. T_b_bbar_1(10650)+ WIDTH T_b_bbar_1(10650)+ PDG2026:observable:M208:M208W:348076	11.5+-2.2 MeV	u_+ : 2.2000000000000000 02 u_- : 2.2000000000000000 02 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
322. T_c_cbar_1(3900) WIDTH T_c_cbar_1(3900)+ PDG2026:observable:M210:M210W:347939	29.5+-2.4 MeV	u_+ : 2.42574928658747 7 u_- : 2.41740148443531 49 Type: NONE Confidence: ND	COMP-MESON E04 / M08
323. psi_2(3823) WIDTH psi_2(3823) PDG2026:observable:M212:M212W:346101	< 2.9 MeV Display: <2.9	u_+ : NA u_- : NA Type: U Confidence: 90	COMP-MESON E04 / M08
324. T_c_cbar(4020) WIDTH T_c_cbar(4020)+ PDG2026:observable:M213:M213W:347977	13+-5 MeV	u_+ : 5.27421046109626 52 u_- : 5.27421046109626 52 Type: NONE Confidence: ND	COMP-MESON E04 / M08
325. T_c_cbar_0(4240)+ WIDTH T_c_cbar_0(4240)0 PDG2026:observable:M216:M216W:348023	220+120-90 MeV	u_+ : 117.783700060746 99 u_- : 87.6641317757724 61 Type: NONE Confidence: ND	COMP-MESON E04 / M08
326. T_c_cbar(4055)+ WIDTH T_c_cbar(4055)+ PDG2026:observable:M223:M223W:348001	45+-13 MeV	u_+ : 12.5299640861416 69 u_- : 12.5299640861416 69 Type: NONE Confidence: ND	COMP-MESON E04 / M08
327. D_s3*(2860)+ WIDTH D_s3^*(2860)+ D_s3^*(2860)- PDG2026:observable:M226:M226W:338536	53+-10 MeV	u_+ : 9.89949493661166 54 u_- : 9.89949493661166 54 Type: NONE Confidence: ND	COMP-MESON E04 / M08
328. a_0(1950) WIDTH a_0(1950)0 PDG2026:observable:M227:M227W:335071	270+-40 MeV	u_+ : 36.4005494464025 93 u_- : 36.4005494464025 93 Type: NONE Confidence: ND	COMP-MESON E04 / M08
329. D_2(2740)0 WIDTH D_2(2740)0 Dbar_2(2740)0 D_2(2740)+ D_2(2740)- PDG2026:observable:M228:M228W:337858	88+-19 MeV	u_+ : 19.4352180211727 09 u_- : 19.4352180211727 09 Type: NONE Confidence: ND	COMP-MESON E04 / M08
330. D_2*(3000)0 WIDTH D_2^*(3000)0 Dbar_2^*(3000)0 D_2^*(3000)+ D_2^*(3000)- PDG2026:observable:M229:M229W:337873	190+-80 MeV	u_+ : 81.4125297481904 79 u_- : 81.4125297481904 79 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
331. T_c_cbar_1(4200)+ WIDTH T_c_cbar_1(4200)+ PDG2026:observable:M231:M231W:348013	310+-40 MeV	u_+ : 35.2609658006496 29 u_- : 36.5870573754017 27 Type: NONE Confidence: ND	COMP-MESON E04 / M08
332. T_b_sbar(5568)+ WIDTH T_b_sbar(5568)+ PDG2026:observable:M232:M232W:348037	19+9-7 MeV	u_+ : 8.64060183089118 01 u_- : 7.18679344353237 86 Type: NONE Confidence: ND	COMP-MESON E04 / M08
333. chi_c1(4274) WIDTH chi_c1(4274) PDG2026:observable:M233:M233W:346625	54+-8 MeV	u_+ : 8.19685254949333 24 u_- : 7.79338377392468 21 Type: NONE Confidence: ND	COMP-MESON E04 / M08
334. chi_c0(4500) WIDTH chi_c0(4500) PDG2026:observable:M234:M234W:346756	87+30-24 MeV	u_+ : 29.9271604050614 39 u_- : 24.3996524356650 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
335. chi_c0(4700) WIDTH chi_c0(4700) PDG2026:observable:M235:M235W:346817	78+15-11 MeV	u_+ : 14.5397299640840 1 u_- : 10.8529313170942 49 Type: NONE Confidence: ND	COMP-MESON E04 / M08
336. chi_c0(3860) WIDTH chi_c0(3860) PDG2026:observable:M237:M237W:346123	200+180-110 MeV	u_+ : 177.369670462568 11 u_- : 105.891453857240 09 Type: NONE Confidence: ND	COMP-MESON E04 / M08
337. pi_2(2005) WIDTH pi_2(2005)0 PDG2026:observable:M239:M239W:335107	370+16-90 MeV	u_+ : 15.9118526477216 2 u_- : 94.1373162126639 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
338. T_c_cbar(4100)+ WIDTH T_c_cbar(4100)+ PDG2026:observable:M240:M240W:348007	150+80-70 MeV	u_+ : 83.4505841801002 65 u_- : 67.7421582177597 94 Type: NONE Confidence: ND	COMP-MESON E04 / M08
339. psi_3(3842) WIDTH psi_3(3842) PDG2026:observable:M241:M241W:346117	2.8+-0.6 MeV	u_+ : 0.61854668376768 462 u_- : 0.61854668376768 462 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
340. Upsilon(10753) WIDTH Upsilon(10753) PDG2026:observable:M243:M243W:347801	29+-9 MeV	u_+ : 8.88144132446980 81 u_- : 8.88144132446980 81 Type: NONE Confidence: ND	COMP-MESON E04 / M08
341. B_1(5721)+ width B_1(5721)0 Bbar_1(5721)0 B_1(5721)+ B_1(5721)- PDG2026:observable:M244:M244W+:342547	31+-6 MeV	u_+ : 5.84704469385892 u_- : 6.09210420060922 43 Type: NONE Confidence: ND	COMP-MESON E04 / M08
342. B_1(5721)0 width B_1(5721)0 Bbar_1(5721)0 B_1(5721)+ B_1(5721)- PDG2026:observable:M244:M244W0:342548	27.5+-3.4 MeV	u_+ : 3.42227784518650 01 u_- : 3.42227784518650 01 Type: NONE Confidence: ND	COMP-MESON E04 / M08
343. B_2*(5747)+ width B_2^(5747)0 Bbar_2^(5747)0 B_2^(5747)+ B_2^(5747)- PDG2026:observable:M245:M245W+:342564	20+-5 MeV	u_+ : 5.46842291584328 51 u_- : 5.46842291584328 51 Type: NONE Confidence: ND	COMP-MESON E04 / M08
344. B_2*(5747)0 width B_2^(5747)0 Bbar_2^(5747)0 B_2^(5747)+ B_2^(5747)- PDG2026:observable:M245:M245W0:342565	24.2+-1.7 MeV	u_+ : 1.69590866342803 49 u_- : 1.70952643154618 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
345. B_J(5840)+ WIDTH B_J(5840)0 Bbar_J(5840)0 B_J(5840)+ B_J(5840)- PDG2026:observable:M246:M246W+:342577	220+-80 MeV	u_+ : 83.5224520712844 05 u_- : 83.5224520712844 05 Type: NONE Confidence: ND	COMP-MESON E04 / M08
346. B_J(5840)0 WIDTH B_J(5840)0 Bbar_J(5840)0 B_J(5840)+ B_J(5840)- PDG2026:observable:M246:M246W0:342578	130+-40 MeV	u_+ : 38.0131556174964 23 u_- : 38.0131556174964 23 Type: NONE Confidence: ND	COMP-MESON E04 / M08
347. eta(2370) WIDTH eta(2370)0 PDG2026:observable:M247:M247W:335262	148+80-28 MeV	u_+ : 77.2222437006731 61 u_- : 28.3671777206377 8 Type: NONE Confidence: ND	COMP-MESON E04 / M08
348. B_J(5970)+ WIDTH B_J(5970)0 Bbar_J(5970)0 B_J(5970)+ B_J(5970)- PDG2026:observable:M248:M248W+:342589	62+-20 MeV	u_+ : 20.6480926301544 19 u_- : 20.2215257441381 8 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
349. B_J(5970)0 WIDTH B _J (5970)0 B _{bar} _J (5970)0 B _J (5970)+ B _J (5970)- PDG2026:observable:M248:M248W0:342590	81+-12 MeV	u_+ : 11.5840507320396 3 u_- : 11.4214634346750 6 Type: NONE Confidence: ND	COMP-MESON E04 / M08
350. D₁[*](2760)0 WIDTH D ₁ [*] (2760)0 D _{bar} ₁ [*] (2760)0 D ₁ [*] (2760)+ D ₁ [*] (2760)- PDG2026:observable:M249:M249W:337869	180+-40 MeV	u_+ : 38.2753184180092 77 u_- : 38.2753184180092 77 Type: NONE Confidence: ND	COMP-MESON E04 / M08
351. T_{cbar}_{sbar}0[*](2870)0 WIDTH T _{cbar} _{sbar} 0 [*] (2870)0 PDG2026:observable:M250:M250W:347917	68+-17 MeV	u_+ : 16.2828339322687 69 u_- : 17.2087901693615 Type: NONE Confidence: ND	COMP-MESON E04 / M08
352. T_{cbar}_{sbar}1[*](2900)0 WIDTH T _{cbar} _{sbar} 1 [*] (2900)0 PDG2026:observable:M251:M251W:347923	106+-10 MeV	u_+ : 10.3961590631619 3 u_- : 10.3961590631619 3 Type: NONE Confidence: ND	COMP-MESON E04 / M08
353. D₀[*](2300) WIDTH D ₀ [*] (2300)0 D _{bar} ₀ [*] (2300)0 D ₀ [*] (2300)+ D ₀ [*] (2300)- PDG2026:observable:M252:M252W:337814	229+-16 MeV	u_+ : 16.0148868979557 51 u_- : 16.0148868979557 51 Type: NONE Confidence: ND	COMP-MESON E04 / M08
354. D₁(2420) WIDTH D ₁ (2420)0 D _{bar} ₁ (2420)0 D ₁ (2420)+ D ₁ (2420)- PDG2026:observable:M253:M253W:337822	31.3+-1.9 MeV	u_+ : 1.93989376648816 9 u_- : 1.93151199434054 9 Type: NONE Confidence: ND	COMP-MESON E04 / M08
355. D₂[*](2460) WIDTH D ₂ [*] (2460)0 D _{bar} ₂ [*] (2460)0 D ₂ [*] (2460)+ D ₂ [*] (2460)- PDG2026:observable:M254:M254W:337837	47.3+-0.8 MeV	u_+ : 0.77431347668341 055 u_- : 0.77362957674648 969 Type: NONE Confidence: ND	COMP-MESON E04 / M08
356. X(1750) WIDTH X(1750)0 PDG2026:observable:M255:M255W:334942	120+-10 MeV	u_+ : 9.29012256224712 55 u_- : 9.82220080272760 93 Type: NONE Confidence: ND	COMP-MESON E04 / M08
357. D_{s0}(2590)+ WIDTH D _{s0} (2590)+ D _{s0} (2590)- PDG2026:observable:M256:M256W:338521	89+-20 MeV	u_+ : 20 u_- : 20 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
358. B_sJ(6063)0 WIDTH B _s J(6063)0 Bbar _s J(6063)0 PDG2026:observable:M257:M257W:343154	26+-6 MeV	u_+ : 5.65685424949238 06 u_- : 5.65685424949238 06 Type: NONE Confidence: ND	COMP-MESON E04 / M08
359. B_sJ(6114)0 WIDTH B _s J(6114)0 Bbar _s J(6114)0 PDG2026:observable:M258:M258W:343158	66+-28 MeV	u_+ : 27.6586333718786 61 u_- : 27.6586333718786 61 Type: NONE Confidence: ND	COMP-MESON E04 / M08
360. T_c cbar sbar 1(4000) WIDTH T _c cbar sbar 1(4000)0 PDG2026:observable:M259:M259W:347965	5 -- 150 MeV Display: 5--150	u_+ : NA u_- : NA Type: R Confidence: ND	COMP-MESON E04 / M08
361. T_c cbar sbar 1(4220)+ WIDTH T _c cbar sbar 1(4220)+ PDG2026:observable:M260:M260W:348019	230+110-90 MeV	u_+ : 110.059075046086 09 u_- : 89.6270048590267 27 Type: NONE Confidence: ND	COMP-MESON E04 / M08
362. chi_{c1}(4685) WIDTH chi _{c1} (4685) PDG2026:observable:M261:M261W:346811	180+-50 MeV	u_+ : 49.8689922145262 23 u_- : 51.6825310377275 5 Type: NONE Confidence: ND	COMP-MESON E04 / M08
363. X(4630) WIDTH X(4630) PDG2026:observable:M262:M262W:346762	350+140-100 MeV	u_+ : 137.971656296596 31 u_- : 102.528169867227 7 Type: NONE Confidence: ND	COMP-MESON E04 / M08
364. a₀(1710) WIDTH a ₀ (1710)0 PDG2026:observable:M263:M263W:334918	107+-15 MeV	u_+ : 14.8250071785796 8 u_- : 14.8250071785796 8 Type: NONE Confidence: ND	COMP-MESON E04 / M08
365. f₀(1810) Breit-Wigner WIDTH f ₀ (1810)0 PDG2026:observable:M264:M264W:334993	137+15-17 MeV	u_+ : 15.0273935810542 31 u_- : 16.785983289291 Type: NONE Confidence: ND	COMP-MESON E04 / M08
366. T_c c(3875)+ WIDTH T _c c(3875)+ PDG2026:observable:M265:M265W:347935	0.048+0.0020-0.0141 MeV	u_+ : 0.002 u_- : 0.01414213562373 0951 Type: NONE Confidence: ND	COMP-MESON E04 / M08

Identity and species	Value and unit	Uncertainty	Sector and status
367. $f_0(2470)$ WIDTH $f_0(2470)0$ PDG2026:observable:M266:M266W:335274	75+14-12 MeV	u_+ : 14.2126704035519 u_- : 12.0415945787922 99 Type: NONE Confidence: ND	COMP-MESON E04 / M08
368. $\eta_1(1855)0$ WIDTH $\eta_1(1855)0$ PDG2026:observable:M267:M267W:335029	188+-19 MeV	u_+ : 18.2482875908946 59 u_- : 19.6977156035922 11 Type: NONE Confidence: ND	COMP-MESON E04 / M08
369. $T_{cc} \bar{c} \bar{c} (6900)0$ WIDTH $T_{cc} \bar{c} \bar{c} (6900)+$ PDG2026:observable:M268:M268W:348045	161+-26 MeV	u_+ : 27.0003842452260 8 u_- : 25.4601545852605 79 Type: NONE Confidence: ND	COMP-MESON E04 / M08
370. $T_{cs} \bar{s} 0^*(2900)0$ WIDTH $T_{cs} \bar{s} 0^*(2900)+$ PDG2026:observable:M269:M269W:347928	119+-29 MeV	u_+ : 29.0688837074972 71 u_- : 29.0688837074972 71 Type: NONE Confidence: ND	COMP-MESON E04 / M08
371. $T_{cs} \bar{s} 0^*(2900)++$ WIDTH $T_{cs} \bar{s} 0^*(2900)+$ PDG2026:observable:M269:M269W:347929	140+-40 MeV	u_+ : 36.2353418639868 68 u_- : 36.2353418639868 68 Type: NONE Confidence: ND	COMP-MESON E04 / M08
372. $\omega(2220)0$ WIDTH $\omega(2220)0$ PDG2026:observable:M270:M270W:335222	105+-34 MeV	u_+ : 33.8072676977683 03 u_- : 33.8072676977683 03 Type: NONE Confidence: ND	COMP-MESON E04 / M08
373. $T_{cc} \bar{c} \bar{c} (6600)$ WIDTH $T_{cc} \bar{c} \bar{c} (6600)+$ PDG2026:observable:M272:M272W:348041	440+-70 MeV	u_+ : 74.6769884366600 39 u_- : 68.9661894166342 36 Type: NONE Confidence: ND	COMP-MESON E04 / M08
374. $T_{cc} \bar{c} \bar{c} (7100)$ WIDTH $T_{cc} \bar{c} \bar{c} (7100)+$ PDG2026:observable:M273:M273W:348049	100+50-40 MeV	u_+ : 49.4064773081424 99 u_- : 38.9486841883008 88 Type: NONE Confidence: ND	COMP-MESON E04 / M08
375. t-quark DECAY WIDTH $t \bar{t}$ PDG2026:observable:Q007:Q007W:333618	1.42+0.19-0.15 GeV	u_+ : 0.19051342973542 351 u_- : 0.15256041171179 069 Type: NONE Confidence: ND	QUARK_EXTENDED E04 / M09

Identity and species	Value and unit	Uncertainty	Sector and status
376. eta WIDTH eta PDG2026:observable:S014:S014W:333730	1.31+-0.05 keV	u_+ : 0.04519890807919 2503 u_- : 0.04519890807919 2503 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
377. Sigma_b+ WIDTH Sigma_b()- Sigमार_b()+ Sigma_b()0 Sigमार_b() b()0 Sigma_b()+ Sigमार_b()- PDG2026:observable:S026:S026W+:351720	5.0+-0.5 MeV	u_+ : 0.47919801016756 203 u_- : 0.47660443013974 058 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
378. Sigma_b- WIDTH Sigma_b()- Sigमार_b()+ Sigma_b()0 Sigमार_b() b()0 Sigma_b()+ Sigमार_b()- PDG2026:observable:S026:S026W-:351721	5.3+-0.5 MeV	u_+ : 0.55180022333948 409 u_- : 0.54475371590490 007 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
379. W WIDTH W+ W- PDG2026:observable:S043:S043W:332468	2.14+-0.05 GeV	u_+ : 0.05365622500314 9413 u_- : 0.05365622500314 9413 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
380. Z WIDTH Z0 PDG2026:observable:S044:S044W:332514	2.4955+-0.0023 GeV	u_+ : 0.0023 u_- : 0.0023 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
381. Sigma_b*+ WIDTH Sigma_b^*()- Sigमार_b^*()+ Sigma_b^*()0 Sigमार_b^*()0 Sigma_b^*()+ Sigमार_b^*()- PDG2026:observable:S062:S062W+:351729	9.4+-0.5 MeV	u_+ : 0.52801285527312 969 u_- : 0.52651649982212 589 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
382. Sigma_b*- WIDTH Sigma_b^*()- Sigमार_b^*()+ Sigma_b^*()0 Sigमार_b^*()0 Sigma_b^*()+ Sigमार_b^*()- PDG2026:observable:S062:S062W-:351730	10.4+-0.8 MeV	u_+ : 0.84589824638254 64 u_- : 0.84310730525944 599 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05
383. D_s*+- WIDTH D_s^*()+ D_s^*()- PDG2026:observable:S074:S074W:338419	<1.9 MeV	u_+ : NA u_- : NA Type: U Confidence: 90	ELEMEN TARY_OR_ EXTENDED E04 / M05
384. H DECAY WIDTH H PDG2026:observable:S126:S126W:332734	3.0+1.5-0.7 MeV	u_+ : 1.45297369001768 31 u_- : 0.71501191940991 071 Type: NONE Confidence: ND	ELEMEN TARY_OR_ EXTENDED E04 / M05

5 Provenance and record correspondence

This key preserves literal codes, hashes and the ordinals of both records for every route. Links return to the corresponding record. The following three keys are common to all 324 routes; their meaning is explained in the reading guide.

route_status

TARGET_FREE_INTERNAL_ROUTE

join_status

EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

tower_forward_status

ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE

Classification equivalents: charged_lepton = charged lepton neutrino = neutrino quark_down = down-type quark quark_up = up-type quark lepton = lepton quark = quark G0_center = G0, centre True = yes False = no YES = yes NO = no

r1c1_h-1v-1 Unified / memory ordinal: 1 / 4

Genealogy	effeba5464fe6134ce328276caffda457de9d85e22f3932aa8ce593a521c1040
Reader	32867437928eda79ef832344d5cde5886a4c715e968ad77746df4b890958246e
Ledger at 108	—

r1c1_h-1v+1 Unified / memory ordinal: 2 / 3

Genealogy	7cc1cc9f41e40afb8776ff845297b2b0b8b440b8724f822cc6fca90c7639f2df
Reader	85737ea39e08488f4972aa61178a8ccbedcd6662f8b923a8009c6ffeba7c5e7b
Ledger at 108	5e13be751c11cc8bf7ef794982848590e3ae4f7273685430f5f6144bfc16ebd7

r1c1_h+1v-1 Unified / memory ordinal: 3 / 2

Genealogy	aabfcc6099652e23f82f1a50e4ad66a28960ccdb0db840370dfdf043e39dd2d5
Reader	ecd7e3616b9a9821dc08227b0714d924426ca0d13867b735ffb5c614b0ed3390
Ledger at 108	—

r1c1_h+1v+1 Unified / memory ordinal: 4 / 1

Genealogy	cb1834fa28511686ade62a6327bd1ef9daabe103193e9e71583b231c288f5657
Reader	ce86abc79b701a0ec9ae2123d730f8c5eb0e88daf5bfa926378543688333fd17
Ledger at 108	—

r1c2_h-1v-1 Unified / memory ordinal: 5 / 8

Genealogy	7334be27187ca745a3c7e70d9f4fea4fb6d8c31f990e2382f7f91263fb447d0c
Reader	7677699b2a6f6153d139cc02ebcac4a1ff7f1894b6da0a7b1e46b74f55110f6a
Ledger at 108	e589bca4743ae3a8412d609f38a1cbca9ab5e81521c621f651183be2730acc06

r1c2_h-1v+1 Unified / memory ordinal: 6 / 7

Genealogy	56149b04962ed2f1a098af0dcf0fc19ff01c1fa5825b32f899d69dea426f1ddb
Reader	474a5b8cbea67b1d3d0ba8e289cb7ef30ed5d908887d6ba46fa80433e81f6993
Ledger at 108	—

r1c2_h+1v-1 Unified / memory ordinal: 7 / 6

Genealogy	ccb61e50f720be11f86bf136d02b36587f92d52c35df35bf4b86ef5d779f7bcf
Reader	e7c7e699830c4f17e88719d151c262a775f5cf6afccadf3b9ce34e426bc57b55
Ledger at 108	—

r1c2_h+1v+1 Unified / memory ordinal: 8 / 5

Genealogy	72b9df1397f2987ac2c70e5f7163572e0b8854825eee3b61db397f28ba9b9ba6
Reader	a1c4553cb5d2563a0a9a4d4d4659a3d9c9316bb9eb97544670b4f84225f0a4f5
Ledger at 108	—

r1c3_h-1v-1 Unified / memory ordinal: 9 / 12

Genealogy fae367f8fb866c76b0300f5bd2cb0dbb059ab3e163b6df3c5e46e685e4d9fa74
Reader 716a4c36600c27b509736fc515d1943728ac458d23500627b24d70bd413dd168
Ledger at 108 —

r1c3_h-1v+1 Unified / memory ordinal: 10 / 11

Genealogy 1120c2a5d9a6516e9138153d5365283c7af167d9e5b3e46cf7ab8d55a1974cfc
Reader cc519544fb45c5f0387e629c3a067329125f213d0292e9935aa46d4bcf1027b8
Ledger at 108 b1cc79b354042ce1877e1c4cf5347fc4f509c23577711204cf7cddec6beea6dc

r1c3_h+1v-1 Unified / memory ordinal: 11 / 10

Genealogy d8f175e57deb40bea01359579d16ae6e1fdd0b458ebd8b41608e5eba1782d060
Reader 29304938102d297060297c362f9d81380336ecff5a81f6affe85c413e0670568
Ledger at 108 —

r1c3_h+1v+1 Unified / memory ordinal: 12 / 9

Genealogy 13e342fa83db93f3f7dc13163653112faecbb1dd8aa9f872d9121b96007e8839
Reader 5aa200a36959917378a478532c886d1ad9e4958c7cd2fe476a7797fc5306366a
Ledger at 108 —

r1c4_h-1v-1 Unified / memory ordinal: 13 / 16

Genealogy db632f763af301d246516155fafabc14d1d401a29c7c7b631471a8d80c95591b
Reader 5f1e54557f63ee41bdd234e20df6fb367b00a65097cb906eb634fce766250a20
Ledger at 108 —

r1c4_h-1v+1 Unified / memory ordinal: 14 / 15

Genealogy aa7327659272d3089edbc1506b9b161ae47d6711976c8a2a4a11676939d76776
Reader cb535d05a86f529c6b30d46bf87f9924e83bfc069b9f421b49a8bed1d576b346
Ledger at 108 —

r1c4_h+1v-1 Unified / memory ordinal: 15 / 14

Genealogy de3386f8b28db7e8c79bad44dd9f558fb5ce61496e054c2ab24a7188cc309d4
Reader e6136c800ce4e9ca12e80a5e90dbb68278b0c098aafaeab826ff704d06ce3b1d1
Ledger at 108 —

r1c4_h+1v+1 Unified / memory ordinal: 16 / 13

Genealogy 1c8b5cdaa63332d1e8f0ae42fd515992f5d9e41e72975eaae772af08588c657a
Reader 338da10d5c8906ec03bdfade56c89b7fd502a0b0682074063adb644b01f77606
Ledger at 108 ce501e32503d7df09e2c5ae61ff9543fcb85dba7afa7a5396fb0f36909af84dc

r1c5_h-1v-1 Unified / memory ordinal: 17 / 20

Genealogy e753e7cf912988efa9df9eda0afb8e532a8b8f33f921722587fb354551c5342c
Reader 9d043f03a6f5102411e0e8ecbc080c073548ea8ce98deaf70e4bb5ae9fbed35e
Ledger at 108 —

r1c5_h-1v+1 Unified / memory ordinal: 18 / 19

Genealogy 959d954a11c429bdcfe3a0db7b21c1810828fff0868a1d0ac26ba7ef3e127805
Reader b90ff070e1aae58cfca71a9ae8253522544bb269f70b257e4e6afd6c0c3c24f8
Ledger at 108 —

r1c5_h+1v-1 Unified / memory ordinal: 19 / 18

Genealogy 49c3be2eb41e01bb1a1f27f1b1ec147e4db99f167f982dc729a2d4993527c5f8
Reader 1b071d274a26233a2a95e7db4a91c4d3e75f2fa81ab184e1dbc739d938ccd981
Ledger at 108 8c0e7e1dd0b049c9330530f4b68eb1be6a328e08bd9c0f293df7e20ea4956ab2

r1c5_h+1v+1 Unified / memory ordinal: 20 / 17

Genealogy 1b217fd91d203aeae19f2a5a5a90fcadb55a5bb548ca02e51e4e20ec78b6444c
Reader 78e79216b3802312a31bc26703b19549f845de0e5c1403ce0bfc7e4278fd13d9
Ledger at 108 —

r1c6_h-1v-1 Unified / memory ordinal: 21 / 24

Genealogy d52d9bc58081da1a7b2f9e5d23e9d5a6d67401f824eb66ed733498c2fd0dab4c
Reader 560f193427fddf83fb498c6ec7a6b2e4a683ed18b5f9d3a632bb97435ce13545
Ledger at 108 b19c718762e5299b6bd2d56cb241cd2cdef7df5b1d224f6951364c8e46e57ade

r1c6_h-1v+1 Unified / memory ordinal: 22 / 23

Genealogy 0af1af6e61e27f82e9c6658e03d303a9801f5cf6609402e5810dcbc944b8c00b
Reader a28a41d6785403d9b38e510f4c5203a2344cac70366f151a6d143c8093c59464
Ledger at 108 —

r1c6_h+1v-1 Unified / memory ordinal: 23 / 22

Genealogy 3629501596c2b8538e13e1e97d2b0c69f585dbc0f13f6f9f5d7b012d7082ea
Reader f992e4a2a1b53de9a0d7533c2a673b02d28083e7f6aaaa45587ef6104b63db2f
Ledger at 108 —

r1c6_h+1v+1 Unified / memory ordinal: 24 / 21

Genealogy c5b01f4a1d0acc94193d804a45897f489ee02521a35214a9b86086717d71cfdc
Reader 0987471451aca8dbdc0c7324674e2ffa4237909412e431862c124cac78f45504
Ledger at 108 —

r1c7_h-1v-1 Unified / memory ordinal: 25 / 28

Genealogy a911ce23dfdeecccc43cb29fe5898ecd2fabdd3f994ca72814b4e1b698537ef02
Reader 62e6f1b10122e3761b7f39574d07f715c69636c58df0f10af2f06d2c4289e08b
Ledger at 108 —

r1c7_h-1v+1 Unified / memory ordinal: 26 / 27

Genealogy f580dc8247645d3812ad5b4931a5bf89ed066bd7a6cd3321f709a411f3996c80
Reader 07b8d450b7f6e3998bfcc5fdf7dc71c50982ba491d4c5dd5bfe6236c6a11645f
Ledger at 108 111ca40e5cb5dd4e1469c8db516ceb95b20b75ec5dfc93b840a6cce91c92b62c

r1c7_h+1v-1 Unified / memory ordinal: 27 / 26

Genealogy 342d7eac14c30ee19c5129f270b31c29fa591198d94f5d42f5c23629fb102c38
Reader e5e14191acc501f24a061fb7067050ef0a2efd5d86cf532156226da818a11e9
Ledger at 108 —

r1c7_h+1v+1 Unified / memory ordinal: 28 / 25

Genealogy e65938ec108f08f67b38c1bfdd3f600db58f9a17d9c59a82b75ccff06d19ce60
Reader 9905fffabaeddac886538ee34f7320c7891f879fe8876c470d8a790de73b4d13
Ledger at 108 —

r1c8_h-1v-1 Unified / memory ordinal: 29 / 32

Genealogy 72789b3efdc97b1d767a5aa73046d084583fa5aba373f9a07a6fc16b1c1f04df
Reader 53e27ee134cda7bb20474d1c7af7cc8921838c094a2dd9bc724b2d30cff5a10a
Ledger at 108 —

r1c8_h-1v+1 Unified / memory ordinal: 30 / 31

Genealogy f612d6ef200a1ff54e74c4ed7c7a510893ca34048d21381466db682cca3cc661
Reader 48191dd8270ba6e68ab41300a7daa22a6daa2102e4aff594b72c5fbc01440d85
Ledger at 108 —

r1c8_h+1v-1 Unified / memory ordinal: 31 / 30

Genealogy 48fc9a27e8d42d250d4ab52cb98cf105f32507281dc8824f92a8ce07cfd13aeb
Reader ab4343ca0f3279dd31d470d4b27e63604203982226a40d2196f33fdb10b85cc1
Ledger at 108 012b2f08ceb6741389c89bacc896d7b2769a75fa04b4753135b68e4509b3ad20

r1c8_h+1v+1 Unified / memory ordinal: 32 / 29

Genealogy 3aa3fba36ac1b06364a3368dfd7a9486ddc4de29c64dc93a5936873261d07e18
Reader f0c45d1809914b63308aa054a401a3933c1b9671f0dce46e834b5c6a8afb2263
Ledger at 108 —

r1c9_h-1v-1 Unified / memory ordinal: 33 / 36

Genealogy ed985a2ed540799150c7e349e2cc896d9aac716d704d265388865e6038a3165e
Reader 66eb3e8ed039dc74f989a1af5a5acbe386ed98412b13ef3dcedb1c6487e9571
Ledger at 108 —

r1c9_h-1v+1 Unified / memory ordinal: 34 / 35

Genealogy a02045b2e206e07f7236c381210f0d3e9dc3fcfa430a0b349f8488eec7605f89
Reader fdda7f9e6057445988cd077dc481c836df8a633e66f5cfad3ab15b2996ff488f
Ledger at 108 —

r1c9_h+1v-1 Unified / memory ordinal: 35 / 34

Genealogy a1fe11c1f44fee339c8dfe318ad4ce23565bd11d206e5e5b0a3f8136f94cbcff
Reader 409959c2ff367348163db758c5659d70ab997d5cb215f7ff22fc64d0db140a39
Ledger at 108 —

r1c9_h+1v+1 Unified / memory ordinal: 36 / 33

Genealogy 5decd17fa25c6e87374e54b0f6b6fa4c85317267086884d1e9b3cc6c2626241b
Reader e8213bede9eb8c189e246a8aac6855898eb38d7c085789b8527ff022f9311b9f
Ledger at 108 7d1f109e1e33bf84ddf9384cf13c342f63017fe9faecd3c73d8063eca5fa5462

r2c1_h-1v-1 Unified / memory ordinal: 37 / 40

Genealogy 42121291b4db1b251482aaa600938754c4e97069468ccdcdb0d654a28f9b0873
Reader 7e147ace8f35cc0e74360e296f6f57ce18196c7f81af3828e0e513a78d821461
Ledger at 108 —

r2c1_h-1v+1 Unified / memory ordinal: 38 / 39

Genealogy e3a4eca88fb523b4cd27f0dcbc45a95e626a0d4fae70ca9085350d36de129e86
Reader f06f943577be9d476fa92b2cb82203fd77be6d41f9d8efd58303786f07a013ef
Ledger at 108 —

r2c1_h+1v-1 Unified / memory ordinal: 39 / 38

Genealogy 6cd735054b3d285158e101710ab447ea6f8236d0455b76b5aa102be68ac85002
Reader efe6e221b6809bf84456087db0f756cc315eba6205271dfe72062438b4952f78
Ledger at 108 —

r2c1_h+1v+1 Unified / memory ordinal: 40 / 37

Genealogy 7654e0aa6d4ec6e5959f770679686d0fe3c0dafd21ff7cb94ed4ea8505f460a2
Reader 39a510c4fa875ca5fc2c73e482279f495f57505ac7d2e9986bf08753fb317a28
Ledger at 108 355e92375547ac1e37c63290e00e0b13115df4812b567558c2432472731b13de

r2c2_h-1v-1 Unified / memory ordinal: 41 / 44

Genealogy 9eb4e897dd2417fdd7ef230e161ee02a73430f2cd4c165e0b5e03f4cd9432a7c
Reader 7390d3cce6527e3250eccb9cd6d58a5d3548fa0525ab0024b7f93b24d292216
Ledger at 108 —

r2c2_h-1v+1 Unified / memory ordinal: 42 / 43

Genealogy 1fe3a911aaa125a9905aeadc38d1dbb63d1759764b7429b039228794908d53b8
Reader 779e5b698afcdc0910f6d9451f0ee72b8327bc1e94ac4096d006fec2406ecf3c
Ledger at 108 e34326b49595fe5ff46f40ddd0b7b0a511a0f57953ba2953b50c9eced0884e64

r2c2_h+1v-1 Unified / memory ordinal: 43 / 42

Genealogy d8cfd857e63aa9d9c90169d1594aaa1ad2e6b70c6d138a85ea226813977a322a
Reader 46e898cdaf66938151ec8b48a75efafe4eba7615becf22c89a22d7a09b274062
Ledger at 108 —

r2c2_h+1v+1 Unified / memory ordinal: 44 / 41

Genealogy bf3e75cc7e977bbf8e883ce1215b4ab850affa3242a704089c1350ee55fde93f
Reader bec293e8a9352a0c9e5b56fb6c7d30c876b80cec47727a2f34d40d9a3e0cd3ca
Ledger at 108 —

r2c3_h-1v-1 Unified / memory ordinal: 45 / 48

Genealogy 21648712dd5dcff54d6151e85c6297f497049ea18f546275a253f506f95483d2
Reader 6140ac8712fe06046683f5fd5d240eb62685661ad9a8244e3cbe742c5051fd66
Ledger at 108 6c3c21ffd2893c3662b538c017a94d623d66cc85b623c55cf8d08a67b4f70c93

r2c3_h-1v+1 Unified / memory ordinal: 46 / 47

Genealogy 2729489d2f4f122b773697f8b3eef3fa11b69d50f3606c4a5418b66587cb1a4b
Reader f610796820399955f3b317dc077df3c856ff4c75f80da841ce0a72214efd06d0
Ledger at 108 —

r2c3_h+1v-1 Unified / memory ordinal: 47 / 46

Genealogy c5f5a7cc8a3f497d2781fcbabb97bede9beb4139194e9cd554320a8d4612024
Reader e8c645b238014406bd74ee9d49df13703368a36d854493d99ef355058b2a9977
Ledger at 108 —

r2c3_h+1v+1 Unified / memory ordinal: 48 / 45

Genealogy 698f0a4ed1e2762225b33b8cf29ee3696219e67444846032e3f9212d53466a3
Reader 816cbe815adfede468d9709ec0485910767d82097a66de13f0984f609f5bd113
Ledger at 108 —

r2c4_h-1v-1 Unified / memory ordinal: 49 / 52

Genealogy 2597b1458c7c6d27655670533a40d2325d645709b9f9e060260b7cc2573da835
Reader 1a459f0dff0ec31f6beadf5400457dcc3102209e4b1f863b0749dcffb5f87ade
Ledger at 108 —

r2c4_h-1v+1 Unified / memory ordinal: 50 / 51

Genealogy 679c8850da30d692454f00039b4e73d58c5eff335d81553221a5a0b3159a5624
Reader ace3d8034260254a3bb93657b2e2123e63e67d16904e7f0b632fdcbb671332da7
Ledger at 108 —

r2c4_h+1v-1 Unified / memory ordinal: 51 / 50

Genealogy 6055f66a8741e1140af22cd2d82c9b62c55c4e32ae9162cad35bad23d3cbdcc2
Reader 9078ab563bb6e210453e86e13e18dd71d892a6b914ff5fc65482a49004bcd077
Ledger at 108 00c21eea4789a8c06b37c22441f75093ea0f0184ff1bc2955403772490155a81

r2c4_h+1v+1 Unified / memory ordinal: 52 / 49

Genealogy 9b29c8580391aae7b3dbcfefd966550abd79054c5e2e9520b547d06b227fe840
Reader 9a3f104e10a55934eddfadbb190a4e8d0d79fb55eba7bc4403c29aac08923eec
Ledger at 108 —

r2c5_h-1v-1 Unified / memory ordinal: 53 / 56

Genealogy d1fd49040e524fea8052b4b0aaf0181192155662059c836370670d5c2814748b
Reader 83380600e5404d7e8880238796855cf64cf95c1fb93d17d2320ce187bf98369d
Ledger at 108 —

r2c5_h-1v+1 Unified / memory ordinal: 54 / 55

Genealogy 2708ff6618e713f0d4900bb9c13096961b5c2b854459b699c5a469f133917fe1
Reader 02b48980193d7b3666a37a97c6cc611e2af419c8f78bacae9bb362a79e43767e
Ledger at 108 —

r2c5_h+1v-1 Unified / memory ordinal: 55 / 54

Genealogy 2133fe30ab9b2c2fda2fe370cf35fabec9776902ae7f1bf5dd7a68db23c22035
Reader 7e4da1b56102a95309b04e40b636fc437722155b1ed9b43bc63ed9a46c0f9c84
Ledger at 108 —

r2c5_h+1v+1 Unified / memory ordinal: 56 / 53

Genealogy 7f85598f2d0761cdb1daa591c78b22b937d0b0cf9020ac23e6f9e943fd562608
Reader fb7d8d4d5037cd9ce900cca6f805cb4785c5e6dc4618e4b3fa59c593bf78eebf
Ledger at 108 1523ad7232fb11fc5646852a3d2a3f1122cadd040d9709643b2c332d82848b2a

r2c6_h-1v-1 Unified / memory ordinal: 57 / 60

Genealogy 455c2b00848873039e0506315671ab73333699750588879e8ae6a9762ba3af54
Reader 36eab0f217dc9a30cf9ca2da7ac99f6c2e67517347899cf3bee9182c711ace32
Ledger at 108 —

r2c6_h-1v+1 Unified / memory ordinal: 58 / 59

Genealogy 6f127fd9f25200c9da395c6482cb10ba06991f9a2b311706e05753d88f848809
Reader 97916a8721973f594ba41fb3ce987a18682cae23e96d3ad81ad92f70aa3ffda7
Ledger at 108 bfc92621bc7d75ba4ede7c52fd53f1a32c441577160adc4a69490655db8b7ea3

r2c6_h+1v-1 Unified / memory ordinal: 59 / 58

Genealogy a820aa46633c948e12bb41b7b41a621ad730d970e797f93ad58c70877be24f53
Reader f5bcac8cac9dad8ee01f13e7f30b39e2243232b4d9a8da8a03297e1638b16cb4
Ledger at 108 —

r2c6_h+1v+1 Unified / memory ordinal: 60 / 57

Genealogy ea9202399ce61bc1abe3e3bbf0faed082c6e4d11f9e0ce94fb1a343e0f60b2c5
Reader 9fdef0c8e1e5027d0dcdddc3848d78073d21e8d15c0ce90a2b53a7de4ed3563b
Ledger at 108 —

r2c7_h-1v-1 Unified / memory ordinal: 61 / 64

Genealogy 93f4d12e988e9b7d84464bea475814d6748da20159cd95ca1872639502d4e22e
Reader 4f6bcfea35a49b57dd08fe0ab2f398016e0a872931ce0832af3f2f6f123be521
Ledger at 108 —

r2c7_h-1v+1 Unified / memory ordinal: 62 / 63

Genealogy fd3fc6ae2ff2d25bc3c572dc5362bf82ba08155cab1135cecc3e4ea8c6cabd14
Reader 97b2ab0747f9cae836c873b69558f370f37393dc00bc1b475ff3498099cf3f6a
Ledger at 108 4ce890da3360de04fce789c2ce40701aacf357aa4d27239b5d03dfa52c8c9f85

r2c7_h+1v-1 Unified / memory ordinal: 63 / 62

Genealogy 6b038b1a3fa41286fe974ac63c5d90cb981ae568eab928b77b8abbe73506531d
Reader 5abbe501c8adf01bdeab4ceef778d25879f7df343cf8deab50624ce7852d47c
Ledger at 108 —

r2c7_h+1v+1 Unified / memory ordinal: 64 / 61

Genealogy 7690711bd23b740afdbe1c4fa7b76bb78b49dfea24072771baed5540881ba560
Reader 2e84b322980e807e05210dea2a1a280f6589f24b5d8d2b88a0a309b40ad8d5a1
Ledger at 108 —

r2c8_h-1v-1 Unified / memory ordinal: 65 / 68

Genealogy 59979b339515cffd98571eb86bd694e33ea0f1973905ec0bf1c9caa74ef63de7
Reader 524796ca2925cfb1d109f28a8a8378d778d802cf51f272b5cbd5bd45fcaa368a
Ledger at 108 d88dd1b6168ecd8f1b4de529649f5ba73c37ee39b4c64650605497dd1d21e3f3

r2c8_h-1v+1 Unified / memory ordinal: 66 / 67

Genealogy 354c6d9af4e6a921677a2e1ca86c2615967aadbbba749be2fb9cbc9471af9ef30
Reader 835f1e89a2323718255fdf26105c31d0c239439f72826178a5dfe560851db501
Ledger at 108 —

r2c8_h+1v-1 Unified / memory ordinal: 67 / 66

Genealogy 5908d499c7cb58ca1019ee6fa049bb035b537bb28190db7680139634d9fbe00b
Reader be902208821a34b042d5e11f1d0fcc5e10dbf5a2f9625040dbc02e416483f054
Ledger at 108 —

r2c8_h+1v+1 Unified / memory ordinal: 68 / 65

Genealogy 84615c8802e6b1fd96b339d0d1d76101fbbba5fc64863032e5be621338e563f29
Reader feeef18fcf009a4833205b5075b749373cf28f0b113b0376d55a8f6231fe5d8b9
Ledger at 108 —

r2c9_h-1v-1 Unified / memory ordinal: 69 / 72

Genealogy 10af243bdfdbc6b04a7e1da71d5e5ca47143ea9930de740045291154fd1ca26
Reader 46188c4764cb2c18ad5e28f7d7701b20516f47767883a606fd272ab0ce1b6e7f
Ledger at 108 —

r2c9_h-1v+1 Unified / memory ordinal: 70 / 71

Genealogy 78a741ee3c7bfbe7ae8641015a71bd34f1888a67bce6ff009113e38ec348d112
Reader fc684626be7c75ff48c0e5350e38d10c764d4db51dbaede36942994403aa60f0
Ledger at 108 4f85ae5a226daca7170f68760b433575e50c60755178c473034d1496819aa0a7

r2c9_h+1v-1 Unified / memory ordinal: 71 / 70

Genealogy c0b14bfdd9f53d4228d9411817166fe0e3268f7dad8c9988d4776186e04fe7f
Reader 78ccb54b24e9defbe6c1315c36f064a2e031e90af5bc9cb69113c1ba082b2a72
Ledger at 108 —

r2c9_h+1v+1 Unified / memory ordinal: 72 / 69

Genealogy bb93b38f6a11bd24fe0d784b0382327e0e9577378da33db4ff0c239336b2076c
Reader 0323a24467e3bb3b19fb86edbb720c35efac9bd2c76f856a02bbfaa6be1de3fb
Ledger at 108 —

r3c1_h-1v-1 Unified / memory ordinal: 73 / 76

Genealogy b49a35c80200681a5cd1c24a3af33bdc3f5dfc15693d461d3c7c6d1a094106a
Reader 9cd3c1cdd9f24b93ee41db3b5db0405bd95787b27d0b9563cb941f3bcf9ec663
Ledger at 108 —

r3c1_h-1v+1 Unified / memory ordinal: 74 / 75

Genealogy aa18d9482f30ff1d1e8a4a2a55cb565061e18e03d89d65c183f098f5ecfa8627
Reader 9f05196446d319f0bee5611cef354231f598d0c69a54da6b36803c840ef9d155
Ledger at 108 4f280c51aedd8c9f5e7c17101e369b774441e690405561db94125e58da5ec0c1

r3c1_h+1v-1 Unified / memory ordinal: 75 / 74

Genealogy f93f9cc3c26fe793e3a4e63395e34edd0c959cb9bc132a5815d675d46041d5b6
Reader 8343e0ea9628938a9aa892c593f5925d0eace614709da84194ad5609bd12b447
Ledger at 108 —

r3c1_h+1v+1 Unified / memory ordinal: 76 / 73

Genealogy b5ee1bee5453ebdf2ca8a16f35b7700fe788e77916965f33ae801dd014d7a637
Reader 6aee9e8b24e2230ffb355d666b1169ced471e4a3b73faf2ce8a3168a2f909cb2
Ledger at 108 —

r3c2_h-1v-1 Unified / memory ordinal: 77 / 80

Genealogy 5fbc72ff0a96e82d4616ba1a6ef070f7a682551f464c637ad43a97a11f6e8a97
Reader 92a383d82ac125a374465a812287a783220c4993af802ee322a89c2c20798cbd
Ledger at 108 —

r3c2_h-1v+1 Unified / memory ordinal: 78 / 79

Genealogy 69e2bd91cf69da875a43f97523596d231b3e64e5813e5f26c0c2804e11812077
Reader 3f2554e27e8f08e18bce3badea9cd8ff21745b26de177ed608864d591051a07e
Ledger at 108 —

r3c2_h+1v-1 Unified / memory ordinal: 79 / 78

Genealogy c11bf3a55ac78a0a7ac79598bfd55908c204e4a60ec21403b684c336250918b2
Reader 4f92d4ea7024f5501e8b0a882bbe09736d0169b49bfb21ad0504f8c1d37c43a
Ledger at 108 —

r3c2_h+1v+1 Unified / memory ordinal: 80 / 77

Genealogy 84b3ba3095f1140b146b108fd484619332fad0f9436e0ecc1ecd0a443acd145b
Reader 6cd46053b4aac7eb750510ca7ff5aa4ff8e13be097058396eee32adb341196d1
Ledger at 108 3364070d4d0d99825f096a5b3dd87451aa757f0f85aa6ab974218271ed7418df

r3c3_h-1v-1 Unified / memory ordinal: 81 / 84

Genealogy 90be6b645dcb7c34479da4ddb58dc6b331678651b48f48bfa32fa2f1cacbf2d2
Reader 0a266f2de19b3916bf94b0203007d4c4edc9fba6891708948e1cd7c8d352827d
Ledger at 108 —

r3c3_h-1v+1 Unified / memory ordinal: 82 / 83

Genealogy 970cf8e420061a4c2e86d72e44ab8e14d201e17d5273d819f231ca3c228b99cc
Reader e52e1aa490feb02a80eacd19facbd502f5a2bead0ea1c7ba6885899eb000c242
Ledger at 108 —

r3c3_h+1v-1 Unified / memory ordinal: 83 / 82

Genealogy fb908c9b7d05c8740756c16ce54c6e7890b67ab3adc021ba17bc80d1580a6a7e
Reader 540ae414abfa6f646b2b1e35635c7d7fe2834db6ead0ec1142e9fb0004b694aa
Ledger at 108 8fc1d1a017faec792451c97d492516e442f2ce08857f8c2aa8930ac01f5a0821

r3c3_h+1v+1 Unified / memory ordinal: 84 / 81

Genealogy f04730d7e2e26b7754d2c540697c1ccdc7a274255c88f943077621c5e09b74ec
Reader 268d8ac49e67cc1679252262900b5b9df37d5c423422bb7f2b05970fb9a860da
Ledger at 108 —

r3c4_h-1v-1 Unified / memory ordinal: 85 / 88

Genealogy fe19794ade36889f09cc94a77d8a97228b8530f2fc302f96a3cae6b71d488126
Reader 4fb57798b1f66825757136b691435da4276f1d7475fb32d37ada397c6bc2cc10
Ledger at 108 47361123860c471c5c3e3660c7e98011f4bdf752d76ffd84b3957ffe686a0fef

r3c4_h-1v+1 Unified / memory ordinal: 86 / 87

Genealogy 735a2f2a02b51b9b28d0c9661b965464ea38522629179c9792287ffeedf17733
Reader 0ecdaa6df15a6f0f268fb041bac3c594cca56daae8f41abdc8b4d3edd12e479b
Ledger at 108 —

r3c4_h+1v-1 Unified / memory ordinal: 87 / 86

Genealogy 1931926d8929382c703b3c3efe5ee8d5ec8768aaba2fdc568564d21d05a242f4
Reader 2dda1905a5534096e9ad9a427c6d8048fc9d39c0c5825306d0203b1a12572285
Ledger at 108 —

r3c4_h+1v+1 Unified / memory ordinal: 88 / 85

Genealogy 44ed44030e829f7ef3f2d2539c0ecae1af3389fcb03b6af443c7515688564f46
Reader 1ee0bda62536a4e364a3618e5ba602ded8f28109b25f42ce1150ff315daeb5f9
Ledger at 108 —

r3c5_h-1v-1 Unified / memory ordinal: 89 / 92

Genealogy e953714a3f1290a221bb6aaa7ffb0b8c7ba40f3de4825cb231af24f88c67487c
Reader 516ac20b777e49c49141c04f9435195b0a72bf8fd107bf1131f9797a550996f1
Ledger at 108 —

r3c5_h-1v+1 Unified / memory ordinal: 90 / 91

Genealogy b34cbe3d2a71fe4fb1ebe4194d4a87718d2bd15ed5ffc2b7222167761d430287
Reader 647352489028a5eb55e737d1ba0d6c5267b15924d267d7995e4d3e1f03269bfd
Ledger at 108 2bdf6193049e9d8f1fc6952673cd5a6d3ece96f280eea32513f93207a69269c1

r3c5_h+1v-1 Unified / memory ordinal: 91 / 90

Genealogy 257cfe72d342b8330026aca6da1825c935e6890f3fa068db6795c936086cae39
Reader fceaa5895efdf740e8a408a65e4bd72b828faadd8731c947705e6375e4f6895d
Ledger at 108 —

r3c5_h+1v+1 Unified / memory ordinal: 92 / 89

Genealogy 5144d95329eaf9c2e01d5312581dfa5ab24fac78ab7b2ea124db8d95db7a998c
Reader 0ac1f885d966f22dfb79a602b576a688b351e32d455ddca04b3f46b937f49b0e
Ledger at 108 —

r3c6_h-1v-1 Unified / memory ordinal: 93 / 96

Genealogy 8e5c94763ee2ea8ffb481fc1fefbf17f6c771f7914fcfcf180fc6ba384a48110
Reader e9907b108eaf6db8390e7129e3c3d3f5c497165928e17a4b9f231a7ba1fa7d0
Ledger at 108 —

r3c6_h-1v+1 Unified / memory ordinal: 94 / 95

Genealogy e11e9cb2d964364d36ac21f76809e6d1f0b312143e898c584c45d1ef8bf497ca
Reader 0b1b5faa852ad0e6071d95786097ecab1e741a4f64d3bdd1d778f85c74963cbd
Ledger at 108 406b35c85cc7381b8d89efa82a29daccaa3ddcc24246e6c04702d9023088d3ad

r3c6_h+1v-1 Unified / memory ordinal: 95 / 94

Genealogy b69e41e3adf88e397196c6a6a46da69641203067a7118c65e4b64e8aa46b34d1
Reader 783dfdedcf8fe84bdabe1f1b52581403e35dfe49f318adb41101ada8b61c501c
Ledger at 108 —

r3c6_h+1v+1 Unified / memory ordinal: 96 / 93

Genealogy 6288242c4687f7c0ef85b06a62c64390d926911da179628087bd63b7387a992d
Reader 9055dce23724fb2ee036867b505c820f79933f0a7d4ecd3d289c0b969ca46da2
Ledger at 108 —

r3c7_h-1v-1 Unified / memory ordinal: 97 / 100

Genealogy 6589a27f5db15e2faedf8e0bf0f4b48887925aa96b94b118940ab4b275b4f51e
Reader a00262e0cf95aa0f99b91ef081a1ac39c8c5f9c7ba5b825eb0d5f9a881b4f01e
Ledger at 108 —

r3c7_h-1v+1 Unified / memory ordinal: 98 / 99

Genealogy 74fca7a065152326f2680e83c4eb79eddfceaf255870015f6dc3f0dcd4da23d4
Reader d4d5eba82dff0efd18906a4c20b1c052b04b63460dea63d1c3908efab54ca465
Ledger at 108 —

r3c7_h+1v-1 Unified / memory ordinal: 99 / 98

Genealogy e56e3ff292c8e2520d3cecaa4e8d296324ef1d8f8f682120e853d2eb302e843b
Reader 809858a6f732a2d7201ea65668fee3960f55de4fc745fd45395c2dd5e8199cd5
Ledger at 108 —

r3c7_h+1v+1 Unified / memory ordinal: 100 / 97

Genealogy 77168e1c5f252e653c32d6b3899be2f09ce151e231c9bec5285c3e8baa931507
Reader 78e3b7b481481612e0e698c15bce3bb6731d92e345512472bbfd23257191308
Ledger at 108 6124227ce899ccb2df636930314e07ff8fc23be0d2d973f8128c25ed2e55be47

r3c8_h-1v-1 Unified / memory ordinal: 101 / 104

Genealogy a3c4b2e42baf237a5e4c2f9b552ebcd29e8bfa6d4980de4980f828b377f1771b
Reader f96905bcd7205151dd5dacf1f6525762d535f2dd174b1c3e19f351bde335b8
Ledger at 108 —

r3c8_h-1v+1 Unified / memory ordinal: 102 / 103

Genealogy 30b5ef0c47761242ea250c0bd063668a9403357735c2c59fab1ad3ef3b33a931
Reader 11b41f4221d7ba5285c94c7305abf3242a05aad86ea35723202c943b48489ea1
Ledger at 108 0dd6b250296cb9be5eb2e1d19659528b7de17fdc8796cf0ec3cc88e3aabfa877

r3c8_h+1v-1 Unified / memory ordinal: 103 / 102

Genealogy d954a2a9083ac81224a21b4c21d85c168f7e0b0f95f774163b3907a8c521f3ce
Reader bce6affb972de83392f296ec01c79cf65f1662441464abdb46015ef5f9986e86
Ledger at 108 —

r3c8_h+1v+1 Unified / memory ordinal: 104 / 101

Genealogy 03881732293dbdb0dd5bde4d431381e4c1078b1ad88c6594eb82b5ddd3f00398
Reader c585e17a9f62ed44ecbcc253a8758fe1c652c8a544d158754d2ae6fddf6df97
Ledger at 108 —

r4c3_h-1v-1 Unified / memory ordinal: 117 / 120

Genealogy dfa8889ea2e308db35da1a1e3a470cb901f5b1b448eecfa8b072910f423cf30
Reader afd3630152262dd724d21cfe75259c892dc768015259d63a8c73a387e703340c
Ledger at 108 —

r4c3_h-1v+1 Unified / memory ordinal: 118 / 119

Genealogy b9a0e830c5b79e19fac92114fb87e7d73c89c8ab1412847c40d4194edfafd295
Reader 45d58d9381a000cc9d767cfaadb64e412d11aefedf7a5b16fdf39ebb2427bb4f
Ledger at 108 —

r4c3_h+1v-1 Unified / memory ordinal: 119 / 118

Genealogy f18a5f3bb67d88d2241efee93e0869d0669b4af613f1efc2a3340b822993e9f0
Reader 2fa6335747d64259eacd9cd2cfd7362a3d7b22ed15a290e43fc0aa7db9ebb8ee
Ledger at 108 —

r4c3_h+1v+1 Unified / memory ordinal: 120 / 117

Genealogy d562cf51d4b94895d64814ea9d20c3894b082a0c3fdbcdd529750f0f92dc6249
Reader 46e7971fd3cde9a3330a90fd8d65631f09d3233dbe7e2bec0108983ff0938898
Ledger at 108 0445e18674117bf622efd6fbfa1d6e4c6164b9238247bb7c762c152bb9686738

r4c4_h-1v-1 Unified / memory ordinal: 121 / 124

Genealogy 77c77bedc6c70e8f673bba3f1a38ebbbad0e9216da5a1135bf40b23f327bb557
Reader 71932081a5e57d25037ce0d5b5db38479d4de5b79809ce8e2c0b6c8aa771cdfc
Ledger at 108 —

r4c4_h-1v+1 Unified / memory ordinal: 122 / 123

Genealogy 287327ac0b7a67951ed95a00dceb2cc4b73e9aae63f89cf3f375f2e14608c87d
Reader 015d0ecdc3d3681a13d3a6dfb1164ab55a41a27e6f103f07f68ac1e94b8ffcd2
Ledger at 108 282e55192bd3edf96e14e9bf0b50e3479922204db1e0fa80756865f131f8c5ed

r4c4_h+1v-1 Unified / memory ordinal: 123 / 122

Genealogy b39d7e1604512deef70b3918e486fa58fffe93c5ba2477a09df55b64831ca74b
Reader 8f35c9aaa9f1b19875c81199d648323876ea87ed664f20cdb441f73adcdccb12
Ledger at 108 —

r4c4_h+1v+1 Unified / memory ordinal: 124 / 121

Genealogy a20fae0de320526914318b1432bf0b0eb2e61de22cd563a0e5a17b0ea9128bc0
Reader 57d09cebf5b2a4289c6b1055b0fc67ad75d3af3cdcf8d84d685d0e8828573d
Ledger at 108 —

r4c5_h-1v-1 Unified / memory ordinal: 125 / 128

Genealogy 79fb9a845241446b08dd24c605f6b1b7bd4981c5e6fa6e0067ea36ce0e7baef1
Reader fb4bcab39bf089e5c421bf352caad6674c9c54f04a9d95494401d5b5db852e28
Ledger at 108 —

r4c5_h-1v+1 Unified / memory ordinal: 126 / 127

Genealogy 0c0060901278e986f81b5b522e367331377866c4eec0a6c9a5fd33b1702579f7
Reader 7359aaa714e722cf55306e7389c0a7821640a976ddca2652c3e9de161d84c014
Ledger at 108 —

r4c5_h+1v-1 Unified / memory ordinal: 127 / 126

Genealogy 9866ca016d61a93c8605f9d393f5c7d1ccbf95b39e0be07eda5205d177bbe0a3
Reader 37ed6a7926a4b489b076fae2ae21dc04dd4b9ca8e9edd354076bf251fcd73744
Ledger at 108 58bd109118dc402143446db43b056afbcc8d2a5991cace6f6ec3fb4670b5eff4

r4c5_h+1v+1 Unified / memory ordinal: 128 / 125

Genealogy 4fcd9bfc143fe82bee933b825b840060bc9cc4cb3164418cc30da5c55f863c5
Reader ba031ff902fef28f8cc55107fa551870981ae35f912431d3e4c6b06e9fd726fa
Ledger at 108 —

r4c6_h-1v-1 Unified / memory ordinal: 129 / 132

Genealogy 87efef613d4e87ec97c29419f7de6f6ea09a79bc753ae3e404b221d8ede75b4a
Reader 6e6b7dd414d89eb1aaaf5a8ea14fac7861011753e2349a9cc217300fd1b80edf
Ledger at 108 —

r4c6_h-1v+1 Unified / memory ordinal: 130 / 131

Genealogy 055c55a6583303aac67730b62377ba8a1d0049ac7aed4debc98aad5881bc9bb5
Reader 8fa03549920be213176b7e0c881c6482be1ed34ccd91ebfb0760198297de4e2b
Ledger at 108 —

r4c6_h+1v-1 Unified / memory ordinal: 131 / 130

Genealogy 406e5327b2546eaa3992ae56c6688e014239f25ef3712139e37360e70307a0cf
Reader c31d14fcf53e490e7b58ce903300244767c13da20e17866554a53b7f1e8aa2ca
Ledger at 108 —

r4c6_h+1v+1 Unified / memory ordinal: 132 / 129

Genealogy edc4254630dee8eed6f09c01001d7e5ff9ae88a52b26f6c84b34889e95b4b44f
Reader 8e63a835c6a45d8cdf6ec7615e7954f6d06dba153209bd739e0e94665079019
Ledger at 108 ac355fc3c7d67e23747af693446ce5973090a1288c89b61ed5e84c164421d478

r4c7_h-1v-1 Unified / memory ordinal: 133 / 136

Genealogy 4ca3456697b5beafb2684f346ef4fe8a825e4e9b1923a0354b9e170e0a544faa
Reader e241f838e36aaa931f974504e5963e774300dbe4095f540812bdb1be522d47ca
Ledger at 108 —

r4c7_h-1v+1 Unified / memory ordinal: 134 / 135

Genealogy 01c95731317c519b15d30504737dbf0755697c530400d12b6dbb06b6fddb26df
Reader 0775e69aa78c62aeb37e0d803ae05fdb216bf35b3648f5cbd17c92719739b065
Ledger at 108 4acc79e1bddf0a8446aca263fca69bd39879caa1e6469c33d703da9172c27cd9

r4c7_h+1v-1 Unified / memory ordinal: 135 / 134

Genealogy 782a318cc548e949b83af90597ea3aa0715c06a57efc8cd4c09cc84eaf42ad4c
Reader f4d3eea43826eed3747c080d0cdd48c809877dc7fc409585fd0274bc331de472
Ledger at 108 —

r4c7_h+1v+1 Unified / memory ordinal: 136 / 133

Genealogy 54436c1307ec79e76d82397f2b17691ba5e8eca04e1f5c1135c5a6f07fdd51cf
Reader 073fbdcdcb0df31cbcc689c539e57777a6ea89a0771eb07238bc113fd3c52f129
Ledger at 108 —

r4c8_h-1v-1 Unified / memory ordinal: 137 / 140

Genealogy 2b804e0cfd9c913aa79e7b2529b115c91f296d3cf84b6837e5e05321bfc2a6c0
Reader 98e82256254f483db3a5842d1f7fd6a569044ecf3c542b941705855559737516
Ledger at 108 bf56e7e7a44142053df74d41b3498a6c84055d812f97946ed2c0674099511566

r4c8_h-1v+1 Unified / memory ordinal: 138 / 139

Genealogy 03804c454b95d8b93e9f278969bffd8bf62a88488eefa4f44c9253afb6c9aa4
Reader 87681b4a0c073f38f2a84e8a65f543f2c195b938cd71606b35a72691d777d3ce
Ledger at 108 —

r4c8_h+1v-1 Unified / memory ordinal: 139 / 138

Genealogy b6ebfc8618552fdd72e29c86168f9f6e2a46c3fe8d74fb9e06ff8d9ef4d8355b
Reader e8e9508b8406628379069d3f617b9beaee92634d170e0a1cf2a20c31d62a42e6
Ledger at 108 —

r4c8_h+1v+1 Unified / memory ordinal: 140 / 137

Genealogy b1167825dc88e346b1033aa368e2c68a7ce87d854b37460a00c312e0b818fce5
Reader b7f40bf3d4a01f15bdaa23dc2ebd2cdd551eca43991086c52426a372930a411b
Ledger at 108 —

r4c9_h-1v-1 Unified / memory ordinal: 141 / 144

Genealogy 1fac69dafa70b07993139f7a846f20abb465238f4558212f54421745cb5e3918
Reader f8d6c21dad31131fc4c2fc3d70185a1f888c237d6474a021870b65196e6d416e
Ledger at 108 —

r4c9_h-1v+1 Unified / memory ordinal: 142 / 143

Genealogy 18959d06bbdaac3d5d1132981d981df9bd28a79c6da33756e8d2149cb51db61
Reader 4c038f821f5bd14ceca5c1f4829b72b372c1563b32e0fcc5b34cfb9eff6992ee
Ledger at 108 1e448780f5a741d6641ffc8e2ae465ac249cdecbbd7f941e44db167f2a18cb1b

r4c9_h+1v-1 Unified / memory ordinal: 143 / 142

Genealogy e624c172e0c76c77cc5995ae25a89b2e951081d6310567419586143808bab0aa
Reader b10d5887020993ff069a374c70846c7db2da53820e650c37b95231a6b241a3cb
Ledger at 108 —

r4c9_h+1v+1 Unified / memory ordinal: 144 / 141

Genealogy 555a0b57737e22122d0fd734f5aa5516d6461c185bde5c94fab7062044c81b31
Reader 40c90b3041a8d94ca52a895335732ae51dc7c7f9837d5ada66d7bc12643ede6a
Ledger at 108 —

r5c1_h-1v-1 Unified / memory ordinal: 145 / 148

Genealogy 44128f77694e22cdb74ce801d17ef58cb5f75058dd488325f94b8db3aef4c2f3
Reader 8e414ca1b56fa4e2e54dc83e480cd2bdb5cfc2052546af04082e0f017d24c4e5
Ledger at 108 —

r5c1_h-1v+1 Unified / memory ordinal: 146 / 147

Genealogy 0f38c6958afd01ba27b6bc49a6893e4853cbeff8b05159d512fb8b00bcb73007
Reader d9b335838b663c211a4ec0d10a405d917b276d2ca4cd5695fbf07de3d36643e7
Ledger at 108 —

r5c1_h+1v-1 Unified / memory ordinal: 147 / 146

Genealogy aaffc169d0598de597a3a2882b457981816b63fb5a1853f5546db1a985aea4ba
Reader deaf6877f8a7c8880ba1417fe43eee1c66d2cde0af9e5961bbf415f2f0787f8e
Ledger at 108 ff9f223faafbe5f7d396cb50240d6d52e53eaab1a745f7719f6f1503b398f6a5

r5c1_h+1v+1 Unified / memory ordinal: 148 / 145

Genealogy 6bb6953ee6341f42515e8490c33965ad839bbc946cbc777dd1d832248ba36e66
Reader 9743965c5f547f81a6eb1b5bfc13cd5ee9140eeb402f7500f7a52149283e0a47
Ledger at 108 —

r5c2_h-1v-1 Unified / memory ordinal: 149 / 152

Genealogy 7cfcd1776e4330be5d0164d074d9db7ebbc9ed421fdd9625dbdae32b33ae0ef6
Reader b369d9f2359a2a99a416d16f936b4222912fe42aa504fb2ca9cbc185be1d8c70
Ledger at 108 8754d795eb4696cdb9a16e289f13d7d0caa88f179c1ebb009c21bcc236cb922c

r5c2_h-1v+1 Unified / memory ordinal: 150 / 151

Genealogy ce452c82855b0221b7ba9f0c035bcb7309bb6bcd57b4db0e3abe86148d83512e
Reader b1388ecfbf1e3638e950a7e90fed725e51d428bc70bb666d2eba9ccc3a4e8599
Ledger at 108 —

r5c2_h+1v-1 Unified / memory ordinal: 151 / 150

Genealogy 2e1dadcad246fee281f87bf135901e695efb2a4a05df738f6c07002407a0b1f2
Reader 184e6040f83f204c85f176971d97ea9040bf8f9263d3f91b694e0a6d3fdcf152
Ledger at 108 —

r5c2_h+1v+1 Unified / memory ordinal: 152 / 149

Genealogy b277f51b0e0d2ae224e9987f04a99ed88a84fd081c5d0ef6662e6eb32b5cde7c
Reader 664bc72329b1dfa461d6b8711673127382f8d4cf08e87ae3a09df6274e4b9cb3
Ledger at 108 —

r5c3_h-1v-1 Unified / memory ordinal: 153 / 156

Genealogy 628aa1329a1d669f1d699ac59afb3d24a5b55af8129e6ebb54860dbb0a921a82
Reader 8d0e0a7c9c083d59bd7752c12edada4283a1227f99644dd99fd5236203276d59
Ledger at 108 —

r5c3_h-1v+1 Unified / memory ordinal: 154 / 155

Genealogy e1edda897425f41c7b0d4061df3db1cc7e64b9f28d3db5f30063789b0434a10d
Reader 3dc097d1af9e277259c79d8cf836e831c425072950eb258b5eadfec90704df5e
Ledger at 108 f686fd28355504421f5166a67fe6521747f0e1dc54b72eb39e0a7a80b67ba3f9

r5c3_h+1v-1 Unified / memory ordinal: 155 / 154

Genealogy 5ebf99eb3286cd3e668d1881ac5a6d25c3fed2315f47a970c3ffd75b4e986591
Reader c5b7a860701e0f33931847efb2a992c31bef6f0d9da926e574e7ab3e68fb470f
Ledger at 108 —

r5c3_h+1v+1 Unified / memory ordinal: 156 / 153

Genealogy 0e0d1f793f8fb5b455383130d2c9912e5b2700a0c2fb4ce1bd1b83c54267422c
Reader 997312cb2b9d722895dd8de8b30b9501c2d380ddb4849906bc9cd2595449bc9
Ledger at 108 —

r5c4_h-1v-1 Unified / memory ordinal: 157 / 160

Genealogy 698e05b31e9a26229e96d506b2415dfbf00df56146f44d71fcbea9b78701e56b
Reader 4920659c510a6cf90e8944ecb1a49dff1bfe16ad3249fd0b5fb8a10930a6547
Ledger at 108 —

r5c4_h-1v+1 Unified / memory ordinal: 158 / 159

Genealogy a3dc2fcf597df676434ea90b519e5540b275620b60073073f9ecf51745655b0d
Reader 2bc38a4fc594251923f006d0b3ea2789df5cf2da0f7d89bab5c40e7dfa00c9bd
Ledger at 108 —

r5c4_h+1v-1 Unified / memory ordinal: 159 / 158

Genealogy 8944ea77aea3c31eb47da1692c7b6e0a3c809102f8b9002ba05b291baa3233c1
Reader 2d9f1de9bfa0686dcfd7cd7ac00fc0b912d37ab73f9fb467dccacfe55a9eddb
Ledger at 108 23031ced1c53e3421cc8604cb7d3e60373a144d81c02cf0c7307d379831633c3

r5c4_h+1v+1 Unified / memory ordinal: 160 / 157

Genealogy 79e4be3ad5ebc6753342e4284adf56b71d921b04cea8f219c842fa9f052e874f
Reader dc5c02bcdaf96e12535891f9aa10bc7bd9659e01354a2a2d5c6e9dc6ea2e751
Ledger at 108 —

r5c5_h-1v-1 Unified / memory ordinal: 161 / 164

Genealogy a4e3cc2eab44f4bb51b119700b1b5cae5f22edbd8263484731a154bda1c77f6b
Reader 98b8a25724d3a874f2368cb2dc819bc6ce0d16adf27017ad5f13936124662d5a
Ledger at 108 —

r5c5_h-1v+1 Unified / memory ordinal: 162 / 163

Genealogy dafa95692d1f9384c65e0afd29106c64a3e62d6f2d545d4a6b6b404ea1b2d70b
Reader 2bdd7331f202139ea68552f27c7eb9dcf445fdc5c70d95f9e3ec7c6558dbce1d
Ledger at 108 —

r5c5_h+1v-1 Unified / memory ordinal: 163 / 162

Genealogy c6b14aedd3f88fa17e0478cc431b63deca8ee71bb8e6d79c4c811add87d03720
Reader 6e152edf9b94425a404b2c13793f91e688091f69eff6453c8efda5d69fa8861c
Ledger at 108 —

r5c5_h+1v+1 Unified / memory ordinal: 164 / 161

Genealogy 57d24c5dc0a69cd8111933d8bc40cc260571c0ef4f8f3008f7197910ff07c7dc
Reader 514b4f56201d42c1181a4a14961f9abe4a328f7751e0d100a75bfb101ea22f75
Ledger at 108 9bf67dd6633e13d03463463e6d5880978f1459c3ba4e9352c82c30f6f5db9d13

r5c6_h-1v-1 Unified / memory ordinal: 165 / 168

Genealogy a03908be0f96c38774c03f0b83c3acb96e9e98b083d11b555c03fb4c6e686faf
Reader a2e862996bf61364ed0e6b6b37873be95744ffabe1b4e86973b0917d492ddcf0
Ledger at 108 —

r5c6_h-1v+1 Unified / memory ordinal: 166 / 167

Genealogy 81927e69032d8dda6c89f7e0c51591bb74a0922c58dc63b96cc20c6508c171f8
Reader b51c5e796ab9a08f438f09085e623965205e7f4be2ed57eaebc1afa16d1504ce
Ledger at 108 598cf2540b6a078e4972cb09c4a957ff723cc9c3e7d7797de08ffcafb6666426

r5c6_h+1v-1 Unified / memory ordinal: 167 / 166

Genealogy a75845a7c9ce451a84ec8de2149e4578724b29820b25582178837475acd97e29
Reader 52f8b4627b04a1a925e56d9021b47cdbf4e172ca6df3500072c1a8cfd00d329
Ledger at 108 —

r5c6_h+1v+1 Unified / memory ordinal: 168 / 165

Genealogy 09134b2a60678acc4e4d7fe41328fa475073a90714988713b330280c61906435
Reader 3a010d9493ed73b62d23d3baf729585440be9fb9b19ad1cfa7ca16b2970beb82
Ledger at 108 —

r5c7_h-1v-1 Unified / memory ordinal: 169 / 172

Genealogy f0ba7ed797a68175311757b8030cdec682fb29e4eed2cdaf8cfd3832be16d4c
Reader 04477201ed2fa53846f5f328151d60bccff05c49e4724a3237f24a7402a084ab
Ledger at 108 —

r5c7_h-1v+1 Unified / memory ordinal: 170 / 171

Genealogy b6376735f9acad08bb8f9c39f4ca3e69be62405d3627943796f664a8a9cec6d6
Reader 54983701597aea6396a38cb59a14122950ea0616556c5847dc64e8bee3fec53f
Ledger at 108 —

r5c7_h+1v-1 Unified / memory ordinal: 171 / 170

Genealogy b34a9417a09638f6bc02ffe98471b0e2b627116ee13284a7ad74bf5074b0f4d2
Reader b8df04a23c90b57616a1bf8334a7f7757313f38e95a5c4540aa36d3e166549dd
Ledger at 108 —

r5c7_h+1v+1 Unified / memory ordinal: 172 / 169

Genealogy 1a5415204e85bcd7ffabce87c95eff5fc47e613ceddd713d8f05b93718293d41
Reader 3f7c2ace630141906ef94299c92c3d34f74897919e7ec93e685fd0b6210dd9f3
Ledger at 108 f764349cd2ba2f9d1ee7db9a6f0bd775d595c4480bce2698061803100c3f18e5

r5c8_h-1v-1 Unified / memory ordinal: 173 / 176

Genealogy 701966ab31f24a2e0ac479f747422728dbafc44021012aef5bc7700e331d65b8
Reader 5d22412c39797827c404556ad608c217b66ea943a89b289f458610a70c151ad7
Ledger at 108 —

r5c8_h-1v+1 Unified / memory ordinal: 174 / 175

Genealogy a03329e40257d5ede551debc7365527a24878f84197e9298100b0b55af57184b
Reader b12e9f54798f62c19f474b63d68348823a829225e5a25e308905bb4e98e7ff7d
Ledger at 108 e8f3e4639e854424d2658b469d5722453265869a56f8e553445bc0037bd9882f

r5c8_h+1v-1 Unified / memory ordinal: 175 / 174

Genealogy fe0e3cc0a84e143289d0a34c2824d4fe4fddc9921c05852eb70f36277c4ab0b0
Reader 41a91226e39febcefa36198283195e1151884a5438e330530b3217ad9b811153
Ledger at 108 —

r5c8_h+1v+1 Unified / memory ordinal: 176 / 173

Genealogy 9f17a032ddfb1ca89f4d34e1fc106e59e07a18fc1dbada65e5061f6b0974791e
Reader 5cc037f3f593ef27c0b04dd60612b27c3ee3fb217cc66ae495e7c6eb2c3f5800
Ledger at 108 —

r5c9_h-1v-1 Unified / memory ordinal: 177 / 180

Genealogy d13b10758f72225bc8a3bb0f29e7117f1e568076f05c67d10e59f0ecd85a0d8d
Reader 3a627fb4a5c98662723a034a9f86de710c9cb3abc8f8d035e035b6e579396d47
Ledger at 108 242498278d713b1359912fec0f87443b9461fd3dd582fd9889d8a0f5668c22be

r5c9_h-1v+1 Unified / memory ordinal: 178 / 179

Genealogy 05d5c4bf566052d2f9fb3f06980ca8d3107fcf46ffefa36a99432ea51df4ffaf
Reader fed584860b86d5c5a263e0095165b1b1f3766fe83937baad4fe826a798377b6b
Ledger at 108 —

r5c9_h+1v-1 Unified / memory ordinal: 179 / 178

Genealogy faa1ebd578b981daa2f34f8d42e7ddcf87be1072ab7569891f0566f3aa0fcc75
Reader 59f7718146a7eed58f86cef869bc7e00ac61585cb84db3ca040a23ab561ec874
Ledger at 108 —

r5c9_h+1v+1 Unified / memory ordinal: 180 / 177

Genealogy a2caaa1885e5e1ec3a77b1b1476830d4cdea1d83ef5de1043297f20d5b040bae
Reader 33b2af69d9d7267f506c5047bd415adc3eb103237644181ff6e49fcadb855252
Ledger at 108 —

r6c1_h-1v-1 Unified / memory ordinal: 181 / 184

Genealogy 5723836945af498aee1f7524fa9f3ca625afae9a8ad2778875e0f116ddf54bd0
Reader 26c999154dc98e67ef1eb0cbe3c4af5dc49cc63ff9acd7220b474c1d35deb661
Ledger at 108 e29c6e53cf987fd4cd0791723fa331bd6c54a8c0e6be144199dae0d5aa759d38

r6c1_h-1v+1 Unified / memory ordinal: 182 / 183

Genealogy e39ec08bbe801ae536649422bf79b77a37f016ba9a4bf092521046f28fab1bc6
Reader 68b45d01d606e5eaf1737c7f7d9c0e216111306976f593bf8cb7006a1f0989a7
Ledger at 108 —

r6c1_h+1v-1 Unified / memory ordinal: 183 / 182

Genealogy 2262a4ead8d34652790e70bcf39dabf4c6da0042f349313dcd38adb241f7266b
Reader e348c32836c7d0ef2ec19bed6383e9257ee192429d72bf57548f615529d8ae53
Ledger at 108 —

r6c1_h+1v+1 Unified / memory ordinal: 184 / 181

Genealogy 7c76e8e8f5478ad536bb1cb8f675ca9aeceb07541dc9051b6af6607d9276aa67
Reader 86da2f8ba7211912d067c9008753f8e115b2fbcc6eca36f42bee9c89352fd976
Ledger at 108 —

r6c2_h-1v-1 Unified / memory ordinal: 185 / 188

Genealogy 7b821c471bf4e01e7dcfca1341b325491b18cf5adbe433f3054bb10ad081a842
Reader a37372d4b23fd8937236a0d2357362801e9e90140f10eb0752028f0835fa0d80
Ledger at 108 —

r6c2_h-1v+1 Unified / memory ordinal: 186 / 187

Genealogy 77bf36177edee2e331d458ba93651e87788985b6fbdf5b735256fadb6b05411
Reader 3be69c77c702f3b17d2f920c2531cf2d4e4c212e395a71905205f20c05c4892e
Ledger at 108 9485907bc9b83fa7aabb744a6a88d1d3223272ea89792d7558c057884f8e9f0

r6c2_h+1v-1 Unified / memory ordinal: 187 / 186

Genealogy a039b8fda9943425c307ca97bc18ad161e61a4a12c0516fe66793b765e700217
Reader cd0552218d548e6a3dd4332197268fca2c06a5b81c36d69aaae3d2c717d9f0a2
Ledger at 108 —

r6c2_h+1v+1 Unified / memory ordinal: 188 / 185

Genealogy 8615c240bf77cdf6ceadf5b08213f6265f4ca61423b297ef17ac1b43ddd3c99
Reader bd032ab996bb19de9c1074953658e061c757ebca4281f4b089db87ed939370c9
Ledger at 108 —

r6c3_h-1v-1 Unified / memory ordinal: 189 / 192

Genealogy cd011cd5866d118889051f57ac2a0d2b5ca6c7abbc4b7d733a0cc35515322e9b
Reader 91deae21588e4bbf1be6e39288698ebdcfc2cde1320d3f0ef090293f311c428
Ledger at 108 —

r6c3_h-1v+1 Unified / memory ordinal: 190 / 191

Genealogy 7b28dd5daca27177a2ca44f0b080b42407a6023d8a10ce5d6e034058ddead35
Reader ccb618b1bf7c78519fb2a0fb4e8ce96be3fd471eb2a4f12cbfb2237fdf4a50c4
Ledger at 108 59349e8798fb69e2e5ae6c2cf1ca8ec3053390b207b9dc18e92a5351bba14dc5

r6c3_h+1v-1 Unified / memory ordinal: 191 / 190

Genealogy 41e436cbeb2b64cfecd0594ddb877df5834695c8c3b033742d8bdd9e04c2774c
Reader 766c9065f7080b17e85ed5f8029fb902f2ab1b563deba91f4bc3ecd230c44c7e
Ledger at 108 —

r6c3_h+1v+1 Unified / memory ordinal: 192 / 189

Genealogy 2b62e065285f7e9dff5a3ecd8ef42eb73412a1ae0bd6ecfaa0f70240b91fb5e9
Reader 102c4214f537ef17f9428490ac06f432b8c2bb4e74669c284d556fbef2f3c983
Ledger at 108 —

r6c4_h-1v-1 Unified / memory ordinal: 193 / 196

Genealogy 978a72f4858eb2d567ab4b95f6eebbde0c0eff2fc96c5e0355718f6086eb7ebd
Reader 4bf0490cb60c31703dd122be074181958c4e70fb96f4d3f4ce35715f4e54679a
Ledger at 108 09534891acc2eaf69f6c6965a3dcbf124972602e60fb8fa02e0909e20d5411b

r6c4_h-1v+1 Unified / memory ordinal: 194 / 195

Genealogy dcdcf9eb3b8abc3fea552a6845f15c06b3f51f76b9bd638f95c451fd3cf68655
Reader 52d4bbb40854b103ab52b35ea76890da19bf740bb9db8f784ae8f256fc3c4d04
Ledger at 108 —

r6c4_h+1v-1 Unified / memory ordinal: 195 / 194

Genealogy 78f3ffaa921e46c61e0a5e8f0098d060fca37d87139b6541dce6069a8bd8a27f
Reader 32872bf64adafd685ad864babb2ca0d2e7551b61ccfecec145ac607cffe25d0d
Ledger at 108 —

r6c4_h+1v+1 Unified / memory ordinal: 196 / 193

Genealogy 77cf5c939793f9ffefd542cdfd63c67547932703378b78382725fd396edf7833
Reader 28764df7e51b08e0cf0a4e1231218c25be1d6f46c66294bcd19eed1f6b285cce
Ledger at 108 —

r6c5_h-1v-1 Unified / memory ordinal: 197 / 200

Genealogy b6badb1f3717c85f8eab31262b0fde338dbc49197b8eab9820df39cfd1d22827
Reader 17bd49a7de09fc63fbf85730b09b8c69da5258cd3a24e4c11552fad8eb796267
Ledger at 108 —

r6c5_h-1v+1 Unified / memory ordinal: 198 / 199

Genealogy 7493844ee9e55ec3fa4463553c671c3bfe04eb94006e369b09aa1c2fe3fd8152
Reader d6f20c0a7858c0752246770ffbfd66432cef12d31a555564cd4c76c2195800ee
Ledger at 108 247db8a58d592a8cdd0ba517321e06c1f88eff82416294586fe44cb507cd1737

r6c5_h+1v-1 Unified / memory ordinal: 199 / 198

Genealogy 0b8c5e26fd15e5487404c126325089f3744921e3c70229aa7c4efd9598c80084
Reader bb50d7c6b121d5b9b27c2a8df96ae10f4164b1fc3d2ae8687629f5ba15b9d059
Ledger at 108 —

r6c5_h+1v+1 Unified / memory ordinal: 200 / 197

Genealogy 459225a8c62810d92ebb16df6c2841855fdcc7973f5669549e10bdb7f48ba668
Reader 3207260be401417eccf8d9f756c5e76d13ba51eeefe672feec28d8ab04ef1ed5
Ledger at 108 —

r6c6_h-1v-1 Unified / memory ordinal: 201 / 204

Genealogy 856f2c4ecede7c9fad89891b9a493ce336be2d4b5153ce9ecce9540fcd10311c
Reader a8e2d94175cce86b4b4ebf52a6722aaa5895cc5b77acfab8bfc923d376d84f57
Ledger at 108 —

r6c6_h-1v+1 Unified / memory ordinal: 202 / 203

Genealogy 362984c5834759f48f4db727a5e95cba6a614bb5b737be4ba5aa7e0092099305
Reader 54b6507550fde869f152d24026fee643979e44fefccb5a54fd5374cede0c4178
Ledger at 108 —

r6c6_h+1v-1 Unified / memory ordinal: 203 / 202

Genealogy ed910a023c24aa3e30b58383d60bafc5d58eb8a1f08e0253b59d1c804d53f35f
Reader 1c9ca174be7648f8a2b7e60e181a19b3b34f7c4e0d98e91bac5a4cbacc8c9a20
Ledger at 108 —

r6c6_h+1v+1 Unified / memory ordinal: 204 / 201

Genealogy ea448f4eea6376a58f482d676495495f0c7b3c6bb76ccae40998ce2547269821
Reader 7d817835fbd8223ed54da24836deff3fe6b2f2a2c237caefadfa287194b6ac4
Ledger at 108 1204da29afa516cae33ccda64944005b00ffe568f4826a585823f1bb2e720ae3

r6c7_h-1v-1 Unified / memory ordinal: 205 / 208

Genealogy 88f473d2cdc82fd9a950b8951b6621d785eb978f2df99dbc0cf48b60503f0407
Reader 0f39551e760683333bc943e7a463b426ae514492913154ad8c8a50a579ac0f4b
Ledger at 108 —

r6c7_h-1v+1 Unified / memory ordinal: 206 / 207

Genealogy 616fe1c153e8405fb9fdd203ddc30c405976b9f53a6b887a78eb2deb09ff3019
Reader 0f506299138488a5bbb95a41ea186f877f691de62c43341f65029d52a295d822
Ledger at 108 e35a87178dee05ba4b834fccd04d9151723185209b3bf306de515f71c318bc40

r6c7_h+1v-1 Unified / memory ordinal: 207 / 206

Genealogy 4ff06f4f62859461a846f4d3090bec6874de8d404dc28e4c85c0039524ebce3f
Reader 82db73e13fb33b93832d00a5d3ffae2a3ce893d4059d410f530239bd20944a08
Ledger at 108 —

r6c7_h+1v+1 Unified / memory ordinal: 208 / 205

Genealogy 44eb46a7747c4aef1e300e344b95cc35662ea5f5e7be767c3e9e1d8d753058fc
Reader d37dee388ad27b72a09d85bd19587c613c7ccaf6048eeba353d31d5f1b642289
Ledger at 108 —

r6c8_h-1v-1 Unified / memory ordinal: 209 / 212

Genealogy 46a7097d2858df9176ba9586b269471dd295cb011c04f97ee1f4da54cdd7a43e
Reader 1a456767b18a5ec986b3b3fb3c3856267b3045e9d69a809c24d95845977eb71f
Ledger at 108 —

r6c8_h-1v+1 Unified / memory ordinal: 210 / 211

Genealogy c662417af4f02cbb24da0df8ddc65daf21c785dcda7367f499a615d5f1c13acc
Reader 0bf138f401402e507cb05c5292252e90d8a251263e47152404d52b831e9fce30
Ledger at 108 —

r6c8_h+1v-1 Unified / memory ordinal: 211 / 210

Genealogy 6244f8a5a54cd67bbef735c4354e23b07ca1ae58e21214f4a6ff3516629e19c1
Reader 65c53fc392fa3dbf000bcb9c9539167b8ce081300211ef00c2eee82e0d9ff65c
Ledger at 108 —

r6c8_h+1v+1 Unified / memory ordinal: 212 / 209

Genealogy 4f8136461cebd5d0e1aae0b94a1310952fec8ed32458cf20ae000c9ad96abc39
Reader 478ca35df2cf0d4564142043734baaef374d7159a3c7775cb731a590a615a0db
Ledger at 108 ec0c7b1546af4e293324a0e6fc89aa4cd160e3c70eacc9a9b3494baf329435e3

r6c9_h-1v-1 Unified / memory ordinal: 213 / 216

Genealogy d29410c367743e1611b951c179305418a0c0787034e642488d40b8421a4f1b20
Reader 05a266cce4affce309ae74ef598dad5069603262543c2e64d874382a83755978
Ledger at 108 —

r6c9_h-1v+1 Unified / memory ordinal: 214 / 215

Genealogy 383fc3c8eab5e689c371ebbb9e5a07b1b4c5c0c869c53d1e34f80c0966fe4683
Reader eca848d3437105f4e3992a55c435b0ebb5e5b25c7918788fba77fb97fa69eb3e
Ledger at 108 —

r6c9_h+1v-1 Unified / memory ordinal: 215 / 214

Genealogy 5f4e2b04f8b80b45dc0a27e968364702c50f95e4243d4268177f10928526aee6
Reader cfa3576154e8cf220eeb6ace24ec313ec4d442e4d1da02a4bf007f30bd2c6621
Ledger at 108 80250ff7621218a03a2e2ca39a7a6e60556d81f62485cf81b5b4de276a81d550

r6c9_h+1v+1 Unified / memory ordinal: 216 / 213

Genealogy 79b913bc6b44d4d200a69fd92eccc9936a66c4d289739af388dbe597c8ed9751
Reader 32027225b3900396db67d96992263b16a9b0766eb91a7c939ac99ad9a97d3bf9
Ledger at 108 —

r7c1_h-1v-1 Unified / memory ordinal: 217 / 220

Genealogy 213e50266085d4cf0de7fe5583f14cd6e9fd4d8201a2fca7bbb3795d68988de6
Reader bcca332d749bd269e02e92f330430569c4525cd5e249720db77549f966a52809
Ledger at 108 —

r7c1_h-1v+1 Unified / memory ordinal: 218 / 219

Genealogy d62df3d96add1bcb3616655b4c5ab36dd6a959ad7680ed61f0bf567035e46c10
Reader 04f3932501d1ad67facab5d82019498af3313df6e346a07adcb84e7c755faee0
Ledger at 108 c10b18e3f00c1efeab677c684d1a7bd005d2ee4a89b23fa3bc838923718a08fa

r7c1_h+1v-1 Unified / memory ordinal: 219 / 218

Genealogy ea1bc7bb3c188b9a7e9dd91a90b29f4b3fe86da770ec0c1a40183a53264492b5
Reader ad40326e62bbf34eabd692806fb96c19a8fd8520056d702c51a3c115dc8ce1fb
Ledger at 108 —

r7c1_h+1v+1 Unified / memory ordinal: 220 / 217

Genealogy 3a495db2e5a0a6707f8a97f724743b33c5696f6ea59a145f59858765279c3c3b
Reader 22a43ddff0542b8baedc3feb41c9e088a990090a8a95fc8d101d973f3a8fed9
Ledger at 108 —

r7c2_h-1v-1 Unified / memory ordinal: 221 / 224

Genealogy de97a58186c1b724ae969ef3d0b46d0b58439e4a07374877722147645eed901a
Reader b5cbdb2acc289475e1adbffda63c9cb08aa1f829e2f904b2bb353276a27933bf
Ledger at 108 —

r7c2_h-1v+1 Unified / memory ordinal: 222 / 223

Genealogy d0d3560438d493017163b00e45d3935896eb8554e0dddda5b8a0856cc7a13b34
Reader 1de0ff8e3d24aca90f7f4d432eafa71c5d5aab1bc65d749f8cd1dc602a5a7a65
Ledger at 108 c97a50bb057d87c6a9915463f589846d13ce5d25a64e07f2625114cb93e204f5

r7c2_h+1v-1 Unified / memory ordinal: 223 / 222

Genealogy a8a820b2e3341e508860df8fa52e8ba6be5fca90c44b0b918cfb3e75fd28d176
Reader 3c31186b3fac2aa2f596f047ea1756c734f0ddc59694e37a98f47d9a3bdc8475
Ledger at 108 —

r7c2_h+1v+1 Unified / memory ordinal: 224 / 221

Genealogy 3bac38d14f028f874e98149b7274261bf05c7219f5381cfc4cbad9f6c1c850b9
Reader d0f373bab545802ead72e058f77728e20bf2d03ce412db3853130c4b1a38f22
Ledger at 108 —

r7c3_h-1v-1 Unified / memory ordinal: 225 / 228

Genealogy 50dd3ab85f947dea000293ba18e7e5d236ed2a2482880fce8a55b876402a96d6
Reader f8cfb75a9cb25ac6b8cbb42e3fc084e2337939344c2a990541bae640dcee3531
Ledger at 108 —

r7c3_h-1v+1 Unified / memory ordinal: 226 / 227

Genealogy 97b0c7b8352c824ea503a1ec527396cfc0d0b44900be367dd4b03945006857bf
Reader b7b4d75304d95a6a60aba23861edd1112aad3aed5ee0e9673c71017ccc3594bd
Ledger at 108 —

r7c3_h+1v-1 Unified / memory ordinal: 227 / 226

Genealogy 34dff48532c73a280d1550ed80b44a2baf487d486dd06bec76436e8e94d65f93
Reader 502b383cae291d41508f616ac645f9bcdef36a9a45477b31c70d5cb7ee016951
Ledger at 108 —

r7c3_h+1v+1 Unified / memory ordinal: 228 / 225

Genealogy 2c8a0ec9e3c50967fd90be0a53359711cda37fe96004fa0dd2d2c109d3ff8203
Reader e7e5c8dda4d5d1f9b878e9cd1a1c64fb8345513eb16b32b91933ab571fbdabb5d
Ledger at 108 d232cdd08b31d355345b67402bf59eda9ec2f460c0b2fde0aabb2dc8b105a72c

r7c4_h-1v-1 Unified / memory ordinal: 229 / 232

Genealogy 61986b6175f7b937676573337389fee3569c31355092cf947843a9d930ba1bc2
Reader 5924cc1f56504b4c33f30badd28f71c226336afe924e078101dceaa8e836decf
Ledger at 108 —

r7c4_h-1v+1 Unified / memory ordinal: 230 / 231

Genealogy c59fd018875bb4727075a352424cec29d327972ce0436e0d6620d4c581058224
Reader e049714ffb2d3eea8b496d146d72cdeb379c2764679bbf9137027cbadddd41f3
Ledger at 108 c7e0c373385cd688c8ff7aafd12c63d3af22800c9ae593a7a3bc5952d7512202

r7c4_h+1v-1 Unified / memory ordinal: 231 / 230

Genealogy 7a6c6470c763c81771cd0d8889250edb8a8675dd0391a21b31385cba4993e652
Reader be05cd14c4f089027d45446ae145d5bbd383576b0d091e1660097007ebf4fc04
Ledger at 108 —

r7c4_h+1v+1 Unified / memory ordinal: 232 / 229

Genealogy 3b0c0dbfa4786567c9e927a3d9255385f88a8bf3e4ee10b39a8990045dbb8251
Reader 7e34fe3bd8853b363e1c7195129b813c582bbfbb75ebd6651f739c23d7b1b85a
Ledger at 108 —

r7c5_h-1v-1 Unified / memory ordinal: 233 / 236

Genealogy f6ed0ad96ec6366aed3b499144349882d3ffcd7cd44284510c3270821d0056c
Reader 25d3f4ddb064b47c9a608e17a712bb09d854df59edc3903e31a2673b50b1ae0a
Ledger at 108 68cfa4635763ee620dde045d3406cd27b8ada4eb740d3dd02cfcde7e4775e236

r7c5_h-1v+1 Unified / memory ordinal: 234 / 235

Genealogy 99ab406ecdac6329ab5ce5eb6325504ecb11f2d04c9d8b346db7499d7f567b1e
Reader de03114ba14ea4662b556a0e8594ed8b86378685d02ee66d90537006e76e8ab8
Ledger at 108 —

r7c5_h+1v-1 Unified / memory ordinal: 235 / 234

Genealogy 0386e9c2055bfd8ab8494edc4b8a16f00671f216a461804424aa45aab64b22d5
Reader 66b6919786b672f07cdf87be4aefcb18b9cac24fa6eddfae27e4ddf26caba935
Ledger at 108 —

r7c5_h+1v+1 Unified / memory ordinal: 236 / 233

Genealogy b7a1b0a7f7d9328fa3ce85e1fff27ea994c202f1387e665f13646fe39a6f9594
Reader d7423e62f53bddcb400b3cf3d5485932f1df7a7edb0186d4e92c189a60b8a179
Ledger at 108 —

r7c6_h-1v-1 Unified / memory ordinal: 237 / 240

Genealogy bdd0f510981aa2bd14e11cc9e772a40c0fa9629c6c33a6c863e128543bb3c064
Reader 094ff62832260c1076787d26e6f33ddfede703d5dedbf9b86f08912179c0158e
Ledger at 108 —

r7c6_h-1v+1 Unified / memory ordinal: 238 / 239

Genealogy 3b43feb29ef42944b5c6da5d1493e4641332fe290cc733c94b55588185f490d0
Reader 77fffc5e08a1d1d22e2d56a9caa4f75016d1c69261b9bac28a2e8cedd44d0ddb
Ledger at 108 4261f837a3e329d199831ee764123ea51bb952b25504d3f1b96372b344872b11

r7c6_h+1v-1 Unified / memory ordinal: 239 / 238

Genealogy 231bf7dd55d8a438f107de8c3e14075106170fc968fbf990a0d8a6f5ffd71b41
Reader 573249f714bc3cb98d64797e90987918abae980680a392c41e65c25c1bba6d94
Ledger at 108 —

r7c6_h+1v+1 Unified / memory ordinal: 240 / 237

Genealogy 3e939392091ce7df7511a0eeda421eb0299c2ba4582a4adbf492bef6aebffe3f
Reader b6443f7d3f0c44922d917e3268c496cff426301296ffd17b39c8f71c592309c7
Ledger at 108 —

r7c7_h-1v-1 Unified / memory ordinal: 241 / 244

Genealogy f7adee5b3133fb43299c7e7f57cfe9d2104d12951c24558e004b0d1046825410
Reader ee9f7a74820c460ffd5606aa30bddd40bb6875427bf381c1bb6883d6bf2cb2d
Ledger at 108 2b5fcf4ee156f8e7f3d9ef4b48b12761781d58740c7f25b350ada993fdffb43da

r7c7_h-1v+1 Unified / memory ordinal: 242 / 243

Genealogy e5a3a65c351de531521f5eb2e8a895c938d33cc4b6092ae7a236bfde174f67e6
Reader f152871599c0f372a009df5e0bf5d2e89a435b88231a923e31fd5fc32cbfc7f8
Ledger at 108 —

r7c7_h+1v-1 Unified / memory ordinal: 243 / 242

Genealogy b139139d502bd446d91e7d84ef6378829caef959c57c068a87604b10a6c051f8
Reader d3747a41b0c5e19bf084c3f39ce497e825be56b559a223f00b148f1580acb8ed
Ledger at 108 —

r7c7_h+1v+1 Unified / memory ordinal: 244 / 241

Genealogy 97a8e38d1e0bc29075a19475862f706d23c2cee0e4509a10167a321562108b24
Reader 113a5319d2bebd3924e86aff9c2f661818e1d5da017520c4f5a248750b426e5c
Ledger at 108 —

r7c8_h-1v-1 Unified / memory ordinal: 245 / 248

Genealogy 93e2a454a1c5de5f4f4e4b0eec1903cc6a51d31d19260e98a023f085f717f2c5
Reader 5e1722d01279d898e0e53f37eed749fe741983d8db78b09039733d0e70de18
Ledger at 108 —

r7c8_h-1v+1 Unified / memory ordinal: 246 / 247

Genealogy ff6632f4595185491ccf0bb1ee6fb09651fc7ee9d444f4a36b2434921c63e845
Reader c689f380d0c3f1f13830f345827ac079bfef21a0290c3319a4e040b66452df42
Ledger at 108 —

r7c8_h+1v-1 Unified / memory ordinal: 247 / 246

Genealogy 572bed3ec2da43b588e8f03c20b85deb3a3238787c652806bdaedb67e8a2e65d
Reader f6d6251c2a3a0bc51ba9e9d062b640aae53badbca7baa4e87ead4bf0ae639226
Ledger at 108 2bc42341df8c62244bc513d9a683f0a2cd6caf97ec8b8d54ada93d9ac2a0f146

r7c8_h+1v+1 Unified / memory ordinal: 248 / 245

Genealogy 5468c23aaa5fe88b8dfc71dca843b9098b3fad83d80745052c981c396d08d09d
Reader e7b0197ca2c34747ea501fc5fc951f0eabb2154bfe7d1ca632f931071fa14d07
Ledger at 108 —

r7c9_h-1v-1 Unified / memory ordinal: 249 / 252

Genealogy ba5642d596d97f27564669dc98515c905e1973cd764f435ee5c853fcd049beed
Reader 268ca6785b1f8117df7ac1724f8f4b30ac60083edec1102641a6611adc1ffb9c
Ledger at 108 —

r7c9_h-1v+1 Unified / memory ordinal: 250 / 251

Genealogy da00f1d555eb9be25c1ceca2ebc88d96b55a1c8c2011788d814fa5857a504b9f
Reader 183882de6128614ad3eef875eb974e8c55c951558c213f7eb721348c9354f824
Ledger at 108 —

r7c9_h+1v-1 Unified / memory ordinal: 251 / 250

Genealogy ad5ee82a87bb9a61ce4360a2b14bf22c5c60f40e7e75f290ec2dde4d0e7c90fc
Reader ed489aa0c8e0c09e4604ad7f7f9ebf49ec343a76ed5ea60a3369f1411b16ea84
Ledger at 108 —

r7c9_h+1v+1 Unified / memory ordinal: 252 / 249

Genealogy 88eff8f2d71641e554b89b656e2885567c18727bad0128f42fa25e7bdf6a031d
Reader 6c78548094bcd8b058a3bcd868ff2cf83e42197164de9e58aa3dab4b8edc6ad7
Ledger at 108 743b8088ceffb1b382bb7dc86b4e63b91a87688168e208f2080f8a040aefcfd

r8c1_h-1v-1 Unified / memory ordinal: 253 / 256

Genealogy 7f20d693558c4c770e8c14a2edb53682418d3af88e8147e4e46a45b584d0af32
Reader d463a30437186dc5a632dbe1fe2eb6c5c462cdb58acc7d2d24faaa4123025187
Ledger at 108 —

r8c1_h-1v+1 Unified / memory ordinal: 254 / 255

Genealogy c88b93580f9451e0e63048f3f329ae52082f45e65a9494acb1942445b74dbd7d
Reader 06f382a4af0319fc434d2d5109c92a77f6ff4e9eb6a30cdd9620be4b2c25677f
Ledger at 108 —

r8c1_h+1v-1 Unified / memory ordinal: 255 / 254

Genealogy 210c219bbaaf3bf455eaa92f45a017dd9f816ca3b46f2b08b272a9c06868cdc0
Reader 8b66a9561a72fc2cf9d44aad0a4ab9280d6872fd24ae8ade2437e7c71c5da763
Ledger at 108 2abe5fb770abbac72934c668102c504b01e8b32c6bee43d0a62e45a1cbb3f5cd

r8c1_h+1v+1 Unified / memory ordinal: 256 / 253

Genealogy aab792bb556e0bed999eb313b4d4b26ea63b3bad901ed8130f94fb1ec4554219
Reader 7236233560f551425afe01a27858b2d7979db3983540ff9b4e007b9f9b5e0d04
Ledger at 108 —

r8c2_h-1v-1 Unified / memory ordinal: 257 / 260

Genealogy 4d09dc43af4a59915c1a6038f5c475cbd194cfef4bf1d5e357df23d9754d395f
Reader 4128cbf79b80a7af93f10ded32d0960eb5da56cbf47a110bab03f10d446c2316
Ledger at 108 —

r8c2_h-1v+1 Unified / memory ordinal: 258 / 259

Genealogy c10ac37f6c4294f84a5c4eecb9e4d1374086b7aae332df72101d20aff5ff88c8
Reader aa9f45f6e48948c38e93e1da669a8afdc2bfc8e251959f2c2d2ac248ac767317
Ledger at 108 —

r8c2_h+1v-1 Unified / memory ordinal: 259 / 258

Genealogy 4581338ce438211bda03b46bdd171051cbdf4527bfe09385ab84b222943678e0
Reader 30e7d80418eef9d9188a136774e1e1ba1b467b366375105e556e2595fda118ab
Ledger at 108 —

r8c2_h+1v+1 Unified / memory ordinal: 260 / 257

Genealogy 37e66f95874dba7502936c0625906398abd462421242a711a1cebe0da9742e20
Reader a066e554a21d08baafef29692f0e0500bdd858876f9bbc234b52fb9a24bea27f
Ledger at 108 d7f810dc8cc7b1298f7e1c92b9f22e705a1b79fa32aaf5a18405a2e5fbd7422f

r8c3_h-1v-1 Unified / memory ordinal: 261 / 264

Genealogy ad652dbfa6c5c813518166ce23d4443cfd6cc6a3f22f5d7f723510f50c448b8
Reader 3ed4894181a15c6c37003af6a3bffeefbd5b6c91a65783ebe3a130457ab9a8aa
Ledger at 108 —

r8c3_h-1v+1 Unified / memory ordinal: 262 / 263

Genealogy 541a3c42518d601b58431822a3572a2be049c02a300aa4feacd5345ef948f454
Reader ad8c50495acdfc9708f3bf9b44a935a2051df21767326d50c52b3dfeae00d3fb
Ledger at 108 e9f4ee18c9a8859c65aa766dbde0bd196a242a2f3dacd16a052d90accc021dd1

r8c3_h+1v-1 Unified / memory ordinal: 263 / 262

Genealogy cd28c157b1840882849d580e4bfbce448d65ae5f45fe448971e9270dacf4fbc6
Reader 6d4c84d97a1f41ddb2b2bd27d0c2afc77d646b03c697190d46d00c87f92617a1
Ledger at 108 —

r8c3_h+1v+1 Unified / memory ordinal: 264 / 261

Genealogy e3bc25b606504b6c91007699ff3655c139fba11cd7689c98c705d3f9d12f38cd
Reader fea2f4b68d20b684e22997fc21a30bb39fa2a3c31dbf3cf0fef68224900afdb0
Ledger at 108 —

r8c4_h-1v-1 Unified / memory ordinal: 265 / 268

Genealogy d773d886b771b4e9d58210f91db044e3d5fa76dcdbba2d144b4aa6a7816e2a74
Reader 2129600f36965fdc11a791126a3ad1e38346cb4d0a010c35bc87bab5442f815f
Ledger at 108 —

r8c4_h-1v+1 Unified / memory ordinal: 266 / 267

Genealogy aa4a5949c8430a0ec7f118f6522a38b241e22dd0f172a9ad8b2e7d04e3373bbb
Reader 00391297614e66ca2afd3f79601a5f0dec10705608f060644783235dbd3fa513
Ledger at 108 —

r8c4_h+1v-1 Unified / memory ordinal: 267 / 266

Genealogy cbe1adfccc203f0f6a54465ba638e1513bb7f3f92188c76ac22a231f8fc83757
Reader 8e4f93fa6fc6ecbbf4f2d29f80a71639352a11bbbd98c66c5b0772c8df05d749
Ledger at 108 —

r8c4_h+1v+1 Unified / memory ordinal: 268 / 265

Genealogy ad5595794bce61df813886f136cf73b6d5b7712e8f1009a2125b13242b4e8566
Reader cf06adfa5665f171a8b8dbe54798e56195555e71f498639258299d96de416911
Ledger at 108 573c1bd90e4644c8d653dbd9a8229181186b709ffd121a7408c8373b4384376c

r8c5_h-1v-1 Unified / memory ordinal: 269 / 272

Genealogy 7ef6525bcd2aeb751a6f5b43b6d6a27622407710aad04f8f412b15957971689
Reader 37d699ceb70f1c3bf1a9991ce52e71739128ddd2096da7733ba3750a74627b60
Ledger at 108 —

r8c5_h-1v+1 Unified / memory ordinal: 270 / 271

Genealogy 5168b13c9ef8c5df5479cfd86a3b5810444636458700009b99542fd47675d314
Reader 3b12ffa16f7d92e65eccdd73152a1897c1e1bb8b22b621b66ec88ce2035f57c6
Ledger at 108 b139e53ebbb2939e05b997684309cd5bb321796c24a70f00063466350ad16680

r8c5_h+1v-1 Unified / memory ordinal: 271 / 270

Genealogy eedb99f7a53750e7da1c4cfb65caff5bc7d6f7fe2d17ac33bd698f5ff5cced70
Reader 3660859f8e83b5299df231cf992060bcc7cafcad6c8bb559d671dcec38ae9521
Ledger at 108 —

r8c5_h+1v+1 Unified / memory ordinal: 272 / 269

Genealogy 623aeb8e00cd78e8b74b3da659368cebc64b59b20668d849c58cc8e800f3c668
Reader 9f01aaf11034da5b3d5ff388ae07367265257113f331b94b7080b8d05e61d946
Ledger at 108 —

r8c6_h-1v-1 Unified / memory ordinal: 273 / 276

Genealogy 545d27ebe80f7a1b5d6e4faf6f6614e59a75427fdc5db81a0eff641785391008
Reader df6973d8dba7d543ad4991d2fe5c1ebf3eb9a1803d7e933f0074ba080793ff48
Ledger at 108 165e97906c2b634f80f723d3b6db0f07d486840bdcf88079defcfc71121b3293

r8c6_h-1v+1 Unified / memory ordinal: 274 / 275

Genealogy f71b59ca66689f8406ea25f29129a8bb6aad6447707eafc3f076a9d7ed74c3f
Reader 45ad6fbdb7517bd4fc0760b8fc1596fbc582e6077f671d37ea2a5a9582a424
Ledger at 108 —

r8c6_h+1v-1 Unified / memory ordinal: 275 / 274

Genealogy b2394c0d8b4f94d4d3db2058caf65f5cba5037faaba86ab8b741d7614066f5b9
Reader b35fffd6d573fe8075155a84df6a2ed2e98d9bf9fe9e09c1e0624fb83395b24
Ledger at 108 —

r8c6_h+1v+1 Unified / memory ordinal: 276 / 273

Genealogy 1fa8ca368f05cacbb63847f3d7c151b4b61b01f5663f9d4f2e416c66bb475d96
Reader 7958f0da3b1c71e56248b77b0bd939b0279b68ff68cb145e28e5e06d5a3f6e
Ledger at 108 —

r8c7_h-1v-1 Unified / memory ordinal: 277 / 280

Genealogy 3e4f2795854d36045b55509bd916303c028de5ba24d97eeb42176631fad5420
Reader 4e3bc25cd834c5379d7cd435b106743384133c2f09dc9ced7d99880bed745a3e
Ledger at 108 —

r8c7_h-1v+1 Unified / memory ordinal: 278 / 279

Genealogy fd2d7320f5b5383ca26d8e6d224b7df36ee61afb23e51402ebf35ae0ee13a4f9
Reader 3840df2d6dfa4c48e51ce48ca841735ca0b7eaab8b9a9eaaa62c6ba1a7e40f39
Ledger at 108 —

r8c7_h+1v-1 Unified / memory ordinal: 279 / 278

Genealogy a15a54cb8e5aeadb9a14d62ee241c1b33911588d46d105d0ba47dc82b7959ba
Reader 0919f3a4d99a776867a28a6d334ec1b717ff3b3a449176686ee79e4603cead5a
Ledger at 108 f768a45c6e4830319012025088381e9a82811497d99ad157d5b1c623be89b25b

r8c7_h+1v+1 Unified / memory ordinal: 280 / 277

Genealogy 86715e7eccf63f421cf277f3abd516b226f7eb2d97fcf24bd37912af78d1d866
Reader 622a5691a81501fa895d6474a6d9e5ae5acb0ef96649d46875be617dbfe36c68
Ledger at 108 —

r8c8_h-1v-1 Unified / memory ordinal: 281 / 284

Genealogy 0a0cf9df7293b410440f3fb087c85c25fa5039d0fc789b9ab6bad4521fb837b5
Reader 111950fefe357c0f7efaef1ee7794ea3fc454e01de39489ecfd6e18be9fdef9
Ledger at 108 —

r8c8_h-1v+1 Unified / memory ordinal: 282 / 283

Genealogy 44c68f8b3fce4afc7439b1cd4add9d73ea2d68ae57cd4025f427e0da591dda73
Reader 4453d1913b855774dce1ec81650654f10fd4b0c3e26ed21e8dff264854b464b5
Ledger at 108 —

r8c8_h+1v-1 Unified / memory ordinal: 283 / 282

Genealogy 6ccdebcaaf311843d1d894e5d1c8dee1f317bb40cfb7fef1455cd39b357a6807
Reader cee239d95a64fd32c72002799f66be6cd6167d2f7d01c473970156608f61bd1a
Ledger at 108 —

r8c8_h+1v+1 Unified / memory ordinal: 284 / 281

Genealogy 1fef105f9462f0c629ad4505f2fa1a8a2d248638113913dbd37ae66791ac19f1
Reader 735623ac3cc96b608c56fc456cff66daf8b0193b82645553f1fa88efb1ba6bed
Ledger at 108 45951d9d048937aae4c2cfb3a4823a1ff75bbfb0f359f391421e6990f5bba610

r8c9_h-1v-1 Unified / memory ordinal: 285 / 288

Genealogy 8569908f3825cdd46d2aeb615280f04f11d02c6e3d9b945b706a1953b6b91832
Reader 9d065f9157694f9edc555f5e00871f36ffdfbf96d1f47fe11deda4e3a183b594
Ledger at 108 —

r8c9_h-1v+1 Unified / memory ordinal: 286 / 287

Genealogy eff03ec9f7fc2cef5cb21ba6fa45a82da8267d3092637150b953e4aa980bbe87
Reader 70dd7c0eb393147851d6fd2d32d16fd7e393a5d353010311ff0ef60aa076d668
Ledger at 108 c7dfe852a4958d07d62f965178f30c16579632ac1d2a4764b1535ee56b4f89ba

r8c9_h+1v-1 Unified / memory ordinal: 287 / 286

Genealogy 678d6ab8d97c50772b92a1c0c11ae3a7bc3fa03378f49a8a322f7db0e7a20776
Reader 738d138b0027417ccb40bd19ddb10609eb8c8531dffbf0c1903a319e2767add0
Ledger at 108 —

r8c9_h+1v+1 Unified / memory ordinal: 288 / 285

Genealogy 81affe3c95425c70af6e783d072ff89234b1446d1055fed31ee305192280fcb3
Reader 41036aca4f9f8ad135146507fe31266d0452840b897d87b1a5f15e6639516828
Ledger at 108 —

r9c1_h-1v-1 Unified / memory ordinal: 289 / 292

Genealogy a0d3b90a93737ad6bd06c49cbe1e379f6e67a39e7b60dbb4e0e0b5480d1c9730
Reader dd19ac7c0ec61adbef57b6ad840a051fb3b46e587098f2afe5101daba8f867b1
Ledger at 108 c0580ece103707e22c2b73720bc58713206cce80bca99c633743212984199b55

r9c1_h-1v+1 Unified / memory ordinal: 290 / 291

Genealogy 8483a8df880a8c77b15bbbdcd777ed2f7cc79e2b9d4e199dc642f2a9012041b9
Reader ae1cc25e4cd5b1b87552d23b208c33dbc0ec0121a8bb290c101337e779bb5e1d
Ledger at 108 —

r9c1_h+1v-1 Unified / memory ordinal: 291 / 290

Genealogy 52b846cd7eaeff31abb00f1ae440a22d9b65cc73fa2cc099196a0ef20d97df3c
Reader 3db7a708d8ae2238a749a75623f90c80ccd6b52fa0c36042a443240c6f9ebfc0
Ledger at 108 —

r9c1_h+1v+1 Unified / memory ordinal: 292 / 289

Genealogy 905ded93d9b0764d207049712252c1fffb8506d948be0960f3b6e398caa4a8e25
Reader ddb8562334314015284c5247ec81419d909c46c43781c41745de130f32a0e193
Ledger at 108 —

r9c2_h-1v-1 Unified / memory ordinal: 293 / 296

Genealogy da6667267b4ff85fea2f0ff5b28c8962cba978a6c2329bf58403d88aaab3b865
Reader 47731741df561e96ed208b391e19d518430489a7ef9c6d05607c164114f5d9a2
Ledger at 108 —

r9c2_h-1v+1 Unified / memory ordinal: 294 / 295

Genealogy 46e22be8f2160920042cedaf06032c82685a879d1f441907912b9a00df9114a3
Reader a7ff5decdbda0dc9f2f3c09ec8cdeac01d3a3ad66cb4dfed656bf970900d4bb
Ledger at 108 5f8ebc4 added94691deee35c6d4dcacdeec958fdd4a3daff598d7a1a11d1360c0

r9c2_h+1v-1 Unified / memory ordinal: 295 / 294

Genealogy c3161556f17c0edfadbc95693b722fde3bda92e61c398a497d9ea6f4f45c21ff
Reader 0b2ce98332ac4c69c985a795b31e585c57e037a8fbbe9508bf72fe5917ac5e28
Ledger at 108 —

r9c2_h+1v+1 Unified / memory ordinal: 296 / 293

Genealogy 463b5be423eaf1c50aa2a7f5f851d35daf48aacc82ab25714a83f2840170d00c
Reader 3e4d219163e182c8401f7d7dfe984bfad7fb28bba4b15fda36d17bddb01bcd23
Ledger at 108 —

r9c3_h-1v-1 Unified / memory ordinal: 297 / 300

Genealogy 8044e5917e13ccfbda2ceb1f21613f7ce52eaa0853ec71fd75f3220376d19cc0
Reader bb47bdf5cf32f87265f7ab2c767c46440e85eefa678090e2f952f4504fe98c1
Ledger at 108 —

r9c3_h-1v+1 Unified / memory ordinal: 298 / 299

Genealogy b1f5d9047f6748335b6f26efffc458e4baa8c0f8a8dfa8801b0cd5071ee377e9
Reader e138662cf0ef5ed672f884e44662fd4346e0c345559c2c5845dc6dc30a15b0db
Ledger at 108 —

r9c3_h+1v-1 Unified / memory ordinal: 299 / 298

Genealogy 3a549e8092847f10e437f1add78c27b8119bea4b98466f1a495c7fe837e526ab
Reader 75ff441007b14e7d209d0846ef9fadf49bdbc245749a8b5bcddeb11c723234236
Ledger at 108 —

r9c3_h+1v+1 Unified / memory ordinal: 300 / 297

Genealogy 36d1117b7f4f744e498155bbc7dc3f337e038bc6a59df945b16955b65617a24f
Reader 2822864e6d668c0d559901e93691fe2ff2784b9d82824b2d5fe3c593e5cc387d
Ledger at 108 55065291299c2ae2d06f2baa5791c0319edfbd413c3e9c7a8b949c3352304579

r9c4_h-1v-1 Unified / memory ordinal: 301 / 304

Genealogy ee798204ea997ed7ed2199e7b3ab7d6137f5ac35f07e503f6c0a8edae955b848
Reader 18fd69ac1d8dc23d374c9d4b781e339c278b6935a74bb76c1a2baddacb2b6fea
Ledger at 108 —

r9c4_h-1v+1 Unified / memory ordinal: 302 / 303

Genealogy 642cd12482d3d647cdaec0acdd19e9c1f2a7429675270183ed630caea3cf535a
Reader e7f63cff972e6706fe7c8ecd667eb3e076106f380840daddbcaf4c59501a2e78
Ledger at 108 11028c389e35d609ddb9aa1248b79b77987b41783a58974a3c6e6873c14e1f88

r9c4_h+1v-1 Unified / memory ordinal: 303 / 302

Genealogy ff0027584baf795fea0705bea171be417920f7ab7cb5f378d17b7e40adbb917
Reader 7117e0fdd957f42a5d46ef727105788a73ce6a863be2fd32f11bab0adf65bbcb
Ledger at 108 —

r9c4_h+1v+1 Unified / memory ordinal: 304 / 301

Genealogy b5fc9a01e3980db070dd783cd2fcc157585ffca7f68f54eac0c828e4aeedf880
Reader 5b3951f3240c6ee934749d6dc143cce4f650ab607b64271814be5fd67595c351
Ledger at 108 —

r9c5_h-1v-1 Unified / memory ordinal: 305 / 308

Genealogy 77908e472c16ed99b259dbc397c2623e67f0871ad4764417423d1c62bd7e5dfa
Reader 3d68025ffbb2f16ffc10253200612155c3edd2545ed08c9acf932f34b8417dea
Ledger at 108 ee7f57b43ff6bbe2b8b762aec3be88c2f3eec4cac943953bf26b06adfc1e2088

r9c5_h-1v+1 Unified / memory ordinal: 306 / 307

Genealogy 6d009acb2e4a9062d353d9d3d3b9fabbdd58df3dffff48ff543302954972595
Reader b47145e5c4d4c54c6e7bcbdeb698559de66f30511a1ceb50d00e80be87055bf3
Ledger at 108 —

r9c5_h+1v-1 Unified / memory ordinal: 307 / 306

Genealogy b685f1413fffb6cc5e0b9a15517eac78ab70bc66c1db71b3ab13da31d17dc60
Reader db33053c3c7df39bb195f28eadc06b927e42eeddc3e9b8a8f84cf04b4d72ac5b
Ledger at 108 —

r9c5_h+1v+1 Unified / memory ordinal: 308 / 305

Genealogy 75c7d91367c880ed973e5c536d4f7b3ddeb4d6a6c0478a776fb9250721a22979
Reader 8a5cc992862f3e41a477ea869ccbabd19b444a8876e6e3b1b084264cab2e4db1
Ledger at 108 —

r9c6_h-1v-1 Unified / memory ordinal: 309 / 312

Genealogy ca419a4fb89079e264213c9b485a320a311c24d208068ee41866ec2ac1b43c2c
Reader a382f2951d5ce06beded8af0111b7528cf000d7719f0431752c134cdd274a383
Ledger at 108 —

r9c6_h-1v+1 Unified / memory ordinal: 310 / 311

Genealogy 106b15fef86d47d98a2151f4c2319629333008c764daa083780d58ab178d9ec0
Reader 97e31ef7a8d5ddaf45317072ce7a9bc90b1be5f32591931c1267467013eddfa2
Ledger at 108 —

r9c6_h+1v-1 Unified / memory ordinal: 311 / 310

Genealogy 59a1b925f37b198a9b8d28bf47609299f18ecacece1e78e313294726444a07ed
Reader 82e2ed9e32b01eb9363063f95aa0e97a797114c21c1a7b3ae4bdd8f720fbfc4f
Ledger at 108 915f24481eb9d51115cf664e358374d4a4585b514ac886dda764c433b1d9a456

r9c6_h+1v+1 Unified / memory ordinal: 312 / 309

Genealogy 6159b0ea3949ae42d6ac44d167a4d71dd9d0ca4f91305deda987869aff7a5301
Reader 2952410fbf2ee021f5736f19d28673e6bcf3f0867bc4b89a48f77673b413ea76
Ledger at 108 —

r9c7_h-1v-1 Unified / memory ordinal: 313 / 316

Genealogy 8684eaf7f9ac68574b1177dd46056f16497b29a210b4fc305b69f129f3516cf0
Reader e45675ccb2ead11c246854fd23eab5479d5768fdb30d0a82cc741ae6be60f35
Ledger at 108 23ef2c5cd56b7ff35c013077bee1cb8dbd0ee658e4c9ec2393eb87b4c365f550

r9c7_h-1v+1 Unified / memory ordinal: 314 / 315

Genealogy 0392a0bd824d4db3872ee7e55a8e5e59e3b21f749916b615dad01451654991d0
Reader 93e0fc9408e34fa48b346d2aa179d966af4513b405ea24a268f6ebd48d236116
Ledger at 108 —

r9c7_h+1v-1 Unified / memory ordinal: 315 / 314

Genealogy 15c25fad37a466a9092234de4e51eff214bb454130f89e920ac3eabd0af86e23
Reader aeafcc7a88835b5ec7f511281b2c9269e769f3ff86a4680e2e29a9866f3cabca
Ledger at 108 —

r9c7_h+1v+1 Unified / memory ordinal: 316 / 313

Genealogy 8e4b3fc25915492780bec7f316245b548a5a96f4f38c5037b6ad8f9460591d64
Reader d78bc1389070695902c2455878b04823455e0d80836529f9e88079404ea54b23
Ledger at 108 —

r9c8_h-1v-1 Unified / memory ordinal: 317 / 320

Genealogy ce9297be674940a68f160eee97674c669a846625821f53c9ca30b05bd0b56292
Reader 6ef27d7457d2c0a151c7c2cc86034c0ab1ce15e774f84ed5c64f617e44ec520f
Ledger at 108 —

r9c8_h-1v+1 Unified / memory ordinal: 318 / 319

Genealogy 16995ba1149d7ba0bc3cfc7743494696d33ba118e19e0ecc76f644d31711ba9f
Reader bf91d1ae5ba472dde0f327f9602e69918e8799f0e537555d5094f8ab0aac97ca
Ledger at 108 d2a88d330a7badcf0a410fef82f9b474dec91ba1dbe5a7b22a91f36905d51f7

r9c8_h+1v-1 Unified / memory ordinal: 319 / 318

Genealogy 2231f178c6f579c0407e70bcbff9ed750acd6efb3a4dd6381f3268e815826311
Reader 810d7ebd6b5ee7a72a6dab271c40aef23989480802df5c3c6064b00f893231ca
Ledger at 108 —

r9c8_h+1v+1 Unified / memory ordinal: 320 / 317

Genealogy 7c9b0456e1d3dbd9f89b99611a50f337eefa938f02f234c4a7ad065fd62a6692
Reader ec678d84ce11dc61e52a24438e4f523b4d0ea94b94d28d2c805f94c560a94223
Ledger at 108 —

r9c9_h-1v-1 Unified / memory ordinal: 321 / 324

Genealogy 3e3bf9f3b35f1316f0d242822cae4fe211eb31595576f51d063c36e241c4cc99
Reader 1f648323cbeef43af2c169aee5b6a999302aca3e106e2780e370b1480e3ee9e3
Ledger at 108 —

r9c9_h-1v+1 Unified / memory ordinal: 322 / 323

Genealogy 5ca5179e96a7a217149fc629598911284a679d874c335483825169c08628742b
Reader a1fd3c287cd8f05a77fc26d86500b68a6f6d45a515b1bfc32c7201ea592f831a
Ledger at 108 373f8cf1d27efd3b616a88ca8eefef4203f804fab97be9f4d60cd5a3f094d407

r9c9_h+1v-1 Unified / memory ordinal: 323 / 322

Genealogy e847061b4af737e22cde7449fd8c7399b3568176df3655b09f776b95b8d09a80
Reader 1f6b0ca611906a4828e0616ba7456a15244cef42f9db847f10419b48f959a5cd
Ledger at 108 —

r9c9_h+1v+1 Unified / memory ordinal: 324 / 321

Genealogy 8bd3961b89885d9bc5b76d4d18cbbc1bf791e769482c1949918d507488e02251
Reader 871508bdd95895803d299de36cd7cd3e9eade5f22157f1599f294895163b56ce
Ledger at 108 —